EE 501 — Linear System Theory I

Lecture Notes

Based on the handwritten notes by Elif Topaloğlu
of Umut Orguner's lectures (Fall 2024)

LaTeX transcription, corrected and annotated; organized along the Fall 2026–2027 syllabus

How to read these notes. **Bold blue** marks a term at the place where it is defined; **bold** marks a key statement.

- Framed boxes: definitions, theorems, examples.
- Yellow "Explanation" boxes and ▷ *italic lines* are additions that are not in the original handwritten notes.
- The gray *p. N* in the margin gives the page of the original handwritten notes.
- Corrections to the original are listed in the Corrections Log (Appendix A).

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Course Outline

Tentative schedule of EE501 (Fall 2026–2027) and the corresponding sections of these notes.

Week	Subject	Notes
1	Algebraic structures defined on sets, fields and vector spaces	§1.1–1.2
2	Subspaces, linear independence, basis	§1.3–1.7
3	Normed vector spaces, sequences, series and convergence in normed spaces, matrix norms	§2
4	Inner product spaces, orthogonality, Gram–Schmidt orthonormalization	§3
5	Linear transformations, null space, range space, matrix representations	§4
6	Direct sum decompositions, projection theorem, orthogonal projections	§5
7	Systems of linear equations of the form $Ax = b$	§6
8	Spectral analysis of linear transformations, eigenvalues and eigenvectors	§7.1–7.3
9	Invariance under operators, block diagonal forms	§7.4–7.5
10	Cayley–Hamilton theorem, characteristic and minimal polynomials	§8
11–12	Jordan canonical forms	§9
13	Hermitian matrices, positive definite matrices, positive semi-definite matrices	§10
14	Functions of a matrix	§11

References.

- S. Axler, *Linear Algebra Done Right*, Springer, 1997.
- C. A. Desoer, *Notes for a Second Course on Linear Systems*, Van Nostrand Reinhold, 1970.
- C. D. Meyer, *Matrix Analysis and Applied Linear Algebra*, SIAM, 2000.
- G. Strang, *Linear Algebra and its Applications*, 4th ed., Brooks/Cole, 2006.

1 Fields and Vector Spaces

Syllabus: Weeks 1–2 — algebraic structures, fields, vector spaces, subspaces, linear independence, basis

1.1 Fields

Definition 1.1. A **field** is a set $\mathcal{F}$ together with two mappings called **addition** and **multiplication** shown as:

$$\left.\begin{aligned}\oplus : \mathcal{F}\times\mathcal{F}\to\mathcal{F}\\ \odot : \mathcal{F}\times\mathcal{F}\to\mathcal{F}\end{aligned}\right\} \quad \mathcal{F} \text{ should be closed on both } \oplus \text{ and } \odot.$$

$\oplus$ and $\odot$ have the following properties (axioms):

Addition.

(A1) $a\oplus b = b\oplus a \quad \forall a,b\in\mathcal{F}$ (commutativity)

(A2) $a\oplus(b\oplus c) = (a\oplus b)\oplus c \quad \forall a,b,c\in\mathcal{F}$ (associativity)

(A3) $\exists$ (there exists) $0_{\mathcal{F}}\in\mathcal{F}$ such that $a\oplus 0_{\mathcal{F}} = a \quad \forall a\in\mathcal{F}$ (additive identity)

(A4) For each $a\in\mathcal{F}$, $\exists -a\in\mathcal{F}$ such that $a\oplus(-a) = 0_{\mathcal{F}}$ (additive inverse)

Multiplication.

(M1) $a\odot b = b\odot a \quad \forall a,b\in\mathcal{F}$ (commutativity)

(M2) $a\odot(b\odot c) = (a\odot b)\odot c \quad \forall a,b,c\in\mathcal{F}$ (associativity)

(M3) $\exists 1_{\mathcal{F}}\in\mathcal{F}$ such that $a\odot 1_{\mathcal{F}} = a \quad \forall a\in\mathcal{F}$ (multiplicative identity)

(M4) For each $a\in\mathcal{F}$, $a\neq 0_{\mathcal{F}}$, $\exists a^{-1}\in\mathcal{F}$ such that $a\odot a^{-1} = 1_{\mathcal{F}}$ (multiplicative inverse)

Distributivity. The operations $\oplus$ & $\odot$ satisfy

(D) $a\odot(b\oplus c) = (a\odot b)\oplus(a\odot c) \quad \forall a,b,c\in\mathcal{F}$ (distributivity)

Example 1.2. $\mathcal{F}=\{0,1\}$ with $\oplus = +$ and $\odot = \cdot$.

$$\oplus : \mathcal{F}\times\mathcal{F}\to\mathcal{F}, \qquad 1+1 = 2\notin\mathcal{F} \;\Rightarrow\; \text{not closed under } \oplus$$

Also no additive inverse exists for 1 (A4): $1+a=0$ has no solution $a\in\{0,1\}$ (we would need $a=-1\notin\mathcal{F}$).

Remark

With arithmetic *modulo 2* ($1\oplus 1 = 0$, $\odot$ the ordinary product) the set $\{0,1\}$ *is* a field, usually denoted GF(2) or $\mathbb{Z}_2$: here $-1=1$ and $1^{-1}=1$. So whether a set is a field depends on the operations, not only on the set.

Example 1.3. Set of real numbers $\mathbb{R}$ with the standard addition & multiplication.

$$\mathcal{F}=\mathbb{R}, \qquad \oplus = +, \qquad \odot = \cdot \qquad \Rightarrow \text{ A field}$$

Example 1.4. Let $\mathcal{F} = \mathbb{R} \times \mathbb{R} = \mathbb{R}^2$ with $x = (x_1, x_2) \in \mathcal{F}$, $y = (y_1, y_2) \in \mathcal{F}$ and p. 2

$$x \oplus y = (x_1 + y_1,\ x_2 + y_2)$$
$$x \odot y = (x_1 y_1 - x_2 y_2,\ x_2 y_1 + x_1 y_2)$$

▷ *Go over all axioms and check whether they are satisfied.*
Additive identity and inverse:

$$0_{\mathcal{F}} = (0, 0)$$
$$-(x_1, x_2) = (-x_1, -x_2)$$

Assume as example (complex-number notation):

$$(x_1, x_2) = x_1 + j x_2$$
$$(y_1, y_2) = y_1 + j y_2$$
$$x \oplus y = x_1 + y_1 + j(x_2 + y_2)$$
$$x \odot y = x_1 y_1 - x_2 y_2 + j(x_2 y_1 + x_1 y_2)$$

▷ *The multiplication rule is exactly complex multiplication:* $(x_1 + j x_2)(y_1 + j y_2) = x_1 y_1 - x_2 y_2 + j(x_2 y_1 + x_1 y_2)$ *since* $j^2 = -1$.
Multiplicative identity and inverse:

$$1_{\mathcal{F}} = (1, 0)$$
$$(x_1, x_2)^{-1} = \left(\frac{x_1}{x_1^2 + x_2^2},\ \frac{-x_2}{x_1^2 + x_2^2} \right), \qquad \frac{1}{x_1 + j x_2} = \frac{x_1 - j x_2}{x_1^2 + x_2^2}$$

∴ **This is a field.**

Explanation

Check: $(x_1, x_2) \odot (x_1, x_2)^{-1} = \left(\frac{x_1^2 + x_2^2}{x_1^2 + x_2^2},\ \frac{x_2 x_1 - x_1 x_2}{x_1^2 + x_2^2} \right) = (1, 0) = 1_{\mathcal{F}}$, defined for every $x \neq 0_{\mathcal{F}} = (0, 0)$. The remaining axioms (commutativity, associativity, distributivity) are inherited from those of complex numbers; this field is simply $\mathbb{C}$ written as pairs of reals.

Example 1.5. Let $\mathcal{F} = \mathbb{R}$ with $x \oplus y = xy$ and $x \odot y = x + y$. Is $\mathcal{F}$ a field?

$$x \oplus y = xy = yx = y \oplus x \quad \forall x, y \in \mathcal{F} \qquad \text{(A1) } \checkmark$$
$$x \oplus (y \oplus z) = x(yz) = (xy)z = (x \oplus y) \oplus z \quad \forall x, y, z \in \mathcal{F} \qquad \text{(A2) } \checkmark$$

Identities and inverses.

$$0_{\mathcal{F}} = 1: \qquad 0 \oplus a = 0_{\mathcal{F}} \ \Rightarrow\ \text{not satisfied } (0 \cdot a = 1\ \times)$$
$$1_{\mathcal{F}} = 0: \qquad a \in \mathcal{F} \Rightarrow a^{-1} = -a$$

▷ *The additive identity must satisfy* $a \oplus 0_{\mathcal{F}} = a \cdot 0_{\mathcal{F}} = a$ *for all* a, *so* $0_{\mathcal{F}} = 1$. *Then the element* 0 *has no additive inverse, since* $0 \oplus a = 0 \cdot a = 0 \neq 1$ *for every* a *(A4 fails). Likewise* $1_{\mathcal{F}} = 0$ *(as* $a + 0 = a$*), and* $a \odot (-a) = 0 = 1_{\mathcal{F}}$, *so (M4) holds.*

Distributivity.

$$x \odot (y \oplus z) \stackrel{?}{=} (x \odot y) \oplus (x \odot z)$$
$$x + yz \neq (x + y)(x + z) = x^2 + xz + xy + yz \quad \text{(in general; e.g. } x = y = z = 1\text{: } 2 \neq 4)$$

∴ **Not a field, as (A4) & (D) fail.**

Example 1.6. $\mathbb{R}^{2\times 2}$ with standard matrix addition & multiplication: in general $AB \neq BA$, i.e. there is no multiplicative commutativity (M1).

$$\mathbb{R}^n : \begin{cases} n = 1, 2 & \Rightarrow \text{a field can be defined} \\ n = 3, 4, \ldots & \Rightarrow \text{no field can be defined.} \end{cases}$$

Precise meaning

Here "define a field on $\mathbb{R}^n$" means: componentwise addition and a multiplication that is compatible with real scaling (bilinear), so that $\mathbb{R} \subset \mathcal{F}$ sits inside as $(a, 0, \ldots, 0)$. For $n = 1$ we get $\mathbb{R}$, for $n = 2$ we get $\mathbb{C}$ (previous example). For $n \geq 3$ no such field exists (Frobenius/Hopf); e.g. for $n = 4$ the quaternions satisfy every axiom except commutativity (M1), exactly like $\mathbb{R}^{2\times 2}$ fails (M1) (and in addition (M4), as singular matrices have no inverse).

1.2 Vector Spaces

Definition 1.7. A **linear space** (**vector space**) $\mathcal{V}$ is a set whose elements are called **vectors**, associated with a field $\mathcal{F}$ whose elements are called **scalars** and two operations $(\mathcal{V}, \mathcal{F})$: addition $\oplus_{\mathcal{V}}$ and scalar multiplication $\odot_{\mathcal{V}}$ (the subscript is dropped when no confusion arises). p. 3

$$\left.\begin{aligned} \oplus &: \mathcal{V} \times \mathcal{V} \to \mathcal{V} \\ \odot &: \mathcal{F} \times \mathcal{V} \to \mathcal{V} \end{aligned}\right\} \quad \mathcal{V} \text{ should be closed on addition \& multiplication (closedness)}$$

The following axioms should hold for $\oplus$ and $\odot$:

Addition.

(A1) $x \oplus y = y \oplus x \quad \forall x, y \in \mathcal{V}$ (commutativity)

(A2) $x \oplus (y \oplus z) = (x \oplus y) \oplus z \quad \forall x, y, z \in \mathcal{V}$ (associativity)

(A3) $\exists\, 0_{\mathcal{V}} \in \mathcal{V}$ such that $x \oplus 0_{\mathcal{V}} = x \quad \forall x \in \mathcal{V}$ (additive identity)

(A4) For every $x \in \mathcal{V}$, $\exists\, -x \in \mathcal{V}$ such that $x \oplus (-x) = 0_{\mathcal{V}}$ (additive inverse)

Scalar multiplication.

(SM1) $a \odot_{\mathcal{V}} (b \odot_{\mathcal{V}} x) = (a \odot_{\mathcal{F}} b) \odot_{\mathcal{V}} x \quad \forall \underbrace{a, b \in \mathcal{F}}_{\text{scalars}} \quad \forall \underbrace{x \in \mathcal{V}}_{\text{vector}}$ (associativity)

(SM2) $a \odot (x \oplus y) = (a \odot x) \oplus (a \odot y) \quad \forall a \in \mathcal{F} \quad \forall x, y \in \mathcal{V}$ (distributivity over $\oplus_{\mathcal{V}}$)

(SM3) $(a \oplus_{\mathcal{F}} b) \odot x = (a \odot x) \oplus (b \odot x) \quad \forall a, b \in \mathcal{F} \quad \forall x \in \mathcal{V}$ (distributivity over $\oplus_{\mathcal{F}}$)

(SM4) $1_{\mathcal{F}} \odot x = x \quad \forall x \in \mathcal{V}$, where $1_{\mathcal{F}}$ is the multiplicative identity of $\mathcal{F}$ (identity)

Example 1.8. Prove $0_{\mathcal{F}} \odot x = 0_{\mathcal{V}} \quad \forall x \in \mathcal{V}$.
Let $y \triangleq 0_{\mathcal{F}} \odot x$; then $y \in \mathcal{V}$ (as multiplication is closed).

$$\begin{aligned} y \oplus y &= (0_{\mathcal{F}} \odot x) \oplus (0_{\mathcal{F}} \odot x) \\ &\overset{\text{(SM3)}}{=} (0_{\mathcal{F}} \oplus 0_{\mathcal{F}}) \odot x \\ &\overset{\text{(A3) of } \mathcal{F}}{=} 0_{\mathcal{F}} \odot x = y \end{aligned}$$

▷ *(SM3) read from right to left; then $0_{\mathcal{F}} \oplus 0_{\mathcal{F}} = 0_{\mathcal{F}}$ is (A3) of the field $\mathcal{F}$.*
$\Rightarrow \; y \oplus y = y$ is proven.

$$\begin{aligned} \text{(A4)} &\Rightarrow \exists\, -y \text{ such that } y \oplus (-y) = 0_{\mathcal{V}} \\ \text{add } -y \text{ to both sides of } y \oplus y = y &\Rightarrow (y \oplus y) \oplus (-y) = y \oplus (-y) \\ \text{(A1), (A2)} &\Rightarrow (y \oplus (-y)) \oplus y = y \oplus (-y) \\ \text{(A4)} &\Rightarrow y \oplus 0_{\mathcal{V}} = 0_{\mathcal{V}} \\ \text{(A3)} &\Rightarrow \underline{y = 0_{\mathcal{V}}} \end{aligned}$$

Explanation

Since $y = 0_{\mathcal{F}} \odot x$, this proves $0_{\mathcal{F}} \odot x = 0_{\mathcal{V}}$. The same trick ($y \oplus y = y \Rightarrow y = 0_{\mathcal{V}}$) also gives $a \odot 0_{\mathcal{V}} = 0_{\mathcal{V}}$ for every $a \in \mathcal{F}$, and with (SM3) one gets $(-1_{\mathcal{F}}) \odot x = -x$, because $x \oplus ((-1_{\mathcal{F}}) \odot x) = (1_{\mathcal{F}} \oplus (-1_{\mathcal{F}})) \odot x = 0_{\mathcal{F}} \odot x = 0_{\mathcal{V}}$.

Example 1.9. Linear space $(\overset{\mathcal{V}}{\mathcal{F}^n}, \overset{\mathcal{F}}{\mathcal{F}})$; i.e., the vector space of n tuples in $\mathcal{F}$ over the field $\mathcal{F}$. p. 4

$$x = (x_1, x_2, \ldots, x_n), \qquad y = (y_1, y_2, \ldots, y_n), \qquad \text{where } x_i, y_i \in \mathcal{F} \; \forall i$$

Addition and scalar multiplication:

$$\begin{aligned} x \oplus_{\mathcal{V}} y &= (x_1 \oplus_{\mathcal{F}} y_1,\; x_2 \oplus_{\mathcal{F}} y_2,\; \ldots,\; x_n \oplus_{\mathcal{F}} y_n) \\ a \odot_{\mathcal{V}} y &= (a \odot_{\mathcal{F}} y_1,\; a \odot_{\mathcal{F}} y_2,\; \ldots,\; a \odot_{\mathcal{F}} y_n) \end{aligned}$$

$$(\mathbb{R}^n, \mathbb{R}) \checkmark \qquad (\mathbb{C}^n, \mathbb{R}) \checkmark \qquad (\mathbb{C}^n, \mathbb{C}) \checkmark \qquad (\mathbb{R}^n, \mathbb{C}) \times$$

▷ $(\mathbb{C}^n, \mathbb{R})$*: a real multiple of a complex vector is still complex, so it is a vector space (of dimension $2n$ over $\mathbb{R}$).*
▷ $(\mathbb{R}^n, \mathbb{C})$*: closure under scalar multiplication is violated: multiplying a real vector by a complex scalar gives a complex vector, which is $\notin \mathcal{V}$.*

Example 1.10. Set of all real valued functions $t \mapsto f(t)$ defined over the real line.

$$\begin{aligned} \mathcal{F} &= \mathbb{R} \\ \mathcal{V} &= \text{real valued functions} \\ (f \oplus_{\mathcal{V}} g)(t) &= f(t) \oplus_{\mathcal{F}} g(t) \\ (a \odot_{\mathcal{V}} f)(t) &= a \odot_{\mathcal{F}} f(t) \end{aligned}$$

Example 1.11. Set of all polynomials with degree $n = 2$

$$\mathcal{V} = \{a_0 t^2 + a_1 t + a_2 \mid a_0, a_1, a_2 \in \mathbb{R} \ \& \ a_0 \neq 0\}$$

$\oplus$ and $\odot$ are standard polynomial addition and scalar multiplication.
$\therefore$ Not a vector space!

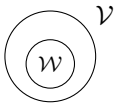
$$\begin{aligned} P_1(t) &= t^2 + 2t + 3 \in \mathcal{V} \\ P_2(t) &= -t^2 + 2t + 3 \in \mathcal{V} \\ (P_1 \oplus P_2)(t) &= 4t + 6 \notin \mathcal{V} \end{aligned}$$

Example 1.12. Set of all polynomials with degree $n \leq 2$ is a vector space.

$$\mathcal{V} = \{a_0 t^2 + a_1 t + a_2 \mid a_0, a_1, a_2 \in \mathbb{R}\}$$

1.3 Subspaces

Definition 1.13. Let $\mathcal{V}$ be a vector space over the field $\mathcal{F}$. A subset $\mathcal{W}$ of $\mathcal{V}$ is a **subspace** of $\mathcal{V}$ if and only if $(\mathcal{W}, \mathcal{F})$ is itself a vector space.



$$\left[\text{only show that} \quad \left.\begin{array}{l} \oplus : \mathcal{V} \times \mathcal{V} \to \mathcal{V} \\ \odot : \mathcal{F} \times \mathcal{V} \to \mathcal{V} \end{array}\right\} \text{ holds} \right]$$

Remark. We can check whether a (nonempty) subset $\mathcal{W}$ is a subspace or not by checking **only closure under addition and scalar multiplication**:

(S1)	$w_1 \oplus w_2 \in \mathcal{W}$	$\forall w_1, w_2 \in \mathcal{W}$	(closure under addition)
(S2)	$a \odot w_1 \in \mathcal{W}$	$\forall w_1 \in \mathcal{W},\ \forall a \in \mathcal{F}$	(closure under scalar multiplication)

Why closure is enough

All axioms that are identities ((A1), (A2), (SM1)–(SM4)) hold in $\mathcal{W}$ automatically because they hold in the bigger set $\mathcal{V}$. The two existence axioms follow from (S2): taking $a = 0_{\mathcal{F}}$ gives $0_{\mathcal{F}} \odot w = 0_{\mathcal{V}} \in \mathcal{W}$ (A3), and taking $a = -1_{\mathcal{F}}$ gives $(-1_{\mathcal{F}}) \odot w = -w \in \mathcal{W}$ (A4). In particular, **every subspace contains** $0_{\mathcal{V}}$; a quick test is: if $0_{\mathcal{V}} \notin \mathcal{W}$, then $\mathcal{W}$ is not a subspace.

Example 1.14. $\mathcal{V} = \mathbb{R}^2$, $\mathcal{F} = \mathbb{R}$, $\quad \mathcal{W} = \left\{ \begin{bmatrix} a \\ 0 \end{bmatrix} \,\middle|\, a \in \mathbb{R} \right\}$ p. 5

$$w_1 \in \mathcal{W}: \quad w_1 = \begin{bmatrix} a \\ 0 \end{bmatrix} \qquad w_1 \oplus w_2 = \begin{bmatrix} a+b \\ 0 \end{bmatrix} \in \mathcal{W}$$

$$w_2 \in \mathcal{W}: \quad w_2 = \begin{bmatrix} b \\ 0 \end{bmatrix} \qquad c \odot w_1 = \begin{bmatrix} ca \\ 0 \end{bmatrix} \in \mathcal{W}$$

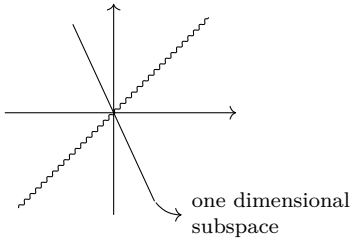
$\therefore$ $\mathcal{W}$ is a subspace of $\mathcal{V}$.

Example 1.15. $\mathcal{V} = \mathbb{R}^2$, $\mathcal{F} = \mathbb{R}$, $\quad \mathcal{W} = \left\{ \begin{bmatrix} a \\ 1 \end{bmatrix} \,\middle|\, a \in \mathbb{R} \right\}$

$$w_1 = \begin{bmatrix} a \\ 1 \end{bmatrix}, \quad w_2 = \begin{bmatrix} b \\ 1 \end{bmatrix} \qquad \Rightarrow \qquad w_1 \oplus w_2 = \begin{bmatrix} a+b \\ 2 \end{bmatrix} \notin \mathcal{W}$$

Not closed under vector addition or scalar multiplication (e.g. $0 \odot w_1 = (0,0)^T \notin \mathcal{W}$: $\mathcal{W}$ does not contain the zero vector).
$\therefore$ $\mathcal{W}$ is not a subspace.

▶ $\{0_\mathcal{V}\}$: zero dimensional vector space $\Rightarrow$ subspace.



$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \in \mathcal{W}$$
$$-\begin{bmatrix} 1 \\ 1 \end{bmatrix} \in \mathcal{W}$$

$a \odot x = 0_\mathcal{V}$ with $a = 0_\mathcal{F}$

▷ *E.g. the line* $\mathcal{W} = \text{sp}\{(1,1)^T\}$ *(the line* $x_2 = x_1$*): with* $(1,1)^T$ *it also contains* $-(1,1)^T$*, and* $0_\mathcal{F} \odot x = 0_\mathcal{V}$ *must lie in* $\mathcal{W}$*.*

$\Rightarrow$ **All subspaces should cross origin.**
$\Rightarrow$ For two dimensional subspace, include all $\mathbb{R}^2$.

Subspaces of $\mathbb{R}^2$

The subspaces of $\mathbb{R}^2$ are exactly: $\{0_\mathcal{V}\}$ (dimension 0), every line through the origin (dimension 1), and $\mathbb{R}^2$ itself (dimension 2). A line not passing through the origin (such as $\{[a,\,1]^T\}$ above) is never a subspace.

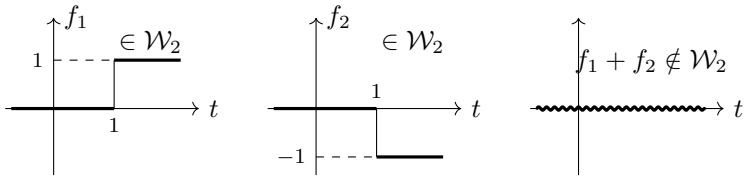
Example 1.16. $\mathcal{V} = \mathbb{R}^{2\times 2}$, $\mathcal{F} = \mathbb{R}$, $\quad \mathcal{W} = \{\text{set of } 2 \times 2 \text{ real valued symmetric matrices}\}$

$$\begin{bmatrix} a_1 & b_1 \\ b_1 & c_1 \end{bmatrix} \oplus \begin{bmatrix} a_2 & b_2 \\ b_2 & c_2 \end{bmatrix} = \begin{bmatrix} a_1+a_2 & b_1+b_2 \\ b_1+b_2 & c_1+c_2 \end{bmatrix} \in \mathcal{W}$$

$$\alpha \begin{bmatrix} a_1 & b_1 \\ b_1 & c_1 \end{bmatrix} = \begin{bmatrix} \alpha a_1 & \alpha b_1 \\ \alpha b_1 & \alpha c_1 \end{bmatrix} \in \mathcal{W}$$

$\therefore$ $\mathcal{W}$ is a subspace of $\mathcal{V}$.

Example 1.17. $\mathcal{V}$: set of real valued functions defined over $\mathbb{R}$
$\mathcal{W}_1$: set of real valued continuous functions
$\mathcal{W}_2$: set of all real valued functions which are discontinuous at $t = 1$
$\therefore \mathcal{W}_1$ is a subspace. For $\mathcal{W}_2$:



▷ *f_1 and f_2 both jump at $t = 1$, but $f_1 + f_2$ is the zero function, which is continuous.*
$\therefore$ Not closed under vector addition & scalar multiplication (multiply with zero: $0 \cdot f_1$ is the zero function, which is continuous, so $\notin \mathcal{W}_2$). Hence $\mathcal{W}_2$ is <u>not</u> a subspace.

Explanation

$\mathcal{W}_1$ is a subspace because sums and scalar multiples of continuous functions are continuous (and the zero function is continuous). $\mathcal{W}_2$ fails already the quick test: the zero vector of $\mathcal{V}$ (the zero function) is not in $\mathcal{W}_2$.

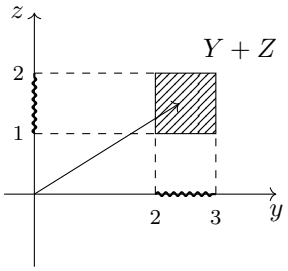
1.4 Sums, Intersections and Products of Subspaces

Definition 1.18. The **sum** $Y + Z$ of two subsets Y & Z of a linear space is the set of all vectors $y + z$ where $y \in Y$ and $z \in Z$. p. 6

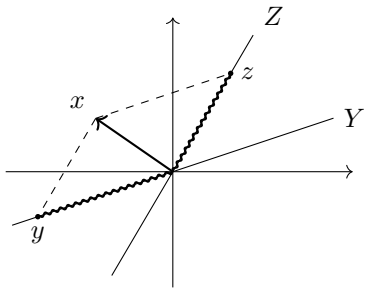
$$Y + Z \triangleq \{y + z \mid y \in Y,\ z \in Z\}$$

Example 1.19. $\mathcal{V} = \mathbb{R}^2$,

$$Y = \left\{ \begin{bmatrix} a \\ 0 \end{bmatrix} \;\middle|\; 2 \le a \le 3 \right\}, \qquad Z = \left\{ \begin{bmatrix} 0 \\ b \end{bmatrix} \;\middle|\; 1 \le b \le 2 \right\}$$



If instead Y and Z are two lines through the origin (subspaces), every vector x can be decomposed as $x = y + z$ with $y \in Y$, $z \in Z$:



The same can be done for every vector, so $Y + Z = \mathbb{R}^2 \Rightarrow$ the sum of two subspaces is again a subspace.

Remark. If Y & Z are both subspaces, then $Y + Z$ **is a subspace**.

Proof. Let $x_1, x_2 \in Y + Z$:

$$x_1 = \underset{\in Y}{y_1} + \underset{\in Z}{z_1}, \qquad x_2 = \underset{\in Y}{y_2} + \underset{\in Z}{z_2}$$

$$x_1 + x_2 = y_1 + z_1 + y_2 + z_2 = \underbrace{y_1 + y_2}_{\in Y} + \underbrace{z_1 + z_2}_{\in Z}$$

▷ *As Y and Z are subspaces, they are closed under addition.*

Similarly, for any scalar α:

$$\underbrace{\alpha x_1}_{\in Y+Z} = \alpha(y_1 + z_1) = \underbrace{\alpha y_1}_{\in Y} + \underbrace{\alpha z_1}_{\in Z}$$

Closed under scalar multiplication.

$\therefore Y + Z$ is a subspace. □

For the two lines Y, Z of the previous example:

$$Y \cap Z = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\} \text{ is a subspace} \Rightarrow \text{not a coincidence!}$$

Remark. If Y & Z are subspaces, then $Y \cap Z$ **is a subspace**.

($Y \cup Z$ is closed under scalar multiplication but not closed under addition, which is the case in the previous example) $\Rightarrow Y \cup Z$ **is not a subspace generally**.

Explanation

Proof for $Y \cap Z$: if $x_1, x_2 \in Y \cap Z$, then $x_1 + x_2 \in Y$ and $x_1 + x_2 \in Z$ (both are subspaces), so $x_1 + x_2 \in Y \cap Z$; similarly $\alpha x_1 \in Y \cap Z$.

Why $Y \cup Z$ fails: for the two lines, $y \in Y$, $z \in Z$ nonzero give $y + z$, which lies on neither line. In fact $Y \cup Z$ is a subspace only if $Y \subseteq Z$ or $Z \subseteq Y$. The sum $Y + Z$ is the **smallest subspace containing** $Y \cup Z$.

Definition 1.20 (Product space)**.** Let $(\mathcal{V}, \mathcal{F})$ & $(\mathcal{W}, \mathcal{F})$ be two linear spaces defined over the same field $\mathcal{F}$. The **product** of $(\mathcal{V}, \mathcal{F})$ & $(\mathcal{W}, \mathcal{F})$ is defined as:

$$\begin{aligned} \mathcal{V} \times \mathcal{W} &= \{(v, w) \mid v \in \mathcal{V},\ w \in \mathcal{W}\} \\ (v_1, w_1) \oplus_{\mathcal{V}\times\mathcal{W}} (v_2, w_2) &= (v_1 \oplus_{\mathcal{V}} v_2,\ w_1 \oplus_{\mathcal{W}} w_2) \\ \alpha \odot_{\mathcal{V}\times\mathcal{W}} (v_1, w_1) &= (\alpha \odot_{\mathcal{V}} v_1,\ \alpha \odot_{\mathcal{W}} w_1) \end{aligned}$$

$$\Rightarrow \quad (\underset{\mathcal{V}\times\mathcal{W}}{\mathbb{R}^2}, \underset{\mathcal{F}}{\mathbb{R}}) = (\underset{\mathcal{V}}{\mathbb{R}}, \underset{\mathcal{F}}{\mathbb{R}}) \times (\underset{\mathcal{W}}{\mathbb{R}}, \underset{\mathcal{F}}{\mathbb{R}})$$

▷ *In general* $\dim(\mathcal{V} \times \mathcal{W}) = \dim \mathcal{V} + \dim \mathcal{W}$*; e.g.* $\mathbb{R}^n \times \mathbb{R}^m = \mathbb{R}^{n+m}$.

1.5 Linear Combinations, Span and Linear Independence

Definition 1.21. A **linear combination** of n vectors $x_1, x_2, \ldots, x_n$ of a linear space $\mathcal{V}$ is a vector of the form p. 7

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = \sum_{i=1}^{n} a_ix_i \qquad \text{where } a_i \in \mathcal{F} \text{ for } i = 1, \ldots, n$$

The set of all linear combinations of $x_1, \ldots, x_n$, i.e.

$$\left\{ \sum_{i=1}^{n} a_ix_i \;\middle|\; a_i \in \mathcal{F},\ i = 1, \ldots, n \right\}$$

is called the **span** of $\{x_1, \ldots, x_n\}$, denoted $\text{sp}\{x_1, \ldots, x_n\}$.

For $\mathcal{V} = \mathbb{R}^2$:

$$x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix},\ x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}: \qquad a_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + a_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \qquad \Rightarrow\ \text{sp}\{x_1, x_2\} = \mathbb{R}^2$$

$$x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix},\ x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix},\ x_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}: \qquad \Rightarrow\ \text{sp}\{x_1, x_2, x_3\} = \mathbb{R}^2$$

▷ *x_3 is redundant: $x_3 = x_1 + x_2$ is already a linear combination of x_1, x_2, so adding it does not enlarge the span.*

Definition 1.22. Vectors $x_1, \ldots, x_n \in \mathcal{V}$ are said to be **linearly independent** if and only if

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0_{\mathcal{V}} \quad \text{implies} \quad a_1 = a_2 = \cdots = a_n = 0_{\mathcal{F}}$$

If $\exists\, a_1, \ldots, a_n$, not all equal to $0_{\mathcal{F}}$, such that

$$a_1x_1 + \cdots + a_nx_n = 0_{\mathcal{V}},$$

then $x_1, x_2, \ldots, x_n$ are **linearly dependent**.

Example 1.23. $x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$:

$$a_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + a_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \quad \Rightarrow \quad a_1 = a_2 = 0 \quad \Rightarrow \quad \text{linearly independent}$$

Example 1.24. $x_1 = \begin{bmatrix}1\\0\end{bmatrix}$, $x_2 = \begin{bmatrix}0\\1\end{bmatrix}$, $x_3 = \begin{bmatrix}1\\1\end{bmatrix}$:

$$a_1\begin{bmatrix}1\\0\end{bmatrix} + a_2\begin{bmatrix}0\\1\end{bmatrix} + a_3\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix} \;\Rightarrow\; \begin{matrix}a_1 + a_3 = 0\\ a_2 + a_3 = 0\end{matrix} \;\Rightarrow\; \text{e.g. } a_1 = a_2 = 1,\ a_3 = -1$$

$\therefore$ linearly dependent.

Example 1.25. Consider the linear space of polynomials of degree $n \le 2$, and

$$P_1(t) = 1, \quad P_2(t) = 1 + t, \quad P_3(t) = 1 + t + t^2, \qquad P = \{P_1(t), P_2(t), P_3(t)\}.$$

Is P linearly independent?

$$\begin{aligned} &a_1 \cdot 1 + a_2(1 + t) + a_3(1 + t + t^2) = 0(t) && \text{(zero polynomial)}\\ \Rightarrow\ &(a_1 + a_2 + a_3) + (a_2 + a_3)t + a_3t^2 = 0(t) \;\Rightarrow\; a_1 = a_2 = a_3 = 0 \end{aligned}$$

▷ *A polynomial is the zero polynomial only if all its coefficients vanish: $a_3 = 0$ (from t^2), then $a_2 + a_3 = 0 \Rightarrow a_2 = 0$ (from t), then $a_1 + a_2 + a_3 = 0 \Rightarrow a_1 = 0$.*
$\therefore$ P is linearly independent.

Example 1.26. $\mathcal{V}$: set of all continuous functions defined over the real line. p. 8

$$S = \{\cos t,\ \sin t,\ \cos(t - \tfrac{\pi}{3})\}. \quad \text{Is } S \text{ linearly independent?}$$

$$a\cos t + b\sin t + c \underbrace{\cos(t - \tfrac{\pi}{3})}_{\cos t \cdot \cos\frac{\pi}{3} + \sin t \cdot \sin\frac{\pi}{3}} = 0(t) \quad \forall t, \qquad \cos\tfrac{\pi}{3} = \tfrac{1}{2}, \quad \sin\tfrac{\pi}{3} = \tfrac{\sqrt{3}}{2}$$

Collecting terms:

$$(a + \tfrac{c}{2})\cos t + \left(b + \tfrac{\sqrt{3}}{2}c\right)\sin t = 0(t),$$

which holds for $a = -\frac{c}{2}$, $b = -\frac{\sqrt{3}}{2}c$ with any c. Choosing $a = -\frac{1}{2}$, $b = -\frac{\sqrt{3}}{2}$, $c = 1$ gives the zero function.
$\therefore$ S **is <u>not</u> linearly independent.**

Linear independence of the columns of a matrix $A = [c_1 \mid c_2 \mid \cdots \mid c_n]$: consider $Ax = 0$, i.e.

$$[c_1 \mid c_2 \mid \cdots \mid c_n]\begin{bmatrix}x_1\\x_2\\\vdots\\x_n\end{bmatrix} = \underbrace{c_1x_1 + c_2x_2 + \cdots + c_nx_n}_{\text{zero}} = \begin{bmatrix}0\\0\\\vdots\\0\end{bmatrix}$$

▷ *Ax is a linear combination of the columns of A with coefficients x_i. Hence:* ***the columns*** *$c_1, \ldots, c_n$* ***are linearly independent*** $\iff$ *$Ax = 0$* ***has only the solution*** *$x = 0$; any nonzero solution x gives a dependence relation among the columns.*

1.6 Basis, Dimension and Coordinates

Definition 1.27. Let $\mathcal{V}$ be a linear space and set of vectors $B = \{x_1, \ldots, x_n\}$ be a subset of $\mathcal{V}$. B is said to be a **basis** for $\mathcal{V}$ if and only if

(i) $\operatorname{sp} B = \mathcal{V}$, and

(ii) B is a linearly independent set.

For $\mathbb{R}^2$:

$$B_1 = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\} \qquad \times \text{ not a basis}$$

$$B_2 = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\} \qquad \checkmark$$

$$B_3 = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} \qquad \times \text{ not linearly independent}$$

$$B_4 = \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\} \qquad \checkmark$$

$$B_5 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\} \qquad \checkmark$$

▷ *The two basis vectors should not be on the same line through the origin (their spans intersect only at the origin); there are infinitely many bases and the number of vectors is always 2.*

▷ *E.g. for B_3: $(0,1)^T = \frac{1}{2}(1,1)^T - \frac{1}{2}(1,-1)^T$. For B_1: $\operatorname{sp} B_1$ is only the line $x_2 = x_1$, not $\mathbb{R}^2$.*

Remark. There are infinitely many basis sets for a linear space. **All of these basis sets have the same number of elements.** That number is called **dimension** of $\mathcal{V}$. If the dimension is finite, $\mathcal{V}$ is said to be **finite dimensional**. Otherwise, $\mathcal{V}$ is **infinite dimensional**.

Example 1.28. $\mathcal{V}$ is the set of 2×2 real valued symmetric matrices:

$$\begin{bmatrix} a & b \\ b & c \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$B = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

▷ *3 elements (dimension 3).*

Infinite dimensional linear space ⇒ set of continuous functions.

▷ *For every n, the functions $1, t, t^2, \ldots, t^n$ are continuous and linearly independent, so no finite set can span this space.*

Definition 1.29. If $y \in \mathcal{V}$ and $B = \{x_1, x_2, \ldots, x_n\}$ is a basis set for $\mathcal{V}$, then there is a **unique** n-tuple of scalars $(a_1, a_2, \ldots, a_n)$ such that p. 9

$$y = \sum_{i=1}^{n} a_i x_i$$

The scalars $(a_1, a_2, \ldots, a_n)$ are called **components** (**coordinates**), **representations** of y with respect to (w.r.t.) basis B. We denote the representation of y w.r.t. B by $[y]_B$.

With B the basis of symmetric matrices above:

$$y = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \in \mathcal{V}, \qquad [y]_B = \underbrace{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}_{\text{unique}}, \qquad y \in \mathcal{V}, \quad [y]_B \in \mathbb{R}^3$$

▶ Why should representation be unique?

Proof. $[y]_B$ is unique. (Proof by contradiction)
Suppose $[y]_B$ is <u>not</u> unique:

$$[y]_B = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix}, \qquad [y]_B = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix}, \qquad \exists\, i \text{ such that } \alpha_i \neq \beta_i$$

By the representation definition,

$$y = \sum_{i=1}^{n} \alpha_i x_i, \qquad y = \sum_{i=1}^{n} \beta_i x_i \qquad \Rightarrow \qquad 0_{\mathcal{V}} = \sum_{i=1}^{n} (\alpha_i - \beta_i) x_i$$

▷ *A linear combination of the basis vectors equal to zero with some of the coefficients nonzero, as* $\alpha_i - \beta_i \neq 0$.

$\therefore x_i$ are linearly dependent $\Rightarrow$ contradiction. □

Example 1.30. $\mathcal{V} = \text{sp}\{\cos t, \sin t\}, \quad B = \{\cos t, \sin t\}$

$$y \triangleq \cos(t - \tfrac{\pi}{3}) = \tfrac{1}{2}\cos t + \tfrac{\sqrt{3}}{2}\sin t \qquad \Rightarrow \qquad [y]_B = \begin{bmatrix} 1/2 \\ \sqrt{3}/2 \end{bmatrix}$$

1.7 Change of Basis

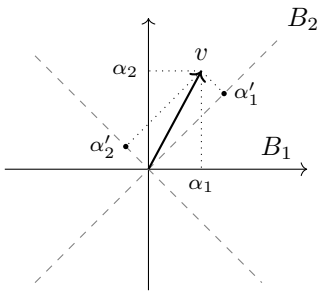
Theorem 1.31 (Change of basis)**.** Let $\mathcal{V}$ be an n-dimensional linear space over $\mathbb{R}$. Let B_1 and B_2 be two basis sets for $\mathcal{V}$. Then, there exists an $n \times n$ **real invertible matrix** P such that

$$[v]_{B_1} = P\,[v]_{B_2} \quad \forall v \in \mathcal{V}$$

$$B_1 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}, \qquad B_2 = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$$

$$v = \alpha_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$[v]_{B_1} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}, \qquad [v]_{B_2} = \begin{bmatrix} \alpha_1' \\ \alpha_2' \end{bmatrix}$$



Finding P for this example

Write v in terms of B_2:

$$v = \alpha_1' \begin{bmatrix}1\\1\end{bmatrix} + \alpha_2' \begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}1 & 1\\1 & -1\end{bmatrix} \begin{bmatrix}\alpha_1'\\\alpha_2'\end{bmatrix}.$$

Since the B_1-coordinates are just the entries of v,

$$[v]_{B_1} = \underbrace{\begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}}_{P} [v]_{B_2}, \qquad P^{-1} = \frac{1}{2}\begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}.$$

In general, **the j-th column of P is $[\,j$-th vector of $B_2\,]_{B_1}$.** P is invertible because its columns are coordinates of a basis, hence linearly independent. For the drawn vector $v = (0.7, 1.3)^T$: $[v]_{B_2} = P^{-1}[v]_{B_1} = (1, -0.3)^T$.

Example 1.32. Consider $\mathcal{V}$: polynomials with degree $n \le 2$.

$$B_1 = \{\overset{v_1}{1},\ \overset{v_2}{1+t},\ \overset{v_3}{1+t+t^2}\}, \qquad B_2 = \{\overset{w_1}{1},\ \overset{w_2}{t},\ \overset{w_3}{t^2}\}$$

Find P such that $[v]_{B_1} = P\,[v]_{B_2}\ \forall v \in \mathcal{V}$. p. 10

$$v = w_1: \quad [w_1]_{B_2} = \begin{bmatrix}1\\0\\0\end{bmatrix} \Rightarrow P \begin{bmatrix}1\\0\\0\end{bmatrix} = \text{1st column of } P = [w_1]_{B_1} = \begin{bmatrix}1\\0\\0\end{bmatrix}$$

$$v = w_2: \quad [w_2]_{B_2} = \begin{bmatrix}0\\1\\0\end{bmatrix} \Rightarrow P \begin{bmatrix}0\\1\\0\end{bmatrix} = \text{2nd column of } P = [w_2]_{B_1} = \begin{bmatrix}-1\\1\\0\end{bmatrix}$$

$$v = w_3: \quad [w_3]_{B_2} = \begin{bmatrix}0\\0\\1\end{bmatrix} \Rightarrow P \begin{bmatrix}0\\0\\1\end{bmatrix} = \text{3rd column of } P = [w_3]_{B_1} = \begin{bmatrix}0\\-1\\1\end{bmatrix}$$

▷ *The coordinates come from writing B_2 in terms of B_1: $w_1 = 1 = v_1$, $w_2 = t = (1+t) - 1 = v_2 - v_1$, $w_3 = t^2 = (1+t+t^2) - (1+t) = v_3 - v_2$.*

Collecting the three columns:

$$P = \begin{bmatrix}1 & -1 & 0\\0 & 1 & -1\\0 & 0 & 1\end{bmatrix}$$

▷ *Upper triangular matrix with diagonal entries 1. The diagonal entries are the eigenvalues $\Rightarrow \det(P)$, the product of the eigenvalues, is $1 \neq 0$, so P is invertible.*

In the opposite direction:

$$[v]_{B_2} = Q\,[v]_{B_1} \ \Rightarrow\ Q = \begin{bmatrix}1 & 1 & 1\\0 & 1 & 1\\0 & 0 & 1\end{bmatrix}, \qquad P = Q^{-1}$$

▷ *Columns of Q are $[v_j]_{B_2}$: $v_1 = 1$, $v_2 = 1+t$, $v_3 = 1+t+t^2$ give $(1,0,0)^T$, $(1,1,0)^T$, $(1,1,1)^T$.*

Check

$$PQ = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

E.g. $v = 3 + 2t + t^2$: $[v]_{B_2} = (3, 2, 1)^T$ and $P[v]_{B_2} = (1, 1, 1)^T$, i.e. $v = 1 \cdot 1 + 1 \cdot (1 + t) + 1 \cdot (1 + t + t^2) = 3 + 2t + t^2$ ✓.

2 Normed Vector Spaces

Syllabus: Week 3 — normed spaces, sequences and convergence, matrix norms

2.1 Norms

Definition 2.1. Let $(\mathcal{V}, \mathcal{F})$ be a vector space, a **norm** of $\mathcal{V}$ is a function shown as

$$\|\cdot\| : \mathcal{V} \to \mathbb{R}_{\geq 0}$$

satisfying the following properties:

(N1) $\|v\| \geq 0 \quad \forall v \in \mathcal{V}, \quad \|v\| = 0 \iff v = 0_{\mathcal{V}}$ (positive definiteness)

(N2) $\|\alpha v\| = |\alpha| \, \|v\| \quad \forall v \in \mathcal{V} \;\&\; \forall \alpha \in \mathcal{F}$ (absolute homogeneity)

(N3) $\|v_1 + v_2\| \leq \|v_1\| + \|v_2\| \quad \forall v_1, v_2 \in \mathcal{V}$ (triangle inequality)

The triplet $(\mathcal{V}, \mathcal{F}, \|\cdot\|)$ is called a **normed space**.

$\|v\| = \|v - 0_{\mathcal{V}}\|$: size of v

$\|v_1 - v_2\|$: distance between v_1 & v_2

We can define a sphere in $\mathcal{V}$:

$$S = \{v \in \mathcal{V} \mid \|v - v_0\| \leq r\} \qquad \text{sphere radius } r, \quad v_0 : \text{center vector}$$

▷ *Strictly, with "$\leq$" this is the (closed) **ball**; the sphere is its boundary $\|v - v_0\| = r$. The "unit spheres" drawn below are the sets $\|x\| = 1$.*

Example 2.2. $\mathcal{V} = \mathbb{R}^n$, $\mathcal{F} = \mathbb{R}$

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \qquad \|x\|_1 \stackrel{?}{=} \sum_{i=1}^{n} |x_i|$$

(N1). $|x_i| \geq 0 \; \forall i \Rightarrow \|x\|_1 \geq 0$ ✓. $0_{\mathcal{V}} = (0, \ldots, 0)^T$; if $\|x\|_1 = 0$:

$$\|x\|_1 = \sum_{i=1}^{n} |x_i| = 0 \;\Rightarrow\; |x_i| = 0 \; \forall i \;\Rightarrow\; x = 0_{\mathcal{V}}$$

(N2). $\alpha x = (\alpha x_1, \alpha x_2, \ldots, \alpha x_n)^T$:

$$\|\alpha x\|_1 = \sum_{i=1}^{n} |\alpha x_i| = \sum_{i=1}^{n} |\alpha| \cdot |x_i| = |\alpha| \cdot \underbrace{\sum_{i=1}^{n} |x_i|}_{\|x\|_1} = |\alpha| \, \|x\|_1$$

(N3). $x + y = (x_1 + y_1,\, x_2 + y_2,\, \ldots,\, x_n + y_n)^T$: p. 11

$$\|x+y\|_1 = \sum_{i=1}^{n} |x_i + y_i| \le \sum_{i=1}^{n} (|x_i| + |y_i|) = \underbrace{\sum_{i=1}^{n} |x_i|}_{\|x\|_1} + \underbrace{\sum_{i=1}^{n} |y_i|}_{\|y\|_1}$$

▷ *The inequality uses the scalar triangle inequality $|x_i + y_i| \le |x_i| + |y_i|$ term by term. Hence $\|x+y\|_1 \le \|x\|_1 + \|y\|_1$ and all three properties hold: $\|\cdot\|_1$ is a norm.*

2.2 The ℓ_p Norms and Unit Spheres

► $\|\cdot\|_1$ is called ℓ_1 **norm**.

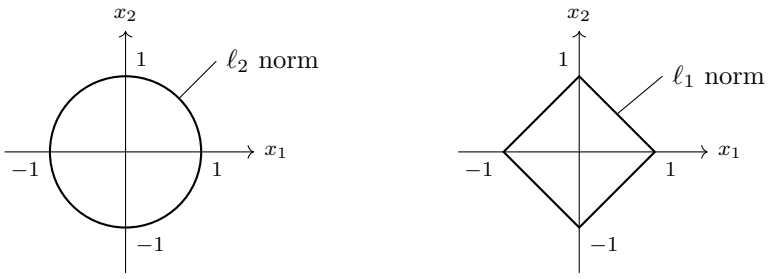
Example 2.3 (ℓ_2 norm)**.**

$$\|x\|_2 = \sqrt{\sum_{i=1}^{n} |x_i|^2} = \left(\sum_{i=1}^{n} |x_i|^2\right)^{1/2}$$

Unit spheres for $x = (x_1, x_2)^T$: $\|x\|_2 = 1$ and $\|x\|_1 = 1$.
For ℓ_1 norm: $|x_1| + |x_2| = 1$

- 1st quadrant ($x_1, x_2 \ge 0$): $x_1 + x_2 = 1$
- 2nd quadrant ($x_1 \le 0 \le x_2$): $-x_1 + x_2 = 1$
- 3rd quadrant ($x_1, x_2 \le 0$): $-x_1 - x_2 = 1$
- 4th quadrant ($x_2 \le 0 \le x_1$): $x_1 - x_2 = 1$

▷ *Four line segments joining $(\pm 1, 0)$ and $(0, \pm 1)$: the ℓ_1 unit sphere is a diamond (a square rotated by 45°), while the ℓ_2 unit sphere $x_1^2 + x_2^2 = 1$ is the circle.*



ℓ_p norm. The ℓ_p **norm** is

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p} \quad \text{where } p \ge 1$$

ℓ_∞ norm. The ℓ_∞ **norm** is

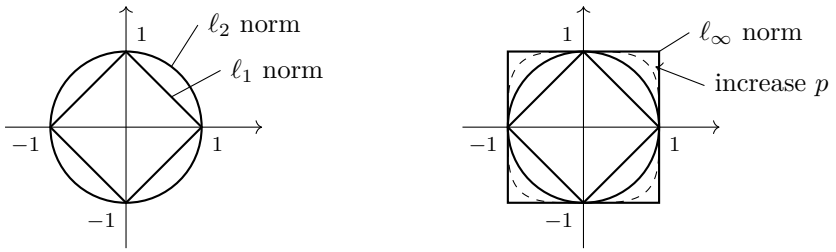
$$\|x\|_\infty = \max_i |x_i| = \lim_{p \to \infty} \|x\|_p$$

For $\mathcal{V} = \mathbb{R}^2$: $x = (x_1, x_2)^T$, $\|x\|_p = (|x_1|^p + |x_2|^p)^{1/p}$. For $x_1 = 2$ & $x_2 = 3$:

$$\begin{aligned}\left\|\begin{bmatrix}2\\3\end{bmatrix}\right\|_p &= (2^p + 3^p)^{1/p} \\ &= (3^p)^{1/p} \cdot \left(1 + \left(\tfrac{2}{3}\right)^p\right)^{1/p} \\ &= 3\left(1 + \left(\tfrac{2}{3}\right)^p\right)^{1/p} \\ \lim_{p\to\infty}\left\|\begin{bmatrix}2\\3\end{bmatrix}\right\|_p &= 3 = \max\left(|x_1|, |x_2|\right) = \left\|\begin{bmatrix}2\\3\end{bmatrix}\right\|_\infty\end{aligned}$$

▷ $\left(\frac{2}{3}\right)^p \to 0$, *so the factor* $\left(1 + (\frac{2}{3})^p\right)^{1/p} \to 1$. *In general the largest* $|x_i|$ *dominates the sum:* $\max_i |x_i| \leq \|x\|_p \leq n^{1/p} \max_i |x_i|$ *and* $n^{1/p} \to 1$.

► Unit spheres $\|x\|_p = 1$, i.e. $\|x - 0_\mathcal{V}\|_p = 1$. The ℓ_∞ unit sphere is $\max(|x_1|, |x_2|) = 1$.



▷ *Dashed curve:* $|x_1|^p + |x_2|^p = 1$ *for an intermediate* p *(drawn for* $p = 4$*); as* p *increases the unit sphere moves from the* ℓ_2 *circle towards the* ℓ_∞ *square.*

For $p = 1/2$:

$$\|x\|_{1/2} = \left(|x_1|^{1/2} + |x_2|^{1/2}\right)^2$$

▷ *It satisfies (N1) & (N2) but need not satisfy the triangle inequality (N3).*

Counterexample:

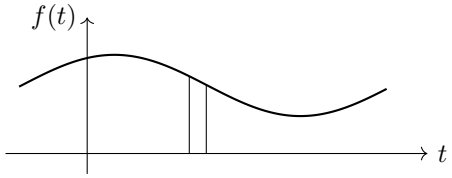
$$x = \begin{bmatrix}9\\0\end{bmatrix}, \quad y = \begin{bmatrix}0\\4\end{bmatrix}, \quad x + y = \begin{bmatrix}9\\4\end{bmatrix}$$

$$\begin{aligned}\|x + y\|_{1/2} &= \left(\sqrt{9} + \sqrt{4}\right)^2 = 25 \\ \|x\|_{1/2} &= 9, \qquad \|y\|_{1/2} = 4 \\ \|x + y\|_{1/2} &\not\leq \|x\|_{1/2} + \|y\|_{1/2}\end{aligned}$$

$$p < 1 \;\Rightarrow\; \textbf{no norms!}$$

▷ *Here* $25 > 9 + 4 = 13$. *Geometrically, for* $p < 1$ *the "unit sphere"* $|x_1|^p + |x_2|^p = 1$ *is star-shaped (curved inwards) and the unit ball is not convex;* ***a norm's unit ball is always convex.***

2.3 Norms on Function Spaces

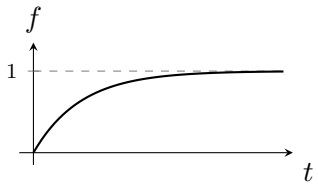


$$\mathcal{L}_p = \left\{ f(\cdot) \;\middle|\; f : \mathbb{R} \to \mathbb{R} \text{ such that } \int_{-\infty}^{\infty} |f(t)|^p \, dt < \infty \right\}, \qquad 1 \le p < \infty$$

$$\begin{aligned}
\|f(t)\|_p &= \left(\int_{-\infty}^{\infty} |f(t)|^p \, dt \right)^{1/p} && \mathcal{L}_p \text{ norm} \\
\|f(t)\|_1 &= \int_{-\infty}^{\infty} |f(t)| \, dt && \mathcal{L}_1 \text{ norm} \\
\|f(t)\|_2 &= \left(\int_{-\infty}^{\infty} |f(t)|^2 \, dt \right)^{1/2} && \mathcal{L}_2 \text{ norm} \\
\|f(t)\|_\infty &= \sup_t |f(t)| && \mathcal{L}_\infty \text{ norm}
\end{aligned}$$

▷ *These are the continuous analogues of the ℓ_p norms: the sum over i becomes an integral over t.*

$$f(t) = 1 - e^{-t}, \quad t \ge 0$$



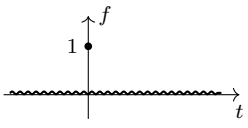
▷ *Value 1 is never reached. A maximum would have to be attained at a finite t, so $\max_t |f(t)|$ does not exist here.*

The supremum, however, does exist:

$$\sup_t |f(t)| = 1 \quad \text{(least upper bound)}$$

▷ *Therefore $\|f\|_\infty$* ***is defined with*** sup *(it always exists for bounded f), which equals* max *whenever the maximum is attained.*

Integral norms do not satisfy $\|x\| = 0 \iff x = 0_{\mathcal{V}}$!



$$f(t) = \begin{cases} 1, & t = 0 \\ 0, & \text{otherwise} \end{cases} \qquad \Rightarrow \qquad \int_{-\infty}^{\infty} |f(t)| \, dt = 0$$

▷ *Such functions are equivalent to the zero function, as they take nonzero values only at finitely many points.*

How this is fixed

$f \neq 0$ but $\|f\|_1 = 0$, so property (N1) fails on the set of all functions. The standard remedy: regard two functions as *equal* in $\mathcal{L}_p$ if they differ only on a set of zero length (e.g. finitely many points), i.e. if $\|f - g\|_p = 0$. With this identification $\|\cdot\|_p$ is a genuine norm on $\mathcal{L}_p$ for $p \geq 1$ (the triangle inequality is Minkowski's inequality).

2.4 Matrix Norms

$$A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}, \qquad \|A\| = ?$$

2.4.1 Entrywise Norms (Frobenius)

1st method.

$$\mathrm{Vec}(A) \triangleq \begin{bmatrix} 1 \\ 3 \\ -2 \\ 4 \end{bmatrix}, \qquad A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$\|A\|_p = \|\mathrm{Vec}(A)\|_p = \left\| \begin{bmatrix} 1 \\ 3 \\ -2 \\ 4 \end{bmatrix} \right\|_p = (|1|^p + |3|^p + |-2|^p + |4|^p)^{1/p}$$

$$= \left(\sum_{i,j} |a_{ij}|^p \right)^{1/p}, \qquad p \geq 1$$

Numbers for this A

$\|A\|_1 = 1 + 2 + 3 + 4 = 10$, $\|A\|_2 = \sqrt{1 + 4 + 9 + 16} = \sqrt{30} \approx 5.48$, $\|A\|_\infty = \max_{i,j} |a_{ij}| = 4$

The case $p = 2$ is the **Frobenius norm** $\|A\|_F$. Since $\mathbb{R}^{m \times n}$ is a vector space that is identified with $\mathbb{R}^{mn}$ through Vec, every vector norm gives a matrix norm in this way.

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1st method (Frobenius norm).

$$\begin{aligned} \|A\|_F &= \Big(\sum_{i,j} |a_{ij}|^2 \Big)^{1/2} && \text{(Frobenius norm)} \\ &= \sqrt{\mathrm{tr}(A^T A)} && \text{(real)} \\ &= \sqrt{\mathrm{tr}(A^H A)} && \text{(complex)} \end{aligned}$$

For example, with

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \qquad A^T A = \underbrace{\begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix}}_{A^T} \underbrace{\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}}_{A} = \begin{bmatrix} a_{11}^2 + a_{21}^2 & \\ & a_{12}^2 + a_{22}^2 \end{bmatrix}$$

$$\begin{aligned}\operatorname{tr}(A^T A) &= a_{11}^2 + a_{21}^2 + a_{12}^2 + a_{22}^2 \\ &= \sum_{i,j} a_{ij}^2\end{aligned}$$

▷ *Only the diagonal of $A^T A$ matters for the trace, so its off-diagonal entries ($a_{11}a_{12} + a_{21}a_{22}$) are left blank.*

Explanation

In general $(A^H A)_{jj} = \sum_i \overline{a_{ij}}\, a_{ij} = \sum_i |a_{ij}|^2$ is the squared length of the j-th column of A, so $\operatorname{tr}(A^H A) = \sum_j \sum_i |a_{ij}|^2$. The Frobenius norm is simply the Euclidean norm of A viewed as a long vector in $\mathcal{F}^{mn}$; we write it $\|A\|_F$ to distinguish it from the *induced* 2-norm of the 2nd method below (which is in general different, e.g. $\|I_n\|_F = \sqrt{n}$ but the induced norm of I_n is 1).

2.4.2 Induced Matrix Norms

2nd method (induced norm). Measure A through its action on vectors.

Definition 2.4. Let $A : \mathbb{R}^n \to \mathbb{R}^m$ be an $m \times n$ matrix. Let $\|\cdot\|_{\mathbb{R}^n}$ and $\|\cdot\|_{\mathbb{R}^m}$ denote the norms in $\mathbb{R}^n$ & $\mathbb{R}^m$ respectively. The **induced matrix norm** is defined as

$$\|A\| \triangleq \max_{\substack{x \in \mathbb{R}^n \\ x \neq 0_{\mathcal{V}}}} \frac{\|Ax\|_{\mathbb{R}^m}}{\|x\|_{\mathbb{R}^n}}$$

$$\underbrace{\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}}_{y}$$

Why second method?

$$\begin{aligned} |\langle x, y \rangle| &\leq \|x\|\,\|y\| \\ \|Ax\| &\leq \|A\|\,\|x\| \\ \|Ax\| &= \frac{\|Ax\|}{\|x\|} \cdot \|x\| \leq \left(\max \frac{\|Ax\|}{\|x\|} \right) \cdot \|x\| = \|A\|\,\|x\| \end{aligned}$$

▷ *"Cauchy–Schwarz"-like inequality for matrices: $\|Ax\| \leq \|A\|\,\|x\|$ **for every** x.*

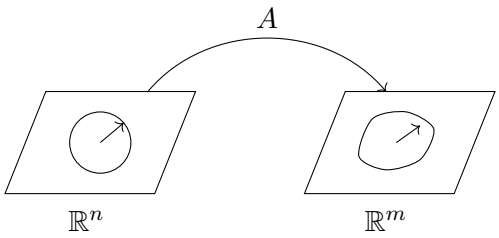
Why the induced norm is useful

The induced norm is the largest "gain" $\|Ax\| \,/\, \|x\|$ of A, so $\|Ax\| \leq \|A\|\,\|x\|$ holds for every x with the smallest possible constant. It is also submultiplicative: $\|ABx\| \leq \|A\|\,\|Bx\| \leq \|A\|\,\|B\|\,\|x\|$, hence $\|AB\| \leq \|A\|\,\|B\|$, which is what makes it convenient for bounding $\|A^k\|$ in system analysis.

$$\begin{aligned} \|A\| &= \max_{\substack{x \in \mathbb{R}^n \\ x \neq 0_{\mathcal{V}}}} \left\| A \frac{x}{\|x\|_{\mathbb{R}^n}} \right\|_{\mathbb{R}^m} \\ &= \max_{\|x\|_{\mathbb{R}^n} = 1} \|Ax\|_{\mathbb{R}^m} \end{aligned}$$

▷ *First step: absolute homogeneity of the norm, $\frac{\|Ax\|}{\|x\|} = \left\| A \frac{x}{\|x\|} \right\|$, and $\frac{x}{\|x\|}$ ranges over all unit vectors.*

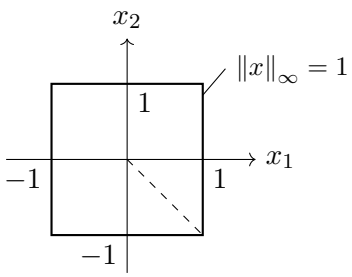
$$\therefore \ \|A\| = \max_{\|x\|_{\mathbb{R}^n}=1} \|Ax\|_{\mathbb{R}^m}$$



Example 2.5. Consider an $m \times n$ matrix with real elements. Choose ℓ_∞-norm $\|\cdot\|_\infty$ for both $\mathbb{R}^n$ & $\mathbb{R}^m$.

$$\|A\|_\infty = \max_{\|x\|_\infty=1} \|Ax\|_\infty, \qquad A = \begin{bmatrix} 1 & -2 \\ -3 & 4 \\ -2 & 2 \end{bmatrix}, \qquad x \in \mathbb{R}^2, \quad y = Ax \in \mathbb{R}^3$$

$$\|x\|_\infty = \max(|x_1|, |x_2|) = 1$$



First row only:

$$A_1 = \begin{bmatrix} 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\max_{\|x\|_\infty=1} |A_1 x| = \max_{\|x\|_\infty=1} |x_1 - 2x_2| = 3$$

▷ *Attained at $x_1 = 1$, $x_2 = -1$ (dashed corner).*

For the whole matrix:

$$Ax = \begin{bmatrix} x_1 - 2x_2 \\ -3x_1 + 4x_2 \\ -2x_1 + 2x_2 \end{bmatrix}$$

$$\|Ax\|_\infty = \max\left(|x_1 - 2x_2|, \ |-3x_1 + 4x_2|, \ |-2x_1 + 2x_2|\right)$$

$$x^* = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \text{or} \quad x^* = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$Ax^* = \begin{bmatrix} 3 \\ -7 \\ -4 \end{bmatrix} \ \Rightarrow \ \|Ax^*\|_\infty = 7$$

$$\underline{\underline{\|A\|_\infty = 7}}$$

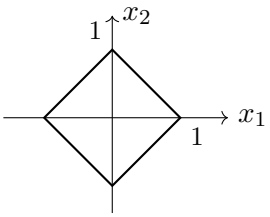
$$\|A\|_\infty = \max_i \sum_{j=1}^{n} |a_{ij}| \qquad \text{(max row sum)}$$

Why the max row sum

If $\|x\|_\infty = 1$ then $|x_j| \le 1$ for all j, so for each row $|(Ax)_i| = \left|\sum_j a_{ij}x_j\right| \le \sum_j |a_{ij}||x_j| \le \sum_j |a_{ij}|$. Hence $\|Ax\|_\infty \le \max_i \sum_j |a_{ij}|$. Equality is reached by taking the row i^* with the largest sum and $x_j = \text{sign}(a_{i^*j})$ (a corner of the square). In the example the row sums are $3, 7, 4$ and row $2 = [-3 \;\; 4]$ gives $x^* = [-1 \;\; 1]^T$ (or its negative).

ℓ_1-norm.

$$\|A\|_1 = \max_{\|x\|_1=1} \|Ax\|_1\,, \qquad \|x\|_1 = |x_1| + |x_2| = 1$$



$$A_1 = \begin{bmatrix} 1 & -2 \end{bmatrix}$$
$$A_1 x = x_1 - 2x_2$$
$$\|A_1 x\| = |x_1 - 2x_2|$$
$$\max_{\|x\|_1=1} \|A_1 x\| = \max_{\|x\|_1=1} |x_1 - 2x_2|$$

▷ *Weighted average of* 1 *and* -2*: on* $\|x\|_1 = 1$, $|x_1 - 2x_2| \le |x_1| + 2|x_2| \le 2(|x_1| + |x_2|) = 2$, *so the best is to put all the weight on the entry of largest magnitude.*

$$x = \begin{bmatrix} 0.5 \\ -0.5 \end{bmatrix} \quad \Rightarrow \quad \tfrac{1}{2} + 1 = \tfrac{3}{2}$$
$$x = \begin{bmatrix} 0.2 \\ -0.8 \end{bmatrix} \quad \Rightarrow \quad \text{larger}$$
$$x = \begin{bmatrix} 0 \\ -1 \end{bmatrix} \quad \Rightarrow \quad \text{largest:} \quad \underline{\|A_1 x\| = 2}$$

Now the whole matrix: p. 15

$$A = \begin{bmatrix} 1 & -2 \\ -3 & 4 \\ -2 & 2 \end{bmatrix}, \qquad Ax = \begin{bmatrix} 1 \\ -3 \\ -2 \end{bmatrix} x_1 + \begin{bmatrix} -2 \\ 4 \\ 2 \end{bmatrix} x_2$$
$$\|Ax\|_1 = \Big\| \underbrace{\begin{bmatrix} 1 \\ -3 \\ -2 \end{bmatrix}}_{6} x_1 + \underbrace{\begin{bmatrix} -2 \\ 4 \\ 2 \end{bmatrix}}_{8\ \checkmark} x_2 \Big\|_1 = 8 \qquad (x_1 = 0,\ x_2 = 1)$$

▷ *Column sums:* $|1| + |-3| + |-2| = 6$ *and* $|-2| + |4| + |2| = 8$, *so* $\|A\|_1 = 8$ *(compare* $\|A\|_\infty = 7$*: different vector norms give different matrix norms).*

$$\|A\|_1 = \max_j \sum_i |a_{ij}| \qquad \text{(max column sum)}$$

Why the max column sum

Write $A = [a_1 \ \cdots \ a_n]$ by columns. If $\|x\|_1 = 1$, $\|Ax\|_1 = \left\|\sum_j x_j a_j\right\|_1 \le \sum_j |x_j| \, \|a_j\|_1 \le (\max_j \|a_j\|_1) \sum_j |x_j| = \max_j \|a_j\|_1$, with equality for $x = e_{j^*}$ (a corner of the diamond), j^* the column of largest absolute sum.

2.5 Convergence of Sequences

Definition 2.6. Let $(\mathcal{V}, \mathcal{F}, \|\cdot\|)$ be a normed space. Let $\{v_n\}_{n=1}^{\infty}$ be a sequence of vectors in $\mathcal{V}$, i.e., $v_n \in \mathcal{V}$ $\forall n$. The sequence $\{v_n\}_{n=1}^{\infty}$ is said to **converge** to the **limit** $\boxed{v \in \mathcal{V}}$ iff

$$\boxed{\|v_n - v\| \to 0 \quad \text{as } n \to \infty}$$

Remark. A sequence is **divergent** if it is not convergent.

Example 2.7. $\mathcal{V} = \mathbb{R}$, $\|v\| = |v|$.

Sequence $v_n = (1/2)^n$. Candidate limit $v = 0 \in \mathcal{V}$:

$$\|v_n - v\| = \left|\frac{1}{2^n} - 0\right| = \frac{1}{2^n}$$
$$\lim_{n\to\infty} \|v_n - v\| = 0 \ \Rightarrow \ v_n \to 0 \text{ as } n \to \infty$$

Sequence $v_n = (-1)^n$. Try $v = 0$:

$$\|(-1)^n - 0\| = |(-1)^n| = 1, \qquad \lim_{n\to\infty} \overbrace{\|v_n - v\|}^{1} = 1 \neq 0$$

▷ *The sequence does not converge to $v = 0$.*
Try $v = 1$:

$$\|(-1)^n - 1\| = |(-1)^n - 1| = \begin{cases} 0, & n \text{ even} \\ 2, & n \text{ odd} \end{cases} \ \Rightarrow \ \text{does not converge}$$

▷ *The distance keeps jumping back to 2, so it does not tend to 0.*
∴ **Sequence is not convergent anywhere.**

Explanation

For an arbitrary candidate $v \in \mathbb{R}$: the even terms are 1 and the odd terms are -1, so by the triangle inequality $2 = |1 - (-1)| \le |v_{2k} - v| + |v - v_{2k+1}|$. Thus at least one of two consecutive distances is ≥ 1, and $\|v_n - v\| \not\to 0$ for every v.

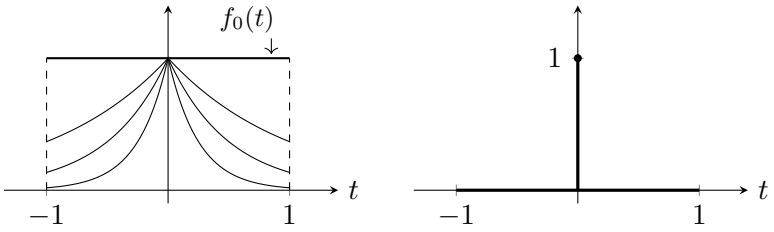
Example 2.8. $\mathcal{V} = \left\{ f : [-1, 1] \to \mathbb{R} \;\middle|\; f(\cdot) \text{ is continuous, } \int_{-1}^{1} |f(t)| \, dt < \infty \right\}$

$$\|f(\cdot)\| = \int_{-1}^{1} |f(t)| \, dt \qquad (\mathcal{L}_1 \text{ norm})$$

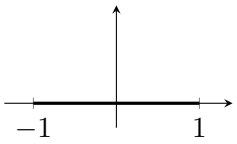
Is f_n convergent?

$$
\begin{aligned}
f_n(t) &= e^{-n|t|}, \quad n \geq 0 \\
f_0(t) &= e^{0t} = 1 \\
f_1(t) &= e^{-|t|} \\
f_2(t) &= e^{-2|t|} \\
&\vdots
\end{aligned}
$$

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▷ *Left:* f_0, f_1, f_2, f_4. *Right: pointwise limit as* $n \to \infty$ *(*0 *for* $t \neq 0$, 1 *at* $t = 0$*)* $\notin \mathcal{V}$ *(not continuous).*



$0(t) \in \mathcal{V}$

Candidate limit: the zero function $0(t)$.
$v \in \mathcal{V}$ is satisfied ✓

$$
\begin{aligned}
\|f_n(t) - 0(t)\| = \|f_n(t)\| &= \int_{-1}^{1} \left| e^{-n|t|} \right| dt \\
&= \int_{-1}^{1} e^{-n|t|}\, dt && \text{(exponential is positive)} \\
&= 2 \cdot \int_{0}^{1} e^{-nt}\, dt && \text{(even function)} \\
&= 2\, \frac{e^{-nt}}{-n} \Big|_0^1 = -2 \left(\frac{e^{-n}}{n} - \frac{1}{n} \right) \\
\|f_n(t) - 0(t)\| &= -\frac{2}{n} (\underbrace{e^{-n} - 1}_{\to -1}) = \frac{2\,(1 - e^{-n})}{n} \leq \frac{2}{n} \\
\lim_{n \to \infty} \|f_n(t) - 0(t)\| &= 0 \;\Rightarrow\; \|v_n - v\| \to 0 \text{ as } n \to \infty \;\checkmark
\end{aligned}
$$

$\therefore$ **It is convergent to the zero function.**

Convergence depends on the distance definition (the norm):

- for the $\mathcal{L}_1$ norm $\Rightarrow$ convergent
- for the $\mathcal{L}_\infty$ (sup) norm $\Rightarrow$ divergent

Explanation

With $\|f\|_\infty = \max_{-1\le t\le 1} |f(t)|$ we get $\|f_n - 0\|_\infty = \max_t e^{-n|t|} = 1$ (at $t = 0$) for every n, so $f_n \not\to 0$. No other continuous g works either: $\|f_n - g\|_\infty \to 0$ would force $g(t) = \lim_n f_n(t)$ pointwise, i.e. $g(0) = 1$ and $g(t) = 0$ for $t \neq 0$, which is not continuous. The $\mathcal{L}_1$ norm only "sees" area, and the area under the narrowing peak vanishes; the $\mathcal{L}_\infty$ (sup) norm sees the peak height, which stays 1.

2.6 Cauchy Sequences and Completeness

Definition 2.9. Let $(\mathcal{V}, \mathcal{F}, \|\cdot\|)$ be a normed space. A sequence $\{v_n\}_{n=0}^\infty$ is said to be a **Cauchy sequence** iff

$$\lim_{n,m\to\infty} \|v_n - v_m\| = 0$$

Claim. Every convergent sequence is a Cauchy sequence. The reverse statement, in general, is not correct.

$$\text{Convergent} \underset{\not\Leftarrow}{\Rightarrow} \text{Cauchy}$$

Proof. Suppose that $\{v_n\}_{n=0}^\infty$ is convergent to $v \in \mathcal{V}$, i.e. $\lim_{n\to\infty} \|v_n - v\| = 0$. Then

$$\begin{aligned}
\lim_{n,m\to\infty} \|v_n - v_m\| &= \lim_{n,m\to\infty} \|v_n - v + v - v_m\| \\
&\le \lim_{n,m\to\infty} \left(\|v_n - v\| + \|v - v_m\| \right) \\
&= \underbrace{\lim_{n\to\infty} \|v_n - v\|}_{=0} + \underbrace{\lim_{m\to\infty} \|v_m - v\|}_{=0} = 0
\end{aligned}$$

▷ *Inequality step: triangle inequality,* $\|a + b\| \le \|a\| + \|b\|$ *with* $a = v_n - v$, $b = v - v_m$.

$$0 \le \lim_{n,m\to\infty} \|v_n - v_m\| \le 0 \;\Rightarrow\; \lim_{n,m\to\infty} \|v_n - v_m\| = 0 \;\Rightarrow\; \{v_n\} \text{ is Cauchy.} \qquad \square$$

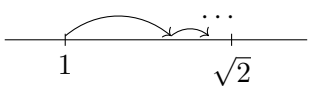
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Example 2.10. $\mathcal{V} = \mathbb{Q}$: set of rational numbers, $\mathcal{F} = \mathbb{Q}$ as well.

$$\sqrt{2} = 1.41421356237310\ldots \;\Rightarrow\; \text{not a rational number}$$

$$\begin{aligned}
v_0 &= 1 = \tfrac{1}{1} \\
v_1 &= 1.4 = \tfrac{14}{10} \\
v_2 &= 1.41 = \tfrac{141}{100} \\
v_3 &= 1.414 = \tfrac{1414}{1000} \\
&\vdots
\end{aligned}$$

We need $v \in \mathcal{V}$ s.t. $\|v_n - v\| \to 0$ as $n \to \infty$, where $v_n \in \mathcal{V}\ \forall n$. But v should be $\sqrt{2}$ for $\|v_n - v\| \to 0$ as $n \to \infty$.



Since $\sqrt{2} \notin \mathcal{V}$, $\therefore$ $\{v_n\}$ is not convergent (in $\mathcal{V}$).

What about Cauchyness?

$$\|v_n - v_m\| \to 0 \text{ as } n, m \to \infty$$

v_n is actually Cauchy as elements are approaching each other.
▷ *Indeed v_n and v_m agree in the first $\min(n,m)$ decimals, so $|v_n - v_m| < 10^{-\min(n,m)} \to 0$.*

$$\therefore \textbf{ Convergent} \Rightarrow \textbf{Cauchy} \quad \text{but} \quad \text{Cauchy} \not\Rightarrow \text{Convergent}$$

▷ *Margin note (Turkish): "ben bi sequence olsam sana converge ederdim."*

If a set does not have holes $\Rightarrow$ **complete** set.

Explanation

The same sequence viewed in $\mathcal{V} = \mathbb{R}$ converges (to $\sqrt{2} \in \mathbb{R}$): convergence depends on the space, not only on the terms. $\mathbb{R}$ is complete (it "fills the holes" of $\mathbb{Q}$), and so is every finite-dimensional normed space over $\mathbb{R}$ or $\mathbb{C}$, e.g. $(\mathbb{R}^n, \|\cdot\|_p)$. Incompleteness is therefore a phenomenon of infinite-dimensional spaces such as function spaces, as the next example shows.

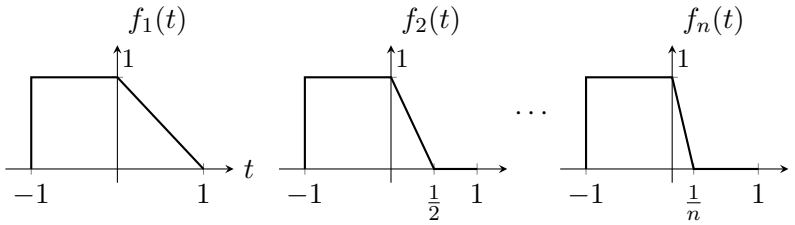
Definition 2.11. A normed space is said to be **complete** iff **every Cauchy sequence is convergent**. A complete normed space is called a **Banach space**.

Example 2.12. A normed space defined as

$$\mathcal{V} = \left\{ f(\cdot) \;\middle|\; f : [-1,1] \to \mathbb{R},\ f \text{ is continuous } \& \int_{-1}^{1} |f(t)|\,dt < \infty \right\}, \qquad \|f\|_1 = \int_{-1}^{1} |f(t)|\,dt$$

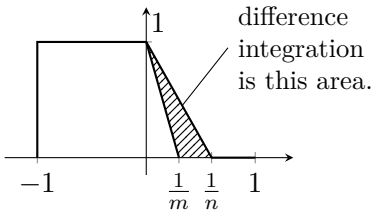
Consider the sequence $\{f_n\}_{n=1}^{\infty}$:

$$f_n(t) = \begin{cases} 1, & -1 \le t \le 0 \\ 1 - nt, & 0 \le t \le \frac{1}{n} \\ 0, & \frac{1}{n} \le t \le 1 \end{cases}$$



Is the sequence $\{f_n\}$ Cauchy? Convergent?

$$\|f_n(t) - f_m(t)\|_1 = \int_{-1}^{1} |f_n(t) - f_m(t)|\,dt$$

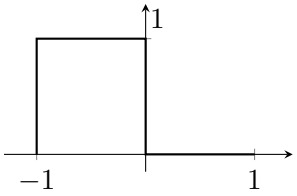


$$\text{Area} = \left|\frac{1}{n} - \frac{1}{m}\right| \cdot \frac{1}{2}$$

▷ *Triangle with vertices* $(0,1)$, $(\frac{1}{m},0)$, $(\frac{1}{n},0)$*: base* $|\frac{1}{n} - \frac{1}{m}|$*, height* 1.

$$\lim_{n,m\to\infty} \left|\frac{1}{n} - \frac{1}{m}\right| \cdot \frac{1}{2} = 0$$

$f_n \xrightarrow{n\to\infty}$



The limit $\notin \mathcal{V}$ (not continuous).
$\Rightarrow \{f_n\}$ is Cauchy but not convergent.
$\therefore (\mathcal{V}, \mathcal{F}, \|\cdot\|_1)$ **is not complete.**

Explanation

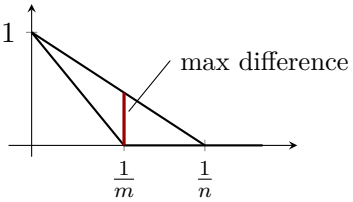
To be precise, "not convergent" means no *continuous* g satisfies $\|f_n - g\|_1 \to 0$. If one did, then on $[-1,0]$ we would need $\int_{-1}^{0} |1-g|\,dt = 0$, so $g = 1$ there; and on every $[\delta, 1]$, $\delta > 0$, $f_n = 0$ eventually, so $g = 0$ there. Then g jumps at $t = 0$, contradicting continuity. The "limit" is the step function, which lies outside $\mathcal{V}$.

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Now with $(\mathcal{V}, \mathcal{F}, \|\cdot\|_\infty)$**.**

$$\|f\|_\infty = \max_{-1\le t\le 1} |f(t)|$$
$$\|f_n(t) - f_m(t)\|_\infty = \max_{-1\le t\le 1} |f_n(t) - f_m(t)|$$

$$f_n(t) = 1 - nt \quad (0 \le t \le \tfrac{1}{n})$$
$$f_n\left(\tfrac{1}{m}\right) = 1 - \frac{n}{m} \quad (m \ge n)$$



$$\|f_n(t) - f_m(t)\|_\infty = f_n\left(\tfrac{1}{m}\right) - \underbrace{f_m\left(\tfrac{1}{m}\right)}_{0} = \underbrace{\left|1 - \frac{n}{m}\right|}_{\neq 0 \text{ unless } n=m} \qquad (m \ge n)$$

e.g. $m = 2n$: $\quad \|f_n - f_{2n}\|_\infty = \frac{1}{2}$ for every n

n ve m'in nasıl ∞*'a gittiği fark etmez, her zaman limit sonucu sıfır olmalı. Bu senaryoda if* $n = m$ *o zaman limit sıfır, onun dışında değil.*

▷ *It does not matter how* n *and* m *go to* ∞*; the limit must always be zero. Here it is zero only if* $n = m$*.*

▷ *The maximum is at* $t = \frac{1}{m}$*: on* $[0, \frac{1}{m}]$ *the difference* $(m-n)t$ *increases, and on* $[\frac{1}{m}, \frac{1}{n}]$ *it equals* $1 - nt$*, which decreases.*

$\therefore \{f_n\}$ **is not Cauchy** in the sup norm.

3 Inner Product Spaces

Syllabus: Week 4 — inner product spaces, orthogonality, Gram–Schmidt

3.1 Inner Products

Definition 3.1. Let $(\mathcal{V}, \mathbb{C})$ be a vector space. An **inner product** is a function from $\mathcal{V} \times \mathcal{V}$ into $\mathbb{C}$,

$$\langle \cdot, \cdot \rangle : \mathcal{V} \times \mathcal{V} \to \mathbb{C}$$

satisfying the following properties:

1) $\langle x, y \rangle = \overline{\langle y, x \rangle} \quad \forall x, y \in \mathcal{V}$ (conjugate symmetry)

2) Linearity with respect to (wrt) the first argument:

   a) $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle \quad \forall x, y, z \in \mathcal{V}$ (additivity)

   b) $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle \quad \forall \alpha \in \mathbb{C}$ and $\forall x, y \in \mathcal{V}$ (homogeneity)

3) $\langle x, x \rangle$ is real & nonnegative, $\langle x, x \rangle \geq 0$, and $\langle x, x \rangle = 0 \Leftrightarrow x = 0_{\mathcal{V}}$ (positive definiteness)

$\therefore$ The triplet $(\mathcal{V}, \mathbb{C}, \langle \cdot, \cdot \rangle)$ is called an **inner product space**.

$$\langle x, y + z \rangle = \overline{\langle y + z, x \rangle} = \overline{\langle y, x \rangle} + \overline{\langle z, x \rangle} = \langle x, y \rangle + \langle x, z \rangle$$

▷ *Additivity wrt second argument* ✓

$$\langle x, \alpha y \rangle = \overline{\langle \alpha y, x \rangle} = \bar{\alpha}\, \overline{\langle y, x \rangle} = \bar{\alpha} \langle x, y \rangle$$

▷ *Homogeneity wrt second argument* × *(α should be real). We have* ***conjugate homogeneity wrt second argument****.*

Explanation

So an inner product is linear in the first argument and *conjugate*-linear in the second: $\left\langle \sum_i \alpha_i x_i, \sum_j \beta_j y_j \right\rangle = \sum_{i,j} \alpha_i \bar{\beta}_j \langle x_i, y_j \rangle$. Over $\mathcal{F} = \mathbb{R}$ the conjugates disappear and the inner product is bilinear and symmetric. Also $\langle x, 0_{\mathcal{V}} \rangle = \langle 0_{\mathcal{V}}, x \rangle = 0$ for all x (take $\alpha = 0$ in homogeneity).

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Example 3.2. $\mathcal{V} = \mathbb{C}^n$, $\mathcal{F} = \mathbb{C}$, $\quad x = [x_1\ x_2 \dots x_n]^T$, $\quad y = [y_1\ y_2 \dots y_n]^T$, $\quad x_i, y_i \in \mathbb{C}$.

$$\langle x, y \rangle = \sum_{i=1}^{n} x_i \bar{y}_i = \bar{y}^T x = y^H x \qquad \left(\begin{bmatrix} \bar{y}_1 & \bar{y}_2 & \cdots \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \end{bmatrix} \right)$$

Check the axioms:

1) $\langle x, y \rangle = \sum_{i=1}^{n} x_i \bar{y}_i = \bar{y}^T x = y^H x$

$$\overline{\langle y, x \rangle} = \overline{x^H y} = \overline{\bar{x}^T y} = \underbrace{x^T \bar{y} = \bar{y}^T x}_{\text{scalar=its transpose}} = y^H x = \langle x, y \rangle$$

2) a) $\langle x+y,z\rangle = z^H(x+y) = z^Hx + z^Hy = \langle x,z\rangle + \langle y,z\rangle$

b) $\langle \alpha x,y\rangle = y^H(\alpha x) = \alpha y^Hx = \alpha\langle x,y\rangle$

3) $\langle x,x\rangle = x^Hx = \sum_{i=1}^n |x_i|^2 \geq 0 \Rightarrow$ real value; $\langle x,x\rangle = 0 \Leftrightarrow x = 0_{\mathcal{V}}$.

Example 3.3. $A, B \in \mathbb{R}^{m\times n}$, $\mathcal{V} = \mathbb{R}^{m\times n}$, $\mathcal{F} = \mathbb{R}$, $A = [a_{ij}]$, $B = [b_{ij}]$:

$$\langle A,B\rangle = \sum_{i,j} a_{ij}b_{ij}$$

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \cdot \begin{bmatrix} b_{11} & b_{21} \\ b_{12} & b_{22} \\ b_{13} & b_{23} \end{bmatrix}}_{A\cdot B^T} = \begin{bmatrix} a_{11}b_{11}+a_{12}b_{12}+a_{13}b_{13} & \times \\ \times & a_{21}b_{21}+a_{22}b_{22}+a_{23}b_{23} \end{bmatrix}$$

$$\begin{aligned} \langle A,B\rangle &\triangleq \operatorname{tr}(AB^T) \\ &= \operatorname{tr}(A^TB) \quad \text{(in the real field)} \end{aligned}$$

▷ *$(AB^T)_{ii} = \sum_j a_{ij}b_{ij}$ (row i of A against row i of B), so the trace adds up all $a_{ij}b_{ij}$; the off-diagonal entries ($\times$) are irrelevant. For complex matrices one uses $\langle A,B\rangle = \operatorname{tr}(AB^H) = \operatorname{tr}(B^HA)$, whose induced norm is the Frobenius norm.*

Example 3.4. $\mathcal{V} = \left\{ f(\cdot) \;\middle|\; f:[0,1]\to\mathbb{C} \text{ s.t. } \int_0^1 |f(t)|^2\,dt < \infty \right\}$

$$\langle f_1,f_2\rangle \triangleq \int_0^1 f_1(t)\overline{f_2(t)}\,dt \qquad (\mathcal{L}_2 \text{ inner product})$$

3.2 The Cauchy–Schwarz Inequality and the Induced Norm

Theorem 3.5 (Cauchy–Schwarz Inequality)**.** Let $\mathcal{V}$ be an inner product space. Let $x, y \in \mathcal{V}$.

$$|\langle x,y\rangle|^2 \leq \langle x,x\rangle\langle y,y\rangle$$

Equality holds iff $x = cy$ for some $c \in \mathbb{C}$ or $y = 0_{\mathcal{V}}$, i.e. **equality iff x and y are linearly dependent**.

Equality check for $x = cy$.

$$\begin{aligned} &\langle x,y\rangle = \langle cy,y\rangle = c\langle y,y\rangle \;\Rightarrow\; |\langle x,y\rangle|^2 = |c|^2\langle y,y\rangle^2 \\ &\langle x,x\rangle = \langle cy,cy\rangle = c\bar{c}\langle y,y\rangle = |c|^2\langle y,y\rangle \qquad (c\bar{c} = |c|^2 \text{ is real}) \end{aligned}$$

▷ *Using the first- and second-argument axioms.*

$$\underbrace{|c|^2\langle y,y\rangle^2}_{|\langle x,y\rangle|^2} = \underbrace{|c|^2\langle y,y\rangle}_{\langle x,x\rangle}\underbrace{\langle y,y\rangle}_{\langle y,y\rangle}$$

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Proof of theorem. **Case 1:** $y = 0_{\mathcal{V}}$.

$$\begin{aligned}
&\langle x, y\rangle = \langle x, 0_{\mathcal{V}}\rangle = \langle x, 0 \cdot 0_{\mathcal{V}}\rangle = 0 \cdot \langle x, 0_{\mathcal{V}}\rangle = 0 \\
&\langle y, y\rangle = 0 \quad \text{(by axioms)} \\
&\underbrace{|\langle x, y\rangle|^2}_{0} \le \underbrace{\langle x, x\rangle}_{\ge 0}\underbrace{\langle y, y\rangle}_{0} \;\Rightarrow\; 0 \le 0 \;\checkmark
\end{aligned}$$

Case 2: $y \neq 0_{\mathcal{V}}$. Then $\langle y, y\rangle \neq 0$ (in fact $\langle y, y\rangle > 0$ by axiom 3, so we may divide by it and multiply inequalities by it). For $\alpha \in \mathbb{C}$ (in field):

$$\begin{aligned}
0 &\le \langle x - \alpha y, x - \alpha y\rangle \\
&= \langle x, x\rangle - \langle x, \alpha y\rangle - \langle \alpha y, x\rangle + \langle \alpha y, \alpha y\rangle \\
&= \langle x, x\rangle - \bar{\alpha}\langle x, y\rangle - \alpha\underbrace{\langle y, x\rangle}_{\overline{\langle x, y\rangle}} + |\alpha|^2\langle y, y\rangle
\end{aligned}$$

Choose $\alpha = \dfrac{\langle x, y\rangle}{\langle y, y\rangle}$:

$$\begin{aligned}
0 &\le \langle x, x\rangle - \frac{\overline{\langle x, y\rangle}}{\langle y, y\rangle}\cdot\langle x, y\rangle - \frac{\langle x, y\rangle}{\cancel{\langle y, y\rangle}}\cdot\cancel{\overline{\langle x, y\rangle}} + \frac{|\langle x, y\rangle|^2}{\cancel{\langle y, y\rangle^2}}\cdot\cancel{\langle y, y\rangle} \\
0 &\le \langle x, x\rangle - \frac{|\langle x, y\rangle|^2}{\langle y, y\rangle} \\
\frac{|\langle x, y\rangle|^2}{\langle y, y\rangle} &\le \langle x, x\rangle \;\Rightarrow\; \therefore\; |\langle x, y\rangle|^2 \le \langle x, x\rangle\langle y, y\rangle
\end{aligned}$$

□

Choice of α and the equality case

αy with $\alpha = \langle x, y\rangle / \langle y, y\rangle$ is the projection of x onto y (see Gram–Schmidt below), so $x - \alpha y \perp y$ and the first line says $\|x - \alpha y\|^2 \ge 0$: the error of the best approximation of x along y has nonnegative length. Equality in the theorem (with $y \neq 0_{\mathcal{V}}$) holds iff $\langle x - \alpha y, x - \alpha y\rangle = 0$, i.e. iff $x = \alpha y$ by axiom 3. This proves the "only if" part of the equality statement; the "if" part is the check for $x = cy$ above.

Remark. An inner product **induces a norm** (the **induced norm**) defined as

$$\|x\|^2 \triangleq \langle x, x\rangle, \qquad \|x\| \triangleq \sqrt{\langle x, x\rangle}$$

Show that this is a norm.

1) $\|x\| \ge 0$, $\|x\| = 0 \Leftrightarrow x = 0_{\mathcal{V}}$ ✓ (by axiom 3 of the inner product) (positive definiteness)

2) $\|\alpha x\| \triangleq \sqrt{\langle \alpha x, \alpha x\rangle} = \sqrt{\alpha \cdot \bar{\alpha}\langle x, x\rangle} = |\alpha|\underbrace{\sqrt{\langle x, x\rangle}}_{\|x\|} = |\alpha|\,\|x\|$ ✓ (absolute homogeneity)

3) $\|x + y\| = \sqrt{\langle x + y, x + y\rangle}$ (triangle inequality)

$$\begin{aligned}
\|x + y\|^2 &= \langle x + y, x + y\rangle = \langle x, x\rangle + \langle x, y\rangle + \underbrace{\langle y, x\rangle}_{\overline{\langle x, y\rangle}} + \langle y, y\rangle \\
&= \langle x, x\rangle + 2\cdot \operatorname{Re}\langle x, y\rangle + \langle y, y\rangle \\
&\le \langle x, x\rangle + 2|\operatorname{Re}\langle x, y\rangle| + \langle y, y\rangle \\
&\le \langle x, x\rangle + 2|\langle x, y\rangle| + \langle y, y\rangle
\end{aligned}$$

By Cauchy–Schwarz, $|\langle x, y\rangle| \le \sqrt{\langle x, x\rangle \langle y, y\rangle}$:

$$\begin{aligned} &\le \langle x, x\rangle + 2\sqrt{\langle x, x\rangle}\sqrt{\langle y, y\rangle} + \langle y, y\rangle \\ &= \left(\sqrt{\langle x, x\rangle} + \sqrt{\langle y, y\rangle}\right)^2 \\ &= (\|x\| + \|y\|)^2 \\ \|x + y\|^2 \le (\|x\| + \|y\|)^2 \;&\Rightarrow\; \|x + y\| \le \|x\| + \|y\| \;\checkmark \end{aligned}$$

▷ *Both sides are nonnegative, so taking square roots preserves the inequality: the triangle inequality holds.*

Angle between vectors. By Cauchy–Schwarz, $|\langle x, y\rangle| \le \|x\|\,\|y\|$; suppose $x \neq 0_{\mathcal{V}}$, $y \neq 0_{\mathcal{V}}$:

$$0 \le \frac{|\langle x, y\rangle|}{\|x\|\,\|y\|} \le 1$$

$$\cos\theta \triangleq \frac{|\langle x, y\rangle|}{\|x\|\,\|y\|} \;\Rightarrow\; \theta = \arccos\left(\frac{|\langle x, y\rangle|}{\|x\|\,\|y\|}\right)$$

▷ *Cauchy–Schwarz is exactly what makes this angle well defined. With the absolute value $\theta \in [0, \frac{\pi}{2}]$; in a real space one often drops it, $\cos\theta = \frac{\langle x,y\rangle}{\|x\|\|y\|} \in [-1, 1]$, which for $\mathbb{R}^2, \mathbb{R}^3$ is the usual geometric angle. $\theta = \frac{\pi}{2} \Leftrightarrow \langle x, y\rangle = 0$ (orthogonal), $\theta = 0 \Leftrightarrow x = cy$.*

Remark. Every inner product space is also a normed space. The converse/reverse is not correct in general.

Explanation

A norm comes from an inner product **iff it satisfies the parallelogram law** $\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$. For example $\|\cdot\|_1$ on $\mathbb{R}^2$ fails it: $x = e_1$, $y = e_2$ give $4 + 4 = 8 \neq 4 = 2 + 2$. So ℓ_1 and ℓ_∞ are norms without an inner product behind them; only ℓ_2 is induced by $\langle x, y\rangle = y^T x$.

Definition 3.6. An inner product space that is complete wrt the induced norm is called a **complete inner product space** / **Hilbert space**.

3.3 Orthogonality and Orthogonal Expansions

Definition 3.7. Two vectors $x, y \in \mathcal{V}$ are said to be **orthogonal** iff $\langle x, y\rangle = 0$. Similarly subsets $S_1, S_2 \subset \mathcal{V}$ are orthogonal iff $\langle x, y\rangle = 0\ \forall x \in S_1, \forall y \in S_2$.

Examples.

1) $\mathcal{V} = \mathbb{R}^n$, $\mathcal{F} = \mathbb{R}$, $\quad \langle x, y\rangle = \sum_{i=1}^n x_i y_i = y^T x = x^T y$

$$\underbrace{B = \{e_1, e_2, \ldots, e_n\}}_{\text{canonical basis set}} \qquad e_1 = \begin{bmatrix}1\\0\\\vdots\\0\end{bmatrix} \quad e_2 = \begin{bmatrix}0\\1\\0\\\vdots\\0\end{bmatrix} \quad \cdots$$

$$\langle e_i, e_j\rangle = \begin{cases}0, & i \neq j\\ 1, & i = j\end{cases} = \delta_{ij}$$

2) $\mathcal{V} = \left\{ f(\cdot) \;\middle|\; f : [-\pi, \pi] \to \mathbb{C} \text{ s.t. } \int_{-\pi}^{\pi} |f(t)|^2\, dt < \infty \right\}$

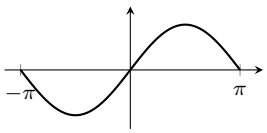
Consider $\{e^{jnt}\}_{n=-\infty}^{\infty}$, $\quad e^{jnt} \in \mathcal{V}$ for all n.

$$\langle f_1(t), f_2(t)\rangle = \int_{-\pi}^{\pi} f_1(t)\, \overline{f_2(t)}\, dt$$

$$\begin{aligned}\left\langle e^{jnt}, e^{jmt}\right\rangle &= \int_{-\pi}^{\pi} e^{jnt}\, e^{-jmt}\, dt = \int_{-\pi}^{\pi} e^{j(n-m)t}\, dt\\ &= \int_{-\pi}^{\pi} \left[\cos((n-m)t) + j\sin((n-m)t)\right] dt\\ &= \int_{-\pi}^{\pi} \cos((n-m)t)\, dt + j\underbrace{\int_{-\pi}^{\pi} \sin((n-m)t)\, dt}_{0}\end{aligned}$$

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$\left.\begin{matrix}\sin t\\ \sin 2t\\ \sin 3t\\ \vdots\end{matrix}\right\}$ all zero due to integral over a period



$$\left\langle e^{jnt}, e^{jmt}\right\rangle = \begin{cases}2\pi, & n = m\\ 0, & n \neq m\end{cases}$$
$$= 2\pi\, \delta_{nm}$$

▷ *For $n = m$: $\int_{-\pi}^{\pi} \cos 0\, dt = 2\pi$. For $k = n - m \neq 0$: $\int_{-\pi}^{\pi} \cos kt\, dt = \frac{2\sin k\pi}{k} = 0$, and $\int_{-\pi}^{\pi} \sin kt\, dt = 0$ since $\sin kt$ is odd.*

3) $S_1 = \{\sin nt\}_{n=1}^{\infty}$, $\; S_2 = \{\cos mt\}_{m=1}^{\infty}$:

$$\begin{aligned}\langle \sin nt, \cos mt\rangle &= \int_{-\pi}^{\pi} \sin nt \cdot \cos mt\, dt\\ &= \int_{-\pi}^{\pi} \frac{1}{2}\Big(\underbrace{\sin((n+m)t)}_{0} + \underbrace{\sin((n-m)t)}_{0}\Big)\, dt = 0\end{aligned}$$

▷ *Product-to-sum: $\sin a \cos b = \frac{1}{2}[\sin(a+b) + \sin(a-b)]$; both terms are odd functions (or zero when $n = m$), so they integrate to 0 over the symmetric interval $[-\pi, \pi]$.*

$$\therefore\; S_1 \perp S_2$$

Expansion in an orthogonal set.

- $z = \alpha_1 z_1 + \alpha_2 z_2$ if $z_1 \perp z_2$:

$$\begin{aligned}\langle z, z_1\rangle &= \langle \alpha_1 z_1 + \alpha_2 z_2, z_1\rangle \\ &= \alpha_1 \langle z_1, z_1\rangle + \alpha_2 \underbrace{\langle z_2, z_1\rangle}_{0} \Rightarrow \alpha_1 = \frac{\langle z, z_1\rangle}{\langle z_1, z_1\rangle}\end{aligned}$$

In general,

$$z = \sum_{i=1}^{n} \alpha_i z_i, \qquad \alpha_i = \frac{\langle z, z_i\rangle}{\langle z_i, z_i\rangle} \quad \forall i$$

▷ *Valid when $z \in \mathrm{sp}\{z_1, \ldots, z_n\}$ and the $z_i \neq 0_{\mathcal{V}}$ are mutually orthogonal: take $\langle \cdot, z_k\rangle$ of both sides and all terms with $i \neq k$ vanish.* ***No linear system has to be solved; each coefficient is computed independently.***

Example (as $z_1 \perp z_2$):

$$\underbrace{\begin{bmatrix}2\\3\end{bmatrix}}_{z} = \beta_1 \underbrace{\begin{bmatrix}1\\1\end{bmatrix}}_{z_1} + \beta_2 \underbrace{\begin{bmatrix}1\\-1\end{bmatrix}}_{z_2}, \qquad \beta_1 = \frac{\langle z, z_1\rangle}{\langle z_1, z_1\rangle} = \frac{5}{2}, \qquad \beta_2 = \frac{\langle z, z_2\rangle}{\langle z_2, z_2\rangle} = \frac{2-3}{2} = -\frac{1}{2}$$

▷ *Check:* $\frac{5}{2}\begin{bmatrix}1\\1\end{bmatrix} - \frac{1}{2}\begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}2\\3\end{bmatrix}$ ✓

- $f(t) = \sum_{n=-\infty}^{\infty} \alpha_n e^{jnt}$:

$$\alpha_n = \frac{\langle f(t), e^{jnt}\rangle}{\langle e^{jnt}, e^{jnt}\rangle} = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t) e^{-jnt}\, dt \qquad \text{(Fourier series coefficients)}$$

▷ *With $\langle f_1, f_2\rangle = \int_{-\pi}^{\pi} f_1(t)\,\overline{f_2(t)}\, dt$.*

3.4 Gram–Schmidt Orthogonalization

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Question. Let $\mathcal{V}$ be an inner product space. Let $S_1 = \{v_1, v_2, \ldots, v_n\}$ be a linearly independent set. Can we obtain another set $S_2 = \{w_1, w_2, \ldots, w_n\}$ s.t.

1) $w_i \perp w_j$ when $i \neq j$
2) $\mathrm{sp}\{S_1\} = \mathrm{sp}\{S_2\}$?

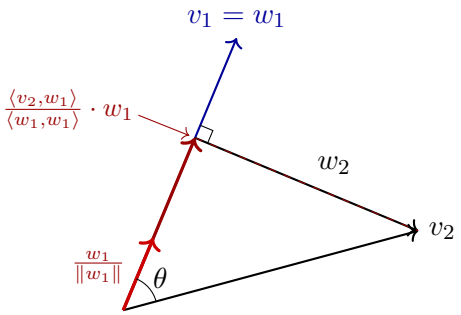
Yes! (constructively, by the following algorithm)

Gram–Schmidt orthogonalization algorithm. The w_k are constructed one at a time.

Step 1. $w_1 = v_1$

Step 2. $w_2 = v_2 - \dfrac{\langle v_2, w_1\rangle}{\langle w_1, w_1\rangle}\cdot w_1$

$$\langle w_2, w_1\rangle = \left\langle v_2 - \frac{\langle v_2, w_1\rangle}{\langle w_1, w_1\rangle}\cdot w_1, w_1 \right\rangle$$
$$= \cancel{\langle v_2, w_1\rangle} - \underbrace{\frac{\cancel{\langle v_2, w_1\rangle}}{\cancel{\langle w_1, w_1\rangle}}}_{\text{scalar}} \cancel{\langle w_1, w_1\rangle} = 0$$



$\langle x, y\rangle = \|x\|\,\|y\|\cos\theta$

$$\underbrace{\frac{\langle v_2, w_1\rangle}{\langle w_1, w_1\rangle}}_{\|w_1\|^2}\cdot w_1 = \frac{\langle v_2, w_1\rangle}{\|w_1\|}\cdot\frac{w_1}{\|w_1\|} = \underbrace{\left\langle v_2, \frac{w_1}{\|w_1\|}\right\rangle}_{\|v_2\|\cos\theta}\frac{w_1}{\|w_1\|}$$

Step 3. $w_3 = v_3 - \dfrac{\langle v_3, w_1\rangle}{\langle w_1, w_1\rangle}\cdot w_1 - \dfrac{\langle v_3, w_2\rangle}{\langle w_2, w_2\rangle}\cdot w_2$

$\dfrac{\langle v_i, w_j\rangle}{\langle w_j, w_j\rangle}\cdot w_j$: **projection** of v_i onto w_j

Step n. $w_n = v_n - \displaystyle\sum_{i=1}^{n-1}\frac{\langle v_n, w_i\rangle}{\langle w_i, w_i\rangle}\cdot w_i$

Why Gram–Schmidt works

Orthogonality (by induction): if $w_1, \dots, w_{k-1}$ are mutually orthogonal, then for $j < k$ $\langle w_k, w_j\rangle = \langle v_k, w_j\rangle - \sum_{i<k}\frac{\langle v_k, w_i\rangle}{\langle w_i, w_i\rangle}\langle w_i, w_j\rangle = \langle v_k, w_j\rangle - \langle v_k, w_j\rangle = 0$, since only the $i = j$ term survives. *Same span*: each w_k is v_k minus a combination of $w_1, \dots, w_{k-1} \in \mathrm{sp}\{v_1, \dots, v_{k-1}\}$, so $\mathrm{sp}\{w_1, \dots, w_k\} = \mathrm{sp}\{v_1, \dots, v_k\}$ for every k. *No division by zero*: $w_k = 0_{\mathcal{V}}$ would make $v_k \in \mathrm{sp}\{v_1, \dots, v_{k-1}\}$, impossible for a linearly independent S_1; so $\langle w_k, w_k\rangle > 0$. In words: w_k **is the component of v_k orthogonal to the span of the previous vectors.**

Example 3.8. $\mathcal{V} = \left\{ f \;\middle|\; f : [-1,1] \to \mathbb{R},\ \int_{-1}^{1}(f(t))^2\,dt < \infty \right\}$, $\mathcal{F} = \mathbb{R}$,

$$\langle f, g\rangle = \int_{-1}^{1} f(t)g(t)\,dt$$

$S_1 = \{v_1, v_2, v_3\}$, $v_1(t) = 1$, $v_2(t) = t$, $v_3(t) = t^2$, i.e. $S_1 = \{1, t, t^2\}$

p. 24

$w_1 = v_1 = 1$

$$w_2 = t - \frac{\overset{0}{\cancel{\langle t, 1\rangle}}}{\langle 1, 1\rangle}\cdot 1 = t, \qquad \langle t, 1\rangle = \int_{-1}^{1} t\,dt = 0$$

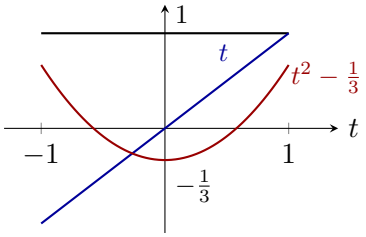
$$w_3 = t^2 - \frac{\langle t^2, 1\rangle}{\langle 1,1\rangle}\cdot 1 - \cancelto{0}{\frac{\langle t^2, t\rangle}{\langle t, t\rangle}}\cdot t = t^2 - \frac{2/3}{2} = t^2 - \frac{1}{3}$$

using

$$\begin{aligned}\left\langle t^2, 1\right\rangle &= \int_{-1}^{1} t^2\,dt = 2\int_0^1 t^2\,dt = \frac{2}{3}\\ \langle 1,1\rangle &= \int_{-1}^{1} 1\,dt = 2\\ \left\langle t^2, t\right\rangle &= \int_{-1}^{1} t^3\,dt = 0\\ \langle t, t\rangle &= \int_{-1}^{1} t^2\,dt = \frac{2}{3}\end{aligned}$$

$$S_2 = \left\{1,\ t,\ t^2 - \tfrac{1}{3}\right\}$$

Legendre polynomials



▷ *Check:* $\left\langle t^2 - \frac{1}{3}, 1\right\rangle = \frac{2}{3} - \frac{1}{3}\cdot 2 = 0$ *and* $\left\langle t^2 - \frac{1}{3}, t\right\rangle = \int_{-1}^{1}(t^3 - \frac{t}{3})\,dt = 0$ *(odd integrand)* ✓. *These are the Legendre polynomials up to scaling: the standard ones are* $P_0 = 1$, $P_1 = t$, $P_2 = \frac{1}{2}(3t^2 - 1) = \frac{3}{2}\left(t^2 - \frac{1}{3}\right)$, *normalised so that* $P_k(1) = 1$.

Linearly dependent vectors. If the vectors are linearly dependent, e.g. $S_1 = \{v_1, v_1\}$:

$$\begin{aligned}w_1 &= v_1\\ w_2 &= v_1 - \cancelto{1}{\frac{\langle v_1, v_1\rangle}{\langle v_1, v_1\rangle}}\cdot v_1 = 0\end{aligned}$$

▷ *When this is the case,* ***remove the redundant vector and resume the process****. In general* $w_k = 0_{\mathcal{V}}$ *exactly when* v_k *depends linearly on* $v_1, \ldots, v_{k-1}$*; discarding such vectors, Gram–Schmidt returns an orthogonal basis of* $\operatorname{sp}\{S_1\}$ *(and its size is* $\dim\operatorname{sp}\{S_1\}$*).*

Gram–Schmidt orthonormalization. Require $\|w_i\| = 1$, i.e. $\langle w_i, w_i\rangle = 1$:

$$\begin{aligned}\bar{w}_1 &= v_1, & w_1 &= \frac{\bar{w}_1}{\|\bar{w}_1\|}\\ \bar{w}_2 &= v_2 - \langle v_2, w_1\rangle\, w_1, & w_2 &= \frac{\bar{w}_2}{\|\bar{w}_2\|}\\ &\vdots\\ \bar{w}_n &= v_n - \sum_{i=1}^{n-1}\langle v_n, w_i\rangle\, w_i, & w_n &= \frac{\bar{w}_n}{\|\bar{w}_n\|}\end{aligned}$$

▷ *Since* $\langle w_i, w_i\rangle = 1$, *the projection coefficients* $\frac{\langle v_k, w_i\rangle}{\langle w_i, w_i\rangle}$ *simplify to* $\langle v_k, w_i\rangle$. *The resulting set is* ***orthonormal****:* $\langle w_i, w_j\rangle = \delta_{ij}$.

4 Linear Transformations

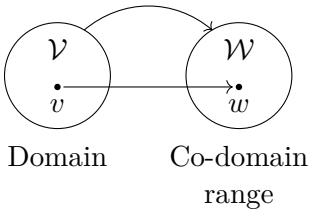
Syllabus: Week 5 — linear transformations, null space, range space, matrix representations

4.1 Definition and Examples

Definition 4.1. Let $\mathcal{V}$ & $\mathcal{W}$ be two vector spaces over the same field $\mathcal{F}$. A **linear transformation** $\mathcal{A}$ is a mapping from $\mathcal{V}$ into $\mathcal{W}$ shown as $\mathcal{A} : \mathcal{V} \to \mathcal{W}$ s.t.

$$\mathcal{A}(\alpha_1 v_1 + \alpha_2 v_2) = \alpha_1 \mathcal{A} v_1 + \alpha_2 \mathcal{A} v_2$$

for all $v_1, v_2 \in \mathcal{V}$ and all $\alpha_1, \alpha_2 \in \mathcal{F}$.



▷ *$\mathcal{V}$ is the **domain** and $\mathcal{W}$ the **co-domain**; the **range** $\mathcal{R}(\mathcal{A}) = \{\mathcal{A}v : v \in \mathcal{V}\}$ is the subset of $\mathcal{W}$ actually reached (in general $\mathcal{R}(\mathcal{A}) \subseteq \mathcal{W}$).*

Explanation

The single condition is equivalent to additivity $\mathcal{A}(v_1 + v_2) = \mathcal{A}v_1 + \mathcal{A}v_2$ together with homogeneity $\mathcal{A}(\alpha v) = \alpha \mathcal{A} v$. Taking $\alpha_1 = \alpha_2 = 0$ gives $\mathcal{A}0_{\mathcal{V}} = 0_{\mathcal{W}}$. Standard examples: $x \mapsto Ax$ for $A \in \mathcal{F}^{m \times n}$ ($\mathcal{V} = \mathcal{F}^n$, $\mathcal{W} = \mathcal{F}^m$, as in the induced-norm section), differentiation $f \mapsto f'$ on polynomials, and $f \mapsto \int_0^1 f(t)\,dt$.

Example 4.2. $\mathcal{V} = \mathcal{W} \triangleq \{\text{polynomials with degree} \le n\}$, $\mathcal{A} = \dfrac{d}{dt}$, p. 25

$$v_1 = \sum_{i=0}^{n} \alpha_i t^i, \qquad v_2 = \sum_{i=0}^{n} \beta_i t^i.$$

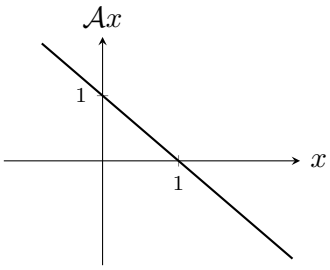
$$\begin{aligned} a_1 v_1 + a_2 v_2 &= \sum_{i=0}^{n} (a_1 \alpha_i + a_2 \beta_i)\, t^i \\ \mathcal{A}(a_1 v_1 + a_2 v_2) &= \sum_{i=1}^{n} (a_1 \alpha_i + a_2 \beta_i) \cdot i\, t^{i-1} \\ &= a_1 \sum_{i=1}^{n} i \alpha_i t^{i-1} + a_2 \sum_{i=1}^{n} i \beta_i t^{i-1} \\ &= a_1 \mathcal{A} v_1 + a_2 \mathcal{A} v_2 \end{aligned}$$

▷ *The constant terms ($i = 0$) differentiate to zero, which is why the sums after differentiation start at $i = 1$. Hence $\frac{d}{dt}$ is linear.*

Example 4.3. $\mathcal{V} = \mathcal{W} = \mathbb{R}^2, \quad x = \begin{bmatrix}\alpha_1\\ \alpha_2\end{bmatrix}, \quad y = \begin{bmatrix}\beta_1\\ \beta_2\end{bmatrix}.$

$$\mathcal{A}x = \begin{bmatrix}\alpha_1\\ \alpha_1+\alpha_2\end{bmatrix}, \qquad ax+by = \begin{bmatrix}a\alpha_1+b\beta_1\\ a\alpha_2+b\beta_2\end{bmatrix}$$
$$\mathcal{A}(ax+by) = \begin{bmatrix}a\alpha_1+b\beta_1\\ a\alpha_1+b\beta_1+a\alpha_2+b\beta_2\end{bmatrix}$$
$$= a\begin{bmatrix}\alpha_1\\ \alpha_1+\alpha_2\end{bmatrix} + b\begin{bmatrix}\beta_1\\ \beta_1+\beta_2\end{bmatrix} = a\mathcal{A}x + b\mathcal{A}y \quad //$$

Example 4.4. $\mathcal{V} = \mathcal{W} = \mathbb{R}, \quad \mathcal{A}x = 1 - x.$



Take $x_1 = 0$, $x_2 = 3$, $\alpha_1 = 1$, $\alpha_2 = 1$:

$$\alpha_1 x_1 + \alpha_2 x_2 = 1\cdot 0 + 1\cdot 3 = 3$$
$$\mathcal{A}(\alpha_1 x_1 + \alpha_2 x_2) = \mathcal{A}3 = 1 - 3 = -2$$
$$\mathcal{A}x_1 = 1, \qquad \mathcal{A}x_2 = -2$$
$$\alpha_1\mathcal{A}x_1 + \alpha_2\mathcal{A}x_2 = -1$$

▷ $-2 \neq -1 \Rightarrow$ ***not a linear transformation.***

The graph should pass through zero: **every linear transformation must satisfy $\mathcal{A}0_V = 0_W$.**

$$\mathcal{A}(\alpha_1 x_1 + \cancel{\alpha_2 x_2}^{0}) = \alpha_1\mathcal{A}x_1 + \cancel{\alpha_2\mathcal{A}x_2}^{0} \qquad \text{(suppose } \alpha_2 = 0\text{)}$$
$$\mathcal{A}(\alpha_1 x_1) = \alpha_1\mathcal{A}(x_1) \qquad \text{(suppose } \alpha_1 = 0\text{)}$$
$$\mathcal{A}(0_V) = 0_W$$

Quick test for linearity

Every linear transformation maps 0_V to 0_W (take $\alpha_1 = 0$ in $\mathcal{A}(\alpha_1 x_1) = \alpha_1\mathcal{A}x_1$). Here $\mathcal{A}0 = 1 - 0 = 1 \neq 0$, so $\mathcal{A}x = 1 - x$ cannot be linear: the graph is a straight line, but it does not pass through the origin (an **affine** map). The converse is false: $\mathcal{A}0 = 0$ alone does not guarantee linearity (e.g. $\mathcal{A}x = x^2$).

Example 4.5. p. 26

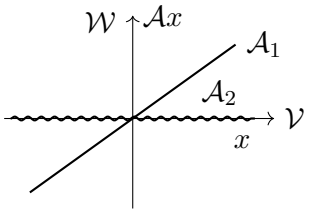
$$\mathcal{V} = \{f \mid f : [0,1] \to \mathbb{R},\ f \text{ is integrable}\}, \qquad \mathcal{W} = \mathbb{R}$$
$$\mathcal{A}f = \int_0^1 f(t)\,dt$$

▷ ✓ *A linear transformation:* $\int_0^1 (af + bg)\,dt = a\int_0^1 f\,dt + b\int_0^1 g\,dt$ *by linearity of the integral.*

4.2 Null Space

Definition 4.6. Let $\mathcal{A} : \mathcal{V} \to \mathcal{W}$ be a linear transformation. The **null space** of $\mathcal{A}$ is defined as

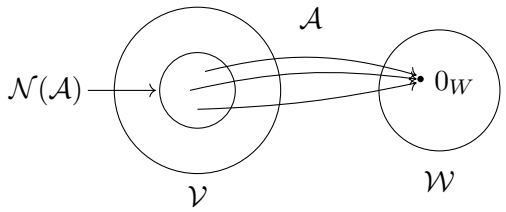
$$\mathcal{N}(\mathcal{A}) \triangleq \{v \in \mathcal{V} \mid \mathcal{A}v = 0_W\}.$$



$$\mathcal{N}(\mathcal{A}_1) = \{0_V\}$$
$$\mathcal{N}(\mathcal{A}_2) = \mathcal{V} = \mathbb{R}$$

▷ *$\mathcal{A}_1$: a line through the origin with nonzero slope (only $x = 0$ maps to 0); $\mathcal{A}_2$: the zero map (every x maps to 0).*

Claim. $\mathcal{N}(\mathcal{A})$ is a subspace of $\mathcal{V}$.



Let $x_1 \,\&\, x_2 \in \mathcal{N}(\mathcal{A})$. Show

$$\left.\begin{aligned} x_1 + x_2 &\overset{?}{\in} \mathcal{N}(\mathcal{A}) \\ \alpha x_1 &\overset{?}{\in} \mathcal{N}(\mathcal{A}) \end{aligned}\right\} \Rightarrow \mathcal{N}(\mathcal{A}) \text{ is a subspace.}$$

We have $\mathcal{A}x_1 = 0_W$, $\mathcal{A}x_2 = 0_W$. Then

$$\mathcal{A}(x_1 + x_2) = \underbrace{\mathcal{A}x_1}_{0_W} + \underbrace{\mathcal{A}x_2}_{0_W} = 0_W \qquad \Rightarrow\ x_1 + x_2 \in \mathcal{N}(\mathcal{A})$$
$$\mathcal{A}(\alpha x_1) = \alpha \mathcal{A}x_1 = \alpha 0_W = 0_W \qquad \Rightarrow\ \alpha x_1 \in \mathcal{N}(\mathcal{A})$$

$\therefore \mathcal{N}(\mathcal{A})$ is closed under addition and scalar multiplication, hence a subspace of $\mathcal{V}$.

Example 4.7. $\mathcal{V} = \{\text{polynomials with degree} \le 2 \text{ on } [0,1]\}, \quad \mathcal{W} = \mathbb{R},$

$$\mathcal{A}f = \int_0^1 f(t)\,dt. \qquad \text{Find a basis for } \mathcal{N}(\mathcal{A}).$$

Let $v = at^2 + bt + c$. Then

$$\begin{aligned} \mathcal{A}v = 0_W \ &\Rightarrow\ \int_0^1 (at^2 + bt + c)\,dt = 0 \\ &\Rightarrow\ \frac{at^3}{3} + \frac{bt^2}{2} + ct\Big|_0^1 = \frac{a}{3} + \frac{b}{2} + c = 0 \\ &\Rightarrow\ c = -\frac{a}{3} - \frac{b}{2}, \qquad \dim \mathcal{N}(\mathcal{A}) = 2 \end{aligned}$$

Hence p. 27

$$\begin{aligned} at^2 + bt + c &= at^2 + bt - \frac{a}{3} - \frac{b}{2} \\ &= a\left(t^2 - \frac{1}{3}\right) + b\left(t - \frac{1}{2}\right) \end{aligned}$$

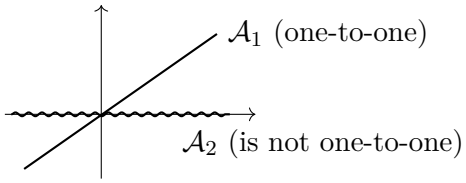
A basis for $\mathcal{N}(\mathcal{A})$: $\left\{t^2 - \frac{1}{3},\ t - \frac{1}{2}\right\}$.

▷ *Check:* $\int_0^1 (t^2 - \frac{1}{3})\,dt = \frac{1}{3} - \frac{1}{3} = 0$, $\int_0^1 (t - \frac{1}{2})\,dt = \frac{1}{2} - \frac{1}{2} = 0$*; the two polynomials have different degrees, so they are linearly independent. Consistent with rank–nullity:* $\dim \mathcal{V} = 3 = \dim \mathcal{N}(\mathcal{A}) + \dim \mathcal{R}(\mathcal{A}) = 2 + 1$.

4.3 One-to-One Transformations

Definition 4.8. A linear transformation $\mathcal{A} : \mathcal{V} \to \mathcal{W}$ is called **one-to-one** iff

$$\mathcal{A}v_1 = \mathcal{A}v_2 \quad \text{implies} \quad v_1 = v_2.$$



Proposition 4.9. A linear transformation $\mathcal{A} : \mathcal{V} \to \mathcal{W}$ is **one-to-one iff** $\mathcal{N}(\mathcal{A}) = \{0_V\}$:

$$\text{one-to-one} \iff \mathcal{N}(\mathcal{A}) = \{0_V\} \iff \dim \mathcal{N}(\mathcal{A}) = \underline{0}.$$

Proof techniques.

$p \Rightarrow q$	(constructive proof)
$\bar{q} \Rightarrow \bar{p}$	(contraposition proof)
$p \wedge \bar{q} = \text{False}$	(contradiction proof)

Proof. **One-to-one** $\Rightarrow \mathcal{N}(\mathcal{A}) = \{0_V\}$**.** We use contraposition ($p \Rightarrow q \ \equiv\ \bar{q} \Rightarrow \bar{p}$), i.e. we show

$$\mathcal{N}(\mathcal{A}) \neq \{0_V\} \ \Rightarrow\ \text{not one-to-one.}$$

$\exists x \neq 0_V$ s.t. $x \in \mathcal{N}(\mathcal{A})$, i.e. $\mathcal{A}x = 0_W$.
Let $v_1 \in \mathcal{V}$ and $v_2 = v_1 + x \in \mathcal{V}$. As $x \neq 0_V \Rightarrow v_2 \neq v_1$.
$\mathcal{A}v_2 = \mathcal{A}v_1 + \underbrace{\mathcal{A}x}_{0_W} = \mathcal{A}v_1 \ \Rightarrow\ \mathcal{A}$ is not one-to-one.

$\mathcal{N}(\mathcal{A}) = \{0_V\} \Rightarrow$ **one-to-one.** Again by contraposition: not one-to-one $\Rightarrow \mathcal{N}(\mathcal{A}) \neq \{0_V\}$.

$$\begin{aligned} &\exists v_1 \neq v_2 \text{ s.t. } \mathcal{A}v_1 = \mathcal{A}v_2 \\ &\Rightarrow\ \mathcal{A}v_1 - \mathcal{A}v_2 = 0_W \\ &\Rightarrow\ \mathcal{A}(v_1 - v_2) = 0_W \\ &\Rightarrow\ 0_V \neq v_1 - v_2 \in \mathcal{N}(\mathcal{A}) \\ &\Rightarrow\ \mathcal{N}(\mathcal{A}) \neq \{0_V\} \end{aligned}$$

□

Why the null space decides injectivity

For a linear map, $\mathcal{A}v_1 = \mathcal{A}v_2 \iff \mathcal{A}(v_1 - v_2) = 0_W \iff v_1 - v_2 \in \mathcal{N}(\mathcal{A})$. So the set of vectors with the same image as v_1 is the shifted copy $v_1 + \mathcal{N}(\mathcal{A})$; images are unique exactly when this copy is the single point v_1, i.e. when $\mathcal{N}(\mathcal{A}) = \{0_V\}$.

Example 4.10. $\mathcal{V} = \mathcal{W} = \{\text{polynomials with degree} \le 2\}$, $\mathcal{A} = \dfrac{d}{dt}$. Is $\mathcal{A}$ one-to-one? p. 28

$$\mathcal{A}(at^2 + bt + c) = 2at + b = 0_W = 0(t) \;\Rightarrow\; a = 0,\; b = 0$$
$$\mathcal{N}(\mathcal{A}) = \{c\} = \{\text{constant polynomials}\}, \qquad \dim \mathcal{N}(\mathcal{A}) = 1$$
$$\mathcal{N}(\mathcal{A}) \neq \{0_V\} \;\Rightarrow\; \textbf{Not one-to-one}$$

▷ *Side note in the notes:* $\mathcal{R}(\mathcal{A}) = \{\text{poly. degree} \le 1\} = \mathrm{sp}\{1, t\}$ *(2 elements,* $\dim \mathcal{R}(\mathcal{A}) = 2$*).*

Example 4.11. $\mathcal{V} = \{f \mid f : [0,1] \to \mathbb{R},\ f \text{ is integrable}\}$, $\mathcal{W} = \mathbb{R}$,

$$\mathcal{A}f = \int_0^1 f(t)\,dt. \qquad \text{Is } \mathcal{A} \text{ one-to-one?}$$

Take

$$f(t) = 2t - 1 \neq 0_V, \qquad \mathcal{A}f = \left(t^2 - t\right)\Big|_0^1 = 0.$$

$\therefore$ Not one-to-one. $(\dim \mathcal{N}(\mathcal{A}) = \infty)$

▷ *E.g.* $t^k - \frac{1}{k+1}$, $k = 1, 2, 3, \ldots$ *all integrate to* 0 *and are linearly independent, so the null space is infinite dimensional.*

Example 4.12. $\mathcal{V} = \mathcal{W} = \mathbb{R}^2$, $x = \begin{bmatrix}\alpha_1 \\ \alpha_2\end{bmatrix}$, $\mathcal{A}x = \begin{bmatrix}\alpha_1 \\ \alpha_1 + \alpha_2\end{bmatrix}$. Is $\mathcal{A}$ one-to-one?

$$\mathcal{A}x = \begin{bmatrix}0 \\ 0\end{bmatrix} = \begin{bmatrix}\alpha_1 \\ \alpha_1 + \alpha_2\end{bmatrix} \;\Rightarrow\; \alpha_1 = \alpha_2 = 0$$
$$\mathcal{N}(\mathcal{A}) = \left\{\begin{bmatrix}0 \\ 0\end{bmatrix}\right\} \checkmark \qquad \cancel{\mathcal{N}(\mathcal{A}) = \begin{bmatrix}0 \\ 0\end{bmatrix}}$$

▷ *Crossed out:* $\mathcal{N}(\mathcal{A})$ *is a subspace (a set), whereas* $\left[\begin{smallmatrix}0 \\ 0\end{smallmatrix}\right]$ *alone is not a set/subset — write it inside braces.*

4.4 Range Space, Onto Transformations and the Rank–Nullity Theorem

Definition 4.13. Let $\mathcal{A} : \mathcal{V} \to \mathcal{W}$ be a linear transformation. The **range space** of $\mathcal{A}$ is defined as

$$\mathcal{R}(\mathcal{A}) \triangleq \{w \in \mathcal{W} \mid \exists v \in \mathcal{V} \text{ s.t. } \mathcal{A}v = w\}.$$

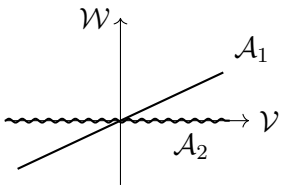
Claim. $\mathcal{R}(\mathcal{A})$ is a subspace of $\mathcal{W}$.

Let $w_1, w_2 \in \mathcal{R}(\mathcal{A})$. By the definition of range space,

$$\exists v_1, v_2 \text{ s.t. } \mathcal{A}v_1 = w_1, \quad \mathcal{A}v_2 = w_2$$
$$\mathcal{A}(v_1 + v_2) = \mathcal{A}v_1 + \mathcal{A}v_2 = w_1 + w_2 \;\Rightarrow\; w_1 + w_2 \in \mathcal{R}(\mathcal{A})$$
$$\mathcal{A}(\alpha v_1) = \alpha \mathcal{A}v_1 = \alpha w_1 \;\Rightarrow\; \alpha w_1 \in \mathcal{R}(\mathcal{A})$$

$\therefore$ $\mathcal{R}(\mathcal{A})$ is closed under addition and scalar multiplication, hence a subspace of $\mathcal{W}$.

Example 4.14. $\mathcal{V} = \mathcal{W} = \mathbb{R}$. p. 29



$$\begin{aligned} \mathcal{N}(\mathcal{A}_1) &= \{0\}, & \dim \mathcal{N}(\mathcal{A}_1) &= 0 \\ \mathcal{N}(\mathcal{A}_2) &= \mathcal{V} = \mathbb{R}, & \dim \mathcal{N}(\mathcal{A}_2) &= 1 \\ \mathcal{R}(\mathcal{A}_1) &= \mathcal{W} = \mathbb{R}, & \dim \mathcal{R}(\mathcal{A}_1) &= 1 \\ \mathcal{R}(\mathcal{A}_2) &= \{0\}, & \dim \mathcal{R}(\mathcal{A}_2) &= 0 \end{aligned}$$

▷ $\dim \mathcal{N}(\mathcal{A}_1) + \dim \mathcal{R}(\mathcal{A}_1) = 0 + 1 = 1, \quad \dim \mathcal{N}(\mathcal{A}_2) + \dim \mathcal{R}(\mathcal{A}_2) = 1 + 0 = 1.$

Definition 4.15. A linear transformation $\mathcal{A} : \mathcal{V} \to \mathcal{W}$ is called **onto** iff $\mathcal{R}(\mathcal{A}) = \mathcal{W}$; otherwise, if $\mathcal{R}(\mathcal{A}) \neq \mathcal{W}$ (i.e. $\mathcal{R}(\mathcal{A}) \subsetneq \mathcal{W}$), $\mathcal{A}$ is called **into**.

Example 4.16. $\mathcal{V} = \mathcal{W} = \{\text{polynomials with degree} \leq 2\}$ $(\to \dim 3)$,

$$\mathcal{A} = \frac{d}{dt}, \qquad \mathcal{A}(at^2 + bt + c) = 2at + b.$$

$$\begin{aligned} \mathcal{N}(\mathcal{A}) &= \{\text{constant polynomials}\}, & \dim \mathcal{N}(\mathcal{A}) &= 1 \\ \mathcal{R}(\mathcal{A}) &= \{\text{polynomials with degree} \leq 1\}, & \dim \mathcal{R}(\mathcal{A}) &= 2 \end{aligned}$$

$\mathcal{R}(\mathcal{A}) \neq \mathcal{W} \Rightarrow \therefore \mathcal{A}$ is not onto.
▷ *Sum:* $1 + 2 = 3$, *the same as* $\dim \mathcal{V}$.

Example 4.17. $\mathcal{V} = \{f \mid f : [0,1] \to \mathbb{R},\ f \text{ is integrable}\}, \quad \mathcal{W} = \mathbb{R}$,

$$\mathcal{A}f = \int_0^1 f(t)\,dt, \qquad \mathcal{R}(\mathcal{A}) = \mathbb{R}.$$

▷ *For any* $c \in \mathbb{R}$, *the constant function* $f(t) = c$ *gives* $\mathcal{A}f = c$, *so* $\mathcal{A}$ *is onto (but not one-to-one, see Example 4.11).*

$$\left.\begin{aligned} \dim \mathcal{N}(\mathcal{A}) &\triangleq \textbf{nullity} \\ \dim \mathcal{R}(\mathcal{A}) &\triangleq \textbf{rank} \end{aligned}\right\} \xrightarrow{+} \text{constant}$$

Theorem 4.18 (Rank–Nullity Theorem)**.** If $\mathcal{V}$ is finite dimensional and $\mathcal{A}$ is a linear transformation from $\mathcal{V}$ into $\mathcal{W}$, then $\mathcal{R}(\mathcal{A})$ is finite dimensional and **nullity** + **rank** $= \dim \mathcal{V}$:

$$\dim \mathcal{N}(\mathcal{A}) + \dim \mathcal{R}(\mathcal{A}) = \dim \mathcal{V}.$$

Proof sketch and consequences

Take a basis $\{u_1, \ldots, u_r\}$ of $\mathcal{N}(\mathcal{A})$ and extend it to a basis $\{u_1, \ldots, u_r, v_{r+1}, \ldots, v_n\}$ of $\mathcal{V}$. Then $\mathcal{R}(\mathcal{A}) = \text{sp}\{\mathcal{A}v_{r+1}, \ldots, \mathcal{A}v_n\}$ (since $\mathcal{A}u_i = 0_W$), and these $n - r$ images are linearly independent: $\sum \beta_j \mathcal{A}v_j = 0_W \Rightarrow \sum \beta_j v_j \in \mathcal{N}(\mathcal{A}) \Rightarrow$ all $\beta_j = 0$. Hence $\dim \mathcal{R}(\mathcal{A}) = n - r$.

Consequences: $\mathcal{A}$ one-to-one $\iff$ $\dim \mathcal{R}(\mathcal{A}) = \dim \mathcal{V}$; and if $\dim \mathcal{V} = \dim \mathcal{W} < \infty$, then one-to-one $\iff$ onto. In the examples above: $\frac{d}{dt}$ on polynomials of degree ≤ 2: nullity + rank $= 1 + 2 = 3 = \dim \mathcal{V}$, so it is neither one-to-one nor onto.

4.5 Matrix Representations

Definition 4.19. Let $\mathcal{V}$ (dim $= n$) and $\mathcal{W}$ (dim $= m$) be two linear spaces over the same field $\mathcal{F}$. Let $\mathcal{A} : \mathcal{V} \to \mathcal{W}$ be a linear transformation. Given the bases $B = \{v_1, v_2, \ldots, v_n\}$ and $C = \{w_1, w_2, \ldots, w_m\}$ for $\mathcal{V}$ and $\mathcal{W}$, the transformation can be represented by a matrix $A = [a_{ij}]$ ($a_{ij} \in \mathcal{F}$), its **matrix representation**, s.t.

$$[w]_C = A\,[v]_B$$

for any w, v satisfying $w = \mathcal{A}v$.

▷ *A is $m \times n$; its jth column is the coordinate vector $[\mathcal{A}v_j]_C$ (procedure below). The matrix depends on the chosen bases B and C.*

Example 4.20. p. 30

$$\mathcal{V} = \{\text{polynomials with degree} \le 3\}, \qquad \mathcal{W} = \{\text{polynomials with degree} \le 2\}$$
$$\mathcal{A} : \mathcal{V} \to \mathcal{W}, \qquad \mathcal{A}f = \frac{df}{dt}$$
$$B = \{\underbrace{1}_{v_1}, \underbrace{1+t}_{v_2}, \underbrace{1+t+t^2}_{v_3}, \underbrace{1+t+t^2+t^3}_{v_4}\}$$
$$C = \{\underbrace{1}_{w_1}, \underbrace{1+t}_{w_2}, \underbrace{1+t+t^2}_{w_3}\}$$
$$[w]_C = A\,[v]_B \qquad \text{e.g. } [v]_B = \begin{bmatrix}1\\0\\0\\0\end{bmatrix}$$

Procedure.

* Choose $v = v_j$ for $j = 1, \ldots, n$:

$$[v]_B = e_j = \begin{bmatrix}0\\0\\\vdots\\1\\\vdots\\0\end{bmatrix} \leftarrow j\text{th element}$$

▷ *$A\,e_j$ picks out the jth column of A: this gives the first column for $j = 1$, and in general the jth column.*

* Find $w = \mathcal{A}v_j$.
* **jth column of A is $[\mathcal{A}v_j]_C$.**

Columns of A.

$$v = v_1 : \quad [v_1]_B = \begin{bmatrix}1\\0\\0\\0\end{bmatrix}, \quad \mathcal{A}v_1 = 0 = 0w_1 + 0w_2 + 0w_3 \qquad \Rightarrow\ 1^{\text{st}} \text{ column of } A \text{ is } \begin{bmatrix}0\\0\\0\end{bmatrix}$$

$$v = v_2: \quad [v_2]_B = \begin{bmatrix}0\\1\\0\\0\end{bmatrix}, \quad \mathcal{A}v_2 = 1 = 1w_1 + 0w_2 + 0w_3 \quad \Rightarrow \ 2^{\text{nd}} \text{ column of } A \text{ is } \begin{bmatrix}1\\0\\0\end{bmatrix}$$

$$v = v_3: \quad [v_3]_B = \begin{bmatrix}0\\0\\1\\0\end{bmatrix}, \quad \mathcal{A}v_3 = 2t + 1 = -w_1 + 2w_2 + 0w_3 \quad \Rightarrow \ 3^{\text{rd}} \text{ column of } A \text{ is } \begin{bmatrix}-1\\2\\0\end{bmatrix}$$

$$v = v_4: \quad [v_4]_B = \begin{bmatrix}0\\0\\0\\1\end{bmatrix}, \quad \mathcal{A}v_4 = 3t^2 + 2t + 1$$

$$= -1w_1 - 1w_2 + 3w_3 \quad \Rightarrow \ 4^{\text{th}} \text{ column of } A \text{ is } \begin{bmatrix}-1\\-1\\3\end{bmatrix}$$

$$\boxed{A = \begin{bmatrix}0 & 1 & -1 & -1\\0 & 0 & 2 & -1\\0 & 0 & 0 & 3\end{bmatrix}}$$

▷ *Coordinates in C: $2t+1 = -1\cdot 1 + 2(1+t)$; $3t^2+2t+1 = 3(1+t+t^2) - 1\cdot(1+t) - 1\cdot 1$ (match the t^2, then t, then constant coefficients).*

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p. 31

Check. Let $v = t^3 + 2t^2 + 3t + 4$. Then

$$[v]_B = \begin{bmatrix}1\\1\\1\\1\end{bmatrix}, \qquad [w]_C = A\,[v]_B = \begin{bmatrix}-1\\1\\3\end{bmatrix}$$

$$\begin{aligned} w &= -w_1 + w_2 + 3w_3 \\ &= -1 + 1 + t + 3 + 3t + 3t^2 = 3t^2 + 4t + 3 \quad // \end{aligned}$$

▷ *Indeed $\frac{d}{dt}(t^3+2t^2+3t+4) = 3t^2+4t+3$. ($[v]_B$: $v = 1\cdot v_1 + 1\cdot v_2 + 1\cdot v_3 + 1\cdot v_4$, since $v_1+v_2+v_3+v_4 = 4+3t+2t^2+t^3$.)*

4.6 Change of Basis and Similar Matrices

1) Basis change in the domain. Let $[v]_B = P\,[v]_{\bar{B}}$.

$$\begin{aligned} [w]_C &= A\,[v]_B \\ [w]_C &= \bar{A}\,[v]_{\bar{B}} \\ [w]_C &= A \cdot P\,[v]_{\bar{B}} \end{aligned}$$

$$\therefore \ \boxed{\bar{A} = A \cdot P}$$

2) Basis change in the co-domain. Let $[w]_C = Q\,[w]_{\bar{C}}$.

$$\begin{aligned} [w]_C &= A\,[v]_B \\ [w]_{\bar{C}} &= \bar{A}\,[v]_B \\ Q\,[w]_{\bar{C}} &= A\,[v]_B \\ [w]_{\bar{C}} &= Q^{-1}A\,[v]_B \end{aligned}$$

$$\therefore \boxed{\bar{A} = Q^{-1} \cdot A}$$

3) Basis change in both domain and co-domain. With $[v]_B = P\,[v]_{\bar{B}}$ and $[w]_C = Q\,[w]_{\bar{C}}$:

$$\begin{aligned} [w]_C &= A\,[v]_B \\ [w]_{\bar{C}} &= \bar{A}\,[v]_{\bar{B}} \\ Q\,[w]_{\bar{C}} &= A\,P\,[v]_{\bar{B}} \\ [w]_{\bar{C}} &= Q^{-1}AP\,[v]_{\bar{B}} \end{aligned}$$

$$\therefore \boxed{\bar{A} = Q^{-1}AP}$$

Special case. If $\mathcal{V} = \mathcal{W}$, $\mathcal{A}$ is a linear transformation from $\mathcal{V}$ into $\mathcal{V}$. Then the basis B can be taken to be the same as C. Then a basis change in both the domain and co-domain would give

$$\therefore \boxed{\bar{A} = P^{-1}AP}$$

▷ *Here $Q = P$ since the same basis change is applied to domain and co-domain; A and $\bar{A}$ are then called* ***similar*** *matrices.*

Example 4.21.

$$\begin{aligned} \mathcal{V} &= \{\text{polynomials with degree} \le 3\}, & B &= \{1,\ 1+t,\ 1+t+t^2,\ 1+t+t^2+t^3\} \\ \mathcal{W} &= \{\text{polynomials with degree} \le 2\}, & C &= \{1,\ 1+t,\ 1+t+t^2\} \end{aligned}$$

$$A = \begin{bmatrix} 0 & 1 & -1 & -1 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

∗ Change the basis in the domain to $\bar{B} = \{1, t, t^2, t^3\}$: p. 32

$$\bar{A} = AP, \qquad [v]_B = P\,[v]_{\bar{B}}$$

$$P = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \bar{A} = AP = \begin{bmatrix} 0 & 1 & -2 & 0 \\ 0 & 0 & 2 & -3 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

▷ *Column j of P is $[\bar{v}_j]_B$: $1 = v_1$, $t = v_2 - v_1$, $t^2 = v_3 - v_2$, $t^3 = v_4 - v_3$. Check of $\bar{A}$: $\frac{d}{dt}t^2 = 2t = -2 \cdot 1 + 2(1+t)$ and $\frac{d}{dt}t^3 = 3t^2 = 3(1+t+t^2) - 3(1+t)$, matching columns 3 and 4.*

∗ Change the basis in the co-domain to the canonical basis $\bar{C} = \{1, t, t^2\}$, while keeping the canonical basis $\bar{B}$ in the domain:

$$\begin{aligned} [w]_{\bar{C}} &= \bar{\bar{A}}\,[v]_{\bar{B}} \\ \bar{\bar{A}} &= Q^{-1}\bar{A}, & [w]_C &= Q\,[w]_{\bar{C}} \end{aligned}$$

$$\bar{\bar{A}} = Q^{-1}\bar{A}, \qquad R\,[w]_C = [w]_{\bar{C}}$$
$$\bar{\bar{A}} = R\bar{A}, \qquad [w]_{\bar{C}} = R\,[w]_C$$

$$R = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = Q^{-1}, \qquad \bar{\bar{A}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

▷ $Q^{-1}\bar{A} \Rightarrow$ *linear combinations of rows of* $\bar{A}$ *(row 1 =* $\text{rows } 1+2+3$*, row 2 =* $\text{rows } 2+3$*, row 3 unchanged). Column* j *of* R *is* $[w_j]_{\bar{C}}$*:* $1,\ 1+t,\ 1+t+t^2$.

Check

$\bar{\bar{A}}$ is the matrix of $\frac{d}{dt}$ in the canonical bases: $1 \mapsto 0$, $t \mapsto 1$, $t^2 \mapsto 2t$, $t^3 \mapsto 3t^2$, which gives exactly the columns $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}$. Altogether $\bar{\bar{A}} = Q^{-1}AP = RAP$.

5 Direct Sums and Orthogonal Projections

Syllabus: Week 6 — direct sum decompositions, projection theorem, orthogonal projections

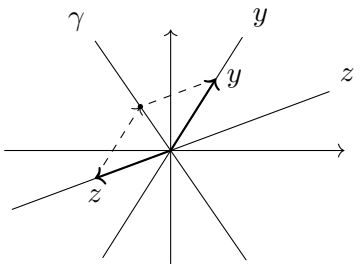
5.1 Direct Sums

Definition 5.1. Let $\mathcal{V}$ be a vector space and let $M_1, \dots, M_k$ be subspaces of $\mathcal{V}$. The **sum** of these subspaces is defined as

$$M = M_1 + M_2 + \cdots + M_k = \{m \in \mathcal{V} \mid m = m_1 + m_2 + \cdots + m_k,\ m_i \in M_i\ \forall i\}.$$

The subspaces are called **linearly independent** iff

$$m_1 + m_2 + \cdots + m_k = 0_V \quad \text{where } m_i \in M_i\ \forall i \quad \Rightarrow \quad m_1 = m_2 = \cdots = m_k = 0_V.$$



$y + z - \gamma = 0$
$\Rightarrow$ not linearly independent

Remark. Let $\mathcal{V} = M_1 + M_2 + \cdots + M_k$ and let $M_1, \dots, M_k$ be linearly independent. Then for $x \in \mathcal{V}$, p. 33

$$x = m_1 + m_2 + \cdots + m_k, \qquad m_i \in M_i\ \forall i,$$

$m_1, m_2, \dots, m_k$ **are unique.**

Proof. Assume $m_1, \dots, m_k$ are not unique, i.e. $\exists i$ s.t. $m_i \neq \bar{m}_i$:

$$\begin{aligned} x &= m_1 + \cdots + m_k \\ x &= \bar{m}_1 + \cdots + \bar{m}_k \\ 0_V &= \underbrace{(m_1 - \bar{m}_1)}_{\in M_1} + \underbrace{(m_2 - \bar{m}_2)}_{\in M_2} + \cdots + \underbrace{(m_k - \bar{m}_k)}_{\in M_k} \end{aligned}$$

$\Rightarrow M_1, \dots, M_k$ are not linearly independent. □

Definition 5.2. Let $M_1, \dots, M_k$ be subspaces of $\mathcal{V}$. Let

i-) $M = M_1 + M_2 + \cdots + M_k$

ii-) $M_1, M_2, \dots, M_k$ are linearly independent.

Then M is said to be the **direct sum** of $M_1, \dots, M_k$. This is shown as

$$M = M_1 \oplus M_2 \oplus \cdots \oplus M_k.$$

Example 5.3. $\mathcal{V} = \mathbb{R}^4, \quad x = (x_1, x_2, x_3, x_4)^T \in \mathbb{R}^4.$

$$M_1 \triangleq \{x \in \mathbb{R}^4 \mid x_4 = x_3 = 0\} : \quad \begin{bmatrix} x_1 \\ x_2 \\ 0 \\ 0 \end{bmatrix}, \quad \dim = 2$$

$$M_2 \triangleq \{x \in \mathbb{R}^4 \mid x_1 = x_2 = 0\} : \quad \begin{bmatrix} 0 \\ 0 \\ x_3 \\ x_4 \end{bmatrix}, \quad \dim = 2$$

$$M_3 \triangleq \{x \in \mathbb{R}^4 \mid x_1 = 0\} : \quad \begin{bmatrix} 0 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \quad \dim = 3$$

$M_1 + M_2 = \mathbb{R}^4$ $\Rightarrow$ direct sum: $M_1 \,\&\, M_2$ linearly independent $\Rightarrow M_1 \oplus M_2 = \mathbb{R}^4$
$M_1 + M_3 = \mathbb{R}^4$ $\Rightarrow$ not direct sum
$M_2 + M_3 = M_3$ $\Rightarrow$ not direct sum ($M_2 \subset M_3$)

For M_1 and M_3:

$$\begin{bmatrix} x_1 \\ x_2 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ x_2' \\ x_3' \\ x_4' \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \qquad \text{e.g.} \quad \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}}_{\in M_1} + \underbrace{\begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \end{bmatrix}}_{\in M_3} = \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}}_{0_V}$$

As the two summands are nonzero, $M_1 \,\&\, M_3$ are linearly dependent. $\therefore$ Not a direct sum.
$M_2 \,\&\, M_3$ are linearly dependent (e.g. $m_2 = (0,0,1,0)^T \in M_2$, $m_3 = -m_2 \in M_3$). $\therefore$ Not a direct sum.

Dimension test

For two subspaces, $M_1 + M_2$ is direct $\iff$ $M_1 \cap M_2 = \{0_V\}$ $\iff$ $\dim(M_1 + M_2) = \dim M_1 + \dim M_2$. Here: $M_1 \cap M_2 = \{0\}$ and $2+2=4$ (direct); $M_1 \cap M_3 = \{(0, x_2, 0, 0)^T\}$, $2+3=5 \neq 4$ (not direct); $M_2 \cap M_3 = M_2$, $2+3 \neq 3$ (not direct).

5.2 Orthogonal Direct Sums and Orthogonal Complements

Definition 5.4. Let $\mathcal{V}$ be an inner product space. Two subspaces $M_1 \,\&\, M_2$ are said to be **orthogonal** iff p. 34

$$\langle m_1, m_2 \rangle = 0 \qquad \forall m_1 \in M_1,\ \forall m_2 \in M_2.$$

We denote this $M_1 \perp M_2$.

Definition 5.5. Let M be equal to $M_1 + M_2 + \cdots + M_k$ and $M_i \perp M_j$ for $i \neq j$. Then M is said to be an **orthogonal direct sum** of $M_1, \ldots, M_k$. We show this as

$$M = M_1 \overset{\perp}{\oplus} M_2 \overset{\perp}{\oplus} \ldots \overset{\perp}{\oplus} M_k.$$

If $M = \mathcal{V}$ then

$$\mathcal{V} = M_1 \overset{\perp}{\oplus} M_2 \overset{\perp}{\oplus} \ldots \overset{\perp}{\oplus} M_k$$

is called an **orthogonal direct sum decomposition** of $\mathcal{V}$.

Orthogonality implies linear independence. Let $\sum_{i=1}^{k} m_i = 0_V, \;\; m_i \in M_i$. Then

$$\left\langle \sum_{i=1}^{k} m_i,\, m_j \right\rangle = \langle 0_V, m_j \rangle = 0$$
$$\sum_{i=1}^{k} \langle m_i, m_j \rangle = 0$$

Since $\langle m_i, m_j \rangle = 0$ when $i \neq j$:

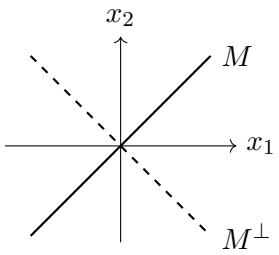
$$\langle m_j, m_j \rangle = 0 \;\Rightarrow\; m_j = 0_V.$$

▷ *The last step uses positive definiteness of the inner product. Since j was arbitrary, all $m_j = 0_V$: orthogonality implies linear independence.*

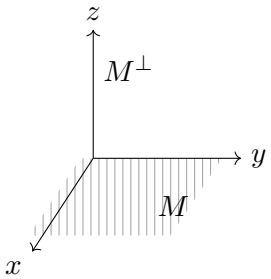
Definition 5.6. Let M be a subspace of an inner product space $\mathcal{V}$. The **orthogonal complement** $M^\perp$ of the subspace is defined as

$$M^\perp = \{x \in \mathcal{V} \mid \langle x, m \rangle = 0 \;\forall m \in M\}.$$

The zero vector is orthogonal to every vector, so $0_V \in M^\perp$ always. Two examples:



$$M = \operatorname{sp}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}, \qquad \begin{bmatrix}-1\\1\end{bmatrix} \in M^\perp, \quad \begin{bmatrix}2\\-2\end{bmatrix} \in M^\perp$$
$$M^\perp = \operatorname{sp}\left\{\begin{bmatrix}-1\\1\end{bmatrix}\right\}, \quad M + M^\perp = M \overset{\perp}{\oplus} M^\perp$$



$$M = \{x\text{-}y \text{ plane}\}$$
$$M^\perp = \{z \text{ axis}\}$$
$$\underset{\dim=2}{M} \overset{\perp}{\oplus} \underset{\dim=1}{M^\perp} = \underset{\dim 3}{\mathbb{R}^3}$$

Orthogonality implies linear independence, thus this is a direct sum; since the summands are also orthogonal, it is an orthogonal direct sum.

Claim. $M^\perp$ **is a subspace of** $\mathcal{V}$**.**

Let $x, y \in M^\perp$, i.e. $\langle x, m\rangle = 0$, $\langle y, m\rangle = 0\ \forall m \in M$. Then

$$\begin{aligned} \langle x+y, m\rangle &= \underbrace{\langle x, m\rangle + \langle y, m\rangle}_{0} \quad (m \in M) \\ &= 0 \quad \forall m \in M && \Rightarrow\ x+y \in M^\perp \\ \langle \alpha x, m\rangle &= \alpha \langle x, m\rangle = 0 \quad \forall m \in M && \Rightarrow\ \alpha x \in M^\perp \end{aligned}$$

$\therefore$ $M^\perp$ is closed under addition and scalar multiplication, hence a subspace of $\mathcal{V}$.

Theorem 5.7 (Orthogonal decomposition)**.** Let $\mathcal{V}$ be an inner product space and M a subspace of $\mathcal{V}$. Then $\mathcal{V}$ **is the orthogonal direct sum of** M **and** $M^\perp$: p. 35

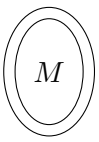
$$\boxed{\mathcal{V} = M \overset{\perp}{\oplus} M^\perp}$$

To prove this we check:

- $\mathcal{V} = M + M^\perp$ (to be shown, proof below);
- $M \,\&\, M^\perp$ are linearly independent ✓;
- $M \perp M^\perp$ ✓.

Proof. **Part I:** $\dim M = n \triangleq \dim \mathcal{V} \Rightarrow M = \mathcal{V}$ ⍰

▷ *Answer to* ⍰*: a basis of* M *consists of* n *linearly independent vectors of* $\mathcal{V}$*; in an* n*-dimensional space any* n *independent vectors form a basis, so they span* $\mathcal{V}$ *and* $M = \mathcal{V}$*.*



$\dim \mathcal{V} = n$

$$\begin{aligned} M^\perp = \mathcal{V}^\perp &= \{0_V\} \\ \mathcal{V} &= M + M^\perp \\ x = \underset{\in M}{x_1} + \underset{\in M^\perp}{x_2} \ &\Rightarrow\ x = \underset{\in M}{x} + \underset{\in M^\perp}{0_V} \quad \checkmark \end{aligned}$$

Part II: $\dim M < n$, with $\dim M = k$. Let

$$\begin{aligned} B &= \{m_1, m_2, m_3, \ldots, m_k\} && \text{(basis for } M\text{)} \\ C &= \{m_1, m_2, \ldots, m_k, m_{k+1}, \ldots, m_n\} && \text{(basis for } \mathcal{V}\text{)} \end{aligned}$$

Apply Gram–Schmidt orthogonalization to C:

$$\bar{C} = \{ \underbrace{\underset{\uparrow m_1}{e_1}, e_2, \ldots, e_k}_{\text{form a basis for } M}, \overbrace{e_{k+1}, \ldots, e_n}^{\text{basis for } M^\perp}\}, \qquad \operatorname{sp}\{e_1, \ldots, e_k\} = \operatorname{sp}\{m_1, \ldots, m_k\}.$$

$\bar{C}$ is a basis for $\mathcal{V}$. For $x \in \mathcal{V}$:

$$x = \sum_{i=1}^{n} \alpha_i e_i = \underbrace{\sum_{i=1}^{k} \alpha_i e_i}_{x_1 \in M} + \underbrace{\sum_{j=k+1}^{n} \alpha_j e_j}_{x_2 \in M^\perp}$$

▷ $x_2 \in M^\perp$ *by construction: Gram–Schmidt makes each* e_j $(j > k)$ *orthogonal to* $e_1, \ldots, e_k$*, which span* M*; hence* $e_j \perp m$ *for all* $m \in M$*, and so is every combination of them.*

$$\Rightarrow\ x = \underset{\in M}{x_1} + \underset{\in M^\perp}{x_2} \ \Rightarrow\ \mathcal{V} = M + M^\perp \qquad \square$$

The remaining two items

The other two items are immediate: $M \perp M^\perp$ by the definition of $M^\perp$, and orthogonality implies linear independence (shown above), equivalently $M \cap M^\perp = \{0_V\}$ since $x \in M \cap M^\perp \Rightarrow \langle x, x\rangle = 0$. Consequently $\dim M^\perp = \dim \mathcal{V} - \dim M$, and $\{e_{k+1}, \dots, e_n\}$ is in fact a basis of $M^\perp$. (Finite dimension of $\mathcal{V}$ is assumed in the proof.)

Example 5.8. $\mathcal{V} = \mathbb{R}^4$,

$$M = \operatorname{sp}\left\{ \begin{bmatrix}1\\2\\3\\4\end{bmatrix}, \begin{bmatrix}4\\3\\2\\1\end{bmatrix} \right\}. \qquad \text{Find a basis for } M^\perp.$$

Complete the basis of M with two canonical vectors to a basis of $\mathbb{R}^4$ and orthogonalize:

$$C = \Big\{ \underbrace{\begin{bmatrix}1\\2\\3\\4\end{bmatrix}}_{m_1}, \underbrace{\begin{bmatrix}4\\3\\2\\1\end{bmatrix}}_{m_2}, \underbrace{\begin{bmatrix}1\\0\\0\\0\end{bmatrix}}_{m_3}, \underbrace{\begin{bmatrix}0\\1\\0\\0\end{bmatrix}}_{m_4} \Big\} \xrightarrow{\text{Gram–Schmidt}} \bar{C} = \{e_1, e_2, e_3, e_4\}$$

$$e_1 = m_1 = \begin{bmatrix}1\\2\\3\\4\end{bmatrix}, \qquad \langle e_1, e_1\rangle = 30$$

$$e_2 = m_2 - \frac{\langle m_2, e_1\rangle}{\langle e_1, e_1\rangle} e_1 = \begin{bmatrix}4\\3\\2\\1\end{bmatrix} - \frac{20}{30}\begin{bmatrix}1\\2\\3\\4\end{bmatrix} = \frac{5}{3}\begin{bmatrix}2\\1\\0\\-1\end{bmatrix} \;\to\; \begin{bmatrix}2\\1\\0\\-1\end{bmatrix}$$

$$e_3 = m_3 - \frac{\langle m_3, e_1\rangle}{\langle e_1, e_1\rangle} e_1 - \frac{\langle m_3, e_2\rangle}{\langle e_2, e_2\rangle} e_2 = \begin{bmatrix}1\\0\\0\\0\end{bmatrix} - \frac{1}{30}\begin{bmatrix}1\\2\\3\\4\end{bmatrix} - \frac{2}{6}\begin{bmatrix}2\\1\\0\\-1\end{bmatrix} = \frac{1}{10}\begin{bmatrix}3\\-4\\-1\\2\end{bmatrix} \;\to\; \begin{bmatrix}3\\-4\\-1\\2\end{bmatrix}$$

$$e_4 = m_4 - \frac{\langle m_4, e_1\rangle}{\langle e_1, e_1\rangle} e_1 - \frac{\langle m_4, e_2\rangle}{\langle e_2, e_2\rangle} e_2 - \frac{\langle m_4, e_3\rangle}{\langle e_3, e_3\rangle} e_3$$

$$= \begin{bmatrix}0\\1\\0\\0\end{bmatrix} - \frac{2}{30}\begin{bmatrix}1\\2\\3\\4\end{bmatrix} - \frac{1}{6}\begin{bmatrix}2\\1\\0\\-1\end{bmatrix} + \frac{4}{30}\begin{bmatrix}3\\-4\\-1\\2\end{bmatrix} = \frac{1}{6}\begin{bmatrix}0\\1\\-2\\1\end{bmatrix} \;\to\; \begin{bmatrix}0\\1\\-2\\1\end{bmatrix}$$

▷ *Scaling a vector does not affect orthogonality, so after each step the vector is rescaled to integer entries ($\to$) and the scaled version is used in the following steps.*

$$\boxed{M^\perp = \operatorname{sp}\{e_3, e_4\} = \operatorname{sp}\left\{ \begin{bmatrix}3\\-4\\-1\\2\end{bmatrix}, \begin{bmatrix}0\\1\\-2\\1\end{bmatrix} \right\}}$$

Verification

$\langle e_3, m_1\rangle = 3 - 8 - 3 + 8 = 0$, $\langle e_3, m_2\rangle = 12 - 12 - 2 + 2 = 0$, $\langle e_4, m_1\rangle = 2 - 6 + 4 = 0$, $\langle e_4, m_2\rangle = 3 - 4 + 1 = 0$, and $\langle e_3, e_4\rangle = -4 + 2 + 2 = 0$. Since $e_3, e_4 \neq 0$, the set C was indeed a basis of $\mathbb{R}^4$.

Alternatively,

$$x \in M^\perp \iff \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{bmatrix} x = 0, \qquad \text{i.e.} \qquad M^\perp = \mathcal{N}\left(\begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{bmatrix}\right),$$

which has dimension $4 - 2 = 2$ and is spanned e.g. by $(1, -2, 1, 0)^T$ and $(0, 1, -2, 1)^T$; note $(3, -4, -1, 2)^T = 3(1, -2, 1, 0)^T + 2(0, 1, -2, 1)^T$, so both answers describe the same subspace. Gram–Schmidt additionally gives an *orthogonal* basis.

5.3 The Projection Theorem

Theorem 5.9 (Hilbert's Projection Theorem)**.** Let H be a Hilbert space and let M be a finite dimensional subspace of H. For any $x \in H$ the following minimization problem has a solution: p. 36

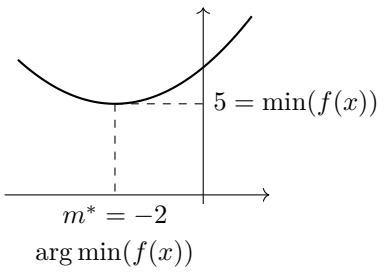
$$m^* = \arg\min_{m \in M} \|x - m\|^2$$

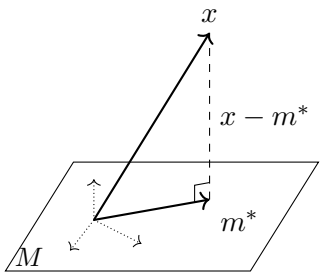
- the exponent 2 does not matter, the norm is positive;
- $\|\cdot\|$ is the norm induced by the inner product.

In other words, we can find the closest vector m^* to x lying in the subspace M. If m^* is a solution, then $x - m^*$ **is orthogonal to** M. Furthermore, **the solution** m^* **is unique**:

$$m^* = \mathrm{Proj}_M x : \text{ “}\textbf{Orthogonal Projection} \text{ of } x \text{ onto } M\text{”}$$

▷ *Reminder on* arg min *vs.* min *(left), and the geometry of the theorem (right).*





⇒ solution exists
⇒ solution is unique
⇒ $x - m^* \perp M$

$$H = M \overset{\perp}{\oplus} M^{\perp}, \qquad x = \underbrace{x_1}_{\in M} + \underbrace{x_2}_{\in M^{\perp}}$$

For any $m \in M$:

$$\begin{aligned}
\|x - m\|^2 &= \langle x - m, x - m \rangle = \langle x_1 + x_2 - m, x_1 + x_2 - m \rangle \\
&= \Big\langle \underbrace{(\underbrace{x_1}_{\in M} - \underbrace{m}_{\in M})}_{\in M} + \underbrace{x_2}_{\in M^{\perp}},\ (x_1 - m) + x_2 \Big\rangle \\
&= \langle x_1 - m, x_1 - m \rangle + \underbrace{\Big\langle \underbrace{x_1 - m}_{\in M},\ \underbrace{x_2}_{\in M^{\perp}} \Big\rangle}_{0} + \underbrace{\langle x_2, x_1 - m \rangle}_{0} + \langle x_2, x_2 \rangle \\
&= \|x_1 - m\|^2 + \|x_2\|^2 \\
&\boxed{m^* = x_1} \text{ to minimize this} \qquad \Longrightarrow \qquad \boxed{\therefore\ \min \|x - m\| = \|x_2\|}
\end{aligned}$$

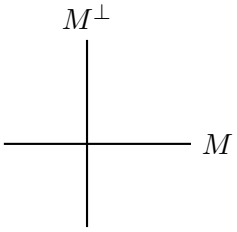
- x_1 exists;
- x_1 is unique, as $M \oplus M^{\perp}$ is a direct sum;
- $x - m^* = x - x_1 = x_2 \in M^{\perp}$.

Why the proof works

The key identity is the Pythagorean theorem $\|x - m\|^2 = \|x_1 - m\|^2 + \|x_2\|^2$, valid because $x_1 - m \in M$ and $x_2 \in M^{\perp}$ are orthogonal. The second term does not depend on m, and the first one is ≥ 0 with equality *only* for $m = x_1$; this gives existence and uniqueness at once. The decomposition $H = M \oplus M^{\perp}$ itself exists because M is finite dimensional: take an orthonormal basis $\{e_1, \ldots, e_k\}$ of M (Gram–Schmidt) and set $x_1 = \sum_i \langle x, e_i \rangle e_i$, $x_2 = x - x_1$; then $\langle x_2, e_j \rangle = \langle x, e_j \rangle - \langle x, e_j \rangle = 0$ for every j, so $x_2 \in M^{\perp}$.

Note. $\mathcal{V} = M^{\perp} \overset{\perp}{\oplus} M, \quad x \in \mathcal{V}, \quad x \notin M.$

$$\mathbb{R}^2 = x \text{ axis} \overset{\perp}{\oplus} y \text{ axis}, \qquad \mathbb{R}^2 = \underbrace{\operatorname{sp}\left\{\begin{bmatrix}1\\0\end{bmatrix}\right\}}_{M} \overset{\perp}{\oplus} \underbrace{\operatorname{sp}\left\{\begin{bmatrix}0\\1\end{bmatrix}\right\}}_{M^{\perp}}$$



$$\begin{bmatrix}1\\2\end{bmatrix} = \underbrace{\begin{bmatrix}1\\0\end{bmatrix}}_{\in M} + \underbrace{\begin{bmatrix}0\\2\end{bmatrix}}_{\in M^{\perp}}$$

If a vector is not in M, it does not directly mean that it should be inside $M^{\perp}$. The vector can be written using M and $M^{\perp}$.

p. 37

$$x \in M \ \Rightarrow\ m^* = \operatorname{Proj}_M x = x, \qquad x = \underbrace{x_1}_{x} + \underbrace{x_2}_{0}: \quad \text{if } x \in M \ \Rightarrow\ \operatorname{Proj}_{M^{\perp}} x = 0$$

Example 5.10. $\mathcal{V} = \mathbb{R}^2, \quad M = \operatorname{sp}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}, \quad \langle x, y\rangle = y^T x, \quad x = \begin{bmatrix}1\\2\end{bmatrix}, \quad m^* = \operatorname{Proj}_M x.$

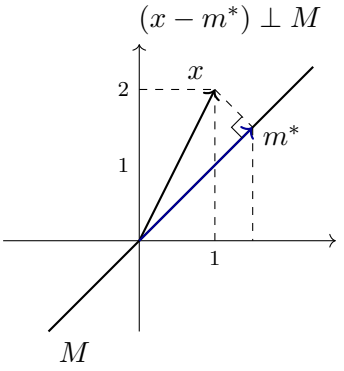
$$m = \alpha \begin{bmatrix}1\\1\end{bmatrix} \;\Rightarrow\; m^* = \alpha^* \begin{bmatrix}1\\1\end{bmatrix}, \qquad (x - m^*) \perp M$$

$$x - m^* = \left(\begin{bmatrix}1\\2\end{bmatrix} - \alpha^* \begin{bmatrix}1\\1\end{bmatrix}\right) \perp \begin{bmatrix}1\\1\end{bmatrix}$$

$$\begin{bmatrix}1-\alpha^*\\2-\alpha^*\end{bmatrix} \perp \begin{bmatrix}1\\1\end{bmatrix}$$

$$1 - \alpha^* + 2 - \alpha^* = 0 \;\Rightarrow\; \alpha^* = \tfrac{3}{2}$$

$$\boxed{m^* = \frac{3}{2}\begin{bmatrix}1\\1\end{bmatrix}}$$



▷ *Check:* $x - m^* = \begin{bmatrix}-1/2 & 1/2\end{bmatrix}^T$ *and* $\begin{bmatrix}-1/2 & 1/2\end{bmatrix}\begin{bmatrix}1 & 1\end{bmatrix}^T = 0$*; the distance from* x *to* M *is* $\|x - m^*\| = 1/\sqrt{2}$.

Example 5.11. **(1)** $\mathcal{V}, \quad M = \operatorname{sp}\{v_1\}.$

$$m^* = \operatorname{Proj}_M x = ? \qquad m^* = \alpha^* v_1, \qquad x - m^* = (x - \alpha^* v_1) \perp v_1$$

Hence

$$\langle x - \alpha^* v_1, v_1\rangle = 0$$

$$\langle x, v_1\rangle - \alpha^* \langle v_1, v_1\rangle = 0$$

$$\Rightarrow \quad \alpha^* = \frac{\langle x, v_1\rangle}{\langle v_1, v_1\rangle}, \qquad m^* = \frac{\langle x, v_1\rangle}{\langle v_1, v_1\rangle} \cdot v_1$$

(2) $\mathcal{V}, \quad M = \operatorname{sp}\{v_1, v_2\}.$

$$m^* = \operatorname{Proj}_M x = ?$$

$$m^* = \alpha_1^* v_1 + \alpha_2^* v_2$$

$$(x - \alpha_1^* v_1 - \alpha_2^* v_2) \perp M \;\Rightarrow\; (x - \alpha_1^* v_1 - \alpha_2^* v_2) \perp v_1 \text{ and } v_2$$

Taking inner products with v_1 and v_2:

$$\langle x - \alpha_1^* v_1 - \alpha_2^* v_2, v_1\rangle = 0 \;\Rightarrow\; \langle v_1, v_1\rangle \alpha_1^* + \langle v_2, v_1\rangle \alpha_2^* = \langle x, v_1\rangle$$

$$\langle x - \alpha_1^* v_1 - \alpha_2^* v_2, v_2\rangle = 0 \;\Rightarrow\; \langle v_1, v_2\rangle \alpha_1^* + \langle v_2, v_2\rangle \alpha_2^* = \langle x, v_2\rangle$$

In matrix form:

$$\underbrace{\begin{bmatrix}\langle v_1, v_1\rangle & \langle v_2, v_1\rangle\\ \langle v_1, v_2\rangle & \langle v_2, v_2\rangle\end{bmatrix}}_{\substack{G\\ \text{Gram matrix}}} \underbrace{\begin{bmatrix}\alpha_1^*\\ \alpha_2^*\end{bmatrix}}_{\alpha^*} = \underbrace{\begin{bmatrix}\langle x, v_1\rangle\\ \langle x, v_2\rangle\end{bmatrix}}_{b} \qquad \Longrightarrow \qquad G \cdot \alpha^* = b, \quad \boxed{\alpha^* = G^{-1} \cdot b}$$

The determinant of G: p. 38

$$\begin{aligned}\det G &= \langle v_1, v_1\rangle \langle v_2, v_2\rangle - \langle v_1, v_2\rangle \underbrace{\langle v_2, v_1\rangle}_{\overline{\langle v_1, v_2\rangle}} \\ &= \langle v_1, v_1\rangle \langle v_2, v_2\rangle - |\langle v_1, v_2\rangle|^2\end{aligned}$$

▷ *Cauchy–Schwarz: equality is satisfied if $v_1 = \alpha v_2$; only in this case* $\det G = 0$.
Hence

$$\det G > 0 \quad \text{if} \quad \underbrace{v_1 \neq \alpha v_2}_{\text{linearly indep.}}, \qquad M = \operatorname{sp}\{v_1, v_2\}, \quad m^* = \alpha_1^* v_1 + \alpha_2^* v_2$$

Explanation

By the Cauchy–Schwarz inequality $|\langle v_1, v_2\rangle|^2 \leq \langle v_1, v_1\rangle \langle v_2, v_2\rangle$, so always $\det G \geq 0$, with equality iff v_1, v_2 are linearly dependent. Hence for a basis $\{v_1, v_2\}$ of M the system $G\alpha^* = b$ has exactly one solution, which is the unique m^* promised by the projection theorem. Note also that G is Hermitian: its (j, i) entry is the complex conjugate of its (i, j) entry.

Example 5.12. $\mathcal{V} = \{f : [0,1] \to \mathbb{R}\}$, $\langle f, g\rangle = \int_0^1 f(t)g(t)\,dt$, $M = \operatorname{sp}\{1, t\}$, $f(t) = t^2$.
Find $m^* = \operatorname{Proj}_M f$, with $v_1(t) = 1$, $v_2(t) = t$:

$$\underbrace{\begin{bmatrix}\langle 1,1\rangle & \langle t,1\rangle \\ \langle 1,t\rangle & \langle t,t\rangle\end{bmatrix}}_{G}\begin{bmatrix}\alpha_1^* \\ \alpha_2^*\end{bmatrix} = b = \begin{bmatrix}\langle t^2,1\rangle \\ \langle t^2,t\rangle\end{bmatrix}$$

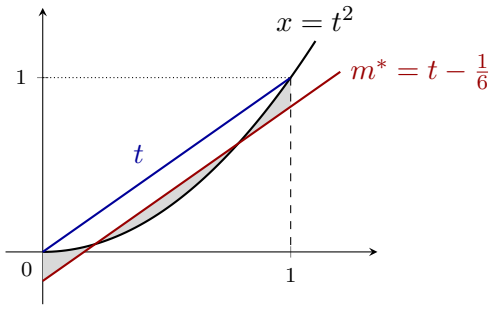
$$\Rightarrow \begin{bmatrix}\int_0^1 dt & \int_0^1 t\,dt \\ \int_0^1 t\,dt & \int_0^1 t^2\,dt\end{bmatrix}\begin{bmatrix}\alpha_1^* \\ \alpha_2^*\end{bmatrix} = \begin{bmatrix}\int_0^1 t^2\,dt \\ \int_0^1 t^3\,dt\end{bmatrix}$$

$$\Rightarrow \underbrace{\begin{bmatrix}1 & 1/2 \\ 1/2 & 1/3\end{bmatrix}}_{G}\begin{bmatrix}\alpha_1^* \\ \alpha_2^*\end{bmatrix} = \begin{bmatrix}1/3 \\ 1/4\end{bmatrix}, \qquad \det G = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

$$\begin{bmatrix}\alpha_1^* \\ \alpha_2^*\end{bmatrix} = \frac{1}{\frac{1}{12}}\begin{bmatrix}1/3 & -1/2 \\ -1/2 & 1\end{bmatrix}\begin{bmatrix}1/3 \\ 1/4\end{bmatrix} = \begin{bmatrix}-1/6 \\ 1\end{bmatrix}$$

$$\boxed{m^* = \operatorname{Proj}_M f = -\frac{1}{6} + t}$$

$$\|f(t)\| = \sqrt{\int_0^1 |f(t)|^2\,dt}, \qquad \min_{m\in M}\left\|\underbrace{t^2}_{x} - m(t)\right\| = \min_{m(t)\in M}\sqrt{\int_0^1 |t^2 - m(t)|^2\,dt}$$



▷ *Shaded: the gap between t^2 and its best approximation m^* in $\mathrm{sp}\{1, t\}$.*

Checking the answer

The error $e(t) = t^2 - t + \frac{1}{6}$ must be orthogonal to 1 and t:

$$\langle e, 1\rangle = \int_0^1 (t^2 - t + \tfrac{1}{6})\, dt = \tfrac{1}{3} - \tfrac{1}{2} + \tfrac{1}{6} = 0,$$
$$\langle e, t\rangle = \int_0^1 (t^3 - t^2 + \tfrac{1}{6}t)\, dt = \tfrac{1}{4} - \tfrac{1}{3} + \tfrac{1}{12} = 0. \quad \checkmark$$

Since $e \perp m^*$, the minimum error is

$$\|e\|^2 = \left\langle e, t^2 \right\rangle = \tfrac{1}{5} - \tfrac{1}{4} + \tfrac{1}{18} = \tfrac{1}{180}, \qquad \text{i.e.} \qquad \min_{m \in M} \left\| t^2 - m \right\| = \tfrac{1}{6\sqrt{5}} \approx 0.075.$$

So $t - \frac{1}{6}$ is the best straight-line (least-squares) fit to t^2 on $[0, 1]$; it crosses t^2 at $t = \frac{1}{2} \pm \frac{1}{2\sqrt{3}}$.

5.4 Computing Projections: the Gram Matrix and Normal Equations

General space. $\mathcal{V}$, n dimensional, $\quad M = \mathrm{sp}\{\underbrace{v_1, v_2, \ldots, v_k}_{\text{linearly indep.}}\}, \quad k < n.$ p. 39

$$m^* = \sum_{i=1}^{k} \alpha_i^* v_i, \qquad x - m^* = x - \sum_{i=1}^{k} \alpha_i^* v_i \;\perp\; v_1, v_2, \ldots, v_k$$

Taking inner products with each v_j:

$$\left\langle x - \sum_{i=1}^{k} \alpha_i^* v_i,\, v_j \right\rangle = 0, \qquad j = 1, 2, \ldots, k$$
$$\langle x, v_j\rangle - \sum_{i=1}^{k} \alpha_i^* \langle v_i, v_j\rangle = 0$$

In matrix form:

$$\underbrace{\begin{bmatrix} \langle v_1, v_1\rangle & \cdots & \langle v_k, v_1\rangle \\ \vdots & \ddots & \vdots \\ \langle v_1, v_k\rangle & \cdots & \langle v_k, v_k\rangle \end{bmatrix}}_{G} \underbrace{\begin{bmatrix} \alpha_1^* \\ \vdots \\ \alpha_k^* \end{bmatrix}}_{\alpha^*} = \underbrace{\begin{bmatrix} \langle x, v_1\rangle \\ \vdots \\ \langle x, v_k\rangle \end{bmatrix}}_{b}$$

G is invertible iff $v_1, \ldots, v_k$ are linearly indep. $\qquad \alpha^* = G^{-1} b$

▷ *These k equations are the **normal equations**; G is the **Gram matrix**. G is Hermitian and positive definite for independent v_i: $\beta^H G \beta = \|\sum_i \beta_i v_i\|^2 > 0$ for $\beta \neq 0$ (written for $\mathcal{F} = \mathbb{C}$ with $G_{ji} = \langle v_i, v_j\rangle$).*

Specific case: orthogonal basis. $v_i \perp v_j$ when $i \neq j$: $\{v_1, \dots, v_k\}$ are orthogonal, so G is diagonal:

$$\begin{bmatrix} \langle v_1, v_1\rangle & & & 0 \\ & \langle v_2, v_2\rangle & & \\ & & \ddots & \\ 0 & & & \langle v_k, v_k\rangle \end{bmatrix} \begin{bmatrix} \alpha_1^* \\ \vdots \\ \alpha_k^* \end{bmatrix} = \begin{bmatrix} \langle x, v_1\rangle \\ \vdots \\ \langle x, v_k\rangle \end{bmatrix}$$

$$G^{-1}b = \begin{bmatrix} \dfrac{1}{\langle v_1, v_1\rangle} & & 0 \\ & \ddots & \\ 0 & & \dfrac{1}{\langle v_k, v_k\rangle} \end{bmatrix} \begin{bmatrix} \langle x, v_1\rangle \\ \vdots \\ \langle x, v_k\rangle \end{bmatrix} = \begin{bmatrix} \dfrac{\langle x, v_1\rangle}{\langle v_1, v_1\rangle} \\ \vdots \\ \dfrac{\langle x, v_k\rangle}{\langle v_k, v_k\rangle} \end{bmatrix}$$

$$\alpha_i^* = \frac{\langle x, v_i\rangle}{\langle v_i, v_i\rangle}, \qquad \boxed{m^* = \sum_{i=1}^{k} \frac{\langle x, v_i\rangle}{\langle v_i, v_i\rangle} \cdot v_i}$$

For an orthonormal set ($v_i \perp v_j$, $i \neq j$, and $\|v_i\| = 1$):

$$\boxed{m^* = \sum_{i=1}^{k} \langle x, v_i\rangle \; v_i}$$

Example 5.13. $\mathcal{V} = \mathbb{R}^{2\times 2}$, $\quad M = \{\text{set of symmetric matrices}\}$

$$x = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \qquad m^* = \operatorname{Proj}_M x, \qquad \langle A, B\rangle = \operatorname{tr}(A^T B)$$

For 2×2 matrices:

$$\operatorname{tr}\left(\begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix}\begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}\right) = \operatorname{tr}\left(\begin{bmatrix} a_{11}b_{11} + a_{21}b_{21} & \times \\ \times & a_{12}b_{12} + a_{22}b_{22} \end{bmatrix}\right)$$
$$= a_{11}b_{11} + a_{21}b_{21} + a_{12}b_{12} + a_{22}b_{22}$$
$$\|A\| = \sqrt{\operatorname{tr}(A^T A)} = \sqrt{\langle A, A\rangle}$$

We look for p. 40

$$\min_{S \text{ sym.}} \left(\operatorname{tr}\left[(x - S)^T (x - S)\right]\right)^{1/2}$$

$$M = \operatorname{sp}\left\{ \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}}_{v_1}, \underbrace{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}_{v_2}, \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}}_{v_3} \right\}$$

▷ *v_1, v_2, v_3 are mutually orthogonal (no common nonzero entries), so the orthogonal-basis formula $\alpha_i^* = \langle x, v_i\rangle \,/\, \langle v_i, v_i\rangle$ applies.*
Hence

$$\begin{aligned} \langle x, v_1\rangle &= a, & \langle v_1, v_1\rangle &= 1 \\ \langle x, v_2\rangle &= b + c, & \langle v_2, v_2\rangle &= 2 \\ \langle x, v_3\rangle &= d, & \langle v_3, v_3\rangle &= 1 \end{aligned}$$

and

$$m^* = a\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix} + \frac{b+c}{2}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} + d\begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix} = \begin{bmatrix}a & \frac{b+c}{2}\\ \frac{b+c}{2} & d\end{bmatrix} = \frac{x+x^T}{2}$$

$$\operatorname{Proj}_{M^\perp} x = x - m^* = x - \left(\frac{x+x^T}{2}\right) = \frac{x-x^T}{2}$$

$$\left.\begin{aligned} M &\to 3 \text{ dim.}\\ M^\perp &\to 1 \text{ dim.}\end{aligned}\right\} \mathcal{V} \Rightarrow 4 \text{ dim.}$$

Explanation

$M^\perp$ is the set of **skew-symmetric** matrices, $M^\perp = \operatorname{sp}\{K_0\}$ with $K_0 = \begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix}$: for $S = S^T$ and $K = -K^T$,

$$\langle K, S\rangle = \operatorname{tr}(S^T K) = \operatorname{tr}(SK) = \operatorname{tr}\left((SK)^T\right) = \operatorname{tr}(K^T S) = -\operatorname{tr}(KS) = -\operatorname{tr}(SK),$$

so $\langle K, S\rangle = 0$. Thus every square matrix splits uniquely as

$$x = \underbrace{\tfrac{x+x^T}{2}}_{\text{symmetric}} + \underbrace{\tfrac{x-x^T}{2}}_{\text{skew-symmetric}},$$

and $\frac{x+x^T}{2}$ is the symmetric matrix closest to x in the Frobenius norm $\sqrt{\operatorname{tr}(A^T A)}$.

5.5 Orthogonal Projections in $\mathbb{C}^n$: Projection Matrices

$\mathcal{V} = \mathbb{C}^n, \quad M = \text{sp}\{m_1, \dots, m_k\}, \quad m_i = [\ \ ] \in \mathbb{C}^n$ (column vectors), $\quad x \in \mathbb{C}^n$.

$$m^* = \text{Proj}_M\, x = ?$$
$$m^* = P_M \cdot x \quad \Rightarrow \quad P_M\text{: } \textbf{projection matrix} \text{ onto } M.$$

Writing m^* in terms of the spanning vectors:

$$m^* = \alpha_1^* m_1 + \alpha_2^* m_2 + \cdots + \alpha_k^* m_k = \underbrace{\left[\, m_1 \vdots m_2 \vdots \cdots \vdots m_k \,\right]}_{B} \underbrace{\begin{bmatrix} \alpha_1^* \\ \alpha_2^* \\ \vdots \\ \alpha_k^* \end{bmatrix}}_{\alpha^*}$$

$$m^* = B\alpha^*, \qquad x - B\alpha^* \perp m_1, m_2, \dots, m_k, \qquad \langle x, y \rangle = y^H x$$

Hence

$$\begin{aligned} \langle x - B\alpha^*, m_j \rangle = 0 \quad \Leftrightarrow \quad & m_j^H (x - B\alpha^*) = 0 \quad \forall j \\ & m_1^H (x - B\alpha^*) = 0 \\ & \vdots \\ & m_k^H (x - B\alpha^*) = 0 \end{aligned}$$

Stacking these k equations:

$$\underbrace{\begin{bmatrix} m_1^H \\ \hline m_2^H \\ \hline \vdots \\ \hline m_k^H \end{bmatrix}}_{B^H} (x - B\alpha^*) = 0 \qquad \Longrightarrow \qquad \begin{aligned} B^H (x - B\alpha^*) &= 0 \\ B^H x - B^H B\alpha^* &= 0 \\ \underbrace{B^H B}_{G}\, \alpha^* &= \underbrace{B^H x}_{b} \end{aligned}$$

Indeed, p. 41

$$\underbrace{\begin{bmatrix} m_1^H \\ \vdots \\ m_k^H \end{bmatrix}}_{B^H} \underbrace{\left[\, m_1 \vdots m_2 \vdots \cdots \vdots m_k \,\right]}_{B} = \begin{bmatrix} m_1^H m_1 & \cdots & m_1^H m_k \\ \vdots & \ddots & \vdots \\ m_k^H m_1 & \cdots & m_k^H m_k \end{bmatrix} = \underbrace{\begin{bmatrix} \langle m_1, m_1 \rangle & \cdots & \langle m_k, m_1 \rangle \\ \vdots & \ddots & \vdots \\ \langle m_1, m_k \rangle & \cdots & \langle m_k, m_k \rangle \end{bmatrix}}_{G}$$

$$B^H x = \begin{bmatrix} m_1^H x \\ m_2^H x \\ \vdots \\ m_k^H x \end{bmatrix} = \begin{bmatrix} \langle x, m_1 \rangle \\ \langle x, m_2 \rangle \\ \vdots \\ \langle x, m_k \rangle \end{bmatrix} = b$$

▷ *So $B^H B\, \alpha^* = B^H x$ is exactly the Gram system $G\alpha^* = b$ of the general case, with $\langle x, y \rangle = y^H x$.*

Solving for α^*:

$$\alpha^* = (B^H B)^{-1} B^H x$$
$$m^* = B\alpha^*$$
$$m^* = \text{Proj}_M\, x = B(B^H B)^{-1} B^H x$$

$$\boxed{P_M = B(B^H B)^{-1} B^H}, \qquad m^* = P_M x$$

For the orthogonal complement:

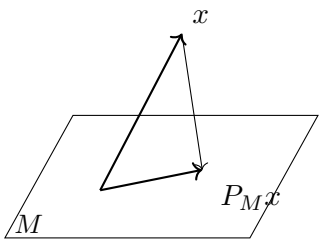
$$\begin{aligned} P_{M^\perp} &= I - P_M \\ \mathrm{Proj}_{M^\perp} x &= x - \mathrm{Proj}_M x \\ &= x - P_M x = (I - P_M)x \end{aligned}$$

Properties of P_M:

$$\begin{aligned} P_M^H &= \left(B(B^HB)^{-1}B^H\right)^H = B((B^HB)^{-1})^H B^H = B(B^HB)^{-1}B^H = P_M \\ &\Rightarrow \; P_M^H = P_M \\ P_M^2 &= P_M \cdot P_M \\ P_M^2 &= B(B^HB)^{-1}\cancel{B^HB(B^HB)^{-1}}B^H = B(B^HB)^{-1}B^H = P_M \end{aligned}$$

$$\boxed{P_M^2 = P_M}$$

▷ *$(B^HB)^{-1}$ is Hermitian because B^HB is:* $\left((B^HB)^{-1}\right)^H = \left((B^HB)^H\right)^{-1} = (B^HB)^{-1}$.



$$P_M \cdot P_M x = P_M x$$
$$\underline{P_M^2 x = P_M x}$$

Remark. An **orthogonal projection matrix** $P \in \mathbb{C}^{n\times n}$ satisfies

i-) $P^H = P$
ii-) $P^2 = P$ $\quad (P^k = P \text{ for every } k \geq 1)$

Also,

$$(I-P)^2 = I - P$$

$$(I-P)(I-P) = I - \cancel{P} - P + \overbrace{P^2}^{P}$$
$$= I - P$$

Remarks on P_M

- Conversely, **any P with $P^H = P$ and $P^2 = P$ is the orthogonal projection onto** $\mathcal{R}(P)$: $x = Px + (I-P)x$ with $Px \in \mathcal{R}(P)$, and $(I-P)x \perp \mathcal{R}(P)$ because
$$(Py)^H(I-P)x = y^H(P-P^2)x = 0 \qquad \text{for all } y.$$
- P_M **depends only on** M, not on the chosen basis: if $B' = BT$ with T invertible, then $B'(B'^HB')^{-1}B'^H = BT\,T^{-1}(B^HB)^{-1}T^{-H}\,T^HB^H = P_M$.
- If the columns of B are orthonormal, $B^HB = I$ and $P_M = BB^H = \sum_i m_i m_i^H$, which is the matrix form of $m^* = \sum_i \langle x, m_i\rangle\, m_i$.
- $P_M m = m$ for $m \in M$ and $P_M y = 0$ for $y \in M^\perp$ (since $B^Hy = 0$); hence $\mathcal{R}(P_M) = M$, $\mathcal{N}(P_M) = M^\perp$.

Example 5.14. $\mathcal{V} = \mathbb{R}^3$, $\quad M = \operatorname{sp}\left\{\begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}1\\-1\\1\end{bmatrix}\right\}$, $\quad x = \begin{bmatrix}1\\0\\0\end{bmatrix}$. $\quad$ Find $\operatorname{Proj}_M x = ?$ p. 42

$$B = \begin{bmatrix}1 & 1\\1 & -1\\1 & 1\end{bmatrix}, \qquad B^H B = B^T B = \begin{bmatrix}1 & 1 & 1\\1 & -1 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\1 & -1\\1 & 1\end{bmatrix} = \begin{bmatrix}3 & 1\\1 & 3\end{bmatrix},$$

$$(B^H B)^{-1} = \begin{bmatrix}3 & -1\\-1 & 3\end{bmatrix}\cdot\frac{1}{8}$$

Then

$$\begin{aligned}\operatorname{Proj}_M x &= B(B^H B)^{-1}B^H x = P_M x\\ &= \frac{1}{8}\begin{bmatrix}1 & 1\\1 & -1\\1 & 1\end{bmatrix}\begin{bmatrix}3 & -1\\-1 & 3\end{bmatrix}\underbrace{\begin{bmatrix}1 & 1 & 1\\1 & -1 & 1\end{bmatrix}\begin{bmatrix}1\\0\\0\end{bmatrix}}_{\begin{bmatrix}1\\1\end{bmatrix}}\\ &= \frac{1}{8}\begin{bmatrix}1 & 1\\1 & -1\\1 & 1\end{bmatrix}\underbrace{\begin{bmatrix}3 & -1\\-1 & 3\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix}}_{\begin{bmatrix}2\\2\end{bmatrix}}\\ &= \frac{1}{8}\cdot\begin{bmatrix}4\\0\\4\end{bmatrix} = \begin{bmatrix}1/2\\0\\1/2\end{bmatrix}\end{aligned}$$

$$x - \operatorname{Proj}_M x = \begin{bmatrix}1\\0\\0\end{bmatrix} - \begin{bmatrix}1/2\\0\\1/2\end{bmatrix} = \begin{bmatrix}1/2\\0\\-1/2\end{bmatrix}$$

▷ *This difference should be orthogonal to the basis vectors ("sağlaması için kullanılabilir" — can be used to check the result).*

▷ *Check, and the full projection matrix (its first column is indeed* $\operatorname{Proj}_M x$*):*

$$\begin{bmatrix}1 & 1 & 1\end{bmatrix}\begin{bmatrix}1/2\\0\\-1/2\end{bmatrix} = 0, \qquad \begin{bmatrix}1 & -1 & 1\end{bmatrix}\begin{bmatrix}1/2\\0\\-1/2\end{bmatrix} = 0 \quad \checkmark$$

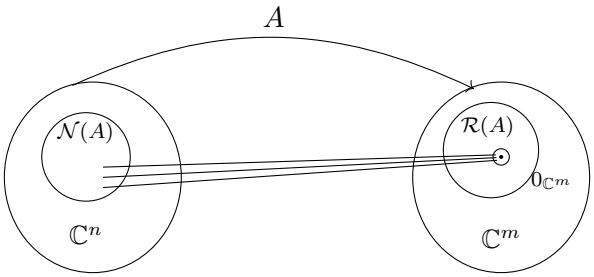
$$P_M = \frac{1}{8}B\begin{bmatrix}3 & -1\\-1 & 3\end{bmatrix}B^T = \frac{1}{2}\begin{bmatrix}1 & 0 & 1\\0 & 2 & 0\\1 & 0 & 1\end{bmatrix}$$

6 Systems of Linear Equations $Ax = b$

Syllabus: Week 7 — systems of linear equations of the form $Ax = b$

6.1 Existence and Uniqueness of Solutions

$$Ax = b, \qquad A \in \mathbb{C}^{m\times n}, \quad x \in \mathbb{C}^n, \quad b \in \mathbb{C}^m$$



▷ *A maps every vector of $\mathcal{N}(A)$ to $0_{\mathbb{C}^m}$.*

The two fundamental subspaces of A:

$$\textbf{range space}\text{: } \mathcal{R}(A) = \{y \in \mathbb{C}^m \mid \exists x \in \mathbb{C}^n \text{ satisfying } y = Ax\}$$
$$\textbf{null space}\text{: } \mathcal{N}(A) = \{x \in \mathbb{C}^n \mid Ax = 0_{\mathbb{C}^m}\}$$

Rank–nullity:

$$\underbrace{\dim(\mathcal{R}(A))}_{\text{rank}} + \underbrace{\dim(\mathcal{N}(A))}_{\text{nullity}} = n = \#\text{ of columns of } A$$

Recall also, for a subspace M of $\mathcal{V}$:

$$\mathcal{V} = M \overset{\perp}{\oplus} M^\perp$$
$$v = v_1 + v_2, \qquad v_1 \in M, \quad v_2 \in M^\perp, \qquad v_1, v_2 \text{ are unique}$$
$$v_1 \perp v_2, \qquad \dim M + \dim M^\perp = \dim \mathcal{V}$$

Two questions:

1-) Is there an exact solution? $(\mathcal{R}(A))$
2-) If there is, is it unique? $(\mathcal{N}(A))$

▷ *Question 1 is answered by $\mathcal{R}(A) \subseteq \mathbb{C}^m$ (the codomain side), question 2 by $\mathcal{N}(A) \subseteq \mathbb{C}^n$ (the domain side).*

★ $\underbrace{\begin{bmatrix}1\\1\end{bmatrix}}_{A} \underset{\in\mathbb{R}}{x} = \underbrace{\begin{bmatrix}2\\2\end{bmatrix}}_{b}, \qquad b \in \mathcal{R}(A)$ p. 43

$\Rightarrow \exists$ an exact solution. $\Rightarrow$ The solution is unique.

★ $\underbrace{\begin{bmatrix}1\\1\end{bmatrix}}_{A} x = \underbrace{\begin{bmatrix}2\\1\end{bmatrix}}_{b}$ $\Rightarrow$ No exact solution! $b \notin \mathcal{R}(A)$

$$y = Ax = \left[\, a_1 \vdots a_2 \vdots a_3 \vdots \cdots \vdots a_n \,\right] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n$$

$\mathcal{R}(A) = \text{sp}\{\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_n\} \triangleq$ **column space** of A

$\mathcal{R}(A) = \text{sp}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}, \qquad \begin{bmatrix}2\\1\end{bmatrix} \notin \mathcal{R}(A)$

$\therefore$ 1-) **There exists an exact solution iff** $b \in \mathcal{R}(A)$**.**

★ $\underbrace{\begin{bmatrix}1 & 2\\1 & 2\end{bmatrix}}_{A} \underset{\in\mathbb{R}^2}{x} = \underbrace{\begin{bmatrix}2\\2\end{bmatrix}}_{b}$: $\mathcal{R}(A) = \text{sp}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}$ & $b \in \mathcal{R}(A) \Rightarrow \therefore \exists$ an exact solution

$$x_1 = \begin{bmatrix}2\\0\end{bmatrix} \quad \text{or} \quad x_2 = \begin{bmatrix}0\\1\end{bmatrix} \quad \text{or} \quad x_3 = \begin{bmatrix}4\\-1\end{bmatrix} \quad \ldots$$

Dimensions in the first example:

$$\left.\begin{array}{l} A = \begin{bmatrix}1\\1\end{bmatrix}, \qquad \dim \mathbb{C} = 1 \\ \mathcal{R}(A) = \text{sp}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}, \qquad \dim \mathcal{R}(A) = 1 \end{array}\right\} \quad \begin{array}{l} \underbrace{\dim(\mathcal{R}(A))}_{1} + \underbrace{\dim(\mathcal{N}(A))}_{0} = 1 \leftarrow \#\text{ of columns} \\ \mathcal{N}(A) = \{0\}, \qquad \dim(\mathcal{N}(A)) = 0 \end{array}$$

and in the third example:

$$A = \begin{bmatrix}1 & 2\\1 & 2\end{bmatrix}, \qquad \mathcal{R}(A) = \text{sp}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}, \qquad \dim \mathcal{R}(A) = 1$$

$$\underbrace{\dim \mathcal{R}(A)}_{1} + \underbrace{\dim \mathcal{N}(A)}_{1} = 2, \qquad \mathcal{N}(A) = \text{sp}\left\{\begin{bmatrix}2\\-1\end{bmatrix}\right\}$$

$$x_1 - x_2 = \begin{bmatrix}2\\-1\end{bmatrix}, \qquad x_1 - x_3 = \begin{bmatrix}-2\\1\end{bmatrix}, \qquad x_2 - x_3 = \begin{bmatrix}-4\\2\end{bmatrix} \qquad \text{all inside null space.}$$

In general:

$$A\hat{x} = b \qquad \hat{x} \text{ is exact}$$

$$\underset{\in\mathcal{N}(A)}{\hat{x} + m} \;\Rightarrow\; A(\hat{x}+m) = A\hat{x} + \underbrace{\overset{\in\mathcal{N}(A)}{Am}}_{0} = b + 0 = b$$

- if $\hat{x}$ satisfies $A\hat{x} = b$ exactly $\Rightarrow$ $\hat{x} + m$ satisfies $A(\hat{x}+m) = b$ exactly $\;(m \in \mathcal{N}(A))$;
- if $m \neq 0 \Rightarrow \hat{x} \neq \hat{x} + m$;
- if $\mathcal{N}(A)$ includes nonzero vectors $\Rightarrow$ **infinitely many exact solutions**.

p. 44

$\therefore$ 2-) If an exact solution $\hat{x}$ exists, then **it is unique** $\iff \mathcal{N}(A) = \{0\}$ or $\dim \mathcal{N}(A) = 0$.

Explanation

Conversely, if $A\hat{x} = b$ and $A\tilde{x} = b$, then $A(\hat{x} - \tilde{x}) = 0$, i.e. any two exact solutions differ by a vector of $\mathcal{N}(A)$. Hence **the full solution set is the affine set** $\{\hat{x} + m : m \in \mathcal{N}(A)\}$. By rank–nullity, $\mathcal{N}(A) = \{0\}$ iff $\operatorname{rank} A = n$ (the columns of A are linearly independent); this requires $m \geq n$.

6.2 Least-Squares Solutions

★ $\begin{bmatrix}1\\1\end{bmatrix} x = \begin{bmatrix}2\\1\end{bmatrix}, \qquad Ax = b, \qquad b \notin \mathcal{R}(A)$

Least squares problem.

$$\min_{x\in\mathbb{C}^n} \underbrace{\|Ax - b\|^2}_{\text{error}}$$

▷ *The exponent 2 doesn't matter; $\|\cdot\|$ is the induced norm of the inner product.*

$$\hat{x} = \arg\min_{x\in\mathbb{C}^n} \|Ax - b\|$$

▷ $\hat{x}$: ***minimum error solution*** / ***least squares solution***.

Since $b \in \mathbb{C}^m$, decompose it:

$$\mathbb{C}^m = M \overset{\perp}{\oplus} M^\perp$$

$$\mathbb{C}^m = \mathcal{R}(A) \overset{\perp}{\oplus} \mathcal{R}^\perp(A)$$

$$b = \underset{\in\mathcal{R}(A)}{b_1} + \underset{\in\mathcal{R}^\perp(A)}{b_2} \;\Rightarrow\; b_1 = \operatorname{Proj}_{\mathcal{R}(A)} b, \quad b_2 = \operatorname{Proj}_{\mathcal{R}^\perp(A)} b \;\Rightarrow\; b_1 \perp b_2$$

Then

$$\begin{aligned}
\|Ax-b\|^2 &= \langle Ax-b, Ax-b\rangle \\
&= \big\langle (\underbrace{Ax}_{\in\mathcal{R}(A)} - \underbrace{b_1}_{\in\mathcal{R}(A)}) - \underbrace{b_2}_{\in\mathcal{R}^\perp(A)},\ (Ax-b_1)-b_2\big\rangle \qquad (\mathcal{R}(A) \text{ is a subspace}) \\
&= \langle Ax-b_1, Ax-b_1\rangle \\
&\quad \cancel{-\langle Ax-b_1, b_2\rangle} \\
&\quad \cancel{-\langle b_2, Ax-b_1\rangle} \qquad (\to 0 \text{ because of orthogonality}) \\
&\quad + \langle b_2, b_2\rangle \\
&= \langle Ax-b_1, Ax-b_1\rangle + \langle b_2, b_2\rangle \\
&= \overbrace{\Big\|A\underbrace{x}_{\text{find an } x} - \underbrace{b_1}_{\in\mathcal{R}(A)}\Big\|^2}^{\text{zero}} + \|b_2\|^2 \geq \|b_2\|^2
\end{aligned}$$

▷ ***The first term can be made zero:*** *$b_1 \in \mathcal{R}(A)$, so some x satisfies $Ax = b_1$ exactly.*

Therefore

$$\boxed{\min_{x\in\mathbb{C}^n} \|Ax-b\| = \|b_2\| = \left\|\mathrm{Proj}_{\mathcal{R}^\perp(A)}\, b\right\|}$$

$$A\hat{x} = b_1, \qquad \hat{x} = \arg\min_{x\in\mathbb{C}^n} \|Ax-b\|, \qquad b_1 = \mathrm{Proj}_{\mathcal{R}(A)}\, b$$

$$\underbrace{\hat{x}}_{\substack{\text{min error /} \\ \text{least squares solution}}} = \arg\min_{x\in\mathbb{C}^n} \|Ax-b\|$$

Normal equations

$A\hat{x} = b_1$ is equivalent to $b - A\hat{x} = b_2 \perp \mathcal{R}(A)$, i.e. $a_j^H(b-A\hat{x}) = 0$ for every column a_j of A:

$$A^H(b-A\hat{x}) = 0 \iff \boxed{A^HA\,\hat{x} = A^Hb}$$

(the same equations as $B^HB\alpha^* = B^Hx$ before, with $B = A$). **These normal equations always have a solution.** If the columns of A are linearly independent, A^HA is invertible and $\hat{x} = (A^HA)^{-1}A^Hb$ is unique; otherwise there are infinitely many least squares solutions, differing by elements of $\mathcal{N}(A^HA) = \mathcal{N}(A)$.

★ $\begin{bmatrix}1\\1\end{bmatrix} x = \begin{bmatrix}2\\1\end{bmatrix}, \qquad b \notin \mathcal{R}(A), \qquad \mathcal{R}(A) = \mathrm{sp}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}$ p. 45

$$P_{\mathcal{R}(A)} = B(B^HB)^{-1}B^H, \qquad B = \begin{bmatrix}1\\1\end{bmatrix}, \qquad B^HB = \begin{bmatrix}1 & 1\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix} = 2$$

$$P_{\mathcal{R}(A)} = \begin{bmatrix}1\\1\end{bmatrix}\frac{1}{2}\begin{bmatrix}1 & 1\end{bmatrix} = \begin{bmatrix}1/2 & 1/2\\ 1/2 & 1/2\end{bmatrix}$$

$$b_1 = P_{\mathcal{R}(A)}b = \begin{bmatrix}1/2 & 1/2\\ 1/2 & 1/2\end{bmatrix}\begin{bmatrix}2\\1\end{bmatrix} = \begin{bmatrix}3/2\\3/2\end{bmatrix}$$

$$\underbrace{\begin{bmatrix}1\\1\end{bmatrix}}_{A}\hat{x} = \underbrace{\begin{bmatrix}3/2\\3/2\end{bmatrix}}_{b_1}, \qquad \hat{x} = \frac{3}{2}$$

▷ *$\hat{x} = \frac{3}{2}$ is the average of the entries of $b = \begin{bmatrix}2 & 1\end{bmatrix}^T$.*

▷ *Same result from the normal equations: $A^HA = 2$, $A^Hb = 3 \Rightarrow \hat{x} = \frac{3}{2}$; $b - b_1 = \begin{bmatrix}1/2 & -1/2\end{bmatrix}^T$, so the minimum error is $\|b - b_1\| = \frac{1}{\sqrt{2}}$.*

★ $\begin{bmatrix}1\\1\end{bmatrix} x = \begin{bmatrix}2\\1\end{bmatrix}, \qquad \mathcal{R}(A) = \text{sp}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}$

$$b_1 = P_{\mathcal{R}(A)}b = \begin{bmatrix}1/2 & 1/2\\1/2 & 1/2\end{bmatrix}\begin{bmatrix}2\\1\end{bmatrix} = \begin{bmatrix}3/2\\3/2\end{bmatrix}$$

Then

$$\begin{aligned} &A\hat{x} = b_1, && \hat{x} = 3/2\\ &\mathcal{N}(A) = \{0\},\ \dim\mathcal{N}(A) = 0 && \Rightarrow\ \hat{x} \textbf{ is unique.}\end{aligned}$$

★ $\begin{bmatrix}1 & 2\\1 & 2\end{bmatrix} x = \begin{bmatrix}2\\1\end{bmatrix}, \qquad \mathcal{R}(A) = \text{sp}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}, \qquad b \notin \mathcal{R}(A)$

$$b_1 = P_{\mathcal{R}(A)}b = \begin{bmatrix}3/2\\3/2\end{bmatrix}$$

$$\begin{bmatrix}1 & 2\\1 & 2\end{bmatrix}\hat{x} = \begin{bmatrix}3/2\\3/2\end{bmatrix} \quad \Rightarrow \quad \hat{x} = \begin{bmatrix}3/2\\0\end{bmatrix}, \quad \hat{x} = \begin{bmatrix}0\\3/4\end{bmatrix}, \quad \hat{x} = \begin{bmatrix}1/2\\1/2\end{bmatrix}$$

$$\mathcal{N}(A) = \text{sp}\left\{\begin{bmatrix}2\\-1\end{bmatrix}\right\} \quad \Rightarrow \quad \hat{x} \textbf{ is not unique!}$$

▷ *All least squares solutions: $\hat{x} = \begin{bmatrix}3/2 & 0\end{bmatrix}^T + t\begin{bmatrix}2 & -1\end{bmatrix}^T$, $t \in \mathbb{R}$ (e.g. $t = -\frac{3}{4}$ gives $\begin{bmatrix}0 & 3/4\end{bmatrix}^T$, $t = -\frac{1}{2}$ gives $\begin{bmatrix}1/2 & 1/2\end{bmatrix}^T$). Each leaves the same error $\|b_2\| = \frac{1}{\sqrt{2}}$.*

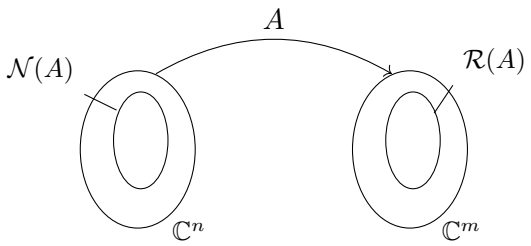
6.3 Minimum-Norm Solutions

Minimum norm solution.

$$b \in \mathcal{R}(A) \ \Rightarrow\ \min_{Ax=b}\|x\|, \qquad \hat{x}_1 = \arg\min_{Ax=b}\|x\|, \qquad x = \hat{x} + \overset{\in\mathcal{N}(A)}{m}$$

$$\min_{m\in\mathcal{N}(A)}\|\hat{x} + m\|, \qquad A\hat{x} = b$$

▷ *When there are infinitely many solutions, pick the one with the smallest norm. Every solution has the form $\hat{x} + m$ for one fixed solution $\hat{x}$, so we minimize over $m \in \mathcal{N}(A)$.*



$$\mathbb{C}^m = \mathcal{R}(A) \overset{\perp}{\oplus} \mathcal{R}^\perp(A)$$

$$\mathbb{C}^n = \mathcal{N}^\perp(A) \overset{\perp}{\oplus} \mathcal{N}(A)$$
$$\hat{x} = \underset{\in\mathcal{N}^\perp(A)}{\hat{x}_1} + \underset{\in\mathcal{N}(A)}{\hat{x}_2}, \qquad \hat{x}_1 = \text{Proj}_{\mathcal{N}^\perp(A)}\,\hat{x}, \quad \hat{x}_2 = \text{Proj}_{\mathcal{N}(A)}\,\hat{x}, \qquad \hat{x}_1 \perp \hat{x}_2$$

Then p. 46

$$\begin{aligned}
\|\hat{x}+m\|^2 &= \langle \hat{x}+m, \hat{x}+m\rangle \\
&= \langle \underset{\in\mathcal{N}^\perp}{\hat{x}_1} + (\underset{\in\mathcal{N}}{\hat{x}_2} + \underset{\in\mathcal{N}}{m}),\ \underset{\in\mathcal{N}^\perp}{\hat{x}_1} + (\underset{\in\mathcal{N}}{\hat{x}_2} + \underset{\in\mathcal{N}}{m})\rangle \\
&= \langle \hat{x}_1, \hat{x}_1\rangle + \langle \hat{x}_2+m, \hat{x}_2+m\rangle \\
&= \|\hat{x}_1\|^2 + \|\hat{x}_2+m\|^2 \\
&\geq \|\hat{x}_1\|^2
\end{aligned}$$

▷ *Equality for* $m^* = -\hat{x}_2 \;\Rightarrow\; \|\hat{x}_2+m\| = 0.$

Hence

$$\min_{\substack{Ax=b \\ m\in\mathcal{N}(A)}} \|\hat{x}+m\| = \|\hat{x}-\hat{x}_2\| = \|\hat{x}_1\|$$

and $\hat{x}_1$ is itself a solution:

$$\begin{aligned}
A\hat{x} &= b \\
A(\hat{x}_1+\hat{x}_2) &= b \\
A\hat{x}_1 + \underbrace{A\hat{x}_2}_{0} &= b \\
\Rightarrow \quad A\hat{x}_1 &= b
\end{aligned}$$

$$\boxed{\hat{x}_1 = \text{Proj}_{\mathcal{N}^\perp(A)}\,\hat{x} = \text{minimum norm solution}}$$

Summary of the method

- $\hat{x}_1$ does not depend on which solution $\hat{x}$ we start from: two solutions differ by an element of $\mathcal{N}(A)$, which the projection onto $\mathcal{N}^\perp(A)$ removes. So **the minimum norm solution is unique**.
- $\mathcal{N}^\perp(A) = \mathcal{R}(A^H)$ (since $Ax = 0 \iff x \perp$ every row of A, i.e. every column of A^H). Hence the minimum norm solution can be written $\hat{x}_1 = A^H z$, and $AA^H z = b$; if A has linearly independent rows, $\hat{x}_1 = A^H(AA^H)^{-1}b$.
- If $b \notin \mathcal{R}(A)$, combine both steps: replace b by $b_1 = \text{Proj}_{\mathcal{R}(A)}\,b$, take any least squares solution and project it onto $\mathcal{N}^\perp(A)$. The result is the **minimum norm least squares solution** (this is what the pseudo-inverse A^+b computes); the example below does exactly this.

Example 6.1.

$$\begin{bmatrix}1 & 2\\ 1 & 2\end{bmatrix} x = \begin{bmatrix}2\\1\end{bmatrix}, \qquad b_1 = \text{Proj}_{\mathcal{R}(A)}\,b = \begin{bmatrix}3/2\\3/2\end{bmatrix}$$

$$\begin{bmatrix}1 & 2\\ 1 & 2\end{bmatrix}\hat{x} = \begin{bmatrix}3/2\\3/2\end{bmatrix} \quad \Rightarrow \quad \hat{x} = \begin{bmatrix}3/2\\0\end{bmatrix}$$

$$\mathcal{N}(A) = \text{sp}\left\{\begin{bmatrix}2\\-1\end{bmatrix}\right\}, \qquad \underbrace{\dim\mathcal{N}(A)}_{1} + \underbrace{\dim\mathcal{N}^\perp(A)}_{1} = 2$$

$$\mathcal{N}^\perp(A) = \operatorname{sp}\Big\{ \underbrace{\begin{bmatrix}1\\2\end{bmatrix}}_{v} \Big\} = \mathcal{R}(A^H)$$

$$P_{\mathcal{N}^\perp(A)} = B(B^HB)^{-1}B^H \overset{B=v}{=} \frac{vv^H}{v^Hv} = \frac{\begin{bmatrix}1\\2\end{bmatrix}\begin{bmatrix}1 & 2\end{bmatrix}}{\begin{bmatrix}1 & 2\end{bmatrix}\begin{bmatrix}1\\2\end{bmatrix}} = \frac{1}{5}\begin{bmatrix}1 & 2\\2 & 4\end{bmatrix}$$

$$\hat{x}_1 = \frac{1}{5}\begin{bmatrix}1 & 2\\2 & 4\end{bmatrix}\begin{bmatrix}3/2\\0\end{bmatrix} = \frac{1}{5}\begin{bmatrix}3/2\\3\end{bmatrix} = \begin{bmatrix}3/10\\3/5\end{bmatrix}$$

Checking the answer

- $A\hat{x}_1 = \begin{bmatrix}3/10+6/5\\3/10+6/5\end{bmatrix} = \begin{bmatrix}3/2\\3/2\end{bmatrix} = b_1$, so $\hat{x}_1$ is a least squares solution.
- $\hat{x}_1 = \frac{3}{10}\begin{bmatrix}1 & 2\end{bmatrix}^T \in \mathcal{N}^\perp(A)$, i.e. $\hat{x}_1 \perp \begin{bmatrix}2 & -1\end{bmatrix}^T$.
- Starting from $\begin{bmatrix}0 & 3/4\end{bmatrix}^T$ or $\begin{bmatrix}1/2 & 1/2\end{bmatrix}^T$ instead gives the same $\hat{x}_1$, e.g.

$$\tfrac{1}{5}\begin{bmatrix}1 & 2\\2 & 4\end{bmatrix}\begin{bmatrix}0\\3/4\end{bmatrix} = \begin{bmatrix}3/10\\3/5\end{bmatrix}.$$

- Norms: $\|\hat{x}_1\|^2 = \frac{9}{20} = 0.45$, smaller than $\left\|\begin{bmatrix}3/2 & 0\end{bmatrix}^T\right\|^2 = 2.25$, $\left\|\begin{bmatrix}0 & 3/4\end{bmatrix}^T\right\|^2 = 0.5625$ and $\left\|\begin{bmatrix}1/2 & 1/2\end{bmatrix}^T\right\|^2 = 0.5$.

6.4 General Procedure and the Fundamental Theorem

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- If $b \in \mathcal{R}(A)$: $\exists$ **an exact solution** to $A\hat{x} = b$.
- If $b \notin \mathcal{R}(A)$:

$$b_1 = \operatorname{Proj}_{\mathcal{R}(A)} b, \qquad A\hat{x} = b_1$$

  - If $\mathcal{N}(A) = \{0\} \Rightarrow \hat{x}$ **is unique**.
  - If $\mathcal{N}(A) \neq \{0\} \Rightarrow \hat{x}_1 = \operatorname{Proj}_{\mathcal{N}^\perp(A)} \hat{x}$.

$\hat{x}$: **minimum error solution** (least squares)
$\hat{x}_1$: **minimum error-norm solution**

Why the procedure works

- The error $b - A\hat{x}$ is smallest when $A\hat{x}$ is the point of $\mathcal{R}(A)$ closest to b, i.e. $A\hat{x} = b_1 = \operatorname{Proj}_{\mathcal{R}(A)} b$ (orthogonality principle: $b - b_1 \perp \mathcal{R}(A)$). Writing this as $A^H(b - A\hat{x}) = 0$ gives the **normal equations** $A^HA\hat{x} = A^Hb$.
- If $\mathcal{N}(A) \neq \{0\}$, every $\hat{x} + n$ with $n \in \mathcal{N}(A)$ gives the same $A\hat{x} = b_1$, so the least-squares solution is not unique. Splitting $\hat{x} = \hat{x}_1 + \hat{x}_2$ with $\hat{x}_1 \in \mathcal{N}^\perp(A)$, $\hat{x}_2 \in \mathcal{N}(A)$ gives $A\hat{x}_1 = A\hat{x} = b_1$ and, by Pythagoras, $\|\hat{x} + n\|^2 = \|\hat{x}_1\|^2 + \|\hat{x}_2 + n\|^2 \geq \|\hat{x}_1\|^2$. Hence $\hat{x}_1$ **is the least-squares solution of minimum norm**, and it does not depend on which $\hat{x}$ we

started from.

Example 6.2. $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}$, $\quad n = 3,\ m = 2 \quad (A \in \mathbb{C}^{m\times n}$: n columns, m rows).

$$\mathcal{R}(A) = \operatorname{sp}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}, \qquad \dim(\mathcal{R}(A)) = 1$$

$$\mathcal{N}(A) = \operatorname{sp}\left\{\begin{bmatrix}3\\0\\-1\end{bmatrix}, \begin{bmatrix}2\\-1\\0\end{bmatrix}\right\}$$

$$\underbrace{\dim(\mathcal{R}(A))}_{1} + \underbrace{\dim(\mathcal{N}(A))}_{2} = 3, \qquad \underbrace{\dim(\mathcal{N}(A))}_{2} + \underbrace{\dim(\mathcal{N}^{\perp}(A))}_{1} = 3$$

For $\mathcal{N}^{\perp}(A)$, a vector $[a\ b\ c]^T$ orthogonal to both basis vectors of $\mathcal{N}(A)$:

$$\begin{matrix} 3a - c = 0 \\ 2a - b = 0 \end{matrix} \quad \Rightarrow \quad \begin{matrix} c = 3a \\ b = 2a \end{matrix} \quad \Rightarrow \quad \begin{bmatrix}a\\b\\c\end{bmatrix} = a \underbrace{\begin{bmatrix}1\\2\\3\end{bmatrix}}_{\text{row of } A} \quad \Rightarrow \quad \mathcal{N}^{\perp}(A) = \operatorname{sp}\left\{\begin{bmatrix}1\\2\\3\end{bmatrix}\right\}$$

$$\mathcal{N}^{\perp}(A) = \mathcal{R}(A^H)$$

▷ $\therefore$ ***Fundamental theorem of linear algebra.***

$$\hat{x}_1 = \operatorname{Proj}_{\mathcal{N}^{\perp}(A)} \hat{x} = \operatorname{Proj}_{\mathcal{R}(A^H)} \hat{x}$$

▷ $\mathcal{R}(A^H)$*: row space of* A.

Dimensions and rank. For $A \in \mathbb{C}^{m\times n}$:

$$\dim(\underbrace{\mathcal{R}(A)}_{k}) + \dim(\underbrace{\mathcal{N}(A)}_{n-k}) = \#\text{ of columns} = n$$

$$\dim(\underbrace{\mathcal{N}(A)}_{n-k}) + \dim(\underbrace{\mathcal{R}(A^H)}_{k}) = n \qquad (\dim(\mathcal{N}(A)) + \dim(\mathcal{N}^{\perp}(A)) = n)$$

$$\left.\begin{matrix} \dim(\mathcal{R}(A)) = \#\text{ of lin. indep. columns of } A = \text{column rank} = k \\ \dim(\mathcal{R}(A^H)) = \#\text{ of lin. indep. rows of } A = \text{row rank} = k \end{matrix}\right\} \ k = \textbf{rank} \text{ of a matrix}$$

$$\#\text{ linearly indep. rows} = \#\text{ linearly indep. columns} \quad \text{as} \quad \dim \mathcal{R}(A) = k = \dim \mathcal{R}(A^H)$$

k: **# of linearly indep. rows = # of linearly indep. columns** (column rank = row rank).

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Theorem 6.3 (Fundamental theorem of linear algebra)**.** $\mathcal{R}(A^H) = \mathcal{N}^{\perp}(A)$

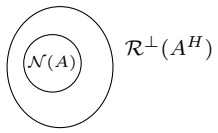
Proof. $\langle x, y\rangle \triangleq y^H x$.

$$\left\langle \underset{\mathbb{C}^m}{Ax}, \underset{\mathbb{C}^m}{y} \right\rangle_{\mathbb{C}^m} = y^H A x = (A^H y)^H x = \left\langle \underset{\mathbb{C}^n}{x}, \underset{\mathbb{C}^n}{A^H y} \right\rangle_{\mathbb{C}^n}$$

Prove $\mathcal{N}(A) = \mathcal{R}^{\perp}(A^H)$, which is easier to prove.

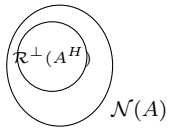
$$\left.\begin{matrix} \mathcal{N}(A) \subset \mathcal{R}^{\perp}(A^H) \\ \mathcal{R}^{\perp}(A^H) \subset \mathcal{N}(A) \end{matrix}\right\} \text{ prove both } \Rightarrow \ \mathcal{N}(A) = \mathcal{R}^{\perp}(A^H)$$

Part 1. $x \in \mathcal{N}(A) \Rightarrow Ax = 0$.



$$
\begin{aligned}
&\langle Ax, y\rangle \quad \text{where } \forall y \in \mathbb{C}^m \\
&\langle Ax, y\rangle = 0 \\
&\left\langle x, A^H y\right\rangle = 0 \quad \forall y \in \mathbb{C}^m \\
\Rightarrow\ & x \perp A^H y \quad \forall y \in \mathbb{C}^m \\
\Rightarrow\ & x \perp \mathcal{R}(A^H) \Rightarrow x \in \mathcal{R}^{\perp}(A^H)
\end{aligned}
$$

Part 2. $x \in \mathcal{R}^{\perp}(A^H)$.

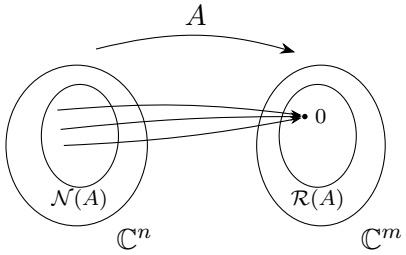


$$
\begin{aligned}
&x \in \mathcal{R}^{\perp}(A^H) \\
\Rightarrow\ & x \perp A^H y \quad \forall y \in \mathbb{C}^m \\
&\left\langle x, A^H y\right\rangle = 0 \quad \forall y \in \mathbb{C}^m \\
&\langle Ax, y\rangle = 0 \quad \forall y \in \mathbb{C}^m
\end{aligned}
$$

▷ *$Ax \in \mathbb{C}^m$, so choose $y = Ax$:*

$$\langle Ax, Ax\rangle = \|Ax\|^2 = 0 \Rightarrow Ax = 0 \Rightarrow x \in \mathcal{N}(A)$$

Both inclusions hold, so $\mathcal{N}(A) = \mathcal{R}^{\perp}(A^H)$. Taking orthogonal complements of both sides (in finite dimensions $(S^{\perp})^{\perp} = S$ for a subspace S) gives $\mathcal{N}^{\perp}(A) = \mathcal{R}(A^H)$. □



$$
\begin{aligned}
&\overset{k}{\dim(\mathcal{R}(A))} + \overset{n-k}{\dim(\mathcal{N}(A))} = n \\
&\dim(\mathcal{N}(A)) + \dim(\mathcal{N}^{\perp}(A)) = n \\
&\underbrace{\dim(\mathcal{R}(A^H))}_{k} + \underbrace{\dim(\mathcal{N}(A))}_{n-k} = n \\
&\underbrace{\dim(\mathcal{R}(A))}_{\text{column rank}} = \underbrace{\dim(\mathcal{R}(A^H))}_{\text{row rank}} = k
\end{aligned}
$$

▷ *k: rank.*

6.5 Special Cases by Rank

6.5.1 Case I: Full Column Rank

A is/has **full column rank**: columns of A are linearly independent.

$$
m\Big\{\underbrace{\begin{bmatrix} \\ \\ \end{bmatrix}}_{\text{tall matrix}}\overset{n}{}\begin{bmatrix} \\ \\ \end{bmatrix}\overset{n}{} = \begin{bmatrix} \\ \\ \end{bmatrix}\Big\}m \qquad m \geq n
$$

- $n = \#$ of unknowns, $\quad m = \#$ of equations
- $m > n$: **overdetermined** system of equations.

Assume

$$\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$$

3 vectors cannot be linearly indep. for $\mathbb{R}^2$.

▷ *Under the columns: "2 dim", "2 dimension". Each column lies in $\mathbb{R}^2$, so at most 2 of them can be linearly independent; full column rank therefore forces $m \geq n$.*

$$\dim(\mathcal{R}(A)) = n$$
$$\dim(\mathcal{N}(A)) = 0 \Rightarrow \mathcal{N}(A) = \{0\} \qquad A \text{ is one-to-one}$$

If we have an exact solution, then the solution is unique.

If $b \notin \mathcal{R}(A)$:

$$P_{\mathcal{R}(A)} = B(B^H B)^{-1} B^H$$

with $B = A$:

$$P_{\mathcal{R}(A)} = A(A^H A)^{-1} A^H$$
$$b_1 = P_{\mathcal{R}(A)} b = A(A^H A)^{-1} A^H b \qquad (A\hat{x} = b_1)$$
$$\Rightarrow \quad A(\underbrace{\hat{x} - (A^H A)^{-1} A^H b}_{\in \mathcal{N}(A)}) = 0$$
$$\Rightarrow \quad \hat{x} - (A^H A)^{-1} A^H b = 0 \;\Rightarrow\; \boxed{\hat{x} = (A^H A)^{-1} A^H b}$$

▷ *Unique solution as $\mathcal{N}(A) = \{0\}$.*

$(A^H A)^{-1} A^H$: **left pseudo inverse** of A.

Explanation

- $A^H A$ **$(n \times n)$ is invertible exactly when A has full column rank**: $A^H Ax = 0 \Rightarrow x^H A^H Ax = \|Ax\|^2 = 0 \Rightarrow Ax = 0 \Rightarrow x = 0$.
- The same formula comes directly from the normal equations $A^H A\hat{x} = A^H b$.
- "Left" inverse: $((A^H A)^{-1} A^H)A = I_n$. If $b \in \mathcal{R}(A)$, it returns the exact solution.

6.5.2 Case II: Full Row Rank

A is/has **full row rank**: all rows of A are linearly indep.

$$m \Big\{ \underbrace{\begin{bmatrix} & n & \\ & & \end{bmatrix}}_{\text{fat matrix}} \begin{bmatrix} \\ \\ \end{bmatrix} = \begin{bmatrix} \\ \end{bmatrix} \qquad m \leq n$$

- m: # of equations, $\quad n$: # of unknowns
- $m < n$: **underdetermined** system of equations.

$$\dim(\mathcal{R}(A^H)) = m$$
$$\dim(\mathcal{R}(A)) = m \;\Rightarrow\; \mathcal{R}(A) = \mathbb{C}^m$$

$b \in \mathcal{R}(A) \quad \forall b \in \mathbb{C}^m \Rightarrow$ **Always has an exact solution** (at least one): A is onto.

$$\hat{x}_1 = \text{Proj}_{\mathcal{N}^\perp(A)}\, \hat{x} = \text{Proj}_{\mathcal{R}(A^H)}\, \hat{x}$$

with $B = A^H$:

$$P_{\mathcal{R}(A^H)} = A^H(AA^H)^{-1} A$$
$$\hat{x}_1 = A^H(AA^H)^{-1} \underbrace{A\hat{x}}_{b} \;\Rightarrow\; \hat{x}_1 = A^H(AA^H)^{-1} b$$

$A^H(AA^H)^{-1}$: **right pseudo inverse** of A.

Explanation

- AA^H $(m \times m)$ is invertible since the rows of A are independent (same argument as for $A^H A$, applied to A^H).
- "Right" inverse: $A(A^H(AA^H)^{-1}) = I_m$, so $A\hat{x}_1 = b$ is indeed an exact solution.
- $\hat{x}_1 = A^H((AA^H)^{-1}b) \in \mathcal{R}(A^H) = \mathcal{N}^\perp(A)$, so **it is the minimum-norm one** among the infinitely many exact solutions $\hat{x}_1 + \mathcal{N}(A)$.

6.5.3 Case III: Full Column and Full Row Rank

A is/has both full column and row rank.

$$\left.\begin{matrix} m \geq n \\ n \geq m \end{matrix}\right\} \; m = n: \quad \left.\begin{matrix} A \text{ is a square matrix} \\ A \text{ is one-to-one} \\ A \text{ is onto} \end{matrix}\right\} \; \begin{matrix} A \text{ has an inverse} \\ A \text{ is invertible.} \end{matrix}$$

▷ *Both pseudo inverses reduce to A^{-1} here: $(A^HA)^{-1}A^H = A^{-1}(A^H)^{-1}A^H = A^{-1}$ and $A^H(AA^H)^{-1} = A^{-1}$.*

Exact solution exists. The solution is unique.

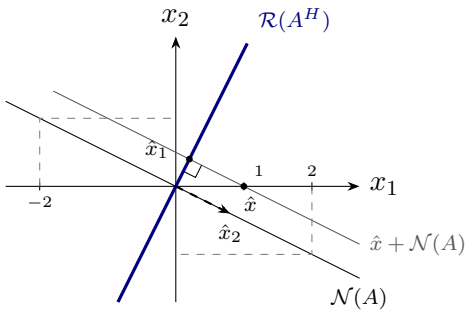
$$Ax = b, \qquad x_1 = \hat{x} = x = A^{-1}b$$

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Example 6.4.

$$\underbrace{\begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{b}, \qquad \mathcal{R}(A) = \operatorname{sp}\left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}, \quad b \in \mathcal{R}(A) \;\Rightarrow\; \text{has exact solution}, \quad \hat{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\dim \mathcal{R}(A) = 1$$
$$\dim \mathcal{N}(A) = 1$$
$$\mathcal{N}(A) = \operatorname{sp}\left\{\begin{bmatrix} -2 \\ 1 \end{bmatrix}\right\}$$
$$\hat{x} + n = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \alpha \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \quad n \in \mathcal{N}(A)$$



▷ *In the sketch: $\mathcal{R}(A^H) = \operatorname{sp}\{[1\ 2]^T\}$; the solution set $\hat{x} + \mathcal{N}(A)$ is the line through $\hat{x} = [1\ 0]^T$ parallel to $\mathcal{N}(A)$; $\hat{x}_1 = \frac{1}{5}[1\ 2]^T$ is its point on $\mathcal{R}(A^H)$, and $\hat{x}_2 = \hat{x} - \hat{x}_1 \in \mathcal{N}(A)$ is the part to "get rid of".*

▷ *Check: $\hat{x}_1 = \frac{1}{5}[1\ 2]^T$ gives $A\hat{x}_1 = [1\ 1]^T = b$, and $\|\hat{x}_1\|^2 = \frac{1}{5} < 1 = \|\hat{x}\|^2$; the right pseudo inverse cannot be used here since $AA^H = [\frac{5}{5}\ \frac{5}{5}]$ is singular (A has rank 1, neither full row nor full column rank), so the projection route is used.*

6.5.4 Case IV: Neither Full Column nor Full Row Rank

A is/has neither full column rank nor full row rank.

Example 6.5.

$$\underbrace{\begin{bmatrix}1&0&1\\1&1&2\\0&1&1\end{bmatrix}}_{A}\underbrace{\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}}_{x}=\underbrace{\begin{bmatrix}1\\0\\3\end{bmatrix}}_{b}$$

2 linearly indep. columns
2 linearly indep. rows
$\dim(\mathcal{R}(A))=\dim(\mathcal{R}(A^H))=2$
$\dim(\mathcal{N}(A))=1$
$\dim(\mathcal{N}^{\perp}(A))=2$

$$\begin{bmatrix}1\\0\\3\end{bmatrix}=\underset{\alpha}{1}\begin{bmatrix}1\\1\\0\end{bmatrix}+\underset{\beta}{3}\begin{bmatrix}0\\1\\1\end{bmatrix}\quad\times\qquad\Rightarrow\qquad b\notin\mathcal{R}(A)\Rightarrow\textbf{No exact solution.}$$

▷ *Column 3 = column 1 + column 2, so* $\mathcal{R}(A)=\mathrm{sp}\{[1\ 1\ 0]^T,[0\ 1\ 1]^T\}$. *Matching the 1st and 3rd entries of b forces* $\alpha=1$, $\beta=3$, *but then the 2nd entry would be* $\alpha+\beta=4\neq 0$.

Method I.

$$P_{\mathcal{R}(A)}=B(B^HB)^{-1}B^H$$

$$B=\begin{bmatrix}1&0\\1&1\\0&1\end{bmatrix},\qquad B^HB=\begin{bmatrix}1&1&0\\0&1&1\end{bmatrix}\begin{bmatrix}1&0\\1&1\\0&1\end{bmatrix}=\begin{bmatrix}2&1\\1&2\end{bmatrix}$$

$$(B^HB)^{-1}=\frac{1}{3}\begin{bmatrix}2&-1\\-1&2\end{bmatrix}$$

$$P_{\mathcal{R}(A)}=\begin{bmatrix}1&0\\1&1\\0&1\end{bmatrix}\frac{1}{3}\begin{bmatrix}2&-1\\-1&2\end{bmatrix}\begin{bmatrix}1&1&0\\0&1&1\end{bmatrix}=\frac{1}{3}\begin{bmatrix}2&1&-1\\1&2&1\\-1&1&2\end{bmatrix}$$

$$b_1=P_{\mathcal{R}(A)}b=\frac{1}{3}\begin{bmatrix}-1\\4\\5\end{bmatrix}$$

$$\underbrace{\begin{bmatrix}1&0&1\\1&1&2\\0&1&1\end{bmatrix}}_{A}\underbrace{\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}}_{\hat{x}}=\underbrace{\frac{1}{3}\begin{bmatrix}-1\\4\\5\end{bmatrix}}_{b_1}\qquad\Rightarrow\qquad\hat{x}=\frac{1}{3}\begin{bmatrix}-1\\5\\0\end{bmatrix}$$

minimum error solution.

▷ *This was method I. With* $x_3=0$: $x_1=-\frac{1}{3}$, $x_1+x_2=\frac{4}{3}\Rightarrow x_2=\frac{5}{3}$, *and the 3rd row* $x_2+x_3=\frac{5}{3}$ *checks. The error is* $b-b_1=\frac{4}{3}[1\ -1\ 1]^T$, *which is orthogonal to both columns of B, as it must be.*

Method II.

$$\underbrace{\begin{bmatrix}1&0\\1&1\\0&1\end{bmatrix}}_{A_1}\underbrace{\begin{bmatrix}x_1\\x_2\end{bmatrix}}_{z}=\begin{bmatrix}1\\0\\3\end{bmatrix}$$

$\Rightarrow\mathcal{N}(A_1)=\{0\}$
$\Rightarrow$ full column rank
$\Rightarrow$ Exact solution is known

▷ *Idea: drop the dependent third column (set* $x_3=0$*); the remaining* A_1 *has full column rank and* $\mathcal{R}(A_1)=\mathcal{R}(A)$, *so Case I applies.*

$$\hat{z}=(A_1^HA_1)^{-1}A_1^Hb=\frac{1}{3}\begin{bmatrix}2&-1\\-1&2\end{bmatrix}\begin{bmatrix}1&1&0\\0&1&1\end{bmatrix}\begin{bmatrix}1\\0\\3\end{bmatrix}=\underbrace{\frac{1}{3}\begin{bmatrix}-1\\5\end{bmatrix}}_{\hat{z}\ \to\ \text{unique}}\qquad\therefore\ \hat{x}=\frac{1}{3}\begin{bmatrix}-1\\5\\0\end{bmatrix}$$

Minimum-norm solution. $A^H = A^T$ as matrix is real.

$$A^H = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$$

$$P_{\mathcal{R}(A^H)} = B(B^H B)^{-1} B^H, \qquad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$(B^H B)^{-1} = \frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

$$P_{\mathcal{R}(A^H)} = \frac{1}{3}\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

$$\hat{x}_1 = P_{\mathcal{R}(A^H)}\hat{x} = \frac{1}{9}\begin{bmatrix} -7 \\ 11 \\ 4 \end{bmatrix}$$

▷ $\therefore \hat{x}_1$***: minimum error, minimum-norm solution.***

Check

$A\hat{x}_1 = \frac{1}{9}[-3 \quad 12 \quad 15]^T = \frac{1}{3}[-1 \quad 4 \quad 5]^T = b_1$, so $\hat{x}_1$ has the same (minimum) error as $\hat{x}$. Moreover $\hat{x} - \hat{x}_1 = \frac{4}{9}[1 \quad 1 \quad -1]^T$ spans $\mathcal{N}(A)$ (indeed $A[1\ 1\ -1]^T = 0$), and $\|\hat{x}_1\|^2 = \frac{186}{81} \approx 2.30 < \frac{26}{9} \approx 2.89 = \|\hat{x}\|^2$.

7 Spectral Analysis of Linear Transformations

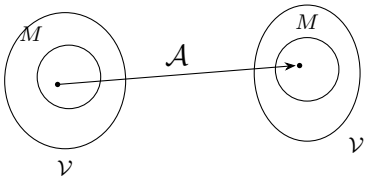
Syllabus: Weeks 8–9 — eigenvalues and eigenvectors, invariance under operators, block diagonal forms

7.1 Invariant Subspaces

For a matrix A, the set of its eigenvalues $\{\lambda_1, \lambda_2, \ldots, \lambda_n\}$ is called the **spectrum** of the matrix.

Definition 7.1. Let $\mathcal{A} : \mathcal{V} \to \mathcal{V}$ be a linear transformation defined over a vector space $\mathcal{V}$. A subspace M of $\mathcal{V}$ is said to be **invariant under $\mathcal{A}$** iff

$$\mathcal{A}x \in M \quad \forall x \in M$$



Example 7.2. $\mathcal{V} = \{f \mid f$ functions defined over $\mathbb{R}$ which are infinitely differentiable$\}$

$$\mathcal{A} = \frac{d}{dt} : \ \mathcal{V} \to \mathcal{V}, \qquad M_1 \triangleq \text{sp}\{\sin t\} \subset \mathcal{V}$$

$$\mathcal{A}(\alpha \sin t) = \alpha \mathcal{A} \sin t = \alpha \cos t \notin M_1$$

$\therefore M_1$ **is not invariant under $\mathcal{A}$.**

$$M_2 \triangleq \text{sp}\{\cos t, \sin t\}$$
$$x = \alpha_1 \cos t + \alpha_2 \sin t$$
$$\mathcal{A}x = -\alpha_1 \sin t + \alpha_2 \cos t \in \text{sp}\{\cos t, \sin t\} = M_2$$

$\therefore M_2$ **is invariant under $\mathcal{A}$.**

$$M_3 = \{0\}, \qquad \mathcal{A}0 = 0 \in M_3$$
$$M_4 = \text{sp}\{e^{2t}\}, \qquad x = \alpha e^{2t}, \qquad \mathcal{A}x = 2\alpha e^{2t} \in M_4$$

$\therefore M_3$ and M_4 are invariant under $\mathcal{A}$ (e^{2t} is mapped to a multiple of itself: it is an "eigenfunction" of $\frac{d}{dt}$ with eigenvalue 2).

Explanation

The trivial subspaces $\{0\}$ and $\mathcal{V}$ are always invariant. **A one-dimensional subspace** sp$\{e\}$ **is invariant iff** $\mathcal{A}e = \lambda e$ for some scalar λ, i.e. iff e is an eigenvector — this is the link between invariant subspaces and the eigenvalues defined below.

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Example 7.3. $\mathcal{N}(\mathcal{A})$ & $\mathcal{R}(\mathcal{A})$ are invariant under $\mathcal{A}$.

$\underline{\mathcal{N}(\mathcal{A})}$: $x \in \mathcal{N}(\mathcal{A})$

$$\mathcal{A}x = 0_{\mathcal{V}} \in \mathcal{N}(\mathcal{A})$$

$\mathcal{N}(\mathcal{A})$ is invariant under $\mathcal{A}$.

$\underline{\mathcal{R}(\mathcal{A})}$: $x \in \mathcal{R}(\mathcal{A})$

$$\mathcal{A}x \in \mathcal{R}(\mathcal{A})$$

$\mathcal{R}(\mathcal{A})$ is invariant under $\mathcal{A}$.

7.2 Powers and Polynomials of a Linear Transformation

Definition 7.4. **Powers** of a linear transformation $\mathcal{A} : \mathcal{V} \to \mathcal{V}$ are defined as

$$\mathcal{A}^k x = \underbrace{\mathcal{A}(\mathcal{A} \cdots \mathcal{A}(\mathcal{A}x) \dots)}_{\mathcal{A} \text{ is applied } k \text{ times}}$$

When the domain and the codomain are the same, we can define the powers of a linear transformation.

- $\mathbb{R}^n \to \mathbb{R}^m$, $m \neq n \Rightarrow$ not defined.
- $\mathcal{A} : \mathcal{V} \to \mathcal{W}$: $\mathcal{A}(\underbrace{\mathcal{A}x}_{\in \mathcal{W}})$ $\times$ not defined, since $\mathcal{A}$ acts on elements $\in \mathcal{V}$.

$\star$ **Polynomials of $\mathcal{A}$** can be constructed using linear combinations of powers of $\mathcal{A}$.

$$\begin{aligned} p(s) &= \alpha_0 s^n + \alpha_1 s^{n-1} + \cdots + \alpha_{n-1} s + \alpha_n \, (s^0) \\ p(\mathcal{A}) &= \alpha_0 \mathcal{A}^n + \alpha_1 \mathcal{A}^{n-1} + \cdots + \alpha_{n-1} \mathcal{A}^1 + \alpha_n \mathcal{A}^0 \\ p(\mathcal{A}) &= \alpha_0 \mathcal{A}^n + \cdots + \alpha_{n-1} \mathcal{A} + \alpha_n I \end{aligned}$$

$$\mathcal{A}^0 x = x = Ix$$
$$\mathcal{A}^0 = I$$

For example, with $p(\mathcal{A}) = 3\mathcal{A}^3 + 2\mathcal{A}^2 + \mathcal{A} + I$:

$$\begin{aligned} p(\mathcal{A})\mathcal{A}x &= (3\mathcal{A}^3 + 2\mathcal{A}^2 + \mathcal{A} + I)\mathcal{A}x = 3\mathcal{A}^4 x + 2\mathcal{A}^3 x + \mathcal{A}^2 x + \mathcal{A}x \\ &= \mathcal{A}(3\mathcal{A}^3 x + 2\mathcal{A}^2 x + \mathcal{A}x + x) \\ &= \mathcal{A}\underbrace{(3\mathcal{A}^3 + 2\mathcal{A}^2 + \mathcal{A} + I)}_{p(\mathcal{A})} x \end{aligned}$$

$$\boxed{p(\mathcal{A})\mathcal{A}x = \mathcal{A}p(\mathcal{A})x \quad \forall x \in \mathcal{V}} \qquad \Rightarrow \qquad p(\mathcal{A})\mathcal{A} = \mathcal{A}p(\mathcal{A})$$

▷ *$\mathcal{A}$ commutes with $p(\mathcal{A})$.*

Example 7.5. $\mathcal{N}(p(\mathcal{A}))$ & $\mathcal{R}(p(\mathcal{A}))$ are invariant under $\mathcal{A}$.
$\underline{\mathcal{N}(p(\mathcal{A}))}$: $x \in \mathcal{N}(p(\mathcal{A}))$. Is $\mathcal{A}x \in \mathcal{N}(p(\mathcal{A}))$?

$$\begin{aligned} p(\mathcal{A})\mathcal{A}x &= \mathcal{A}\underbrace{p(\mathcal{A})x}_{0_\mathcal{V}} = \mathcal{A}0_\mathcal{V} = 0_\mathcal{V} \\ &\Rightarrow \mathcal{A}x \in \mathcal{N}(p(\mathcal{A})) \end{aligned}$$

$\underline{\mathcal{R}(p(\mathcal{A}))}$: $x \in \mathcal{R}(p(\mathcal{A}))$, $x = p(\mathcal{A})y$. p. 53

$$\begin{aligned} \mathcal{A}x = \mathcal{A}p(\mathcal{A})y = p(\mathcal{A})\underbrace{\mathcal{A}y}_{y'} = p(\mathcal{A})y' \\ \Rightarrow \mathcal{A}x \in \mathcal{R}(p(\mathcal{A})) \end{aligned}$$

7.3 Eigenvalues and Eigenvectors

⋆ Matrix representation:

$$\mathcal{A}: \underset{\substack{\downarrow B \\ \text{basis set for } \mathcal{V}}}{\overset{n}{\mathcal{V}}} \to \underset{\substack{\downarrow B \\ \text{basis set for } \mathcal{V}}}{\overset{n}{\mathcal{V}}} \qquad A: \mathbb{C}^n \to \mathbb{C}^n \qquad \begin{array}{cc} y = \mathcal{A}x & \lambda e = \mathcal{A}e \\ \Downarrow & \Downarrow \\ [y]_B = A[x]_B & \lambda[e]_B = A[e]_B \end{array}$$

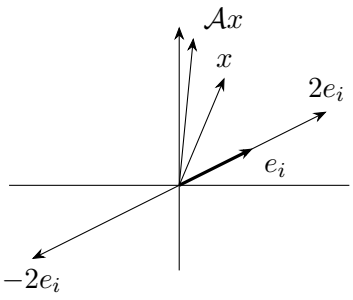
$$\det \mathcal{A} \triangleq \det A, \qquad \operatorname{tr} \mathcal{A} \triangleq \operatorname{tr} A$$

▷ *Well defined: another basis gives* $\bar{A} = P^{-1}AP$, *and* $\det(P^{-1}AP) = \det A$, $\operatorname{tr}(P^{-1}AP) = \operatorname{tr} A$.

Definition 7.6. Let A denote the matrix representation of a linear transformation $\mathcal{A}$ from $\mathcal{V}$ into $\mathcal{V}$ (A is a square matrix). The **eigenvalues** of $\mathcal{A}$ denoted by λ_i are defined as the n roots of the equation

$$\det(sI - A) = 0,$$

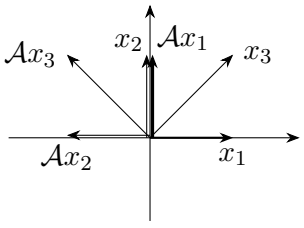
where $d(s) \triangleq \det(sI - A)$ is called the **characteristic polynomial** of A. The vectors $e_i \neq 0$ satisfying $Ae_i = \lambda_i e_i$ are called **eigenvectors** of A.



$$\mathcal{A}e_i = \lambda_i e_i$$
$$\lambda_i = 2$$
$$\lambda_i = -2$$

▷ *A general* x *is turned into a different direction* ($\mathcal{A}x \not\parallel x$); *an eigenvector* e_i *is only scaled, e.g. to* $2e_i$ *if* $\lambda_i = 2$ *or to* $-2e_i$ *if* $\lambda_i = -2$, *so the line* $\operatorname{sp}\{e_i\}$ *is invariant.*

$$A = \underset{\text{rotation matrix}}{\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}} \qquad x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \qquad x_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



$$s^2 + 1 = 0 \qquad \lambda_{1,2} = \pm j$$

$$\underset{\text{real}}{A}\, e_i = \underset{\text{complex}}{\lambda_i}\, e_i \qquad R: \text{ rotation matrix}$$

$$Re_i = \lambda_i e_i$$
$$\|Re_i\| = \|\lambda_i e_i\|$$
$$\cancel{\|e_i\|} = |\lambda_i| \cancel{\|e_i\|} \qquad (e_i: \text{ nonzero vector})$$
$$|\lambda_i| = 1$$
$$\therefore \quad \boxed{\lambda_i = e^{j\omega}}$$

Explanation

No real vector keeps its direction under a 90° rotation, so there are no real eigenvectors; the eigenvectors are complex: $A[1\ -j]^T = [j\ 1]^T = j\,[1\ -j]^T$ and $A[1\ j]^T = -j\,[1\ j]^T$. A rotation preserves length, $\|Re\| = \|e\|$ (also for complex e, since $R^H R = I$), which is what forces

$|\lambda_i| = 1$; here $\lambda = \pm j = e^{\pm j\pi/2}$, i.e. $\omega = \pm\pi/2$ is the rotation angle.

⋆ Existence of eigenvectors:

$$A\overset{\neq 0}{e_i} = \lambda_i e_i \qquad d(\lambda_i) = \det(\lambda_i I - A) = 0$$

$$A = [c_1\ c_2\ \dots\ c_n]$$
$$\exists\ \alpha_1, \alpha_2, \dots, \alpha_n \text{ not all equal to zero satisfying}$$
$$\alpha_1 c_1 + \alpha_2 c_2 + \cdots + \alpha_n c_n = 0$$
$$\underset{A}{[c_1\ c_2\ \dots\ c_n]} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix} = 0$$

▷ *A is not full column rank $\Rightarrow \exists \alpha \neq 0$ s.t. $A\alpha = 0$. Apply this to the singular matrix $\lambda_i I - A$ ($\det(\lambda_i I - A) = 0$ means its columns are dependent):*

$$\exists \alpha \text{ s.t. } (\lambda_i I - A)\alpha = 0$$
$$A\alpha = \lambda_i \alpha \ \Rightarrow\ \text{Eğer bir eigenvalue varsa en az 1 eigenvector vardır}$$

▷ ***If there is an eigenvalue, then there is at least 1 eigenvector**: $\alpha \neq 0$ is an eigenvector for λ_i.*

Example 7.7. $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$:

$$d(s) = \det(sI - A) = \det \begin{bmatrix} s-2 & -1 \\ -1 & s-2 \end{bmatrix}$$
$$= (s-2)^2 - 1 = s^2 - 4s + 3 = (s-3)(s-1) = 0$$
$$\Rightarrow\ \lambda_1 = 3, \quad \lambda_2 = 1$$

$$Ae_i = \lambda_i e_i$$
$$(A - \lambda_i I)e_i = 0$$
$$e_i \in \underbrace{\mathcal{N}(A - \lambda_i I)}_{\textbf{eigenspace}}, \quad \dim \geq 1 \qquad \begin{matrix} p(s) = s - \lambda_i \\ p(A) = A - \lambda_i I \end{matrix}$$

$$A - \lambda_1 I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad e_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$
$$A - \lambda_2 I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \qquad e_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\mathcal{N}(A - \lambda_1 I) = \text{sp}\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$$
$$\mathcal{N}(A - \lambda_2 I) = \text{sp}\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$$

$$\mathcal{N}(A-\lambda_1 I) \overset{\perp}{\oplus} \mathcal{N}(A-\lambda_2 I) = \mathbb{C}^2$$

▷ $\overset{\perp}{\oplus}$: *basis vectors are linearly indep. and orthogonal* ($[1\ 1]^T \perp [1\ -1]^T$); *dimensions* $1+1=2=\dim \mathbb{C}^2$.

$$\alpha \begin{bmatrix}1\\1\end{bmatrix} + \beta \begin{bmatrix}1\\-1\end{bmatrix} = \begin{matrix}\mathbb{R}^2\\ \mathbb{C}^2\end{matrix}$$

$$\alpha \begin{bmatrix}1\\1\end{bmatrix} + \beta \begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix} \quad \text{iff } \alpha=\beta=0 \ \Rightarrow\ \text{subspaces are linearly indep.}$$

▷ *Orthogonality is special here because* $A = A^H$ *is Hermitian; in general eigenspaces are only linearly independent (see p. 57).*

p. 55

⋆ Change of basis:

$$y = Ax, \qquad B = \left\{ \begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix} \right\} \ \rightarrow\ \text{canonical basis set}$$

$$x = \begin{bmatrix}x_1\\x_2\end{bmatrix}, \quad [x]_B, \qquad [y]_B = A[x]_B \ \sim\ [y]_{\bar{B}} = \bar{A}[x]_{\bar{B}}, \qquad \begin{bmatrix}x_1\\x_2\end{bmatrix} = x_1 \begin{bmatrix}1\\0\end{bmatrix} + x_2 \begin{bmatrix}0\\1\end{bmatrix}$$

$$\bar{B} = \left\{ \begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}1\\-1\end{bmatrix} \right\}, \qquad A = \begin{bmatrix}2 & 1\\1 & 2\end{bmatrix}$$

$$x = \begin{bmatrix}1\\1\end{bmatrix}, \qquad [x]_{\bar{B}} = \begin{bmatrix}1\\0\end{bmatrix}, \qquad y = Ax = \begin{bmatrix}3\\3\end{bmatrix} = 3\begin{bmatrix}1\\1\end{bmatrix}, \qquad [y]_{\bar{B}} = \begin{bmatrix}3\\0\end{bmatrix}$$

$$\begin{bmatrix}3\\0\end{bmatrix} = \bar{A}\begin{bmatrix}1\\0\end{bmatrix} \qquad \Rightarrow\ \bar{A} = \begin{bmatrix}3 & \\0 & \end{bmatrix}$$

$$x = \begin{bmatrix}1\\-1\end{bmatrix}, \qquad [x]_{\bar{B}} = \begin{bmatrix}0\\1\end{bmatrix}, \qquad y = Ax = \begin{bmatrix}1\\-1\end{bmatrix} = 1\begin{bmatrix}1\\-1\end{bmatrix}, \qquad [y]_{\bar{B}} = \begin{bmatrix}0\\1\end{bmatrix}$$

$$\begin{bmatrix}0\\1\end{bmatrix} = \bar{A}\begin{bmatrix}0\\1\end{bmatrix} \qquad \Rightarrow\ \bar{A} = \begin{bmatrix}3 & 0\\0 & 1\end{bmatrix}$$

$$P = [e_1 \mid e_2], \qquad \bar{A} = f(A) = ?$$

$$Ae_1 = \lambda_1 e_1$$
$$Ae_2 = \lambda_2 e_2$$

$$[\lambda_1 e_1 \mid \lambda_2 e_2] = [Ae_1 \mid Ae_2] = A[e_1 \mid e_2]$$

$$\underbrace{[e_1 \mid e_2]}_{P} \underbrace{\begin{bmatrix}\lambda_1 & 0\\0 & \lambda_2\end{bmatrix}}_{\bar{A}} = A \cdot P \ \Rightarrow\ P\bar{A} = AP$$

$$\Rightarrow\ \boxed{\bar{A} = P^{-1}AP}$$

▷ P *is invertible (full rank: its columns* e_1, e_2 *are linearly indep.).*

Explanation

The columns of P are the new basis vectors written in the old basis, so $[x]_B = P[x]_{\bar{B}}$. Then $[y]_B = A[x]_B$ becomes $P[y]_{\bar{B}} = AP[x]_{\bar{B}}$, i.e. $[y]_{\bar{B}} = P^{-1}AP\,[x]_{\bar{B}}$. Check:

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad P^{-1} = \tfrac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad P^{-1}AP = \tfrac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 3 & 1 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix},$$

as found above.

$$\mathbb{C}^2 = \underset{1}{\mathcal{N}}(A - \lambda_1 I) \overset{\perp}{\oplus} \underset{1}{\mathcal{N}}(A - \lambda_2 I), \qquad \left[\begin{array}{c|c} \overset{1\times 1}{\lambda_1} & 0 \\ \hline 0 & \lambda_2 \end{array}\right]$$

▷ *Each block is* 1×1 *as* $\dim \mathcal{N}(A - \lambda_1 I) = \dim \mathcal{N}(A - \lambda_2 I) = 1$.

7.4 Block Diagonal Representations

Theorem 7.8 (Block diagonal representation)**.** Consider the linear transformation

$$y = Ax \quad \text{where } A \in \mathbb{C}^{n\times n},\ y, x \in \mathbb{C}^n.$$

Suppose that

i) $\mathbb{C}^n = M_1 \oplus M_2 \oplus \cdots \oplus M_k$.

ii) M_i is invariant under A $\forall i$.

Then, the transformation A has a **block diagonal representation**

$$\bar{A} = \begin{bmatrix} \bar{A}_1 & & & 0 \\ & \bar{A}_2 & & \\ & & \ddots & \\ 0 & & & \bar{A}_k \end{bmatrix}$$

where $\bar{A} = P^{-1}AP$, p. 56

$$P = [P_1 \mid P_2 \mid \cdots \mid P_k]$$
$$P_i = [b_i^1\ b_i^2\ b_i^3\ \ldots\ b_i^{n_i}], \qquad n_i = \dim M_i$$

$b_i^1, \ldots, b_i^{n_i} \in \mathbb{C}^n$ are basis vectors of M_i. $\bar{A}_i$ is of size $n_i \times n_i$.

Sketch of a proof.

$$k = 2, \qquad \mathbb{C}^n = M_1 \oplus M_2, \qquad n = 5$$
$$M_1 = \mathrm{sp}\{b_1^1, b_1^2\}, \qquad \dim M_1 = 2$$
$$M_2 = \mathrm{sp}\{b_2^1, b_2^2, b_2^3\}, \qquad \dim M_2 = 3$$

$$P = [P_1 \mid P_2], \qquad \begin{matrix} P_1 = [b_1^1\ b_1^2]_{5\times 2} \\ P_2 = [b_2^1\ b_2^2\ b_2^3]_{5\times 3} \end{matrix} \qquad \bar{A} = P^{-1}AP \ \Rightarrow\ P\bar{A} = AP$$

$$\underbrace{[b_1^1\, b_1^2 \mid b_2^1\, b_2^2\, b_2^3]}_{P} \underbrace{\overset{2\times 2 \,\to\, \dim M_1 \times \dim M_1}{\left[\begin{array}{cc|ccc} \alpha_1^1 & \beta_1^1 & 0 & 0 & 0 \\ \alpha_1^2 & \beta_1^2 & 0 & 0 & 0 \\ \hline 0 & 0 & \alpha_2^1 & \beta_2^1 & \gamma_2^1 \\ 0 & 0 & \alpha_2^2 & \beta_2^2 & \gamma_2^2 \\ 0 & 0 & \alpha_2^3 & \beta_2^3 & \gamma_2^3 \end{array}\right]}}_{\bar{A} \quad \left(\begin{smallmatrix} 3\times 3 \\ \dim M_2 \times \dim M_2 \end{smallmatrix}\right)} = A \overbrace{[b_1^1\ b_1^2\ b_2^1\ b_2^2\ b_2^3]}^{P}$$

$$= [\underbrace{Ab_1^1}_{\in M_1} \mid \underbrace{Ab_1^2}_{\in M_1} \mid \underbrace{Ab_2^1}_{\in M_2} \mid \underbrace{Ab_2^2}_{\in M_2} \mid \underbrace{Ab_2^3}_{\in M_2}]$$

▷ $\in M_1$: *lin. comb. of* M_1 *vectors;* $\in M_2$: *lin. comb. of* M_2 *vectors.*

Why the off-diagonal blocks are zero

Column j of $P\bar{A}$ is P times column j of $\bar{A}$, i.e. the combination of the basis vectors with the coefficients in that column. Since M_1 is invariant, $Ab_1^1 = \alpha_1^1 b_1^1 + \alpha_1^2 b_1^2$ needs *no* M_2 vectors, so the lower entries of columns 1–2 are zero; likewise $Ab_2^j \in M_2$ gives zeros in the upper part of columns 3–5. The coefficients are unique because the five vectors form a basis of $\mathbb{C}^5$ ($\mathbb{C}^n = M_1 \oplus M_2$), which also makes P invertible. Each block $\bar{A}_i$ is the matrix of A restricted to M_i.

Example 7.9.

$$A = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 3 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$M_1 \triangleq \operatorname{sp}\left\{ \overset{b_1^1}{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}}, \overset{b_1^2}{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}} \right\}, \ \dim M_1 = 2, \qquad M_2 \triangleq \operatorname{sp}\left\{ \overset{b_2^1}{\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}} \right\}, \ \dim M_2 = 1$$

$$Ab_1^1 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix} = 2b_1^1 + 0b_1^2 \in M_1$$

$$Ab_1^2 = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} = 1b_1^1 + 2b_1^2 \in M_1$$

$$\begin{aligned} A(\alpha_1 b_1^1 + \alpha_2 b_1^2) &= \alpha_1 Ab_1^1 + \alpha_2 Ab_1^2 \\ &= \alpha_1(2b_1^1) + \alpha_2(1b_1^1 + 2b_1^2) \\ &= (2\alpha_1 + \alpha_2)b_1^1 + 2\alpha_2 b_1^2 \in M_1 \end{aligned}$$

$\therefore$ M_1 **is invariant under** A**.**

$$\left.\begin{array}{ll} Ab_1^1 = 2b_1^1 & \Rightarrow\ b_1^1 \text{ is an eigenvector} \\ Ab_1^2 = 1b_1^1 + 2b_1^2 & \Rightarrow\ b_1^2 \text{ is not an eigenvector} \end{array}\right\} \ \Rightarrow\ M_1 \text{ is not an eigenspace}$$

$$Ab_2^1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = 1b_2^1 \in M_2 \ \Rightarrow \begin{array}{l} \therefore\ M_2 \text{ is invariant under } A \\ M_2 \text{ is an eigenspace} \end{array}$$

$$\mathbb{C}^3 = \underset{\checkmark}{M_1 \oplus M_2}, \qquad \alpha_1 b_1^1 + \alpha_2 b_1^2 + \beta_1 b_2^1 = 0$$

p. 57

▷ *Linearly independent (only solution $\alpha_1 = \alpha_2 = \beta_1 = 0$: the 3rd entry gives $\beta_1 = 0$, the 1st $\alpha_1 = 0$, then $\alpha_2 = 0$). No orthogonality (no need for it actually).*

$$\bar{A} = \left[\begin{array}{c|c} & \\ \hline & \end{array}\right], \qquad \bar{A} = P^{-1}AP \Rightarrow P\bar{A} = AP, \qquad P = [b_1^1\ b_1^2 \mid b_2^1]$$

$$\underbrace{[b_1^1\ b_1^2\ b_2^1]}_{P} \underbrace{\left[\begin{array}{cc|c} 2 & 1 & 0 \\ 0 & 2 & 0 \\ \hline 0 & 0 & 1 \end{array}\right]}_{\bar{A}} = A[b_1^1\ b_1^2\ b_2^1]$$

$$= [Ab_1^1 \mid Ab_1^2 \mid Ab_2^1]$$

▷ *The columns of $\bar{A}$ are the coefficients found above: $Ab_1^1 = 2b_1^1$, $Ab_1^2 = 1b_1^1 + 2b_1^2$, $Ab_2^1 = 1b_2^1$.*

$$d(s) = \det(sI - A) = (s-2)^{\textcircled{2}}(s-1), \qquad \lambda_1 = 2\ \&\ \lambda_2 = 1$$

▷ *Expanding along the 3rd row: $d(s) = (s-1)\big[(s-1)(s-3)+1\big] = (s-1)(s^2-4s+4) = (s-1)(s-2)^2$; the same follows from $\det(sI - \bar{A})$, since $\bar{A}$ is block triangular.*

$$\underbrace{\mathcal{N}(A - \lambda_1 I)}_{\dim=1} + \underbrace{\mathcal{N}(A - \lambda_2 I)}_{\dim=1} \neq \mathbb{C}^3$$

▷ $\Rightarrow$ *only two linearly indep. eigenvectors ($[1\ 1\ 0]^T$ and $[0\ 1\ 1]^T$) $\Rightarrow$* ***Not diagonalizable*** *as the dimensions do not add up ($1 + 1 < 3$).*

$$\underset{3\ \checkmark}{\mathbb{C}^3} = \underbrace{\mathcal{N}\big((A - \lambda_1 I)^{\textcircled{2}}\big)}_{2} \oplus \underbrace{\mathcal{N}(A - \lambda_2 I)}_{1}$$

Explanation

The circled exponent is the same number in both places: $\lambda_1 = 2$ is a double root of $d(s)$, and the right invariant subspace for it is $\mathcal{N}\big((A-2I)^2\big)$ (a generalized eigenspace), not the eigenspace. Indeed

$$(A-2I)^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \mathcal{N}\big((A-2I)^2\big) = \{x :\ x_3 = 0\} = \mathrm{sp}\{b_1^1, b_1^2\} = M_1.$$

By the result on p. 52 ($\mathcal{N}(p(A))$ is invariant), this subspace is automatically invariant under A. So A cannot be diagonalized, but it is block diagonal in this basis, with an upper-triangular 2×2 block for λ_1 (a Jordan block).

7.5 Distinct Eigenvalues and Diagonalization

Let $A \in \mathbb{C}^{n\times n}$, $\lambda_1 \neq \lambda_2$, $Ae_1 = \lambda_1 e_1$, $Ae_2 = \lambda_2 e_2$.

$$\begin{aligned} & \alpha_1 e_1 + \alpha_2 e_2 = 0 \\ \Rightarrow\ & (A-\lambda_1 I)(\alpha_1 e_1 + \alpha_2 e_2) = 0 \\ \Rightarrow\ & \alpha_1 \underbrace{\cancel{(A-\lambda_1 I)e_1}}_{0} + \alpha_2 \underbrace{(A-\lambda_1 I)e_2}_{\neq 0 \text{ (see below)}} = 0 \ \Rightarrow\ \boxed{\alpha_2 = 0} \end{aligned}$$

where

$$\begin{aligned} (A-\lambda_1 I)e_1 &= Ae_1 - \lambda_1 e_1 = \lambda_1 e_1 - \lambda_1 e_1 = 0 \\ (A-\lambda_1 I)e_2 &= Ae_2 - \lambda_1 e_2 = \lambda_2 e_2 - \lambda_1 e_2 = (\lambda_2 - \lambda_1)e_2 \neq 0 \end{aligned}$$

Then $\alpha_1 e_1 = 0$ with $e_1 \neq 0$ gives $\alpha_1 = 0$ as well, so e_1, e_2 **are linearly independent**.

Eigenvectors of distinct eigenvalues are independent

The same trick works for k distinct eigenvalues: if $\sum_{i=1}^{k} \alpha_i e_i = 0$, apply $\prod_{i\neq j}(A-\lambda_i I)$. Every term except the j-th is annihilated, leaving $\alpha_j \prod_{i\neq j}(\lambda_j - \lambda_i)\, e_j = 0$, so $\alpha_j = 0$ for every j. Consequently, **if $A \in \mathbb{C}^{n\times n}$ has n distinct eigenvalues, it is diagonalizable**: its eigenvectors form a basis, $\mathbb{C}^n = \mathcal{N}(A-\lambda_1 I)\oplus\cdots\oplus\mathcal{N}(A-\lambda_n I)$, and $P^{-1}AP = \operatorname{diag}(\lambda_1,\ldots,\lambda_n)$ with $P = [e_1\ \ldots\ e_n]$. The example above shows this can fail when an eigenvalue is repeated.

Theorem 7.10. Let A be an $n \times n$ matrix with n distinct eigenvalues, $\lambda_1, \lambda_2, \ldots, \lambda_n$ where $\lambda_i \neq \lambda_j$ when $i \neq j$. Then p. 58

$$\begin{array}{ccccccccc} \mathbb{C}^n = & \underset{\operatorname{sp}\{e_1\}}{\mathcal{N}(A-\lambda_1 I)} & \oplus & \underset{\operatorname{sp}\{e_2\}}{\mathcal{N}(A-\lambda_2 I)} & \oplus\cdots\oplus & \underset{\operatorname{sp}\{e_n\}}{\mathcal{N}(A-\lambda_n I)} & & \\ & 1 & + & 1 & +\cdots+ & 1 & = & n \end{array}$$

Since $\mathcal{N}(A-\lambda_i I)$ is invariant under A, A **has a diagonal representation**:

$$\bar{A} = \begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix}, \qquad \bar{A} = P^{-1}AP \quad \text{where } P = \left[\, e_1 \mid e_2 \mid \cdots \mid e_n \,\right].$$

Linear independence of $e_1,\ e_2,\ \ldots,\ e_n$. Suppose

$$\alpha_1 e_1 + \alpha_2 e_2 + \cdots + \alpha_n e_n = 0$$

and use

$$(A-\lambda_i I)e_j = Ae_j - \lambda_i e_j = \lambda_j e_j - \lambda_i e_j = (\lambda_j - \lambda_i)e_j.$$

Multiply with $(A-\lambda_1 I)$, then with $(A-\lambda_2 I)$, and so on:

$$\begin{aligned} \times(A-\lambda_1 I):\quad & 0 + (\lambda_2-\lambda_1)\alpha_2 e_2 + \alpha_3(\lambda_3-\lambda_1)e_3 + \cdots + \alpha_n(\lambda_n-\lambda_1)e_n = 0 \\ \times(A-\lambda_2 I):\quad & 0 + 0 + \alpha_3(\lambda_3-\lambda_1)(\lambda_3-\lambda_2)e_3 + \cdots + \alpha_n(\lambda_n-\lambda_1)(\lambda_n-\lambda_2)e_n = 0 \\ & \vdots \\ & \alpha_n \underbrace{(\lambda_n-\lambda_1)(\lambda_n-\lambda_2)\cdots(\lambda_n-\lambda_{n-1})}_{\neq 0}\ \underbrace{e_n}_{\neq 0} = 0 \quad\Longrightarrow\quad \alpha_n = 0 \Longrightarrow \forall \alpha_i = 0 \end{aligned}$$

▷ *The product is nonzero for distinct eigenvalues (this case), and e_n is nonzero.*

Completing the argument

After the $n-1$ multiplications only the e_n-term survives, so $\alpha_n = 0$. The original relation then has only the terms $e_1, \ldots, e_{n-1}$; repeating the procedure with the factors $(A-\lambda_1 I), \ldots, (A-\lambda_{n-2} I)$ gives $\alpha_{n-1} = 0$, and so on; hence all $\alpha_i = 0$. Thus $\{e_1, \ldots, e_n\}$ is a set of n linearly independent vectors in $\mathbb{C}^n$, i.e. a basis, and each $\mathcal{N}(A-\lambda_i I)$ is exactly one-dimensional (it is at least 1-dimensional, and the dimensions cannot add up to more than n). This proves the direct sum in the theorem.

$$P = \begin{bmatrix} e_1 \mid e_2 \mid \cdots \mid e_n \end{bmatrix}, \qquad \bar{A} = P^{-1}AP \;\Rightarrow\; P\bar{A} = AP$$

$$\begin{bmatrix} e_1 \mid e_2 \mid \cdots \mid e_n \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & \vdots \\ \vdots & & \ddots & \vdots \\ 0 & \cdots & \cdots & \lambda_n \end{bmatrix} = A\begin{bmatrix} e_1 \mid e_2 \mid \cdots \mid e_n \end{bmatrix}$$

$$= \begin{bmatrix} Ae_1 \mid Ae_2 \mid \cdots \mid Ae_n \end{bmatrix}$$

$$= \begin{bmatrix} \lambda_1 e_1 & \lambda_2 e_2 & \ldots & \lambda_n e_n \end{bmatrix}$$

$$\begin{aligned} \therefore \text{ Distinct eigenvalues } &\Rightarrow \text{ Diagonalizable matrix} \\ &\Rightarrow \text{ we can find } P \text{ matrix \& } \bar{A} \\ \text{Distinct eigenvalues } &\not\Leftarrow \text{ Diagonalizable} \end{aligned}$$

Contrapositive ($p \Rightarrow q \;\equiv\; \bar{q} \Rightarrow \bar{p}$): Non-diagonalizable $\Rightarrow$ repeated eigenvalues.

Two diagonal matrices with repeated eigenvalues:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad d(s) = (s-1)^3 \qquad \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

▷ *Diagonal, but repeated eigenvalues.*

Both matrices are already diagonal (so trivially diagonalizable with $P = I$), yet their eigenvalues are repeated: this is why the converse implication fails.

8 Characteristic and Minimal Polynomials

Syllabus: Week 10 — Cayley–Hamilton theorem, characteristic and minimal polynomials

8.1 Similar Matrices and the Characteristic Polynomial

Similar matrices have the same characteristic polynomial: p. 59

$$\begin{aligned}\bar{d}(s) &= \det(sI - \bar{A}) \\ &= \det(sI - P^{-1}AP) \\ &= \det(sP^{-1}P - P^{-1}AP) = \det\left(P^{-1}(sI - A)P\right) \\ &= \det P^{-1}\ \det(sI - A)\ \det P \\ &= \frac{1}{\cancel{\det P}} \cdot \det(sI - A) \cdot \cancel{\det P} \\ &= \det(sI - A) = d(s)\end{aligned}$$

▷ *Uses* $\det(XY) = \det X \det Y$ *and* $\det P^{-1} = 1/\det P$. *Hence* ***similar matrices have the same eigenvalues*** *(with the same algebraic multiplicities); in particular the eigenvalues of* A *are the diagonal entries of* $\bar{A}$.

Example 8.1.

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}, \qquad sI - A = \begin{bmatrix} s-1 & 0 & 0 \\ 1 & s-2 & -1 \\ 0 & 0 & s-3 \end{bmatrix}$$

$$\det(sI - A) = (s-1)\begin{vmatrix} s-2 & -1 \\ 0 & s-3 \end{vmatrix} = (s-1)(s-2)(s-3) = 0 \quad \Rightarrow \quad \lambda = 1, \quad \lambda = 2, \quad \lambda = 3$$

$$A - \lambda_1 I = \begin{bmatrix} 0 & 0 & 0 \\ -1 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}, \qquad \mathcal{N}(A - \lambda_1 I) = \operatorname{sp}\left\{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}\right\}$$

$$A - \lambda_2 I = \begin{bmatrix} -1 & 0 & 0 \\ -1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \mathcal{N}(A - \lambda_2 I) = \operatorname{sp}\left\{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\right\}$$

$$A - \lambda_3 I = \begin{bmatrix} -2 & 0 & 0 \\ -1 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \mathcal{N}(A - \lambda_3 I) = \operatorname{sp}\left\{\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}\right\}$$

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \bar{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} = P^{-1}AP$$

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \qquad \bar{A} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Check for the second choice ($P\bar{A} = AP$):

$$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \overbrace{\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}}^{\bar{A}} = A \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 3 & 2 & 1 \\ 3 & 0 & 0 \end{bmatrix}$$

▷ *Both sides agree, so $P^{-1}AP = \bar{A}$. Reordering the eigenvectors in P only reorders the eigenvalues on the diagonal of $\bar{A}$.*

Characteristic polynomial and equation.

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad d(s) = \det(sI - A) = \left| \begin{bmatrix} s-2 & -1 \\ -1 & s-2 \end{bmatrix} \right|$$

$$\begin{aligned} d(s) &= s^2 - 4s + 3 && \Rightarrow \textbf{ charac. poly.} \\ d(s) &= 0 && \Rightarrow \textbf{ charac. eqn.} \end{aligned}$$

$$\begin{aligned} d(A) = A^2 - 4A + 3I &= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - 4\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} + 3\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} - \begin{bmatrix} 8 & 4 \\ 4 & 8 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \\ &\underline{\underline{d(A) = 0_{n\times n}}} \end{aligned}$$

8.2 The Cayley–Hamilton Theorem

Theorem 8.2 (Cayley–Hamilton, CHT)**. Every $n \times n$ matrix satisfies its characteristic equation.** p. 60

$$\therefore\ d(A) = 0_{n\times n}$$

$$\begin{aligned} d(s) &= s^n + d_1 s^{n-1} + d_2 s^{n-2} + \cdots + d_{n-1}s + d_n \\ d(A) &= A^n + d_1 A^{n-1} + d_2 A^{n-2} + \cdots + d_{n-1}A + d_n I = 0 \\ A^n &= -d_1 A^{n-1} - \cdots - d_{n-1}A - d_n \underbrace{I}_{A^0} \end{aligned}$$

▷ $\Rightarrow$ *among the powers $I, A, A^2, \ldots$ we can have at most n linearly independent matrices.*

Remark. CHT says that **n^{th} and higher order powers of an $n \times n$ matrix can be written as a linear combination of its lower powers.**

For the matrix A above, $d(A) = 0$ gives:

$$\begin{aligned} A^2 - 4A + 3I = 0 \ \Rightarrow\ A^2 &= 4A - 3I \\ A^3 = AA^2 = A(4A - 3I) &= 4A^2 - 3A = 4(4A - 3I) - 3A \\ &= 13A - 12I \end{aligned}$$

The inverse follows in the same way:

$$\begin{aligned} A^2 - 4A + 3I &= 0 \\ A - 4I + 3A^{-1} &= 0 \qquad \text{(multiply by } A^{-1}\text{; } A \text{ is invertible since } d(0) = 3 \neq 0) \\ A^{-1} &= -\frac{A}{3} + \frac{4I}{3} = \frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \end{aligned}$$

In general, every power A^i $(i \geq n)$ reduces to

$$\therefore\ A^i = \alpha_1 A^{n-1} + \alpha_2 A^{n-2} + \cdots + \alpha_{n-1}A + \alpha_n I$$

The 2×2 case. By CHT, for a 2×2 matrix every power is a combination of A and I. To find the coefficients we use eigenvectors. For an eigenpair $Ae_1 = \lambda_1 e_1$:

$$\begin{aligned} Ae_1 &= \lambda_1 e_1 \\ A^2 e_1 &= A(Ae_1) = \lambda_1^2 e_1 \\ A^i e_1 &= \lambda_1^i e_1 \\ p(A)e_1 &= p(\lambda_1)e_1 \end{aligned}$$

▷ *For any polynomial p: $p(A)$ acts on an eigenvector as multiplication by $p(\lambda_1)$.*

$$\boxed{A^i = \alpha_i A + \beta_i I} \qquad \lambda_1 = 3 \ (e_1), \qquad \lambda_2 = 1 \ (e_2)$$

$$\begin{aligned} \underbrace{A^i e_1}_{\lambda_1^i e_1} &= \alpha_i \underbrace{Ae_1}_{\lambda_1 e_1} + \beta_i \underbrace{Ie_1}_{e_1} & A^i e_2 &= \alpha_i A e_2 + \beta_i I e_2 \\ 3^i e_1 &= 3\alpha_i e_1 + \beta_i e_1 & \lambda_2^i e_2 &= \alpha_i \lambda_2 e_2 + \beta_i e_2 \\ 3^i e_1 &= (3\alpha_i + \beta_i) e_1 & \lambda_2^i &= \alpha_i \lambda_2 + \beta_i \end{aligned}$$

Since $e_1, e_2 \neq 0$ we may compare coefficients; with $\lambda_2 = 1$ this gives

$$\begin{aligned} 3\alpha_i + \beta_i &= 3^i \\ \alpha_i + \beta_i &= 1 \qquad (= 1^i) \end{aligned}$$

▷ *Subtracting:* $2\alpha_i = 3^i - 1$.

$$\alpha_i = \frac{3^i - 1}{2}, \qquad \beta_i = 1 - \alpha_i = 1 - \frac{3^i - 1}{2} = \frac{3 - 3^i}{2}$$

$$\boxed{A^i = \frac{3^i - 1}{2} A + \frac{3 - 3^i}{2} I}$$

Check

$i = 0$: $0 \cdot A + 1 \cdot I = I$; $i = 1$: $A + 0 \cdot I = A$; $i = 2$: $4A - 3I$ (as found from CHT); $i = 3$: $13A - 12I$ (as above). The formula holds for every $i \geq 0$, and in fact also for $i = -1$: $\frac{3^{-1}-1}{2} A + \frac{3-3^{-1}}{2} I = -\frac{A}{3} + \frac{4I}{3} = A^{-1}$.

8.3 The Minimal Polynomial

Consider p. 61

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad d(s) = (s-2)^2(s-1)$$

$$\begin{aligned} d(A) &= (A - 2I)^2 (A - I) \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$m(s) = (s-2)(s-1)$$
$$m(A) = (A-2I)(A-I) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0_{n\times n}$$

▷ *Lowest order ⇒ minimal polynomial. (A degree-1 monic $A - cI$ cannot vanish, since A is not a multiple of I.)*

Definition 8.3. A **monic polynomial** is a polynomial whose coefficient of the highest-degree term is unity ($= 1$).

* $s^2 + 2s + 3$ (Monic)
* $2s^2 + 2s + 3$ (Not monic)

Definition 8.4. For an $n \times n$ matrix A, the **minimal polynomial** is the monic polynomial $m(s)$ with the smallest degree/order s.t.

$$m(A) = 0_{n\times n}$$

Theorem 8.5. Given $A \in \mathbb{C}^{n\times n}$ let $m(s)$ be its minimal polynomial. Then

I) $m(s)$ is **unique**;

II) $m(s)$ **divides** $d(s)$ without any remainder: $\exists q(s)$ s.t. $d(s) = m(s)q(s)$;

III) **every root of** $d(s)$ **is also a root of** $m(s)$.

Candidates for $d(s) = (s-1)(s-2)^2$.

- ~~$m(s) = (s-1)$~~, ~~$m(s) = (s-2)$~~, ~~$m(s) = (s-2)^2$~~
  ▷ not possible, property III (a root of $d(s)$ is missing)
- ~~$m(s) = (s-1)(s-3)$~~, ~~$m(s) = (s-2)^3$~~
  ▷ not possible, property II (they do not divide $d(s)$)
- $m(s) = (s-1)(s-2)$ ✓ $m(s) = (s-1)(s-2)^2$ ✓
  ▷ the only possibilities (i.e. $d(s) = \underbrace{m(s)}_{(s-1)(s-2)} \underbrace{q(s)}_{(s-2)}$ or $q(s) = 1$)

In general, for $A \in \mathbb{C}^{n\times n}$:

$$d(s) = (s-\lambda_1)^{r_1}(s-\lambda_2)^{r_2}\cdots(s-\lambda_k)^{r_k}, \qquad r_1 + r_2 + \cdots + r_k = n$$
$$m(s) = (s-\lambda_1)^{m_1}(s-\lambda_2)^{m_2}\cdots(s-\lambda_k)^{m_k}, \qquad \boxed{1 \le m_i \le r_i}$$

▷ *Lower bound ($1 \le m_i$) by property III; upper bound by property II.*

Proof. **I)** $p \to q \equiv p \wedge \bar{q} = \text{False} \overset{\Delta}{=}$ proof by contradiction. p. 62

Suppose $m(s)$ is not unique and minimal polynomial $(p \wedge \bar{q})$:

$$\begin{aligned} m_1(s) &\longrightarrow m_1(A) = 0_{n\times n} \\ m_2(s) &\longrightarrow m_2(A) = 0_{n\times n} \\ m_1(s) &\neq m_2(s) \end{aligned}$$

$$\begin{aligned} m_1(s) &= s^l + a_1 s^{l-1} + \cdots + a_l \\ m_2(s) &= s^l + b_1 s^{l-1} + \cdots + b_l, \qquad \exists i \text{ s.t. } a_i \neq b_i \\ r(s) &= m_1(s) - m_2(s) \\ r(s) &= (a_1 - b_1)s^{l-1} + \cdots + (a_l - b_l) \\ \bar{r}(s) &= \frac{r(s)}{\alpha} \end{aligned}$$

▷ $r(s) \neq 0$ *since* $m_1 \neq m_2$, *and* $\deg r < l$ *since the* s^l *terms cancel;* α*: coefficient of the highest order term* $\Rightarrow \bar{r}(s)$ *is monic.*

$$\deg \bar{r}(s) < \big(\deg m_1(s) = \deg m_2(s)\big)$$

$$\bar{r}(A) = \frac{r(A)}{\alpha} = \frac{m_1(A) - m_2(A)}{\alpha} = \frac{0_{n\times n} - 0_{n\times n}}{\alpha} = 0_{n\times n}$$

$\bar{r}(A) = 0_{n\times n}$: we assumed m_1, m_2 are minimal, but found a monic $\bar{r}(s)$ of lower degree that also annihilates $A \Rightarrow$ contradiction with minimality. Hence $m(s)$ is unique.

II) Divide $d(s)$ by $m(s)$ (polynomial division):

$$d(s) = m(s)q(s) + r(s), \qquad \deg r(s) < \deg m(s).$$

Suppose $r(s) \neq 0(s)$ (proof by contradiction). Then

$$\exists\, \bar{r}(s) = \frac{r(s)}{\alpha} \text{ and } \bar{r}(s) \text{ is monic} \quad (\alpha: \text{ leading coefficient of } r(s)). \qquad \bar{r}(A) = ?$$

$$\underbrace{d(A)}_{0_{n\times n}\ (\text{CHT})} = \underbrace{m(A)}_{0_{n\times n}} q(A) + r(A) \qquad \Rightarrow \qquad r(A) = 0_{n\times n} \Rightarrow \bar{r}(A) = 0_{n\times n}$$

▷ $\bar{r}$ *is monic, annihilates* A *and has lower degree than* m*: contradiction. So* $r(s) = 0$*, i.e.* $m(s)$ *divides* $d(s)$.

The same argument in general

The proof of II only used $d(A) = 0$. Hence: $m(s)$ **divides every polynomial** $p(s)$ **with** $p(A) = 0$**.**

III) Roots of $d(s)$ are eigenvalues of A.

$$\begin{aligned} &d(\lambda_i) = 0 \\ &Ae_i = \lambda_i e_i, \qquad e_i \neq 0 \\ &m(\lambda_i) \overset{?}{=} 0 \\ &m(s) = s^l + a_1 s^{l-1} + \cdots + a_l \\ &m(A) = A^l + a_1 A^{l-1} + \cdots + a_l I = 0_{n\times n} \end{aligned}$$

$$
\begin{aligned}
&(A^l + a_1A^{l-1} + \cdots + a_lI)e_i = 0_{n\times 1}\\
&A^le_i + a_1A^{l-1}e_i + \cdots + a_{l-1}Ae_i + a_le_i = 0_{n\times 1}\\
&(\lambda_i^l + a_1\lambda_i^{l-1} + \cdots + a_{l-1}\lambda_i + a_l)\ \underbrace{e_i}_{\neq 0} = 0_{n\times 1}\\
&\lambda_i^l + a_1\lambda_i^{l-1} + \cdots + a_{l-1}\lambda_i + a_l = 0 \ \Rightarrow\ m(\lambda_i) = 0
\end{aligned}
$$

▷ *$A^je_i = \lambda_i^je_i$; a scalar times a nonzero vector is zero only if the scalar is zero.*

$$\therefore\ \lambda_i \text{ is a root of } m(s). \qquad \square$$

8.4 Minimal Polynomial and Null-Space Decomposition

$$
\begin{aligned}
\therefore\ d(s) &= (s-\lambda_1)^{r_1}(s-\lambda_2)^{r_2}\cdots(s-\lambda_k)^{r_k}\\
m(s) &= (s-\lambda_1)^{m_1}(s-\lambda_2)^{m_2}\cdots(s-\lambda_k)^{m_k}\\
1 \le m_i &\le r_i, \qquad r_1 + r_2 + \cdots + r_k = n
\end{aligned}
$$

p. 63

Distinct eigenvalue case.

$$
\begin{aligned}
d(s) &= (s-\lambda_1)^1(s-\lambda_2)^1\cdots(s-\lambda_n)^1\\
r_i = 1 &\Rightarrow m_i = 1\\
m(s) &= d(s)
\end{aligned}
$$

$$
A = \begin{bmatrix} 1 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & 3 \end{bmatrix}, \qquad d(s) = (s-1)(s-2)(s-3) = m(s)
$$

$$
\mathbb{C}^3 = \underbrace{\mathcal{N}(A-\lambda_1 I)}_{\text{dim}=1} \oplus \underbrace{\mathcal{N}(A-\lambda_2 I)}_{\text{dim}=1} \oplus \underbrace{\mathcal{N}(A-\lambda_3 I)}_{\text{dim}=1}
$$

$$
\mathbb{C}^n = \underbrace{\mathcal{N}\big((A-\lambda_1 I)^1\big)}_{\text{dim}=1} \oplus \cdots \oplus \underbrace{\mathcal{N}\big((A-\lambda_n I)^1\big)}_{\text{dim}=1}
$$

Example 8.6.

$$
A_1 = \begin{bmatrix} 1 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & 2 \end{bmatrix}, \qquad A_2 = \begin{bmatrix} 1 & 0 & 0\\ 0 & 2 & 1\\ 0 & 0 & 2 \end{bmatrix}
$$

Matrix A_1.

$$
\begin{aligned}
d_1(s) &= (s-1)(s-2)^2\\
m_1(s) &= \boxed{(s-1)(s-2)}\ \checkmark\\
&= \cancel{(s-1)(s-2)^2}
\end{aligned}
$$

$$
A_1 - I = \begin{bmatrix} 0 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1 \end{bmatrix}, \qquad \text{rank} = 2,\ \text{dim} = 3,\ \text{null} = 1
$$

$$A_1 - 2I = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \text{rank} = 1,\ \dim = 3,\ \text{null} = 2$$

▷ *Columns 2, 3 of $A_1 - I$: linearly independent vectors.*

$$(A_1 - I)(A_1 - 2I) = 0_{3\times 3}$$

$$\mathcal{N}(A_1 - I) = \text{sp}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$\mathcal{N}(A_1 - 2I) = \text{sp}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

▷ *Linearly independent ⇒ direct sum.*

$$\underbrace{\mathcal{N}\big((A_1 - I)^1\big)}_{\dim=1} \oplus \underbrace{\mathcal{N}\big((A_1 - 2I)^1\big)}_{\dim=2} = \mathbb{C}^3$$

Matrix A_2.

$$\begin{aligned} d_2(s) &= (s-1)(s-2)^2 \\ m_2(s) &= \cancel{(s-1)(s-2)} \\ &= \boxed{(s-1)(s-2)^2}\ \checkmark \end{aligned}$$

$$A_2 - I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \text{rank} = 2,\ \dim = 3,\ \text{null} = 1$$

$$A_2 - 2I = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \neq 0_{3\times 3}, \qquad \text{rank} = 2,\ \dim = 3,\ \text{null} = 1$$

$$(A_2 - 2I)^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(A_2 - I)(A_2 - 2I) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \neq 0_{3\times 3}$$

▷ *Hence $(s-1)(s-2)$ is not the minimal polynomial; only $(A_2 - I)(A_2 - 2I)^2 = 0_{3\times 3}$.*

$$\mathcal{N}\big((A_2 - I)\big) = \text{sp}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$\mathcal{N}\big((A_2 - 2I)\big) = \text{sp}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$$\mathbb{C}^3 \neq \underbrace{\mathcal{N}\big((A_2 - I)\big)}_{\dim=1} \oplus \underbrace{\mathcal{N}\big((A_2 - 2I)\big)}_{\dim=1}$$

$$\mathcal{N}\big((A_2 - 2I)^2\big) = \operatorname{sp}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\mathbb{C}^3 = \underbrace{\mathcal{N}\big((A_2 - I)^1\big)}_{\dim=1} \oplus \underbrace{\mathcal{N}\big((A_2 - 2I)^2\big)}_{\dim=2}$$

What the example shows

In both cases $\mathbb{C}^3$ splits into null spaces of powers of $A - \lambda_i I$, and the power needed for λ_i is exactly the exponent m_i in the minimal polynomial. For A_1 ($m_i = 1$) the eigenvectors alone span $\mathbb{C}^3$, so A_1 is diagonalizable. For A_2 there are only two independent eigenvectors; the missing direction $\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^T$ is not an eigenvector but satisfies $(A_2 - 2I)^2 x = 0$ (a *generalized eigenvector*). This is made precise in the next theorem.

9 Jordan Canonical Form

Syllabus: Weeks 11–12 — Jordan canonical forms

9.1 Generalized Eigenspaces and the Jordan Form

Theorem 9.1 (Null-space decomposition)**.** p. 64

$$d(s) = (s-\lambda_1)^{r_1}(s-\lambda_2)^{r_2}\cdots(s-\lambda_k)^{r_k}$$

$$\mathbb{C}^n = \overbrace{\mathcal{N}\big((A-\lambda_1 I)^{m_1}\big)}^{\dim=r_1} \oplus \overbrace{\mathcal{N}\big((A-\lambda_2 I)^{m_2}\big)}^{\dim=r_2} \oplus \cdots \oplus \overbrace{\mathcal{N}\big((A-\lambda_k I)^{m_k}\big)}^{\dim=r_k}$$

$$m(s) = (s-\lambda_1)^{m_1}(s-\lambda_2)^{m_2}\cdots(s-\lambda_k)^{m_k}$$

$*$ If we form a P matrix out of the columns which are the basis vectors for the null spaces $\mathcal{N}\big((A-\lambda_i I)^{m_i}\big)$ then we can have

$$\begin{aligned} \bar{A} &= P^{-1}AP, \\ P &= \big[\, P_1 \mid P_2 \mid \cdots \mid P_k \,\big]_{n\times n}, \\ P_i &= \big[\, \underbrace{b_1^i \;\; b_2^i \;\; \ldots \;\; b_{r_i}^i}_{\substack{\text{basis vectors for} \\ \mathcal{N}((A-\lambda_i I)^{m_i})}} \,\big]_{n\times r_i} \end{aligned}$$

▷ *P has $r_1 + r_2 + \cdots + r_k = n$ columns.*

$$\bar{A} = \begin{bmatrix} \bar{A}_1 & & & 0 \\ & \bar{A}_2 & & \\ & & \ddots & \\ 0 & & & \bar{A}_k \end{bmatrix}_{n\times n}$$

where the size of $\bar{A}_i$ is $r_i \times r_i$.

$*$ If P_i is selected in a specific way, we can have

$$\bar{A}_i = \begin{bmatrix} \lambda_i & * & & 0 \\ & \lambda_i & * & \\ & & \ddots & * \\ 0 & & & \lambda_i \end{bmatrix}_{r_i\times r_i}, \qquad * \in \{0,1\}$$

In this case $\bar{A}$ is called the **Jordan Canonical Form** of A.

Why $\bar{A}$ is block diagonal

Each subspace $\mathcal{N}\big((A-\lambda_i I)^{m_i}\big)$ is invariant under A: if $(A-\lambda_i I)^{m_i}x = 0$ then, since A commutes with $(A-\lambda_i I)^{m_i}$, also $(A-\lambda_i I)^{m_i}(Ax) = A(A-\lambda_i I)^{m_i}x = 0$. So A maps the columns of P_i into the span of the columns of P_i, and in the basis given by the columns of P the matrix $\bar{A} = P^{-1}AP$ has no entries coupling different blocks. Each $\bar{A}_i$ has the single eigenvalue λ_i (with multiplicity r_i). The "specific way" of choosing P_i is via Jordan chains $x,\ (A-\lambda_i I)x,\ (A-\lambda_i I)^2 x, \ldots$; every chain produces one Jordan block (a λ_i-diagonal with 1's above it).

Example 9.2.

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}, \qquad \left.\begin{matrix} d(s) = (s-1)(s-2)^2 \\ m(s) = (s-1)(s-2)^2 \end{matrix}\right\} \text{ found before}$$

$$\mathbb{C}^3 = \underbrace{\mathcal{N}((A-I))}_{\dim=1} \oplus \underbrace{\mathcal{N}((A-2I)^2)}_{\dim=2}$$

Powers of $A-I$.

$$(A-I) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad \text{rank} = 2, \qquad \dim \mathcal{N}(A-I) = 1, \qquad \mathcal{N}(A-I) = \text{sp}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$(A-I)^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}, \quad \text{rank} = 2, \quad \dim \mathcal{N}((A-I)^2) = 1, \quad \mathcal{N}((A-I)^2) = \text{sp}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$\vdots$$

$$(A-I)^l = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & l \\ 0 & 0 & 1 \end{bmatrix}, \quad \text{rank} = 2, \quad \dim \mathcal{N}((A-I)^l) = 1, \quad \mathcal{N}((A-I)^l) = \text{sp}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$\mathcal{N}(A-I) = \mathcal{N}((A-I)^2) = \cdots = \mathcal{N}((A-I)^l)$$

Powers of $A-2I$. p. 65

$$(A-2I) = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \dim \mathcal{N}(A-2I) = 1, \qquad \mathcal{N}(A-2I) = \text{sp}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$$(A-2I)^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \dim \mathcal{N}((A-2I)^2) = 2, \quad \mathcal{N}((A-2I)^2) = \text{sp}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\therefore \ \mathcal{N}(A-2I) \subsetneq \mathcal{N}((A-2I)^2) \text{ (proper subset) } = \mathcal{N}((A-2I)^3) \cdots$$

▷ $\subsetneq$: $\subset$ & $\neq$, *i.e. subset and not equal.*

$$(A-2I)^3 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \dim \mathcal{N}((A-2I)^3) = 2, \quad \mathcal{N}((A-2I)^3) = \text{sp}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

In general:

$$\begin{matrix} \dim \\ \mathcal{N}(A-\lambda_i I) \end{matrix} \subsetneq \begin{matrix} \dim \\ \mathcal{N}((A-\lambda_i I)^2) \end{matrix} \subsetneq \cdots \subsetneq \underbrace{\begin{matrix} \dim \\ \mathcal{N}((A-\lambda_i I)^{m_i}) \end{matrix}}_{\dim = r_i} = \begin{matrix} \dim \\ \mathcal{N}((A-\lambda_i I)^{m_i+1}) \end{matrix} = \cdots$$

▷ $\dim = r_i$: *largest dimension* $\hookrightarrow$ *exponent* r_i *in the characteristic polynomial.*

Example 9.3.

$$A = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 2 \end{array}\right], \qquad \begin{aligned} d(s) &= (s-1)^3(s-2) \\ m(s) &= ? \quad 3 \text{ possibilities:} \end{aligned}$$

1) $(s-1)(s-2)$

2) $(s-1)^2(s-2)$ $\Leftarrow$ This is the minimal polynomial

3) $(s-1)^3(s-2)$

$$(A-I) = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \dim \mathcal{N}(A-I) = 2$$

$$(A-I)^2 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \dim \mathcal{N}\big((A-I)^2\big) = 3 \ \checkmark$$

▷ $\dim \mathcal{N}\big((A-I)^2\big) = 3 = r_1$ *is reached at power* 2, *so* $m_1 = 2$; *for* $\lambda_2 = 2$, $r_2 = 1$ *forces* $m_2 = 1$. *Hence* $m(s) = (s-1)^2(s-2)$; *indeed* $(A-I)^2(A-2I) = 0$ *but* $(A-I)(A-2I) \neq 0$.

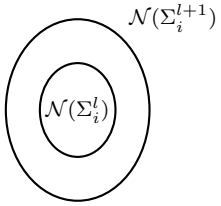
9.2 Chains of Null Spaces

∗ Let $\Sigma_i \triangleq A - \lambda_i I$. Then p. 66

$$\begin{array}{ccccccccccc} \mathcal{N}(\Sigma_i) & \subsetneq & \mathcal{N}(\Sigma_i^2) & \subsetneq & \mathcal{N}(\Sigma_i^3) & \cdots & \subsetneq & \mathcal{N}(\Sigma_i^{m_i}) & = & \mathcal{N}(\Sigma_i^{m_i+1}) & \cdots \\ 1 \leq \dim & < & \dim & < & \dim & \cdots & < & \dim = r_i & = & \dim = r_i & \end{array}$$

Proof.

Claim. $\mathcal{N}(\Sigma_i^l) \subset \mathcal{N}(\Sigma_i^{l+1})$



$$\begin{aligned} x \in \mathcal{N}(\Sigma_i^l) &\Rightarrow \Sigma_i^l x = 0 \\ &\Rightarrow \Sigma_i \cdot \Sigma_i^l x = \Sigma_i \cdot 0 = 0 \\ &\Rightarrow \Sigma_i^{l+1} x = 0 \Rightarrow x \in \mathcal{N}(\Sigma_i^{l+1}) \end{aligned}$$

Claim. If $\mathcal{N}(\Sigma_i^l) = \mathcal{N}(\Sigma_i^{l+1})$ then $\mathcal{N}(\Sigma_i^{l+1}) = \mathcal{N}(\Sigma_i^{l+2})$

Assume $\mathcal{N}(\Sigma_i^l) = \mathcal{N}(\Sigma_i^{l+1})$. Need to show $\mathcal{N}(\Sigma_i^{l+1}) = \mathcal{N}(\Sigma_i^{l+2})$, i.e.

✓ $\mathcal{N}(\Sigma_i^{l+1}) \subset \mathcal{N}(\Sigma_i^{l+2})$

$\mathcal{N}(\Sigma_i^{l+2}) \subset \mathcal{N}(\Sigma_i^{l+1})$

$$\begin{aligned} x \in \mathcal{N}(\Sigma_i^{l+2}) &\Rightarrow \Sigma_i^{l+2} x = 0 \\ &\Rightarrow \Sigma_i^{l+1} \Sigma_i x = 0 \Rightarrow \Sigma_i x \in \mathcal{N}(\Sigma_i^{l+1}) \\ &\Rightarrow \Sigma_i x \in \mathcal{N}(\Sigma_i^l) \\ &\Rightarrow \Sigma_i^l \Sigma_i x = 0 \Rightarrow \Sigma_i^{l+1} x = 0 \Rightarrow x \in \mathcal{N}(\Sigma_i^{l+1}) \end{aligned}$$

□

Consequence

By the two claims the chain of null spaces grows strictly as long as it grows at all; once two consecutive null spaces coincide it is constant forever. Since the dimensions are bounded by n, the chain must stop growing, and it **stops exactly at the power** m_i (the exponent in $m(s)$), where the dimension equals the algebraic multiplicity r_i. Because each step adds at least one dimension, $m_i \le r_i$ again.

Example 9.4.

$$A = \begin{bmatrix} \boxed{\begin{smallmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{smallmatrix}} & & & & & \\ & \boxed{\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}} & & & & \\ & & \boxed{2} & & & \\ & & & \boxed{2} & & \\ & & & & \boxed{\begin{smallmatrix} 3 & 1 \\ 0 & 3 \end{smallmatrix}} & \\ & & & & & \boxed{3} \end{bmatrix}_{10\times 10}, \qquad \begin{aligned} &\dim \mathcal{N}\big((A-\lambda_i I)^l\big) = ? \\ &d(s) = ? \\ &m(s) = ? \end{aligned}$$

▷ *Block diagonal; all entries outside the boxed blocks are zero.*

First block.

$$A_1 = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}, \qquad d_1(s) = (s-2)^3, \qquad m_1(s) = (s-2)^3$$

$$A_1 - 2I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad \operatorname{rank} = 2, \qquad \dim \mathcal{N}(A_1 - 2I) = 1$$

$$(A_1 - 2I)^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \dim \mathcal{N}\big((A_1 - 2I)^2\big) = 2$$

$$(A_1 - 2I)^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad \dim \mathcal{N}\big((A_1 - 2I)^3\big) = 3$$

$$(A_1 - 2I)^4 = 0_{3\times 3}, \qquad \dim \mathcal{N}\big((A_1 - 2I)^4\big) = 3 \ \ldots$$

▷ *For $(A_1 - 2I)^3$: no need to calculate as it can be max 3 and is not equal to 1 or 2.*
▷ *Each Jordan block of size p behaves the same way:* $\dim \mathcal{N}\big((J-\lambda I)^l\big) = \min(l, p)$. *Each block of A contributes to the dimensions of A independently.*

Whole matrix A. p. 67

$$d(s) = (s-2)^7(s-3)^3$$

▷ *$\lambda = 2$ appears in blocks of sizes $3, 2, 1, 1$ $(3+2+1+1=7)$; $\lambda = 3$ in blocks of sizes $2, 1$ $(2+1=3)$.*

$$\begin{aligned}
&\Sigma_1 = A - 2I && \Sigma_2 = A - 3I \\
&\dim \mathcal{N}(\Sigma_1) = 4 && \dim \mathcal{N}(\Sigma_2) = 2 \\
&\dim \mathcal{N}(\Sigma_1^2) = 6 \quad (+2) && \dim \mathcal{N}(\Sigma_2^2) = 3 \quad (+1) \\
&\dim \mathcal{N}(\Sigma_1^3) = 7 \quad (+1) && \Rightarrow (s-3)^2 \quad (+0) \\
&\Rightarrow (s-2)^3 \quad (+0)
\end{aligned}$$

$$\boxed{\therefore\ m(s) = (s-2)^3(s-3)^2}$$

▷ *Exponent 3: largest Jordan block (JB) for $\lambda_i = 2$; exponent 2: largest JB for $\lambda_i = 3$.*

9.3 Counting Jordan Blocks

(JB: Jordan block.)

$$\begin{aligned}
\dim \mathcal{N}(\Sigma_i) &= \#\text{ of JB's for } \lambda_i \\
&= \#\text{ of JB's with size} \geq 1 \text{ for } \lambda_i \\
&= \max \#\text{ of linearly independent eigenvectors for } \lambda_i
\end{aligned}$$

$$\begin{aligned}
\dim \mathcal{N}(\Sigma_i^2) - \dim \mathcal{N}(\Sigma_i^1) &= \#\text{ of JB's with size} \geq 2 \text{ for } \lambda_i \\
\dim \mathcal{N}(\Sigma_i^3) - \dim \mathcal{N}(\Sigma_i^2) &= \#\text{ of JB's with size} \geq 3 \text{ for } \lambda_i \\
&\vdots \\
\dim \mathcal{N}(\Sigma_i^l) - \dim \mathcal{N}(\Sigma_i^{l-1}) &= \#\text{ of JB's with size} \geq l \text{ for } \lambda_i.
\end{aligned}$$

For $l = 1$:

$$\dim \mathcal{N}(\Sigma_i) - \underbrace{\dim \mathcal{N}(\overbrace{\Sigma_i^0}^{I})}_{0} = \#\text{ of JB's with size} \geq 1 \text{ for } \lambda_i$$

▷ *Works for $l = 1$ also.*

m_i is the size of the largest JB.

Reading off the block sizes

Write $\delta_l = \dim \mathcal{N}(\Sigma_i^l) - \dim \mathcal{N}(\Sigma_i^{l-1})$ (the increments, $\delta_l = \#$ JB's of size $\geq l$). Then

$$\#\text{ of JB's of size exactly } l \ = \ \delta_l - \delta_{l+1}.$$

Reason: a block of size p contributes 1 to $\dim \mathcal{N}(\Sigma_i^l)$ for each $l \leq p$, so it raises the dimension by one at each of the steps $l = 1, \ldots, p$ and not after. The increments therefore stop being positive at $l = m_i$, the size of the largest block.

⇒ For diagonalizable matrix, minimal polynomial should have the power at most 1:

JCF is diagonal iff $m_i = 1\ \forall i$

$$\mathbb{C}^n = \mathcal{N}(\Sigma_1^{m_1}) \oplus \cdots \oplus \mathcal{N}(\Sigma_k^{m_k})$$
$$\mathbb{C}^n = \underbrace{\mathcal{N}(\Sigma_1)}_{r_1} \oplus \cdots \oplus \underbrace{\mathcal{N}(\Sigma_k)}_{r_k}$$

▷ $\mathcal{N}(\Sigma_1)$*: at most* r_1 *linearly independent eigenvectors.*

The second decomposition holds exactly when $m_i = 1$ for all i, i.e. when $\dim \mathcal{N}(\Sigma_i) = r_i$ for every eigenvalue.

⇒ A **is diagonalizable iff there exist** n **linearly independent eigenvectors for** A**.**

Example 9.5. p. 68

$$A = \begin{bmatrix} 1 & -1 & 1 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 2 & -2 & -2 & 0 \end{bmatrix} \quad \Rightarrow \quad \begin{array}{l} d(s) = s^4 \\ m(s) = ? \\ \text{JCF} = ? \end{array}$$

▷ $d(s) = s^4$*: Learn to calculate this! (Shortcut:* $A^3 = 0$ *below, so every eigenvalue satisfies* $\lambda^3 = 0$*, i.e.* $\lambda = 0$ *with multiplicity 4.)*

$$\Sigma_1 = A - \cancel{\lambda_1^0} I = A, \qquad \dim \mathcal{N}(\Sigma_1) = 2, \quad \dim \mathcal{N}(\Sigma_1^2) = 3, \quad \dim \mathcal{N}(\Sigma_1^3) = 4$$

$$A^2 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 \end{bmatrix}, \qquad A^3 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

▷ $\operatorname{rank} A = 2$ *(row 4* $= -2 \cdot$*row 1*$+4 \cdot$*row 2, row 3* $= 0$*)* $\Rightarrow \dim \mathcal{N}(A) = 2$*;* $\operatorname{rank} A^2 = 1 \Rightarrow \dim \mathcal{N}(A^2) = 3$*;* $A^3 = 0 \Rightarrow \dim \mathcal{N}(A^3) = 4$*.*

$$\begin{array}{ll} \therefore\ m(s) = s^3 & \\ \#\text{ of JB's} = 2 & (2-0) \\ \#\text{ of JB's with size} \geq 2 = 1 & (3-2) \\ \#\text{ of JB's with size} \geq 3 = 1 & (4-3) \\ \#\text{ of JB's with size} \geq 4 = 0 & (4-4) \end{array}$$

⇒ one JB with size = 3, zero JB with size = 2, one JB with size = 1:

$$\text{JCF} = \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right]$$

⇒ **Not diagonalizable!** Only $\dim \mathcal{N}(A) = 2 < 4$ independent eigenvectors exist; equivalently, $m_1 = 3 \neq 1$.

Example 9.6. Let $A \in \mathbb{R}^{8\times 8}$ have the following subspace dimensions:

$$\begin{aligned} \dim \mathcal{N}(A-3I) &= 5 \qquad (+5)\\ \dim \mathcal{N}((A-3I)^2) &= 7 \qquad (+2)\\ \dim \mathcal{N}((A-3I)^3) &= 8 \qquad (+1)\\ &\qquad\quad (+0) \end{aligned}$$

$d(s) = ?\quad m(s) = ?\quad \text{JCF} = ?$

$$\begin{aligned} m(s) &= (s-3)^3\\ d(s) &= (s-3)^8 \end{aligned}$$

▷ $\Rightarrow$ *the maximum number of linearly independent eigenvectors is 5.*
▷ $\dim \mathcal{N}\big((A-3I)^3\big) = 8 = n$, *so* 3 *is the only eigenvalue* ($r_1 = 8$) *and the chain stops at power* 3 ($m_1 = 3$).
Block sizes from the increments $5, 2, 1, 0$: 5 JB's in total; $5-2=3$ of size 1, $2-1=1$ of size 2, $1-0=1$ of size 3 (check: $3\cdot 1 + 1\cdot 2 + 1\cdot 3 = 8$).

$$\text{JCF} = \left[\begin{array}{ccc|ccc|cc} 3 & 1 & 0 & 0 & 0 & 0 & 0 & 0\\ 0 & 3 & 1 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 3 & 0 & 0 & 0 & 0 & 0\\ \hline 0 & 0 & 0 & 3 & 1 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0\\ \hline 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0\\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 3 & 0\\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 \end{array}\right]$$

Example 9.7. Let $A \in \mathbb{R}^{4\times 4}$. Single eigenvalue at $\lambda = 7$, $m_1 = 2$. Find all possible JCFs the matrix A can have.

$$\begin{aligned} d(s) &= (s-7)^4\\ m(s) &= (s-7)^2 \end{aligned}$$

$$\begin{array}{ccc} \mathcal{N}(A-7I) & \subsetneq & \mathcal{N}\big((A-7I)^2\big)\\ 1 \le\ ? & & 4\\ \text{dim=2 or 3 or 1} & & \end{array}$$

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Case I: $\dim \mathcal{N}(A-7I) = 3$**.**

$$\begin{array}{rccccccc} & \{0\} & \subsetneq & \mathcal{N}(A-7I) & \subsetneq & \mathcal{N}(A-7I)^2 & \subsetneq & \cdots\\ \dim: & 0 & & 3 & & 4 & & 4\\ \text{increment:} & & +3 & & +1 & & +0 & \end{array}$$

$$\begin{array}{cc} \downarrow & \downarrow\\ \text{2 JB's of size 1} & \text{1 JB of size 2} \end{array}$$

$$A = \left[\begin{array}{cc|c|c} 7 & 1 & 0 & 0\\ 0 & 7 & 0 & 0\\ \hline 0 & 0 & 7 & 0\\ \hline 0 & 0 & 0 & 7 \end{array}\right] \qquad \text{① 1st possibility}$$

Case II: $\dim \mathcal{N}(A-7I)=2$.

$$\begin{array}{rccccccc} & \{0\} & \subsetneq & \mathcal{N}(A-7I) & \subsetneq & \mathcal{N}(A-7I)^2 & \subsetneq & \cdots \\ \dim: & 0 & & 2 & & 4 & & 4 \\ \text{increment:} & & +2 & & +2 & & +0 & \end{array}$$

$$A=\left[\begin{array}{cc|cc} 7 & 1 & 0 & 0 \\ 0 & 7 & 0 & 0 \\ \hline 0 & 0 & 7 & 1 \\ 0 & 0 & 0 & 7 \end{array}\right] \qquad \text{② 2nd possibility}$$

Case III: $\dim \mathcal{N}(A-7I)=1$.

$$\begin{array}{rccccccc} & \{0\} & \subsetneq & \mathcal{N}(A-7I) & \subsetneq & \mathcal{N}(A-7I)^2 & \subsetneq & \cdots \\ \dim: & 0 & & 1 & & 4 & & 4 \\ \text{increment:} & & +1 & & +3 & & +0 & \end{array}$$

▷ *These increments must decrease or stay constant ⇒ no such matrix A exists!*
∴ Two possibilities.

Why the increments cannot grow

The increment $\dim \mathcal{N}(A-7I)^k - \dim \mathcal{N}(A-7I)^{k-1}$ equals the number of Jordan blocks of size $\geq k$. Every block of size $\geq k+1$ also has size $\geq k$, so the increments form a non-increasing sequence. In Case III they would be $1,3,0$: there would be one block in total but three blocks of size ≥ 2, which is impossible. Equivalently: with $\dim \mathcal{N}(A-7I)=1$ there is a single Jordan block, so it must be 4×4; then $m(s)=(s-7)^4$ and $\dim \mathcal{N}(A-7I)^2=2$, not 4.

9.4 Generalized Eigenvectors and Jordan Chains

Let $x\in\mathcal{N}(\Sigma_i^{m_i})$, $x\notin\mathcal{N}(\Sigma_i^{m_i-1})$:

$$\Sigma_i^{m_i}x=0, \qquad \Sigma_i^{m_i-1}x\neq 0 \quad \Rightarrow \quad x\text{: } \textbf{generalized eigenvector}$$

$$\{\underbrace{\Sigma_i^{m_i-1}x}_{\text{not zero}},\ \Sigma_i^{m_i-2}x,\ \ldots,\ \Sigma_i x,\ x\} \quad \Rightarrow \quad \text{none of the vectors are equal to zero.} \quad \textit{(Jordan chain)}$$

▷ *If $\Sigma_i^{m_i-2}x$ were zero, then $\Sigma_i(\Sigma_i^{m_i-2}x)=\Sigma_i^{m_i-1}x$ would be zero ⇒ it cannot be.*

Linear independence: suppose

$$\alpha_1\Sigma_i^{m_i-1}x+\alpha_2\Sigma_i^{m_i-2}x+\cdots+\alpha_{m_i-1}\Sigma_i x+\alpha_{m_i}x=0.$$

Multiplying by $\Sigma_i^{m_i-1}$:

$$\alpha_1\underbrace{\cancel{\Sigma_i^{2m_i-2}}}_{0}x+\alpha_2\underbrace{\cancel{\Sigma_i^{2m_i-3}}}_{0}x+\cdots+\alpha_{m_i-1}\underbrace{\cancel{\Sigma_i^{m_i}}}_{0}x+\alpha_{m_i}\Sigma_i^{m_i-1}x=0$$
$$\Rightarrow\ \alpha_{m_i}\Sigma_i^{m_i-1}x=0\ \Rightarrow\ \alpha_{m_i}=0.$$

Multiplying the remaining equation

$$\alpha_1\Sigma_i^{m_i-1}x+\alpha_2\Sigma_i^{m_i-2}x+\cdots+\alpha_{m_i-1}\Sigma_i x=0$$

by $\Sigma_i^{m_i-2}$, every term except the last again contains a power $\geq m_i$ of Σ_i:

$$\alpha_{m_i-1}\Sigma_i^{m_i-1}x=0\ \Rightarrow\ \alpha_{m_i-1}=0.$$

Continuing in this way (multiplying by $\Sigma_i^{m_i-3}, \dots, \Sigma_i, I$) gives $\alpha_{m_i-2} = \cdots = \alpha_1 = 0$.

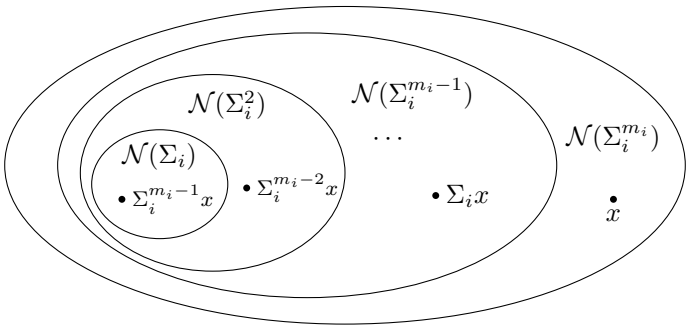
$\therefore$ **All vectors of the chain are linearly independent.**

p. 70

Where the chain vectors lie:

$$\begin{aligned} \Sigma_i(\Sigma_i^{m_i-1}x) = 0 &\quad\Rightarrow\quad \Sigma_i^{m_i-1}x \in \mathcal{N}(\Sigma_i) \\ \Sigma_i^{m_i-1}(\Sigma_i x) = 0 &\quad\Rightarrow\quad \Sigma_i x \in \mathcal{N}(\Sigma_i^{m_i-1}) \\ \Sigma_i^{m_i-2}\Sigma_i x = \Sigma_i^{m_i-1}x \neq 0 &\quad\Rightarrow\quad \Sigma_i x \notin \mathcal{N}(\Sigma_i^{m_i-2}) \end{aligned}$$

▷ *In general $\Sigma_i^k x \in \mathcal{N}(\Sigma_i^{m_i-k})$ but $\Sigma_i^k x \notin \mathcal{N}(\Sigma_i^{m_i-k-1})$: the k-th vector of the chain sits exactly in the "layer" $\mathcal{N}(\Sigma_i^{m_i-k}) \setminus \mathcal{N}(\Sigma_i^{m_i-k-1})$.*



9.5 Computing J and P: Worked Examples

Example 9.8.

$$A = \begin{bmatrix} 1 & 1 & -1 \\ -1 & 3 & 2 \\ 0 & 0 & 1 \end{bmatrix} \qquad d(s) =? \quad m(s) =? \quad \text{JCF} =? \quad P =?$$

$$sI - A = \begin{bmatrix} s-1 & -1 & 1 \\ 1 & s-3 & -2 \\ 0 & 0 & s-1 \end{bmatrix}$$

Expanding along the last row (signs $+\ -\ +$; only the $(3,3)$ entry is nonzero):

$$\begin{aligned} d(s) = \det(sI - A) &= (s-1)\begin{vmatrix} s-1 & -1 \\ 1 & s-3 \end{vmatrix} \\ &= (s-1)\big[(s-1)(s-3)+1\big] \\ &= (s-1)(s^2 - 4s + 4) \\ &\boxed{d(s) = (s-1)(s-2)^2} \qquad \lambda_1 = 2, \quad \lambda_2 = 1 \end{aligned}$$

$$\Sigma_1 = A - 2I = \begin{bmatrix} -1 & 1 & -1 \\ -1 & 1 & 2 \\ 0 & 0 & -1 \end{bmatrix}, \qquad \text{rank} = 2, \quad \dim\mathcal{N}(\Sigma_1) = 1$$

$m(s)$ possibilities:

1- $(s-1)(s-2)$ $\quad\times$

2- $(s-1)(s-2)^2$ $\quad\checkmark$

$$\dim\mathcal{N}(\Sigma_1) = 1 \;\Rightarrow\; \text{not } 2 \quad\Rightarrow\quad \boxed{m(s) = (s-1)(s-2)^2}$$

▷ *If* $m_1 = 1$*, then* $\mathcal{N}(\Sigma_1) = \mathcal{N}(\Sigma_1^{m_1})$ *would have dimension* $r_1 = 2$*; but* $\dim \mathcal{N}(\Sigma_1) = 3 - \operatorname{rank} \Sigma_1 = 1$*, so* $m_1 = 2$.

Further,

$$\Sigma_1^2 = \begin{bmatrix} 0 & 0 & 4 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \qquad \Sigma_2 = A - I = \begin{bmatrix} 0 & 1 & -1 \\ -1 & 2 & 2 \\ 0 & 0 & 0 \end{bmatrix}, \quad \operatorname{rank} = 2, \quad \dim \mathcal{N}(\Sigma_2) = 1$$

	$\mathcal{N}(\Sigma_1)$	$\subsetneq$	$\mathcal{N}(\Sigma_1^2)$	$=$	$\mathcal{N}(\Sigma_1^3)$
dim :	1		2		2
increment:	+1		+1		+0

	$\mathcal{N}(\Sigma_2)$	$=$	$\mathcal{N}(\Sigma_2^2)$
dim :	1		1

$\Rightarrow$ 1 JB of size 1 for λ_2. p. 71

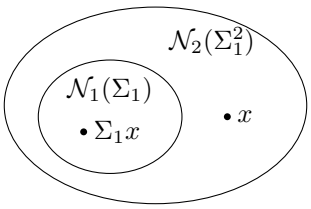
$$(\#\text{ of JB's for } \lambda_1) = \dim \mathcal{N}(\Sigma_1) = 1$$
$$(\#\text{ of JB's with size} \geq 2 \text{ for } \lambda_1) = \dim \mathcal{N}(\Sigma_1^2) - \dim \mathcal{N}(\Sigma_1) = 1$$
$$(\#\text{ of JB's with size} \geq 3 \text{ for } \lambda_1) = \dim \mathcal{N}(\Sigma_1^3) - \dim \mathcal{N}(\Sigma_1^2) = 0$$

▷ *Differences of consecutive counts:* $1 - 1 = 0$ *JB of size exactly* 1, $1 - 0 = 1$ *JB of size exactly* 2.

$\Rightarrow$ 1 JB of size 2 for λ_1:

$$J = \left[\begin{array}{cc|c} 2 & 1 & 0 \\ 0 & 2 & 0 \\ \hline 0 & 0 & 1 \end{array}\right]$$

$x \in \mathcal{N}(\Sigma_1^2), \quad x \notin \mathcal{N}(\Sigma_1)$:



$$x = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \text{or} \quad x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \cdots$$

P **matrix is not unique**, e.g.:

$$J = P^{-1}AP, \qquad P_1 = 2P$$
$$P_1^{-1}AP_1 = \tfrac{1}{2}P^{-1}AP \cdot 2 = P^{-1}AP = J$$

Choosing the first one:

$$x = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad \Sigma_1 x = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad \{\Sigma_1 x, x\}$$

- x: generalized eigenvector; $\Sigma_1 x$**: should be eigenvector.**
- $\{\Sigma_1 x, x\}$: linearly indep., form a basis for $\mathcal{N}(\Sigma_1^2)$.

$$y \in \mathcal{N}(\Sigma_2), \quad y \notin \mathcal{N}(\Sigma_2^0) = \mathcal{N}(I) = \{0\} \quad (y \text{ should be nonzero})$$



$$y = \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix}$$

$$\bar{A} = P^{-1}AP, \qquad P\bar{A} = AP, \qquad P = \left[\begin{array}{cc|c} \Sigma_1 x & x & y \end{array}\right]$$

$$\underbrace{\begin{bmatrix} 1 & 0 & 4 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}}_{P} \underbrace{\left[\begin{array}{cc|c} 2 & 1 & 0 \\ 0 & 2 & 0 \\ \hline 0 & 0 & 1 \end{array}\right]}_{\bar{A}=J} = A \begin{bmatrix} 1 & 0 & 4 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 4 \\ 2 & 3 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

▷ *Check: column by column,* $PJ = [\,2\Sigma_1 x \mid \Sigma_1 x + 2x \mid y\,]$ *and* $AP = [\,A\Sigma_1 x \mid Ax \mid Ay\,]$ *both equal the matrix on the right, and* $\det P = 1 \neq 0$*, so* $P^{-1}AP = J$*.*

For $P = \left[\begin{array}{cc|c} x & \Sigma_1 x & y \end{array}\right]$:

$$J = \left[\begin{array}{cc|c} 2 & 0 & \\ 1 & 2 & \\ \hline & & 1 \end{array}\right] \qquad \times \text{ not accepted}$$

▷ *This is also a JCF, but it is not required in this course.*

Starting the columns of P with y instead: p. 72

$$P = \left[\begin{array}{c|cc} y & \Sigma_1 x & x \end{array}\right]: \quad \bar{A} = \left[\begin{array}{c|cc} 1 & 0 & 0 \\ \hline 0 & 2 & 1 \\ 0 & 0 & 2 \end{array}\right] \checkmark \qquad P = \left[\begin{array}{c|cc} y & x & \Sigma_1 x \end{array}\right]: \quad \bar{A} = \left[\begin{array}{c|cc} 1 & 0 & 0 \\ \hline 0 & 2 & 0 \\ 0 & 1 & 2 \end{array}\right] \times$$

▷ *The second one (crossed out in the notes): this ordering will not be expected in this course.*

Recipe used in this example

1. Factor $d(s)$; for each λ_i form $\Sigma_i = A - \lambda_i I$ and compute $\dim \mathcal{N}(\Sigma_i^k)$, $k = 1, 2, \ldots$ until it reaches r_i; the first such k is m_i.
2. Increments of these dimensions give the number of JBs of size $\geq k$; their differences give the number of JBs of each size.
3. For each block of size p pick a vector $x \in \mathcal{N}(\Sigma_i^p) \setminus \mathcal{N}(\Sigma_i^{p-1})$ and put the chain $\Sigma_i^{p-1}x, \ldots, \Sigma_i x, x$ (eigenvector first) into consecutive columns of P, choosing the chains so that all columns are linearly independent.

The ordering *eigenvector first* is what produces the 1's above the diagonal; reversing a chain puts them below the diagonal.

9.6 Special Cases

Case I: $A \in \mathbb{C}^{n\times n}$, single eigenvalue λ_i. We have $m_i = r_i$.

$$d(s) = (s-\lambda_i)^{r_i}, \qquad m(s) = (s-\lambda_i)^{m_i = r_i} = d(s)$$

$$\underbrace{\mathbb{C}^n}_{\dim n} = \mathcal{N}(\Sigma_i^{m_i}), \qquad \dim \mathcal{N}(\Sigma_i^{m_i}) = r_i = n, \qquad n = r_i = m_i$$

$$\begin{array}{ccccccccc} \mathcal{N}(\Sigma_i) & \subsetneq & \mathcal{N}(\Sigma_i^2) & \subsetneq & \cdots & \subsetneq & \mathcal{N}(\Sigma_i^{m_i-1}) & \subsetneq & \mathcal{N}(\Sigma_i^{m_i}) \\ 1 & \underbrace{<}_{+1} & 2 & < & \cdots & < & m_i - 1 & \underbrace{<}_{+1} & r_i = m_i = n \end{array}$$

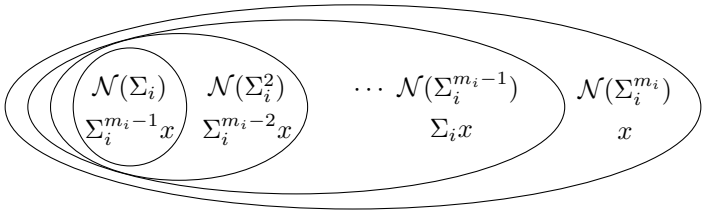
▷ n *dimensions have to be gained in* $m_i = n$ *strict steps, each of at least* $+1 \Rightarrow$ *every increment is exactly* $+1$*, in particular* $\dim \mathcal{N}(\Sigma_i) = 1$*.*

Counting Jordan blocks:

$$\#\text{ of JB's} = 1$$

$$
\begin{aligned}
&\#\text{ of JB's with size} \geq 2 = 1\\
&\#\text{ of JB's with size} \geq 3 = 1\\
&\qquad\qquad\vdots\\
&\#\text{ of JB's with size} \geq m_i = 1 \quad \Rightarrow \textbf{ single JB of size } m_i = r_i = n\\
&\#\text{ of JB's with size} \geq m_i + 1 = 0
\end{aligned}
$$

$$
\text{JB} = \begin{bmatrix}
\lambda_i & 1 & & & \\
 & \lambda_i & 1 & & 0 \\
 & & \lambda_i & \ddots & \\
0 & & & \ddots & 1 \\
 & & & & \lambda_i
\end{bmatrix}_{n\times n}
\qquad\qquad J = P^{-1}AP \;\Rightarrow\; P = ?
$$



$$
\underbrace{\{\Sigma_i^{m_i-1}x,\ \Sigma_i^{m_i-2}x,\ \ldots,\ \Sigma_i x,\ x\}}_{m_i=r_i=n \text{ l. indep. vectors}} \qquad 1 \text{ chain with } n \text{ elements}
$$

Take the chain as the columns of P: p. 73

$$
P = \left[\ \Sigma_i^{m_i-1}x \ \middle|\ \cdots \ \middle|\ \Sigma_i x \ \middle|\ x\ \right], \qquad P\bar{A} = AP
$$

$$
\left[\ \Sigma_i^{m_i-1}x \ \middle|\ \cdots \ \middle|\ \Sigma_i x \ \middle|\ x\ \right]
\begin{bmatrix}
\lambda_i & 1 & 0 & \cdots & 0 \\
0 & \lambda_i & 1 & \cdots & \vdots \\
\vdots & 0 & \lambda_i & \ddots & \vdots \\
\vdots & \vdots & 0 & \ddots & 1 \\
0 & 0 & 0 & \cdots & \lambda_i
\end{bmatrix}
= A\left[\ \Sigma_i^{m_i-1}x \ \middle|\ \cdots \ \middle|\ \Sigma_i x \ \middle|\ x\ \right]
$$

$$
= \left[\ A\Sigma_i^{m_i-1}x \ \middle|\ \cdots \ \middle|\ A\Sigma_i x \ \middle|\ Ax\ \right]
$$

$\Rightarrow x \in \mathcal{N}(\Sigma_i^{m_i})$ & $x \notin \mathcal{N}(\Sigma_i^{m_i-1})$:

$$
\begin{aligned}
&\Sigma_i^{m_i}x = 0, \qquad \Sigma_i^{m_i-1}x \neq 0\\
&\Sigma_i(\Sigma_i^{m_i-1}x) = 0\\
&(A-\lambda_i I)(\Sigma_i^{m_i-1}x) = A\Sigma_i^{m_i-1}x - \lambda_i \underbrace{I}_{1}\Sigma_i^{m_i-1}x = 0\\
&A\Sigma_i^{m_i-1}x = \lambda_i\Sigma_i^{m_i-1}x
\end{aligned}
$$

$\Rightarrow$ Next vector of the chain:

$$
\begin{aligned}
&\Sigma_i(\Sigma_i^{m_i-2}x) = \Sigma_i^{m_i-1}x\\
&(A-\lambda_i I)(\Sigma_i^{m_i-2}x) = \Sigma_i^{m_i-1}x\\
&A\Sigma_i^{m_i-2}x = \lambda_i\Sigma_i^{m_i-2}x + \Sigma_i^{m_i-1}x
\end{aligned}
$$

$\Rightarrow$ and similarly:

$$\Sigma_i(\Sigma_i^{m_i-3}x) = \Sigma_i^{m_i-2}x$$
$$(A-\lambda_i I)(\Sigma_i^{m_i-3}x) = \Sigma_i^{m_i-2}x \;\Rightarrow\; A\Sigma_i^{m_i-3}x = \lambda_i\Sigma_i^{m_i-3}x + \Sigma_i^{m_i-2}x$$

▷ *Hence column k of AP equals $\lambda_i \cdot$ (column k of P) + (column $k-1$ of P), which is exactly column k of PJ: a λ_i on the diagonal and a 1 just above it.*

Altogether:

$$\{\Sigma_i^{m_i-1}x,\ \Sigma_i^{m_i-2}x,\ \ldots,\ \Sigma_i x,\ x\} \quad \text{single chain of size } m_i.$$

Case II: $A \in \mathbb{C}^{n\times n}$, single eigenvalue λ_i, $m_i = 1$.

$$d(s) = (s-\lambda_i)^{r_i}, \qquad m(s) = (s-\lambda_i)$$

$$\mathbb{C}^n = \underset{r_i=n}{\mathcal{N}(\Sigma_i^1)} \qquad \begin{array}{ccccc} \mathcal{N}(\Sigma_i) & = & \mathcal{N}(\Sigma_i^2) & = & \cdots \\ r_i & \underbrace{=}_{+0} & r_i & & \end{array}$$

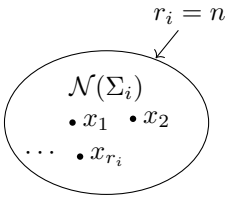
$$\#\text{ of JB's} = r_i = n$$
$$\#\text{ of JB's with size} \geq 2 = 0 \quad \Rightarrow\ n \text{ JB's of size } 1$$

▷ m_i ***is the size of the largest JB for*** λ_i**.**

p. 74

Hence:

$$J = \begin{bmatrix} \boxed{\lambda_i} & & & & \\ & \boxed{\lambda_i} & & & \\ & & \boxed{\lambda_i} & & \\ & & & \ddots & \boxed{\lambda_i} \end{bmatrix}$$



$x_1 \in \mathcal{N}(\Sigma_i), \quad x_1 \notin \underbrace{\mathcal{N}(\Sigma_i^0)}_{I} \Rightarrow x_1 \neq 0$: chain $\{x_1\}$

$x_2 \in \mathcal{N}(\Sigma_i), \quad x_2 \notin \mathcal{N}(\Sigma_i^0)$: chain $\{x_2\}$

$\Rightarrow x_1$ & x_2 are linearly indep.

$\vdots$

$x_{r_i} \in \mathcal{N}(\Sigma_i), \quad x_{r_i} \notin \mathcal{N}(\Sigma_i^0)$: chain $\{x_{r_i}\}$

$\Rightarrow r_i = n$ chains of size 1.

Each of these vectors is an eigenvector:

$$P = \left[\begin{array}{c|c|c|c} x_1 & x_2 & \cdots & x_{r_i} \end{array}\right]$$
$$P\bar{A} = AP$$
$$\left[\begin{array}{c|c|c|c} x_1 & x_2 & \cdots & x_{r_i} \end{array}\right] \underbrace{\begin{bmatrix} \lambda_i & & 0 \\ & \ddots & \\ 0 & & \lambda_i \end{bmatrix}}_{\bar{A}} = \begin{bmatrix} Ax_1 & Ax_2 & \cdots & Ax_{r_i} \end{bmatrix}$$
$$= \begin{bmatrix} \lambda_i x_1 & \lambda_i x_2 & \cdots & \lambda_i x_{r_i} \end{bmatrix} \quad \text{(as they are eigenvectors)}$$
$$J = \lambda_i I$$

$$J = P^{-1}AP \;\Rightarrow\; A = PJP^{-1} = P(\lambda_i I)P^{-1} = \lambda_i PP^{-1} = \lambda_i I$$

Equivalently, from the minimal polynomial:

$$m(s) = s - \lambda_i$$
$$m(A) = 0_{n\times n} \;\Rightarrow\; A - \lambda_i I = 0_{n\times n} \;\Rightarrow\; A = \lambda_i I$$

9.7 Further Examples

Example 9.9. A is a 6×6 matrix, λ_i is the only eigenvalue. p. 75

$$d(s) = (s-\lambda_i)^6, \qquad \dim\mathcal{N}(\Sigma_i) = 2,$$
$$m(s) = (s-\lambda_i)^3, \qquad \dim\mathcal{N}(\Sigma_i^2) = 4.$$

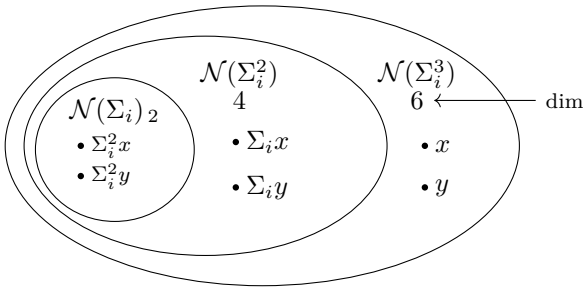
$$\underset{2}{\mathcal{N}(\Sigma_i)} \subsetneq \underset{4}{\mathcal{N}(\Sigma_i^2)} \subsetneq \underset{6}{\mathcal{N}(\Sigma_i^3)} = \underset{6}{\mathcal{N}(\Sigma_i^4)} \qquad \mathbb{C}^6 = \mathcal{N}(\Sigma_i^3)$$

- $m(s) = (s-\lambda_i)$ would be needed for diagonalizability.
- At most 2 linearly indep. eigenvectors.
- $\Rightarrow$ **not diagonalizable** (it should be 6).

Jordan blocks:

$$\left.\begin{array}{l}\#\text{ of JB's} = 2\\ \#\text{ of JB's with size} \geq 2 = 2\\ \#\text{ of JB's with size} \geq 3 = 2\\ \#\text{ of JB's with size} \geq 4 = 0\end{array}\right\} \text{ 2 JB's of size 3} \qquad J = \left[\begin{array}{ccc|ccc} \lambda_i & 1 & 0 & & & \\ 0 & \lambda_i & 1 & & 0 & \\ & & \lambda_i & & & \\ \hline & & & \lambda_i & 1 & 0 \\ & 0 & & & \lambda_i & 1 \\ & & & 0 & & \lambda_i \end{array}\right]$$

▷ *# of JB's with size* $\geq k = \dim\mathcal{N}(\Sigma_i^k) - \dim\mathcal{N}(\Sigma_i^{k-1})$*: here* $2-0$, $4-2$, $6-4$, $6-6$.



Chains:

$$x \in \mathcal{N}(\Sigma_i^3),\; x \notin \mathcal{N}(\Sigma_i^2): \qquad \{\Sigma_i^2 x, \Sigma_i x, x\} \;\Rightarrow\; \text{alone not sufficient to form a basis}$$
$$y \in \mathcal{N}(\Sigma_i^3),\; y \notin \mathcal{N}(\Sigma_i^2): \qquad \{\Sigma_i^2 y, \Sigma_i y, y\}$$

▷ *The two chains together should be linearly indep. to form a basis.*

$\Rightarrow$ Condition: $\Sigma_i^2 x$ **&** $\Sigma_i^2 y$ **should be linearly indep.**

Why this condition suffices

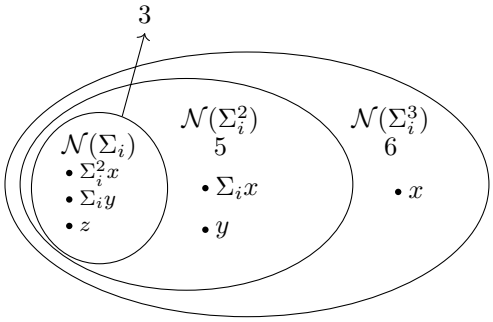
Suppose a combination of the six vectors vanishes. Applying Σ_i^2 kills everything except the coefficients of x and y, leaving $a\Sigma_i^2 x + b\Sigma_i^2 y = 0$, so $a = b = 0$ by the condition. Applying Σ_i to what remains gives the same relation for the coefficients of $\Sigma_i x, \Sigma_i y$, and finally for $\Sigma_i^2 x, \Sigma_i^2 y$. So independence of the eigenvectors at the bottom of the chains implies independence of the whole set (the same argument as for a single chain).

$$P = \left[\begin{array}{ccc|ccc} \Sigma_i^2 x & \Sigma_i x & x & \Sigma_i^2 y & \Sigma_i y & y \end{array}\right]$$

⇒ Continue same example but $\dim \mathcal{N}(\Sigma_i) = 3$, $\dim \mathcal{N}(\Sigma_i^2) = 5$.

$$\begin{array}{rcl} \#\text{ of JB's} = 3 & & m_i = \text{size of largest JB} = 3 \\ \#\text{ of JB's with size} \geq 2 = 2 & & \text{1 JB of size 3} \\ \#\text{ of JB's with size} \geq 3 = 1 & \Longrightarrow & \text{1 JB of size 2} \\ \#\text{ of JB's with size} \geq 4 = 0 & & \text{1 JB of size 1} \end{array}$$

$$J = \left[\begin{array}{ccc|cc|c} \lambda_i & 1 & 0 & & & \\ 0 & \lambda_i & 1 & & & \\ 0 & 0 & \lambda_i & & & \\ \hline & & & \lambda_i & 1 & \\ & & & 0 & \lambda_i & \\ \hline & & & & & \lambda_i \end{array}\right]$$



Chains:

$$\begin{array}{ll} x \in \mathcal{N}(\Sigma_i^3),\ x \notin \mathcal{N}(\Sigma_i^2): & \{\Sigma_i^2 x, \Sigma_i x, x\} \\ y \in \mathcal{N}(\Sigma_i^2),\ y \notin \mathcal{N}(\Sigma_i): & \{\Sigma_i y, y\} \\ z \in \mathcal{N}(\Sigma_i),\ z \notin \mathcal{N}(\Sigma_i^0) \Rightarrow z \neq 0: & \{z\} \end{array}$$

⇒ $\Sigma_i^2 x,\ \Sigma_i y,\ z$ **should be linearly indep.** p. 76

$$P = \left[\begin{array}{ccc|c|c} \Sigma_i^2 x & \Sigma_i x & x & \Sigma_i y & y & z \end{array}\right]$$

Example 9.10.

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 1 & 0 & 2 \end{bmatrix} \qquad \begin{array}{l} d(s) = (s-1)^3(s-2), \qquad \lambda_1 = 1,\ \lambda_2 = 2 \\ m(s)\,?\quad J\,?\quad P\,? \end{array}$$

▷ *You should know how to calculate $d(s)$.*
For $\lambda_1 = 1$:

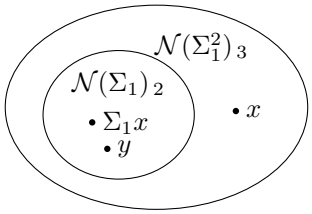
$$\Sigma_1 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 1 \end{bmatrix}, \quad \Sigma_1^2 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{array}{l} \dim \mathcal{N}(\Sigma_1) = 2, \quad \text{rank} = 2 \\ \text{it should increase (to } r_1 = 3) \end{array} \quad \Rightarrow \quad (s-1)^2(s-2) = m(s)$$

$\underset{2}{\mathcal{N}(\Sigma_1)} \subsetneq \underset{3}{\mathcal{N}(\Sigma_1^2)} \Rightarrow$ 1 JB of size 2, 1 JB of size 1 $\qquad \underset{1}{\mathcal{N}(\Sigma_2)} = \underset{1}{\mathcal{N}(\Sigma_2^2)} \Rightarrow$ 1 JB of size 1

$\#$ of JB's for $\lambda_1 = 2$
$\#$ of JB's with size $\geq 2 = 1$
$\#$ of JB's with size $\geq 3 = 0$

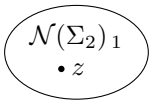
$$\mathbb{C}^4 = \underset{3}{\mathcal{N}(\Sigma_1^2)} \oplus \underset{1}{\mathcal{N}(\Sigma_2)} \qquad J = \left[\begin{array}{cc|c|c} 1 & 1 & & \\ 0 & 1 & & \\ \hline & & 1 & \\ \hline & & & 2 \end{array}\right]$$



$x \in \mathcal{N}(\Sigma_1^2)$, $x \notin \mathcal{N}(\Sigma_1)$;
$y \in \mathcal{N}(\Sigma_1)$ & $y \neq 0$; $\quad y$ & $\Sigma_1 x$ lin. indep.

$$x = \begin{bmatrix}0\\1\\0\\0\end{bmatrix}, \qquad \Sigma_1 x = \begin{bmatrix}1\\0\\0\\1\end{bmatrix}, \qquad y = \begin{bmatrix}0\\0\\1\\0\end{bmatrix} \text{ (eigenvector)}$$

Chains: $\{\Sigma_1 x,\ x\}$ and $\{y\}$.
$z \in \mathcal{N}(\Sigma_2)$, $z \neq 0$:



$$z = \begin{bmatrix}0\\0\\0\\1\end{bmatrix} \qquad P = \left[\begin{array}{c|c|c} \Sigma_1 x & x & y \end{array} \middle| \begin{array}{c} z \end{array}\right]$$

Check

$$P = \begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\1&0&0&1\end{bmatrix}, \qquad AP = PJ = \begin{bmatrix}1&1&0&0\\0&1&0&0\\0&0&1&0\\1&1&0&2\end{bmatrix}, \qquad \det P = 1 \neq 0,$$

so indeed $P^{-1}AP = J$. (Σ_1^2 has rank 1, so $\dim \mathcal{N}(\Sigma_1^2) = 3 = r_1$, confirming $m_1 = 2$.)

10 Hermitian and Positive Definite Matrices

Syllabus: Week 13 — Hermitian, positive definite and positive semi-definite matrices

10.1 Hermitian Matrices and Their Eigenstructure

* An $n \times n$, in general, complex matrix $A \in \mathbb{C}^{n\times n}$ is called **Hermitian** iff p. 77

$$\bar{A}^T = A, \qquad \text{i.e.} \qquad A^H = A,$$

where $^{-T} \triangleq {}^H$: the complex conjugate transpose is denoted as Hermitian.

* If $A \in \mathbb{R}^{n\times n}$:

$$A^T = A: \text{ Symmetric matrices are Hermitian.}$$

* Example:

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \lambda_1 = 3,\ \lambda_2 = 1, \qquad e_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

* In addition to being linearly independent, eigenvectors of Hermitian matrices (for distinct eigenvalues) are **orthogonal**.

* **Eigenvalues are always real** in Hermitian matrices.

Theorem 10.1. Let $A \in \mathbb{C}^{n\times n}$ be Hermitian. Then $\langle x, Ax\rangle$ **is real** $\forall x \in \mathbb{C}^n$.

* $\langle x, y\rangle = y^H x$: ℓ_2-inner product. Properties:

$$\begin{aligned} &\langle \alpha x, y\rangle = \alpha \langle x, y\rangle, && \langle x, x\rangle \text{ is real,} \\ &\langle x, \alpha y\rangle = \bar{\alpha} \langle x, y\rangle, && \langle x, x\rangle \geq 0 \ \ \forall x \in \mathbb{C}^n, \\ &\langle x, y\rangle = \overline{\langle y, x\rangle}, && \langle x, x\rangle = 0 \iff x = 0, \end{aligned}$$

$$\langle x, Ay\rangle = (Ay)^H x = y^H (A^H x) = \left\langle A^H x, y \right\rangle.$$

Proof.

$$\langle x, Ax\rangle = (Ax)^H x = x^H A^H x \overset{A^H = A}{=} x^H A x: \textbf{ Quadratic Form}$$

$$\begin{aligned} \overline{\langle x, Ax\rangle} &= \langle Ax, x\rangle \\ &= \left\langle x, A^H x \right\rangle \qquad \text{(identity above for } A^H \text{, with } y = x) \\ &= \langle x, Ax\rangle \qquad \text{if } A \text{ is Hermitian} \end{aligned}$$

The complex conjugate is equal to itself $\Rightarrow \langle x, Ax\rangle$ is real. □

Theorem 10.2. If A is Hermitian, then **its eigenvalues are all real**. p. 78

Proof. λ_i, e_i: one eigenvalue of A and corresponding eigenvector.

$$\begin{aligned} \langle x, Ax\rangle \Big|_{x=e_i} &= \langle e_i, Ae_i\rangle \\ &= \left\langle A^H e_i, e_i \right\rangle \end{aligned}$$

$$
\begin{aligned}
&= \langle Ae_i, e_i \rangle \\
&= \langle \lambda_i e_i, e_i \rangle \\
&= \lambda_i \underbrace{\langle e_i, e_i \rangle}_{\text{always real}} \quad \Rightarrow \text{ should be real}
\end{aligned}
$$

$$
\Rightarrow \underline{\underline{\lambda_i \text{ is real}}}
$$

▷ *By the previous theorem $\langle e_i, Ae_i \rangle$ is real, and $\langle e_i, e_i \rangle = \|e_i\|_2^2 > 0$ since $e_i \neq 0$; hence $\lambda_i = \langle e_i, Ae_i \rangle / \langle e_i, e_i \rangle$ is real.*

□

Theorem 10.3. Let A be Hermitian, and let λ_i, λ_j be two distinct eigenvalues of A with eigenvectors e_i and e_j. Then e_i **and** e_j **are orthogonal**:

$$
\lambda_i \neq \lambda_j \;\Rightarrow\; \langle e_i, e_j \rangle = 0
$$

Proof.

$$
\begin{aligned}
\langle e_i, Ae_j \rangle &= \langle e_i, \lambda_j e_j \rangle = \lambda_j \langle e_i, e_j \rangle \\
\| \\
\langle Ae_i, e_j \rangle &= \langle \lambda_i e_i, e_j \rangle = \lambda_i \langle e_i, e_j \rangle
\end{aligned}
$$

$$
\begin{aligned}
\lambda_j \langle e_i, e_j \rangle &= \lambda_i \langle e_i, e_j \rangle \\
\underbrace{(\lambda_i - \lambda_j)}_{\neq 0} \underbrace{\langle e_i, e_j \rangle}_{=0} &= 0
\end{aligned}
$$

▷ *Top row: $\langle e_i, \lambda_j e_j \rangle = \bar{\lambda}_j \langle e_i, e_j \rangle = \lambda_j \langle e_i, e_j \rangle$ since λ_j is real. Left equality: $\langle e_i, Ae_j \rangle = \langle A^H e_i, e_j \rangle = \langle Ae_i, e_j \rangle$.*

Hence

$$
\mathcal{N}(A - \lambda_i I) \perp \mathcal{N}(A - \lambda_j I) \;:\; \text{Eigenspaces are also orthogonal.}
$$

□

Theorem 10.4. Let A be Hermitian. Then its minimal polynomial is p. 79

$$
m(s) = (s - \lambda_1)(s - \lambda_2) \cdots (s - \lambda_k)
$$

when A has k distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_k$, i.e. $m_i = 1 \;\forall i$.

In general:

$$
\begin{aligned}
d(s) &= (s - \lambda_1)^{r_1} (s - \lambda_2)^{r_2} \cdots (s - \lambda_k)^{r_k} \\
m(s) &= (s - \lambda_1)^{m_1} (s - \lambda_2)^{m_2} \cdots (s - \lambda_k)^{m_k}, \qquad 1 \leq m_i \leq r_i \text{ in general case.}
\end{aligned}
$$

Proof. Let $\Sigma_i = A - \lambda_i I$. In general

$$
\mathcal{N}(\Sigma_i) \subsetneq \mathcal{N}(\Sigma_i^2) \subsetneq \cdots \subsetneq \mathcal{N}(\Sigma_i^{m_i}) = \mathcal{N}(\Sigma_i^{m_i+1}).
$$

$$
\mathcal{N}(\Sigma_i) = \mathcal{N}(\Sigma_i^2) \;:\; \text{Theorem's Claim.}
$$

We show both inclusions.

$$
\mathcal{N}(\Sigma_i) \subset \mathcal{N}(\Sigma_i^2): \quad \Sigma_i x = 0 \text{ implies } \Sigma_i(\Sigma_i x) = 0 \qquad \checkmark
$$

$\mathcal{N}(\Sigma_i^2) \subset \mathcal{N}(\Sigma_i)$:

$$\begin{aligned} x \in \mathcal{N}(\Sigma_i^2) &\Rightarrow \Sigma_i^2 x = 0 \\ &\Rightarrow \left\langle x, \Sigma_i^2 x \right\rangle = 0 \\ &\Rightarrow \langle x, \Sigma_i \Sigma_i x \rangle = 0 \\ &\Rightarrow \left\langle \Sigma_i^H x, \Sigma_i x \right\rangle = 0 \\ &\Rightarrow \langle \Sigma_i x, \Sigma_i x \rangle = 0 \\ &\Rightarrow \|\Sigma_i x\|_2^2 = 0 \Rightarrow \Sigma_i x = 0 \Rightarrow x \in \mathcal{N}(\Sigma_i) \end{aligned}$$

▷ *Used:* $\Sigma_i^H = (A - \lambda_i I)^H = A^H - \bar{\lambda}_i I^H = A - \lambda_i I = \Sigma_i$ *(since* λ_i *is real).*

Hence $\mathcal{N}(\Sigma_i) = \mathcal{N}(\Sigma_i^2)$, the chain of null spaces stops at the first step, and $m_i = 1$ for every i. □

∗ Direct sum decompositions:

$$\mathbb{C}^n = \mathcal{N}(\Sigma_1^{m_1}) \oplus \mathcal{N}(\Sigma_2^{m_2}) \oplus \cdots \oplus \mathcal{N}(\Sigma_k^{m_k})$$
$$\mathbb{C}^n = \mathcal{N}(\Sigma_1) \overset{\perp}{\oplus} \mathcal{N}(\Sigma_2) \overset{\perp}{\oplus} \cdots \overset{\perp}{\oplus} \mathcal{N}(\Sigma_k)$$

▷ *Orthogonal as every subspace defines a unique eigenspace: since* $m_i = 1$, $\mathcal{N}(\Sigma_i^{m_i}) = \mathcal{N}(\Sigma_i)$ *is the eigenspace of* λ_i*, and eigenspaces of distinct eigenvalues are orthogonal.*

∗ m_i is the size of the largest Jordan block for λ_i.

∗ For a Hermitian matrix: p. 80

$$\underset{\dim r_i}{\mathcal{N}(\Sigma_i)} = \underset{\dim r_i}{\mathcal{N}(\Sigma_i^2)} \quad \Rightarrow \quad \begin{array}{l} (\#\text{ of JB's for } \lambda_i) = r_i \\ (\#\text{ of JB's with size} \geq 2 \text{ for } \lambda_i) = 0 \\ \Rightarrow\ r_i \text{ JB's of size 1 for } \lambda_i \end{array}$$

$$J = \left[\begin{array}{ccc|ccc|c|ccc} \lambda_1 & 0 & 0 & & & & & & & \\ 0 & \lambda_1 & 0 & & & & & & & \\ & & \lambda_1 & & & & & & & \\ \hline & & & \lambda_2 & 0 & 0 & & & & \\ & & & & \lambda_2 & 0 & & & & \\ & & & 0 & & \lambda_2 & & & & \\ \hline & & & & & & \ddots & & & \\ \hline & & & & & & & \lambda_k & 0 & 0 \\ & & & & & & & & \lambda_k & 0 \\ & & & & & & & 0 & & \lambda_k \end{array}\right] : \text{diagonal matrix.}$$

▷ *The* λ_1 *block is* $r_1 \times r_1$*, the* λ_2 *block* $r_2 \times r_2$*, . . . , the* λ_k *block* $r_k \times r_k$*.*

∗ J is diagonal for a Hermitian matrix. **Any Hermitian matrix is diagonalizable**:

$$J = \begin{bmatrix} \lambda_1 I_{r_1 \times r_1} & & & \\ & \lambda_2 I_{r_2 \times r_2} & & \\ & & \ddots & \\ & & & \lambda_k I_{r_k \times r_k} \end{bmatrix}_{n \times n}$$

∗ Any Hermitian matrix has n linearly independent eigenvectors.

∗ For diagonalization with orthonormal eigenvectors (unitary diagonalization), we need to have:

$$N^H N = N N^H \;:\; \textbf{normal matrix} \text{ (outside the scope of the course)},$$
$$N^H N = N^2 = N N^H \text{ for Hermitian matrices (inherently normal)}.$$

Summary.

$$\text{Hermitian} \implies \left.\begin{array}{l} \lambda_i\text{'s are real} \\ e_i \perp e_j \text{ when } \lambda_i \neq \lambda_j \\ m_i = 1 \;\; \forall i \\ J \text{ is diagonal} \\ n \text{ linearly independent eigenvectors} \end{array}\right\} \impliedby \begin{array}{l} A \text{ is normal} \\ \text{(for these 3 statements)} \end{array}$$

▷ *The three bracketed statements are equivalent, and they mean diagonalizability.*

▷ $\not\Longleftarrow$*: the reverse implication (to Hermitian) is not true!*

Hermitian, normal, diagonalizable

- Hermitian ⇒ normal ⇒ diagonalizable; neither arrow can be reversed.
- A is normal $\iff$ A has n *orthonormal* eigenvectors $\iff$ $A = U\Lambda U^H$ with U unitary. Hermitian matrices are exactly the normal matrices with real eigenvalues.
- Counterexamples: $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ is normal (and diagonalizable over $\mathbb{C}$, eigenvalues $\pm j$) but not Hermitian; $\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$ is diagonalizable (two distinct eigenvalues, so $m_i = 1$) but not normal, its eigenvectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ are not orthogonal.

10.2 Unitary Diagonalization

p. 81

Theorem 10.5 (Unitary diagonalization)**.** Let A be an $n \times n$ Hermitian matrix. Then, **there exists a unitary matrix** P s.t.

$$J = P^H A P.$$

What J looks like here

For a Hermitian matrix the Jordan form J is *diagonal* with *real* entries: the eigenvalues of $A = A^H$ are real, and A is always diagonalizable (every eigenvalue has $m_j = 1$ in the minimal polynomial). The theorem adds that the diagonalizing matrix can be chosen *unitary*, i.e. $P^{-1} = P^H$, so that $A = PJP^H$ (unitary diagonalization).

⋆ **Unitary matrix**: $P^{-1} = P^H$.

→ **Orthogonal matrix** (real-valued unitary matrix): $P^{-1} = P^T$.

⋆ $P^{-1}P = PP^{-1} = I$, i.e. $P^H P = PP^H = I$.

Columns of a unitary matrix.

$$P = \left[\, p_1 \,\vdots\, p_2 \,\vdots\, \cdots \,\vdots\, p_n \,\right], \qquad P^H = \begin{bmatrix} p_1^H \\ p_2^H \\ \vdots \\ p_n^H \end{bmatrix}$$

$$P^H P = \begin{bmatrix} p_1^H \\ p_2^H \\ \vdots \\ p_n^H \end{bmatrix} \left[\, p_1 \,\vdots\, p_2 \,\vdots\, \cdots \,\vdots\, p_n \,\right] = \begin{bmatrix} p_1^H p_1 & p_1^H p_2 & \cdots & p_1^H p_n \\ p_2^H p_1 & p_2^H p_2 & \cdots & p_2^H p_n \\ \vdots & & & \vdots \\ p_n^H p_1 & & & p_n^H p_n \end{bmatrix}$$

$$= \begin{bmatrix} \langle p_1, p_1 \rangle & \langle p_2, p_1 \rangle & \cdots & \langle p_n, p_1 \rangle \\ \langle p_1, p_2 \rangle & & & \vdots \\ \vdots & & & \\ \langle p_1, p_n \rangle & & & \langle p_n, p_n \rangle \end{bmatrix}$$

▷ *All entries are inner products, similar to a Gram matrix.*

⇒ **Columns of P are orthonormal!**

$$\langle p_i, p_j \rangle = \delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

p. 82

Rows of a unitary matrix.

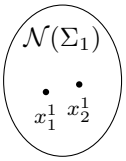
$$P = \left[\begin{array}{c} r_1 \\ \hline r_2 \\ \hline \vdots \\ \hline r_n \end{array}\right], \qquad \underbrace{\begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_n \end{bmatrix}}_{P} \underbrace{\left[\, r_1^H \,\vdots\, r_2^H \,\vdots\, \cdots \,\vdots\, r_n^H \,\right]}_{P^H} = I$$

⇒ **Rows are also orthonormal!**

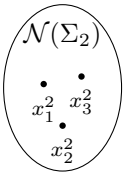
Proof (kind of proof, not an actual one). A is Hermitian, $A \in \mathbb{C}^{5\times 5}$; λ_1, λ_2 are eigenvalues of A, $\lambda_1 \neq \lambda_2$, and $\Sigma_j = A - \lambda_j I$:

$$d(s) = (s - \lambda_1)^2 (s - \lambda_2)^3$$
$$m(s) = (s - \lambda_1)(s - \lambda_2)$$

$$\mathbb{C}^5 = \underbrace{\mathcal{N}(\Sigma_1)}_{\dim 2} \overset{\perp}{\oplus} \underbrace{\mathcal{N}(\Sigma_2)}_{\dim 3}$$



$\{x_1^1\}$ $\{x_2^1\}$: Chains.
x_1^1 & x_2^1 are linearly indep.



$\{x_1^2\}, \{x_2^2\}, \{x_3^2\}$
x_1^2, x_2^2 & x_3^2 are linearly indep.

eigenvectors of A.

$$J = \begin{bmatrix} \lambda_1 & & & & \\ & \lambda_1 & & 0 & \\ & & \lambda_2 & & \\ & 0 & & \lambda_2 & \\ & & & & \lambda_2 \end{bmatrix}$$

$$P = \left[\, x_1^1 \,\vdots\, x_2^1 \,\vdots\, x_1^2 \,\vdots\, x_2^2 \,\vdots\, x_3^2 \,\right]_{5\times 5}$$

P is not directly unitary. The theorem states that there exists a unitary matrix P, which can be found. How can we convert P to a unitary one?

$$\underbrace{x_1^1 \quad x_2^1}\qquad \underbrace{x_1^2 \quad x_2^2 \quad x_3^2}$$

▷ *Vectors are orthogonal between the two sets. Vectors inside the sets are not necessarily orthogonal.*

Why eigenvectors of different eigenvalues are orthogonal

If $Ax = \lambda_1 x$, $Ay = \lambda_2 y$ with $A = A^H$ (so $\lambda_1, \lambda_2 \in \mathbb{R}$), then

$$\lambda_1\, y^H x = y^H A x = (Ay)^H x = \lambda_2\, y^H x \;\Rightarrow\; (\lambda_1 - \lambda_2)\, y^H x = 0 \;\Rightarrow\; y^H x = 0.$$

Gram–Schmidt inside each eigenspace keeps the vectors in that eigenspace (they remain eigenvectors), so the orthonormalized set still diagonalizes A.

⋆ We can apply **Gram–Schmidt orthonormalization to the two basis sets** and obtain the unitary matrix

$$P = \left[\, e_1^1 \,\vdots\, e_2^1 \,\vdots\, e_1^2 \,\vdots\, e_2^2 \,\vdots\, e_3^2 \,\right].$$

□

10.3 Spectral Decomposition

With $J = P^H A P$, we obtain the **spectral decomposition** of A: p. 83

$$\begin{aligned} A &= PJP^H \\ &= \left[\, e_1 \,\vdots\, e_2 \,\vdots\, \cdots \,\vdots\, e_n \,\right] \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix} \begin{bmatrix} e_1^H \\ \hline e_2^H \\ \hline \vdots \\ \hline e_n^H \end{bmatrix} \\ &= \left[\, \lambda_1 e_1 \,\vdots\, \lambda_2 e_2 \,\vdots\, \cdots \,\vdots\, \lambda_n e_n \,\right] \begin{bmatrix} e_1^H \\ \hline e_2^H \\ \hline \vdots \\ \hline e_n^H \end{bmatrix} \\ &= \lambda_1 e_1 e_1^H + \lambda_2 e_2 e_2^H + \cdots + \lambda_n e_n e_n^H \end{aligned}$$

$$\boxed{A = \sum_{i=1}^{n} \lambda_i e_i e_i^H}$$

Explanation

Each $e_i e_i^H$ is the orthogonal projection onto $\mathrm{sp}\{e_i\}$ (rank one, $(e_i e_i^H)^2 = e_i e_i^H$). So a Hermitian matrix is a weighted sum of orthogonal projections onto its eigen-directions, weighted by the (real) eigenvalues. Also $\sum_i e_i e_i^H = PP^H = I$.

10.4 Positive Definite and Positive Semi-Definite Matrices

Definition 10.6. A Hermitian matrix is said to be **positive definite** iff

$$\langle x, Ax\rangle > 0 \quad \forall x \neq 0 \qquad (x^H Ax > 0).$$

A Hermitian matrix is said to be **positive semi-definite** iff

$$\langle x, Ax\rangle \geq 0 \quad \forall x \qquad (x^H Ax \geq 0).$$

Theorem 10.7. If a Hermitian matrix is positive definite, or positive semi-definite, then

$$\underbrace{\lambda_i > 0}_{\text{(positive definite)}} \quad \text{or} \quad \underbrace{\lambda_i \geq 0}_{\text{(positive semi-definite)}} \quad \text{for all } i.$$

Proof (only for the positive-definite case). Let λ_i, e_i be an eigenpair, $e_i \neq 0$.

$$\begin{aligned} \langle e_i, Ae_i\rangle &> 0 \\ \langle e_i, \lambda_i e_i\rangle &> 0 \\ \lambda_i \langle e_i, e_i\rangle &> 0 \\ \lambda_i \|e_i\|^2 &> 0 \end{aligned}$$

▷ $\|e_i\|^2$ *is positive (as* $e_i \neq 0$*)* $\Rightarrow$ λ_i *should be positive.*

The semi-definite case is identical with "$>$" replaced by "$\geq$". □

Converse

The converse also holds: with the spectral decomposition,

$$x^H Ax = \sum_i \lambda_i x^H e_i e_i^H x = \sum_i \lambda_i \left| e_i^H x \right|^2.$$

If all $\lambda_i > 0$ this is > 0 for $x \neq 0$ (not all $e_i^H x$ vanish since $\{e_i\}$ is a basis). Hence **a Hermitian A is positive definite $\iff$ all $\lambda_i > 0$** (semi-definite $\iff$ all $\lambda_i \geq 0$).

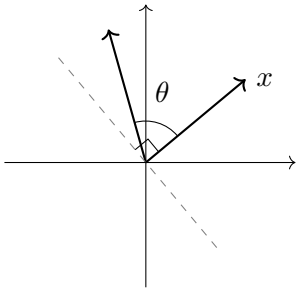
p. 84

Geometric interpretation. $\star$ $\langle x, y\rangle = \|x\|\,\|y\| \cdot \cos\theta$, so

$$\langle x, Ax\rangle = \underbrace{\|x\|}_{>0}\,\underbrace{\|Ax\|}_{>0}\cos\theta > 0 \quad \forall x \neq 0.$$

▷ $\|Ax\| > 0$, *otherwise the inner product would be 0.*

▷ $\cos\theta > 0 \hookrightarrow -90° < \theta < 90°$.


θ
x

11 Functions of a Matrix

Syllabus: Week 14 — functions of a matrix

11.1 Definition via Power Series

Functions of a matrix are *not* applied entrywise:

$$\exp\left(\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}\right) \neq \begin{bmatrix}e^2 & e\\ e & e^2\end{bmatrix} \qquad \text{MATLAB: } \cancel{\texttt{exp(A)}},\ \texttt{expm(A)}\ \checkmark$$

$$\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}^{1/2} \neq \begin{bmatrix}\sqrt{2} & 1\\ 1 & \sqrt{2}\end{bmatrix} \qquad \text{MATLAB: } \cancel{\texttt{sqrt(A)}},\ \texttt{sqrtm(A)}\ \checkmark$$

Motivation:

$$\begin{aligned}&\dot{x} = Ax, && \dot{x} = ax,\\ &x(t) = e^{At}x_0, && x(t) = e^{at}x_0,\end{aligned} \qquad e^{At} = \sum_{i=0}^{\infty} \frac{(At)^i}{i!}.$$

Taylor series.

$$f(s) = \sum_{i=0}^{\infty} \frac{f^{(i)}(s_0)}{i!}(s - s_0)^i, \qquad f^{(i)}(s_0) = \frac{d^i f}{ds^i}(s_0)$$

If $s_0 = 0$:

$$f(s) = \sum_{i=0}^{\infty} \underbrace{\frac{f^{(i)}(0)}{i!}}_{\alpha_i} s^i \quad \Longrightarrow \quad f(s) = \sum_{i=0}^{\infty} \alpha_i s^i$$

For a matrix:

$$(*) \qquad f(A) = \sum_{i=0}^{\infty} \frac{f^{(i)}(s_0)}{i!}(A - s_0 I)^i$$

$$f(A) = \sum_{i=0}^{\infty} \alpha_i A^i$$

Reduction by Cayley–Hamilton.

$$\begin{aligned}d(s) &= (s - \lambda_1)^{r_1} \cdots (s - \lambda_k)^{r_k}\\ d(s) &= s^n + d_1 s^{n-1} + \cdots + d_n\\ d(A) &= 0_{n\times n}\\ A^n &= -d_1 A^{n-1} - d_2 A^{n-2} - \cdots - d_n I\end{aligned}$$

Hence **all powers A^i, $i \geq n$, can be written using $I, A, \ldots, A^{n-1}$:**

$$f(A) = \sum_{i=0}^{\infty} \alpha_i A^i = c_0 I + c_1 A + c_2 A^2 + \cdots + c_{n-1} A^{n-1}$$

▷ *$c_0, c_1, \ldots, c_{n-1}$: n unknown coefficients.*

Using the minimal polynomial instead: p. 85

$$
\begin{aligned}
m(s) &= (s-\lambda_1)^{m_1}\cdots(s-\lambda_k)^{m_k}, \qquad \ell = m_1+m_2+\cdots+m_k \\
m(s) &= s^\ell + g_1 s^{\ell-1} + \cdots + g_\ell, \qquad m(A) = 0_{n\times n} \\
A^\ell &= -g_1 A^{\ell-1} - g_2 A^{\ell-2} - \cdots - g_\ell I \\
f(A) &= c_0 I + c_1 A + \cdots + c_{\ell-1} A^{\ell-1}
\end{aligned}
$$

▷ $c_0, c_1, \ldots, c_{\ell-1}$: *$\ell$ unknowns, $\ell \le n$.*

$$
\left.
\begin{aligned}
f(s) &= \sum_{i=0}^{\infty} \alpha_i s^i, \quad & f(A) &= \sum_{i=0}^{\infty} \alpha_i A^i \\
p(s) &= \sum_{i=0}^{\ell-1} c_i s^i, \quad & p(A) &= \sum_{i=0}^{\ell-1} c_i A^i
\end{aligned}
\right\rangle \boxed{f(A) = p(A)}.
$$

Why a polynomial of degree $\ell - 1$ suffices

Repeatedly substituting $A^\ell = -g_1 A^{\ell-1} - \cdots - g_\ell I$ reduces every power A^i $(i \ge \ell)$ to a combination of $I, A, \ldots, A^{\ell-1}$; summing the series gives $f(A) = \sum_{i=0}^{\ell-1} c_i A^i$. The coefficients c_i depend on f *and* on A (through $m(s)$), and the scalar polynomial $p(s)$ is in general *not* equal to $f(s)$ — only the matrix values agree. The next pages show that the c_i are fixed by matching f and p (and derivatives) at the eigenvalues.

Two methods to find $f(A)$:

- **Method I:** Interpolating polynomial
- **Method II:** Jordan canonical form

11.2 Method I: Interpolating Polynomial (Diagonalizable Case)

Case I: A is diagonalizable. Then $m(s) = (s-\lambda_1)(s-\lambda_2)\cdots(s-\lambda_k)$, $\ell = k$:

$$
p(s) = \sum_{i=0}^{\ell-1} c_i s^i, \qquad \ell = k \text{ is the \# of unknowns.}
$$

Let λ_j, e_j be the eigenpairs, $j = 1, 2, \ldots, k$. First note

$$
\begin{aligned}
A^0 e_j &= \lambda_j^0 e_j = I e_j \\
A e_j &= \lambda_j e_j \\
A^2 e_j &= A(A e_j) = A\lambda_j e_j = \lambda_j A e_j = \lambda_j^2 e_j \\
A^i e_j &= \lambda_j^i e_j
\end{aligned}
$$

Then

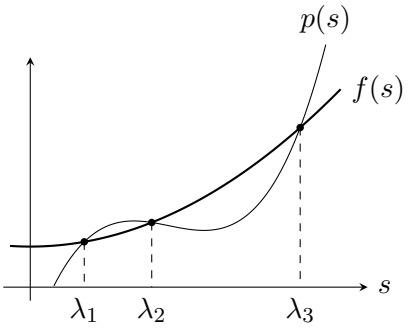
$$
\begin{aligned}
f(A)e_j &= p(A)e_j \\
\sum_{i=0}^{\infty} \alpha_i A^i e_j &= \sum_{i=0}^{\ell-1} c_i A^i e_j \\
\Big(\sum_{i=0}^{\infty} \alpha_i \lambda_j^i\Big) e_j &= \Big(\sum_{i=0}^{\ell-1} c_i \lambda_j^i\Big) e_j \\
f(\lambda_j)e_j &= p(\lambda_j)e_j
\end{aligned}
$$

$$f(\lambda_j)e_j - p(\lambda_j)e_j = 0$$

▷ *e_j is a nonzero vector.*

$$\boxed{f(\lambda_j) = p(\lambda_j)}, \qquad j = 1, 2, \ldots, k$$

$\Rightarrow$ k **equations and** k **unknowns**; we can solve these.



Although $p(s)$ & $f(s)$ are different, they have the same values at the eigenvalues. Thus $p(s)$ is called the **interpolating polynomial**.

$$\left.\begin{matrix} f(s) \neq p(s) \\ f(A) = p(A) \end{matrix}\right\rangle \text{ satisfied only at eigenvalues}$$

$$\underbrace{f(A)\cdot e_j}_{\text{matrix}\times\text{vector}} = \underbrace{f(\lambda_j)\, e_j}_{\text{scalar}\times\text{vector}} \quad \Leftarrow \quad A\cdot e_j = \lambda_j e_j$$

▷ *The eigenvalue is modified by the function f.*

$$\begin{matrix} e^A & e^{\lambda_j} & e_j \\ \log A & \log \lambda_j & e_j \end{matrix}$$

▷ ***Eigenvectors are invariant under functions**: $f(A)$ has the same eigenvectors e_j as A, with eigenvalues $f(\lambda_j)$.*

Example 11.1.

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \Rightarrow \lambda_1 = 3, \quad \lambda_2 = 1, \qquad e_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

▷ *Bunları nasıl buluyorduk? (How did we find these?)* $Ae_1 = \lambda_1 e_1$, $[A - \lambda_1 I]e_1 = 0$.
▷ $d(s) = (s-2)^2 - 1 = (s-3)(s-1)$; $A - 3I = \left[\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right] \Rightarrow e_1 = \left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right]$, $A - I = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right] \Rightarrow e_2 = \left[\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right]$.

$$m(s) = (s-3)(s-1)$$
$$\ell = 2$$
$$p(s) = c_0 + c_1 s$$

Find e^A & $\log A$.

e^A. $f(s) = e^s$; from $f(A) = p(A)$: $f(\lambda_j) = p(\lambda_j)$, $j = 1, 2$:

$$f(3) = e^3 = p(3) = c_0 + 3c_1$$
$$f(1) = e = p(1) = c_0 + c_1$$

$$\begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} e^3 \\ e \end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = -\frac{1}{2} \begin{bmatrix} 1 & -3 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} e^3 \\ e \end{bmatrix} = -\frac{1}{2} \begin{bmatrix} e^3 - 3e \\ -e^3 + e \end{bmatrix}$$

$$\Rightarrow\ c_0 = \frac{3e - e^3}{2}, \qquad c_1 = \frac{e^3 - e}{2}$$

▷ $\det \left[\begin{smallmatrix} 1 & 3 \\ 1 & 1 \end{smallmatrix}\right] = -2$, *so the inverse is* $-\frac{1}{2}\left[\begin{smallmatrix} 1 & -3 \\ -1 & 1 \end{smallmatrix}\right]$.

$$\boxed{e^A = c_0 I + c_1 A = \frac{3e - e^3}{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \frac{e^3 - e}{2}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}} = \frac{1}{2}\begin{bmatrix} e^3 + e & e^3 - e \\ e^3 - e & e^3 + e \end{bmatrix}$$

Check

$$e^A e_1 = \frac{1}{2}\begin{bmatrix} 2e^3 \\ 2e^3 \end{bmatrix} = e^3 e_1, \qquad e^A e_2 = \frac{1}{2}\begin{bmatrix} 2e \\ -2e \end{bmatrix} = e\, e_2,$$

as required by $f(A)e_j = f(\lambda_j)e_j$. Numerically

$$e^A \approx \begin{bmatrix} 11.402 & 8.684 \\ 8.684 & 11.402 \end{bmatrix},$$

which agrees with summing the series $\sum_i A^i/i!$ (MATLAB `expm(A)`), while the entrywise $\left[\begin{smallmatrix} e^2 & e \\ e & e^2 \end{smallmatrix}\right]$ is different.

$\log A$**.** $f(s) = \log s$, $f(\lambda_j) = p(\lambda_j)$, $j = 1, 2$:

$$\begin{aligned} \log 3 &= c_0 + 3c_1 \\ \log 1 &= c_0 + c_1 = 0 \ \Rightarrow\ c_0 = -c_1 \\ \log 3 &= 2c_1 \ \Rightarrow\ c_1 = \frac{\log 3}{2}, \qquad c_0 = -\frac{\log 3}{2} \end{aligned}$$

▷ *Subtracting the second equation from the first gives* $\log 3 = 2c_1$.

$$\boxed{\log A = -\frac{\log 3}{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \frac{\log 3}{2}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}} = \frac{\log 3}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

▷ *Check:* $\log A\, e_1 = \log 3\, e_1$, $\log A\, e_2 = 0 = \log 1\, e_2$; *and* $e^{\log A} = A$.

e^{At}**.** $f(s) = e^{st}$, $f(\lambda_j) = p(\lambda_j)$, $j = 1, 2$:

$$\begin{aligned} e^{3t} &= c_0 + 3c_1 = c_0(t) + 3c_1(t) \\ e^{t} &= c_0 + c_1 = c_0(t) + c_1(t) \end{aligned}$$

▷ *Same linear system as for* e^A *with* $e^3 \to e^{3t}$, $e \to e^t$; *the coefficients now depend on* t.

$$c_0(t) = \frac{3e^t - e^{3t}}{2}, \qquad c_1(t) = \frac{e^{3t} - e^t}{2}$$

$$\boxed{e^{At} = \frac{3e^t - e^{3t}}{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \frac{e^{3t} - e^t}{2}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}} = \frac{1}{2}\begin{bmatrix} e^{3t} + e^t & e^{3t} - e^t \\ e^{3t} - e^t & e^{3t} + e^t \end{bmatrix}$$

This solves

$$\dot{x} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} x, \quad x(0) = x_0 \qquad \Rightarrow \qquad x(t) = e^{At} x_0.$$

▷ *Check: at* $t = 0$, $e^{A\cdot 0} = I$; *and* $\frac{d}{dt} e^{At} = A e^{At}$ *(e.g.* $(1,1)$ *entry:* $\frac{3e^{3t} + e^t}{2} = 2 \cdot \frac{e^{3t} + e^t}{2} + \frac{e^{3t} - e^t}{2}$*).*

11.3 Method I: Non-Diagonalizable and General Case

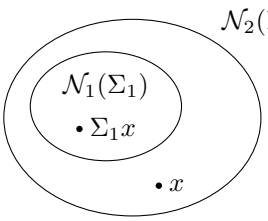
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Case II: A is not diagonalizable.

Example 11.2. Let us have $A \in \mathbb{R}^{3\times 3}$ and

$$m(s) = (s-\lambda_1)^2(s-\lambda_2)$$
$$d(s) = (s-\lambda_1)^2(s-\lambda_2) \qquad \text{as } A \in \mathbb{R}^{3\times 3}$$

$$J = \left[\begin{array}{cc|c} \lambda_1 & 1 & 0 \\ 0 & \lambda_1 & 0 \\ \hline 0 & 0 & \lambda_2 \end{array}\right], \qquad \underbrace{\mathcal{N}(A-\lambda_1 I)}_{1} \subsetneq \underbrace{\mathcal{N}(A-\lambda_1 I)^2}_{2} = \mathcal{N}(A-\lambda_1 I)^3$$



$x \in \mathcal{N}(\Sigma_1^2), \quad x \notin \mathcal{N}(\Sigma_1)$
$\{\Sigma_1 x,\ x\}$



$\mathcal{N}(A-\lambda_2 I), \quad 0 \neq y \in \mathcal{N}(\Sigma_2)$
$\{y\}$

$$P = [\,\Sigma_1 x \,\vdots\, x \,\vdots\, y\,] = [\,e_1 \,\vdots\, f_1 \,\vdots\, e_2\,], \qquad f(A) = p(A)$$

$$f(A)e_1 = p(A)e_1$$
$$f(\lambda_1) = p(\lambda_1)$$
$$f(\lambda_2) = p(\lambda_2)$$

$$\ell = 3, \qquad \boxed{p(s) = c_0 + c_1 s + c_2 s^2}$$

▷ *2 equations, but c_0, c_1, c_2: 3 unknowns ⇒ we need one more.*

Use the generalized eigenvector f_1:

$$f(A)\cdot f_1 = p(A)\cdot f_1, \qquad \sum_{i=0}^{\infty} \alpha_i A^i f_1 = \sum_{i=0}^{\ell-1} c_i A^i f_1$$

$$A^0 f_1 = \lambda_1^0 f_1$$
$$Af_1 = ?$$

$$\Sigma_1 f_1 = e_1, \qquad (A-\lambda_1 I) f_1 = e_1$$
$$\boxed{Af_1 = \lambda_1 f_1 + e_1}$$

▷ *As f_1 is not an eigenvector, Af_1 should include e_1.*

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$$\begin{aligned} A^2 f_1 = A \cdot (A f_1) &= A(\lambda_1 f_1 + e_1) \\ &= \lambda_1 A f_1 + A e_1 \\ &= \lambda_1(\lambda_1 f_1 + e_1) + \lambda_1 e_1 \\ &= \lambda_1^2 f_1 + 2\lambda_1 e_1 \\ A^3 f_1 = A(A^2 f_1) &= A(\lambda_1^2 f_1 + 2\lambda_1 e_1) \\ &= \lambda_1^2(\lambda_1 f_1 + e_1) + 2\lambda_1^2 e_1 \\ &= \lambda_1^3 f_1 + 3\lambda_1^2 e_1 \end{aligned}$$

$$\boxed{A^2 f_1 = \lambda_1^2 f_1 + 2\lambda_1 e_1} \qquad \boxed{A^3 f_1 = \lambda_1^3 f_1 + 3\lambda_1^2 e_1} \qquad \boxed{A^i f_1 = \lambda_1^i f_1 + i\lambda_1^{i-1} e_1}$$

Recall

$$p(s) = \sum_{i=0}^{\ell-1} c_i s^i, \qquad f(s) = \sum_{i=0}^{\infty} \alpha_i s^i, \qquad f(\lambda_1) = p(\lambda_1), \qquad f(\lambda_2) = p(\lambda_2).$$

Then

$$\Big(\sum_{i=0}^{\infty} \alpha_i \lambda_1^i\Big) f_1 + \Big(\sum_{i=0}^{\infty} \alpha_i\, i\lambda_1^{i-1}\Big) e_1 = \Big(\sum_{i=0}^{\ell-1} c_i \lambda_1^i\Big) f_1 + \Big(\sum_{i=0}^{\ell-1} c_i\, i\lambda_1^{i-1}\Big) e_1$$

$$\cancel{f(\lambda_1) f_1} + f'(\lambda_1) e_1 = \cancel{p(\lambda_1) f_1} + p'(\lambda_1) e_1$$

$$\boxed{f'(\lambda_1) = p'(\lambda_1)}$$

3 equations:

$$\begin{aligned} p(\lambda_1) &= f(\lambda_1) \\ p(\lambda_2) &= f(\lambda_2) \\ p'(\lambda_1) &= f'(\lambda_1) \end{aligned}$$

If $m(s) = (s-\lambda_1)^3(s-\lambda_2)$:

$$\begin{aligned} p(\lambda_1) &= f(\lambda_1), \qquad p(\lambda_2) = f(\lambda_2) \\ p'(\lambda_1) &= f'(\lambda_1) \\ p''(\lambda_1) &= f''(\lambda_1) \end{aligned}$$

11.3.1 General Case

$$m(s) = (s-\lambda_1)^{m_1}(s-\lambda_2)^{m_2}\cdots(s-\lambda_k)^{m_k}$$

$$\underbrace{c_0, c_1, \ldots, c_{\ell-1}}_{\ell \text{ unknowns}}, \qquad \ell = m_1 + m_2 + \cdots + m_k$$

$$f^{(t)}(\lambda_j) = p^{(t)}(\lambda_j), \qquad t = 0, \ldots, m_j - 1 \ \ (m_j \text{ equations}), \quad j = 1, \ldots, k$$

Example 11.3. Find $\sin(\pi A)$ in terms of A, where

$$A = \left[\begin{array}{ccc|cc|cc} 0 & 1 & 0 & & & & \\ 0 & 0 & 1 & & & & \\ 0 & 0 & 0 & & & & \\ \hline & & & 0 & 1 & & \\ & & & 0 & 0 & & \\ \hline & & & & & 1 & 1 \\ & & & & & 0 & 1 \end{array}\right].$$

$$
\begin{aligned}
d(s) &= (s-1)^2 s^5 \\
m(s) &= (s-1)^2 s^3, \qquad \ell = 5 \\
p(s) &= c_0 + c_1 s + c_2 s^2 + c_3 s^3 + c_4 s^4 \qquad (4^{\text{th}} \text{ order})
\end{aligned}
$$

Conditions and derivatives:

- $f(0) = p(0)$
- $f'(0) = p'(0)$
- $f''(0) = p''(0)$
- $f(1) = p(1)$
- $f'(1) = p'(1)$

$$
\begin{aligned}
f(s) &= \sin(\pi s) \\
f'(s) &= \pi \cos(\pi s) \\
f''(s) &= -\pi^2 \sin(\pi s)
\end{aligned}
$$

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$$
p'(s) = c_1 + 2c_2 s + 3c_3 s^2 + 4c_4 s^3, \qquad p''(s) = 2c_2 + 6c_3 s + 12c_4 s^2
$$

$$
\begin{aligned}
0 &= f(0) = p(0) = c_0 \\
\pi &= f'(0) = p'(0) = c_1 \\
0 &= f''(0) = p''(0) = 2c_2 \\
0 &= f(1) = p(1) = c_1 + \cancel{c_2} + c_3 + c_4 + \cancel{c_0} = \pi + c_3 + c_4 \\
-\pi &= f'(1) = p'(1) = c_1 + \cancel{2c_2} + 3c_3 + 4c_4 = \pi + 3c_3 + 4c_4
\end{aligned}
$$

▷ $f(0) = \sin 0 = 0,\ f'(0) = \pi \cos 0 = \pi,\ f''(0) = -\pi^2 \sin 0 = 0,\ f(1) = \sin \pi = 0,\ f'(1) = \pi \cos \pi = -\pi.$

$$
\boxed{\begin{aligned} c_0 &= 0 \\ c_1 &= \pi \\ c_2 &= 0 \end{aligned}} \quad \Rightarrow \quad \begin{aligned} c_3 + c_4 &= -\pi \\ 3c_3 + 4c_4 &= -2\pi \end{aligned} \quad \Rightarrow \quad \boxed{\begin{aligned} c_3 &= -2\pi \\ c_4 &= \pi \end{aligned}}
$$

$$
\boxed{\sin(\pi A) = \pi A - 2\pi A^3 + \pi A^4}
$$

We still need to take 3rd (and 4th) powers of A. If A is in JCF, Method II will be much more efficient.

Check

From $c_3 + c_4 = -\pi$: $c_4 = -\pi - c_3$, so $3c_3 - 4\pi - 4c_3 = -2\pi \Rightarrow c_3 = -2\pi$, $c_4 = \pi$. Evaluating $\pi A - 2\pi A^3 + \pi A^4$ blockwise: for the nilpotent blocks N (3×3 and 2×2) $N^3 = 0$, so the result is πN; for the block $\left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$ it gives $\left[\begin{smallmatrix} 0 & -\pi \\ 0 & 0 \end{smallmatrix}\right]$. This matches Method II below:

$$
\sin(\pi N) = \pi N - \frac{\pi^3}{6} N^3 + \cdots = \pi N, \qquad \sin\left(\pi \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\right) = \begin{bmatrix} \sin \pi & \pi \cos \pi \\ 0 & \sin \pi \end{bmatrix} = \begin{bmatrix} 0 & -\pi \\ 0 & 0 \end{bmatrix}.
$$

$$
f^{(t)}(\lambda_j), \qquad t = 0, \ldots, m_j - 1, \quad j = 1, \ldots, k
$$

If any derivative does not exist (∞ or div 0), then $f(A)$ **does not exist**.

Remark. $f(A)$ does not exist when one of the $f^{(t)}(\lambda_j)$, $j = 1, 2, \ldots, k$; $t = 0, \ldots, m_j - 1$ does not exist.

Example 11.4.

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \lambda_1 = 0, \quad \lambda_2 = 1$$

- ~~$\log A = ?$~~, ~~$\log s$~~: gives infinity (at $\lambda_1 = 0$).
- ~~$(I - A)^{-1} = ?$~~, ~~$(1 - s)^{-1} = \frac{1}{1-s}$~~ $\Rightarrow$ for $\lambda_2 = 1$: $\frac{1}{0} = \infty$.
- $A^{-1} \Longrightarrow$ function of a matrix. $s^{-1} = \dfrac{1}{s} \Rightarrow$ if one of the eigenvalues is zero, the inverse does not exist.

Example 11.5. $f(s) = s^{1/2} = \sqrt{s}, \qquad A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad m(s) = s^2.$

$$f(0) = p(0) \qquad \Rightarrow \text{ no problem}$$

$$\text{~~$f'(0) = p'(0)$~~} \qquad \Rightarrow \text{ problem: } \quad f'(s) = \frac{1}{2\sqrt{s}} \to \infty \text{ at } s = 0$$

▷ *The square root function does not apply here: its derivative is not defined at the eigenvalue $s = 0$.*
$\Rightarrow \sqrt{A}$ does not exist for $A = \left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$.

Direct confirmation

Suppose $X^2 = A$ with $A = \left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$. Then $X^4 = A^2 = 0$, so X is nilpotent; a nilpotent 2×2 matrix satisfies $X^2 = 0$ (its minimal polynomial divides s^2). Hence $A = X^2 = 0$, a contradiction. So *no* matrix squares to this A. In contrast, the zero matrix $\left[\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right]$ (with $m(s) = s$, only $f(0)$ needed) has the square root 0: whether $f(A)$ exists depends on which derivatives the minimal polynomial requires, not only on the eigenvalues.

11.4 Method II: Jordan Canonical Form

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$$J = \underbrace{P^{-1}AP}_{\substack{\text{similarity} \\ \text{transformation}}}, \qquad J \text{ is similar to } A.$$

Similar matrices have the same characteristic polynomial:

$$B = P^{-1}AP \qquad \text{(scalar analogue: } b = \cancel{p^{-1}}a\cancel{p} = a\text{)}$$

$$d_B(s) = \det(sI - B) = \det(sI - P^{-1}AP)$$

$$= \det\big(P^{-1}(sI - A)P\big) = \underbrace{\cancel{\det(P^{-1})}}_{\frac{1}{\det P}} \cdot \det(sI - A) \cdot \cancel{\det(P)}$$

$$d_B(s) = d_A(s)$$

Powers of $A = PJP^{-1}$:

$$(PJP^{-1})^0 = I = PP^{-1} = P \cdot J^0 P^{-1}$$

$$(PJP^{-1})^1 = PJP^{-1}$$

$$(PJP^{-1})^2 = PJ\cancel{P^{-1}}\cancel{P}JP^{-1} = PJ^2P^{-1}$$

$$(PJP^{-1})^i = PJ^iP^{-1}$$

Hence

$$f(A) = \sum_{i=0}^{\infty} \alpha_i (PJP^{-1})^i = \sum_{i=0}^{\infty} \alpha_i P(J^i)P^{-1}$$
$$= P \cdot \underbrace{\left(\sum_{i=0}^{\infty} \alpha_i J^i\right)}_{f(J)} P^{-1}$$
$$\boxed{f(A) = Pf(J)P^{-1}}$$

Block structure.

$$J = \begin{bmatrix} J_1 & & & \\ & J_2 & & \\ & & \ddots & \\ & & & J_\sigma \end{bmatrix}, \qquad J_t = \begin{bmatrix} \lambda_j & 1 & & 0 \\ & \lambda_j & 1 & \\ & & \lambda_j & \ddots \\ & & & \ddots & 1 \\ 0 & & & & \lambda_j \end{bmatrix}_{d_t \times d_t}$$

▷ *J_t $(t = 1, \dots, \sigma)$ denotes a generic Jordan block of size d_t belonging to the eigenvalue λ_j.*

$$J^i = \begin{bmatrix} J_1^i & & & \\ & J_2^i & & \\ & & \ddots & \\ & & & J_\sigma^i \end{bmatrix}, \qquad f(J) = \sum_{i=0}^{\infty} \alpha_i J^i = \begin{bmatrix} \sum_{i=0}^{\infty} \alpha_i J_1^i & & & 0 \\ & \sum_{i=0}^{\infty} \alpha_i J_2^i & & \\ & & \ddots & \\ 0 & & & \sum_{i=0}^{\infty} \alpha_i J_\sigma^i \end{bmatrix}$$

$$f(J) = \begin{bmatrix} f(J_1) & & & 0 \\ & f(J_2) & & \\ & & \ddots & \\ 0 & & & f(J_\sigma) \end{bmatrix}$$

One Jordan block. Expanding around $s_0 = \lambda_j$ (advantageous):

$$f(J_t) = \sum_{i=0}^{\infty} \alpha_i J_t^i \underset{s_0=\lambda_j}{=} \sum_{i=0}^{\infty} \frac{f^{(i)}(\lambda_j)}{i!} (J_t - \lambda_j I)^i$$

$$J_t - \lambda_j I = \begin{bmatrix} 0 & 1 & & \\ & 0 & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{bmatrix}_{d_t \times d_t}, \qquad (J_t - \lambda_j I)^2 = \begin{bmatrix} 0 & 0 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & \ddots & 0 \\ & & & & 0 \end{bmatrix}_{d_t \times d_t}$$

▷ *Each power shifts the line of 1's one diagonal up (↗).*

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$$(*) \qquad (J_t - \lambda_j I)^d = 0, \qquad d \geq d_t$$

$$(J_t - \lambda_j I)^{d_t - 1} = \begin{bmatrix} & & 0 & 1 \\ & 0 & & 0 \\ & & & \\ & & & \end{bmatrix}, \qquad (J_t - \lambda_j I)^0 = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$$

$$f(J_t) = \sum_{i=0}^{d_t-1} \frac{f^{(i)}(\lambda_j)}{i!}(J_t - \lambda_j I)^i$$

▷ *Upper limit $d_t - 1$ due to $(*)$.*

$$f(J_t) = \begin{bmatrix} f(\lambda_j) & f'(\lambda_j) & \frac{f''(\lambda_j)}{2!} & \cdots & \frac{f^{(d_t-1)}(\lambda_j)}{(d_t-1)!} \\ & f(\lambda_j) & f'(\lambda_j) & \frac{f''(\lambda_j)}{2!} & \vdots \\ & & \ddots & \ddots & \frac{f''(\lambda_j)}{2!} \\ & 0 & & \ddots & f'(\lambda_j) \\ & & & & f(\lambda_j) \end{bmatrix}$$

- The 0th power (I) forms the main diagonal.
- The 1st power forms the first superdiagonal.

  $\vdots$

Link with Method I

$f(J_t)$ only uses $f(\lambda_j), f'(\lambda_j), \ldots, f^{(d_t-1)}(\lambda_j)$. The largest block of λ_j has size m_j (the exponent in $m(s)$), so Methods I and II use exactly the same data $f^{(t)}(\lambda_j)$, $t = 0, \ldots, m_j - 1$; in particular $f(A)$ exists iff these derivatives exist. Method II avoids solving for $c_0, \ldots, c_{\ell-1}$ and computing powers of A, but needs P and P^{-1}.

Example 11.6.

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \lambda_1 = 3,\ \lambda_2 = 1, \qquad e_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$P = [\, e_1 \vdots e_2 \,], \qquad A = P \underbrace{\left[\begin{array}{c|c} 3 & 0 \\ \hline 0 & 1 \end{array}\right]}_{J} P^{-1} \qquad \Rightarrow \qquad f(A) = P \left[\begin{array}{c|c} f(3) & 0 \\ \hline 0 & f(1) \end{array}\right] P^{-1}$$

With $P = \left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$ and $P^{-1} = \frac{1}{2}\left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$:

$$f(A) = \frac{1}{2}\begin{bmatrix} f(3) + f(1) & f(3) - f(1) \\ f(3) - f(1) & f(3) + f(1) \end{bmatrix}.$$

▷ *For $f(s) = e^s$, $\log s$, e^{st} this reproduces the Method I results e^A, $\log A$, e^{At} above.*

Example 11.7. Find $\sin(\pi A)$ for

$$A = \left[\begin{array}{ccc|cc|c} \frac{1}{2} & 1 & 0 & & & \\ 0 & \frac{1}{2} & 1 & & & \\ 0 & 0 & \frac{1}{2} & & & \\ \hline & & & \frac{3}{2} & 1 & \\ & & & 0 & \frac{3}{2} & \\ \hline & & & & & \frac{5}{2} \end{array}\right] = \begin{bmatrix} J_1 & & \\ & J_2 & \\ & & J_3 \end{bmatrix}.$$

$$A = PJP^{-1}, \quad P = I \text{ chosen} \Rightarrow A = J$$

$$d(s) = \left(s - \tfrac{1}{2}\right)^3 \left(s - \tfrac{3}{2}\right)^2 \left(s - \tfrac{5}{2}\right)$$

$$m(s) = \left(s - \tfrac{1}{2}\right)^3 \left(s - \tfrac{3}{2}\right)^2 \left(s - \tfrac{5}{2}\right)^1$$
$$p(s) = c_0 + c_1 s + c_2 s^2 + c_3 s^3 + c_4 s^4 + c_5 s^5$$

▷ *Method I ↪ too complicated; use Method II blockwise.*

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$$f(s) = \sin(\pi s), \qquad f'(s) = \pi \cos(\pi s), \qquad f''(s) = -\pi^2 \sin(\pi s)$$
$$\sin(\pi J_1) = \begin{bmatrix} 1 & 0 & -\frac{\pi^2}{2} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \sin(\pi J_2) = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, \qquad \sin(\pi J_3) = \begin{bmatrix} 1 \end{bmatrix}$$

▷ J_1 *(*$\lambda = \frac{1}{2}$*, size 3):* $f(\frac{1}{2}) = 1$, $f'(\frac{1}{2}) = \pi \cos\frac{\pi}{2} = 0$, $f''(\frac{1}{2})/2! = -\pi^2/2$.
▷ J_2 *(*$\lambda = \frac{3}{2}$*, size 2):* $f(\frac{3}{2}) = -1$, $f'(\frac{3}{2}) = \pi \cos\frac{3\pi}{2} = 0$.
▷ J_3 *(*$\lambda = \frac{5}{2}$*, size 1):* $f(\frac{5}{2}) = \sin\frac{5\pi}{2} = 1$.

$$\sin(\pi A) = \left[\begin{array}{ccc|cc|c} 1 & 0 & -\frac{\pi^2}{2} & & & \\ 0 & 1 & 0 & & & \\ 0 & 0 & 1 & & & \\ \hline & & & -1 & 0 & \\ & & & 0 & -1 & \\ \hline & & & & & 1 \end{array}\right]$$

Example 11.8 (Square root of a matrix)**.**

$$f(s) = \sqrt{s} = s^{1/2}, \qquad A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad A^{1/2} = P \begin{bmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{1} \end{bmatrix} P^{-1} \;\Rightarrow\; \text{unique}$$

▷ ↪ *unique positive-definite square root of A: MATLAB* ~~`sqrt(A)`~~*,* `sqrtm(A)` ✓.
But if we only require $A^{1/2} \cdot A^{1/2} = A \Rightarrow$ not unique anymore (compare: $x = \sqrt{3}$ vs. $y \cdot y = 3$):

$$A^{1/2} \cdot A^{1/2} = P \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} P^{-1}$$
$$A^{1/2} = P \begin{bmatrix} -\sqrt{3} & 0 \\ 0 & \sqrt{1} \end{bmatrix} P^{-1} \;\Rightarrow\; A^{1/2}A^{1/2} = P \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} P^{-1}$$
$$A^{1/2} = P \begin{bmatrix} \pm\sqrt{3} & 0 \\ 0 & \pm\sqrt{1} \end{bmatrix} P^{-1}$$

▷ *These are all valid for the definition* $A^{1/2}A^{1/2} = A$ *(signs chosen independently: 4 matrices).*

The numbers

With $P = \left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$,

$$P\,\mathrm{diag}(a,b)P^{-1} = \frac{1}{2}\begin{bmatrix} a+b & a-b \\ a-b & a+b \end{bmatrix}, \qquad \left(P\,\mathrm{diag}(a,b)P^{-1}\right)^2 = \frac{1}{2}\begin{bmatrix} a^2+b^2 & a^2-b^2 \\ a^2-b^2 & a^2+b^2 \end{bmatrix},$$

which equals A whenever $a^2 = 3$, $b^2 = 1$. The principal (function) square root is

$$A^{1/2} = \texttt{sqrtm(A)} = \frac{1}{2}\begin{bmatrix} \sqrt{3}+1 & \sqrt{3}-1 \\ \sqrt{3}-1 & \sqrt{3}+1 \end{bmatrix} \approx \begin{bmatrix} 1.366 & 0.366 \\ 0.366 & 1.366 \end{bmatrix},$$

whose eigenvalues $\sqrt{3}, 1$ are positive: it is **the unique positive-definite square root**. The other three choices $(a,b) = (-\sqrt{3},1), (\sqrt{3},-1), (-\sqrt{3},-1)$ also square to A but are not positive definite.

If A is positive definite $\Rightarrow$ the square root is wanted to be positive definite, which is given by the function.

A Corrections Log

The original handwritten notes (which the syllabus itself warns may contain typos) were corrected in the places listed below. Each entry gives the page of the original notes (the gray *p. N* margin tag), the section of these notes where the correction appears, what the handwritten notes had, and the correction together with a brief reason. Purely cosmetic changes (layout, re-ordering, added explanations) are not listed.

Orig. page	Sec.	Notes had	Corrected to / reason
§1 Fields and Vector Spaces			
2	1.1	$x \oplus y = x_1 + y_1 + j(y_1 + y_2)$	$x \oplus y = x_1 + y_1 + j(x_2 + y_2)$: the second component is $x_2 + y_2$.
3	1.2	"$\Rightarrow$" between the members of the chain proving $y \oplus y = y$	"$=$": the steps are equalities.
5	1.3	Middle graph labelled "$\in \mathcal{W}$"	$\in \mathcal{W}_2$ (the set of functions in the example).
6	1.4	"$Y \cup Z$ is not closed under addition but not closed under scalar multiplication"	$Y \cup Z$ is closed under scalar multiplication but (in general) not under addition.
6	1.4	$w_1 \oplus_{\mathcal{V}} w_2$ and $\alpha \odot_{\mathcal{V}} w_1$ in $\mathcal{V} \times \mathcal{W}$	$w_1 \oplus_{\mathcal{W}} w_2$, $\alpha \odot_{\mathcal{W}} w_1$: the second components use the operations of $\mathcal{W}$.
9	1.6	"some of the vectors are nonzero"	Some of the *coefficients* $\alpha_i - \beta_i$ are nonzero.
10	1.7	"$P = [w_1]_{B_1}$", "$P = [w_2]_{B_1}$" and "$P \cdot [w_3]_{B_2}$" on the right	Each is the corresponding column of P, i.e. $[w_j]_{B_1}$.
§2 Normed Vector Spaces			
11	2.1	$\|x_i\|$ in step (1); sum without parentheses in step (3)	$\lvert x_i\rvert$; $\sum_i \big(\lvert x_i\rvert + \lvert y_i\rvert\big)$.
11	2.2	1st quadrant: $x_1 - x_2 = 1$	$x_1 + x_2 = 1$, since $\lvert x_1\rvert + \lvert x_2\rvert = x_1 + x_2$ there; 3rd/4th quadrant cases added.
12	2.3	$\mathcal{L}_p$ and $\mathcal{L}_2$ integrals without dt; $\mathcal{L}_1$ norm without subscript; $\max_t \lvert f(t)\rvert$ for $\mathcal{L}_\infty$	dt and subscript added; $\sup_t \lvert f(t)\rvert$ (the example below shows why the supremum is needed).
13	2.4	$\mathrm{tr}(A)$ on the left-hand side	$\mathrm{tr}(A^T A)$: the trace taken is that of $A^T A$.
15	2.4	Stray "7" written below the "8"	$\|A\|_1 = 8$ (column sums 6 and 8); 7 is $\|A\|_\infty$ from the previous page.
15	2.5	$\|(-1)^n - 1\| = (-1)^n - 1$	$\lvert(-1)^n - 1\rvert$; without the absolute value the expression is negative for odd n.
15	2.5	$\mathcal{L}_1$ integral over $(-\infty, \infty)$	$\int_{-1}^{1}$: functions in $\mathcal{V}$ live on $[-1, 1]$.
16	2.6	Inequality step justified "by absolute convergence"	By the triangle inequality (and $\|v - v_m\| = \|v_m - v\|$ by homogeneity with $\alpha = -1$).
18	2.6	$\lim_{n,m} \|f_n - f_m\|_\infty = \left\lvert 1 - \frac{n}{m}\right\rvert$	The distance itself is $\|f_n - f_m\|_\infty = \left\lvert 1 - \frac{n}{m}\right\rvert$ ($m \geq n$), which does not tend to 0.
§3 Inner Product Spaces			
19	3.1	Row vector $[y_1\ y_2\ \cdots]$ in $y^H x$	$[\bar{y}_1\ \bar{y}_2\ \cdots]$: $y^H = \bar{y}^T$ has conjugated entries.
19	3.1	B^T with rows $b_{11}\, b_{12}$ / $b_{12}\, b_{22}$ / $b_{13}\, b_{23}$	Entry $(1, 2)$ of B^T is b_{21}.
19	3.2	Equality iff $x = cy$	Also $y = 0_{\mathcal{V}}$ (x arbitrary) gives $0 = 0$; equality iff x, y are linearly dependent.
24	3.4	$w_2 = \bar{w}_2 / \|w_2\|$	$w_2 = \bar{w}_2 / \|\bar{w}_2\|$; the last step $w_n = \bar{w}_n / \|\bar{w}_n\|$ (implied by the pattern) added.

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Orig. page	Sec.	Notes had	Corrected to / reason
§4 Linear Transformations			
25	4.1	Sums for v_1, v_2 starting at $i = 1$	$\sum_{i=0}^{n}$: the constant term must be included (the next line already uses $i = 0$).
26	4.2	$\mathcal{N}(\mathcal{A}_1) = 0_{\mathcal{V}}$	$\mathcal{N}(\mathcal{A}_1) = \{0_{\mathcal{V}}\}$: the null space is a set.
28	4.3	Side note labelled $\mathcal{N}(\mathcal{A}) = \{\text{poly. degree} \leq 1\}$	This is $\mathcal{R}(\frac{d}{dt})$ (cf. p. 29); the null space is the constants.
28	4.4	$\alpha w_1 = \mathcal{A}(\alpha v_1) = \alpha \mathcal{A} v_1 = \alpha w_1 \in \mathcal{R}(\mathcal{A})$ (circular chain)	Rewritten as the implication $\mathcal{A}(\alpha v_1) = \alpha w_1 \Rightarrow \alpha w_1 \in \mathcal{R}(\mathcal{A})$.
29	4.4	$\dim \mathcal{R}(\mathcal{A}_1) = 0$ in the last line	$\dim \mathcal{R}(\mathcal{A}_2) = 0$.
29	4.4	$\mathcal{R}(\mathcal{A}) \not\subseteq \mathcal{W}$ (definition of "into")	$\mathcal{R}(\mathcal{A}) \subsetneq \mathcal{W}$: $\mathcal{R}(\mathcal{A})$ is always a subset of $\mathcal{W}$, a proper subset is meant.
31	4.6	$[v]_B$ (no bar) in the second line	$[v]_{\bar{B}}$; the last two lines (combining 1) and 2), giving $\bar{A} = Q^{-1}AP$) added.
32	4.6	First line $\bar{A} = Q^{-1}\bar{A}$	$\bar{\bar{A}} = Q^{-1}\bar{A}$ (double bar on the left).
§5 Direct Sums and Orthogonal Projections			
34	5.2	"M_2 & M_2"	M_1 & M_2.
35	5.2	Un-orthogonalized $[1\ 0\ 0\ 0]^T$, $[0\ 1\ 0\ 0]^T$ labelled e_3, e_4, and $M^\perp = \text{sp}\{e_3, e_4\}$	These are not orthogonal to M (e.g. $\langle [1\ 0\ 0\ 0]^T, m_1 \rangle = 1$); Gram–Schmidt computation added (verified in exact arithmetic). The cramped last vector is read as $[0\ 1\ 0\ 0]^T$.
40	5.4	Third basis matrix labelled v_2	v_3.
41	5.5	$Bx^H = [m_i^H x] = [\langle x, m_i \rangle] =$ [Gram matrix]	$B^H x = [\langle x, m_i \rangle] = b$; the Gram matrix is $B^H B$ (entry $(i, j) = \langle m_j, m_i \rangle$), moved to the line above.
41	5.5	$P_{M^\perp} = (I - P_M)x$	$P_{M^\perp} = I - P_M$; $(I - P_M)x$ is the projected vector.
41	5.5	$B(B^H B^{-1})B^H$ in the last step	$B(B^H B)^{-1}B^H$.
§6 Systems of Linear Equations $Ax = b$			
42	6.1	Second small "$\mathbb{C}^m$" label next to the zero vector in the figure	Read as the subscript $0_{\mathbb{C}^m}$, consistent with $\mathcal{N}(A) = \{x : Ax = 0_{\mathbb{C}^m}\}$.
43	6.1	"$= b + 0 = 0$" at the end; "x is exact"	$= b + 0 = b$; $\hat{x}$ is exact.
44	6.2	$\|Ax - b_1\| + \|b_2\|^2$	$\|Ax - b_1\|^2 + \|b_2\|^2$ (square missing).
45	6.2	Projection applied to $[1\ 1]^T$	Applied to $b = [2\ 1]^T$, giving $b_1 = [3/2\ 3/2]^T$.
49	6.5	$b \in \mathbb{C}^m\ \forall b \in \mathcal{R}(A)$	$b \in \mathcal{R}(A)\ \forall b \in \mathbb{C}^m$ (quantifiers swapped).
49	6.5	$\text{Proj}_{\mathcal{R}(A^H)}\, \hat{x} = \text{Proj}_{\mathcal{R}(A^H)}\, \hat{x}$	$\text{Proj}_{\mathcal{N}^\perp(A)}\, \hat{x} = \text{Proj}_{\mathcal{R}(A^H)}\, \hat{x}$.
50	6.5	$\hat{x} + m$	$\hat{x} + n$, $n \in \mathcal{N}(A)$ (an element of the null space is added).
50	6.5	$z = [x_2\ x_3]^T$	$z = [x_1\ x_2]^T$: A_1 consists of the first two columns of A (consistent with the final $\hat{x}$).
§7 Spectral Analysis of Linear Transformations			
51	7.1	"$\notin M$"	$\notin M_1$ (made explicit).
53	7.3	$\text{tr}\,\mathcal{A} \triangleq \det A$	$\text{tr}\,\mathcal{A} \triangleq \text{tr}\, A$: the trace of the operator is the trace of its matrix.
53	7.3	"R: rotation vectors"	R denotes the rotation matrix.
56	7.4	$M_1 = \text{sp}\{b_1^1, b_2^2\}$, $M_2 = \text{sp}\{b_1^1, b_2^2, b_3^3\}$	$M_1 = \text{sp}\{b_1^1, b_1^2\}$, $M_2 = \text{sp}\{b_2^1, b_2^2, b_2^3\}$ (b_i^j: j-th basis vector of M_i), matching P_1, P_2.
56	7.4	$M_2 = \{b_2^1\}$	$M_2 = \text{sp}\{b_2^1\}$ (span, not the set).

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Orig. page	Sec.	Notes had	Corrected to / reason
57	7.4	$P = [b_1^1\ b_2^1 \mid b_2^1]$	$P = [b_1^1\ b_1^2 \mid b_2^1]$: basis of M_1 followed by that of M_2, as in the next line.
57	7.4	Small mark $(*, +)$ on the circled exponent 2	Dropped (it only linked the exponent to the multiplicity in $d(s)$).
57	7.5	$(A - \lambda_2 I)e_2$ in the second term (which would be 0)	$(A - \lambda_1 I)e_2 = (\lambda_2 - \lambda_1)e_2 \neq 0$.
57	7.5	$A \in \mathbb{C}^n$	$A \in \mathbb{C}^{n\times n}$.
58	7.5	Diagonal matrix labelled A	$\bar{A} = P^{-1}AP$.
§8 Characteristic and Minimal Polynomials			
59	8.1	$A_{31} = 1$	$A_{31} = 0$, consistent with $sI - A$ and all later computations ($A - \lambda_i I$, null spaces, the check of AP).
60	8.2	$A - 4 + 3A^{-1} = 0$	$A - 4I + 3A^{-1} = 0$ (identity matrix missing).
61	8.3	$m(A) = (A - 2)(A - 2)$	$m(A) = (A - 2I)(A - I)$.
61	8.3	$d(s) = m(s)d(s)$	$d(s) = m(s)q(s)$.
61	8.3	Scattered list of crossed-out candidates with check marks (also on crossed-out ones and on $m(s) = s - 1$); last candidate $(s - 1)(s - 2)^1$	Organized by properties II and III: the only possibilities are $(s - 1)(s - 2)$ and $(s - 1)(s - 2)^2$.
61	8.3	Upper bound $m_i \leq k$	$m_i \leq r_i$ (cf. p. 63).
62	8.3	$\bar{r}(s) = r(\alpha)/\alpha$; division $d = mq + r$ only implicit	$\bar{r}(s) = r(s)/\alpha$; the division $d(s) = m(s)q(s) + r(s)$ written out.
62	8.3	e_1 instead of e_i (twice); stray cancelled e_i inside the bracket	e_i; stray term removed.
63	8.4	Exponent 2 on $(s - \lambda_2)$	All exponents are 1 in the distinct-eigenvalue case.
63	8.4	$(A_2 - I)(A_2 - 2I) = 0$	The product is nonzero, which is exactly why $m_2(s) = (s - 1)(s - 2)^2$.
§9 Jordan Canonical Form			
65	9.1	$\mathcal{N}\big((A - 2I)\big)$ in the last column	$\mathcal{N}\big((A - 2I)^3\big)$.
66	9.2	"=" missing between the two null spaces	$\mathcal{N}(\Sigma_i^l) = \mathcal{N}(\Sigma_i^{l+1})$.
66	9.2	A instead of A_1 in the dimension column and last row	A_1.
67	9.3	$\dim \mathcal{N}(\Sigma_i^{l+1}) - \dim \mathcal{N}(\Sigma_i^l) =$ # JB's with size $\geq l$ (off by one)	$\dim \mathcal{N}(\Sigma_i^l) - \dim \mathcal{N}(\Sigma_i^{l-1}) =$ # JB's with size $\geq l$; indices shifted so that $l = 1$ is the first line.
69	9.4	$\Sigma_i^{m_i - 1} \neq 0$ (vector missing)	$\Sigma_i^{m_i - 1}x \neq 0$.
71	9.5	"# of JB w.s. = 2 for λ_2", "# of JB with size = 3 for λ_2"	These counts are for λ_1 and mean size ≥ 2, ≥ 3.
72	9.6	"$1 \leq 1$" under $\mathcal{N}(\Sigma_i)$	$\dim \mathcal{N}(\Sigma_i) = 1$ (each strict step gains exactly $+1$).
73	9.6	$(\Sigma_i^{m_i - 2}x)\Sigma_i$ (factor order); $(\lambda_i + \Sigma_i)$	$\Sigma_i(\Sigma_i^{m_i - 2}x)$; $(\lambda_i I + \Sigma_i)$.
§10 Hermitian and Positive Definite Matrices			
78	10.1	$\lambda_i = \lambda_j \Rightarrow \langle e_i, e_j \rangle = 0$	$\lambda_i \neq \lambda_j$: the hypothesis is distinct eigenvalues.
79	10.1	$1 \leq m_i \leq r_k$	$1 \leq m_i \leq r_i$.
80	10.1	"For diagonalization, we need" normality; $N^2 = N^2$	Normality is needed for *unitary* diagonalization, not diagonalization in general; $N^H N = N^2 = NN^H$ when $N^H = N$.

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Corrections Log (continued)

Orig. page	Sec.	Notes had	Corrected to / reason
80	10.1	"$\Leftrightarrow$ A is normal"	"$\Longleftarrow$": normality implies the three statements, but a diagonalizable matrix need not be normal.
83	10.4	"$\|e_i\|^2$ is non-negative"	Positive (as $e_i \neq 0$): strict positivity is what is needed for $\lambda_i > 0$.
§11 Functions of a Matrix			
84	11.1	$\frac{d^i f}{dt^i}$	$\frac{d^i f}{ds^i}$: the variable is s.
86	11.2	"Eigenvalues are invariant under functions"	Eigenvectors e_j are invariant; the eigenvalues become $f(\lambda_j)$.
86	11.2	$\log 0$	$\log 1 = 0$: the eigenvalue is 1.
86	11.2	$c_2 = -\frac{\log 3}{2}$	$c_0 = -\frac{\log 3}{2}$.
86	11.2	Left-hand side $\log s$	$\log A$.
86	11.2	$f'(\lambda_j) = p(\lambda_j)$ (as written)	$f(\lambda_j) = p(\lambda_j)$.
86	11.2	$\dot{x} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} x_0$	$\dot{x} = Ax$ with $x(0) = x_0$, so $x(t) = e^{At}x_0$.
89	11.3	(Earlier transcription) c_1 struck through in $p(1)$	$c_1 = \pi \neq 0$ is substituted, giving $c_3 + c_4 = -\pi$.
89	11.3	$f^{(t)}(A_j)$	$f^{(t)}(\lambda_j)$.
89	11.3	"Square root function DNA."	Read as "does not apply": $f'(0)$ does not exist (consistent with the remark above).
90	11.4	Generic block labelled J_σ (size d_t) and last block of J^i labelled J^i_t	Made consistent: blocks $J_1, \ldots, J_\sigma$, generic block J_t.
92	11.4	$\mp$ in both diagonal entries of $A^{1/2}$	$\pm\sqrt{3}$, $\pm\sqrt{1}$: the two signs are chosen independently (4 square roots).